THL Series

Industrial Robot

Robot controller TSL3000
Robot controller TSL3000E
Robot controller TS3000
Robot controller TS3000E

INSTRUCTION MANUAL

MAINTENANCE MANUAL

Notice

1. Make sure that this instruction manual is delivered to the final user of this product.
2. Before using this product, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Rev.22
Oct 3, 2022

Shibaura Machine

NUMAZU, JAPAN

This Operation Manual describes the following robots:
THL Series: THL300, THL400, THL500, THL600, THL700, THL800, THL900, THL1000 and THL1200

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WARRANTY

This product is delivered to each customer only after it is inspected very carefully to make sure that it satisfies the SHIBAURA MACHINE's standard. Should it cause an inconvenience, we will guarantee as described below.

1. Warranty period
   SHIBAURA MACHINE agrees to repair or replace as necessary all defective material or workmanship up to the period shown below, whichever comes first.
   1) Eighteen (18) months from the date of dispatch from our plant.
   2) Twelve (12) months from the date of machine installation at customer's job site.
   3) 2,400 running hours from the date of initial machine operation.

2. Contents of warranty
   1) Only the product is subject to SHIBAURA MACHINE's Guarantee. Such Guarantee covers the specifications and functions as defined in the product specifications manual, catalog, instruction manual, etc. In no event does the Guarantee cover any secondary or associated damage caused by malfunction of the product.
   2) SHIBAURA MACHINE repairs the product free of charge only when it malfunctioned after handling or use according to the instruction manual attached to the product within the specified warranty period.

3. Exemption from responsibility
   SHIBAURA MACHINE's Guarantee shall not cover the following cases.
   1) Incorrect use not described in the instruction manual, and trouble or damage caused by negligent use.
   2) Inconvenience caused by aged deterioration or long-term usage (natural fading of coating or painting, deterioration of consumable parts (*1), etc.).
   3) Inconvenience caused by sensuous phenomena (noise generation, etc. which will not affect the function).
   4) Remodeling or disassembly which SHIBAURA MACHINE will not permit.
   5) Trouble and damage caused by insufficient maintenance/inspection or improper repair.

6) Trouble and damage caused by disaster, fire or other external factor.

7) Internal data such as program and point which were created by the customer.

8) When the product purchased in Japan was shipped overseas.

4. Precautions

1) Unless the product was used pursuant to its specifications, SHIBAURA MACHINE will not guarantee the basic performance of the product.

2) If the customer did not observe the warnings and cautions described in this manual, SHIBAURA MACHINE will not assume the responsibility for any consequential accident resulting in injury or death, damage or trouble.

3) Please note that the warnings, cautions and other descriptions stipulated in this manual are only those which can be assumed by SHIBAURA MACHINE as of now.

(*1) The consumable parts signify the replacement parts for maintenance as listed in Section 8 of this manual.

INTRODUCTION

This manual describes the maintenance of this product, THL series, and the TSL3000, TSL3000E, TS3000 and TS3000E controllers.

The maintenance and inspection are essential to maintain the product performance for long years to prevent a trouble and improve the safe work.
Before starting an actual operation, it is strongly recommended to read through this manual and draw up a maintenance schedule.

Safety Precautions

This manual contains important information for the safe use of the robot body and the prevention of injury to the operator and others and the prevention of damage to property. Read and understand the meanings of the following indications and symbols first, and observe these precautions.

[Meanings of Indications]

Indication	Meaning
⚠ DANGER	Indicates the imminent danger that incorrect operation could cause death or serious injury.
⚠ WARNING	Indicates the danger that incorrect operation could cause death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1)] to operator.
NOTICE	Indicates the possibility that incorrect operation could cause property damage[2)] and advice about performing work.

1) Injury indicates injuries such as wounds, burns, and electric shock that do not require hospitalization or long-term outpatient treatment.

2) Damage to property indicates extensive damage related to the destruction of assets or materials.

⚠ DANGER

– In order to ensure robot safety from installation to disposal, be sure to carefully read the separate instruction manual “Safety Manual” before starting the actual work.
Performing an operation different from what is described in the manual could result in harm to people.

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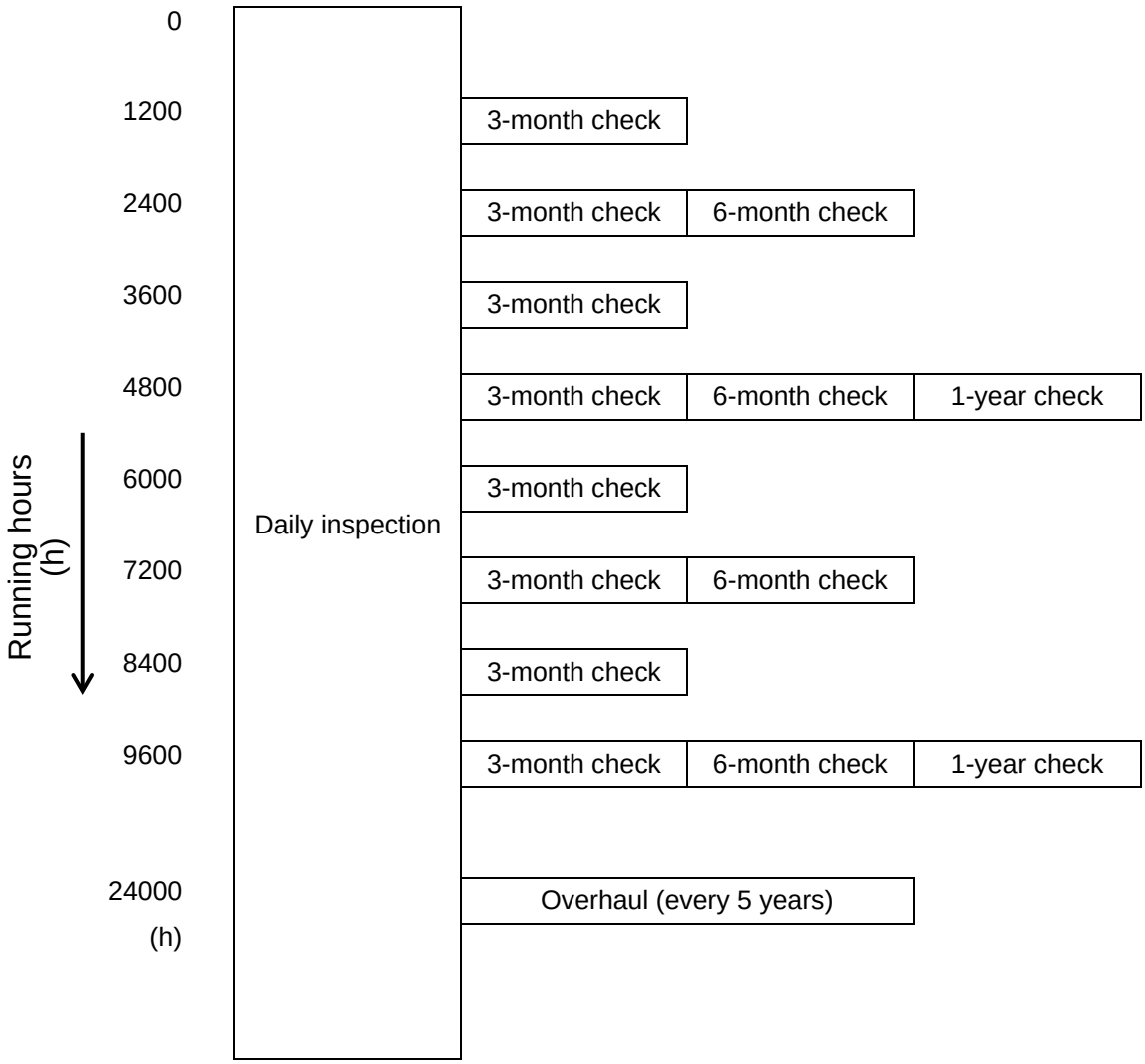
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1. Maintenance

1.1 Maintenance Schedule

Please do maintenance by a qualified person according to laws and regulations of each country. Maintenance comes in the two (2) types; daily inspection, and regular inspection and maintenance. For the regular inspection and maintenance, inspection items are added every 1,200 running hours.



Inspection schedule

- When the system is operated for 16 hours per day:
  1,200hours×16 hours x 25 days = 3 months
- When the system is operated for 24 hours per day:
  1,200hours×24 hours x 25 days = 2 months

If the system is operated for 16 hours per day, it should be inspected every three (3) months. If the number of hours of system operation per day becomes longer, the regular inspection on the system must be done at shorter intervals.

Fig. 1.1 Maintenance Schedule

1.2 Items for Maintenance and Inspection

This section describes the items for maintenance and inspection.
For the executing procedures, see the relevant paragraph listed in the table below.

1.2.1 Inspection at Power OFF (at Non-Operation)

D: Daily inspection Q: Quarterly inspection
S: Semi-annual inspection A: Annual inspection

Table 1.1 Inspection at Power OFF

<Robot>

Description	Check position	D	Q	S	A	Refer to		
						THL300 THL400	THL500 THL600 THL700	THL800 THL900 THL1000 THL1200
Make sure that all bolts are completely tightened and secured. Otherwise, tighten the bolts completely.	Tool set bolts	✓	✓	✓	✓	Para. 2.1.1	Para. 3.1.1	Para. 4.1.1
	Robot installation bolts	✓	✓	✓	✓			
	Motor set bolts				✓			
Make sure that the cable clamps are tightened.	Arm 2, upper base				✓	Para. 2.1.3	Para. 3.1.3	Para. 4.1.3
Make sure that no cut or scratch is found. Clean the contaminant, if any.	Entire robot	✓	✓	✓	✓			
	Arm 2 interior			✓	✓			
Make sure that anti-corrosive is properly applied. If not, apply anti-corrosive.	Ball spline nut section, end face of Ball screw shaft	✓	✓	✓	✓	Para. 2.6.3	Para. 3.6.3	Para. 4.6.3
Make sure that each timing belt is not worn off or cracked. Check each timing belt for tension.	Arm 2 interior			✓	✓	Para. 2.5.5	Para. 3.5.5	Para. 4.5.5
Make sure that the cables and air tubing are not worn out.	Cables and air tubing of each unit	✓	✓	✓	✓	Para. 2.1.2	Para. 3.1.2	Para. 4.1.2
Make sure that each axis can be operated properly by moving the axis by hand.	Entire robot			✓	✓	Para. 2.1.4	Para. 3.1.4	Para. 4.1.4

<Controller>

Description	Check position	D	Q	S	A	Refer to		
						TSL3000	TSL3000E	TS3000 TS3000E
Check of controller air vent holes.	Controller air vent holes	✓	✓	✓	✓	Para. 5.3.1	Para. 6.3.1	Para. 7.3.1

1.2.2 Inspection at Power ON (at Operation)

D: Daily inspection Q: Quarterly inspection
S: Semi-annual inspection A: Annual inspection

Table 1.2 Inspection at Power ON

Description	Check position	D	Q	S	A	Refer to		
						THL300 THL400	THL500 THL600 THL700	THL800 THL900 THL1000 THL1200
Make sure that enough volume of grease is filled. Otherwise, fill the grease.	Ball screw shaft	✓	✓	✓	✓	Para. 2.6.3	Para. 3.6.3	Para. 4.6.3
Push each arm by hand in the servo ON condition to make sure that the arm is secured.	Each joint				✓			
Make sure that no abnormal vibration or noise is caused.	Entire robot	✓	✓	✓	✓			
Check for teaching position	Tip of robot		✓	✓	✓			
Replace the batteries for position detection with new ones.	Battery box				✓	Para. 8.3		
Check of safety devices for function	Safety devices for function	✓	✓	✓	✓	Para. 5.3.2/ Para. 6.3.2/ Para. 7.3.2		

1.2.3 Overhaul

Table 1.3 Overhaul

<Robot>

Description	Maintenance parts	Every 5 years	Refer to		
			THL300 THL400	THL500 THL600 THL700	THL800 THL900 THL1000 THL1200
Check for the consumable parts and replace as necessary.	Motors, belts, ball screws, reduction gears ,Harness	✓	Para. 2.3 to 2.7	Para. 3.3 to 3.7	Para. 4.3 to 4.7

<Controller>

Description	Maintenance parts	Every 5 years	Refer to		
			TSL3000	TSL3000E	TS3000 TS3000E
Replace the battery with a new one.	Backup battery in the controller	✓	Para. 5.3.3	Para. 6.3.3	Para. 7.3.3
Replace the switching power supply unit with a new one.	In the controller	✓	Para. 5.3.4	Para. 6.3.4	Para. 7.3.4

1.3 Cautions on Maintenance and Inspection

Please do maintenance by a qualified person according to laws and regulations of each country. When performing inspection or maintenance of the robot, strictly observe the following precautions to protect yourself and coworkers.

⚠DANGER

– When performing maintenance with the power turned ON in the dangerous range of the robot, be sure to switch the master mode key to “TEACH” to pull out the key and carry it with the teach pendant.
Otherwise, the robot that has operated unpredictably may collide with a worker.
- The worker should receive safety training regarding industrial robot inspection-related work defined by each country’s laws and regulations.
- Display the fact that work is in progress near the controller.

– When performing maintenance with the power turned ON in the dangerous range of the robot, be sure to arrange a work observer outside the dangerous range of the robot.
Otherwise, when a worker in the dangerous range of the robot suffers damage, the worker cannot be rescued.
- The observer should secure safety of the worker and the third party who approaches the dangerous range of the robot.
- The observer should also receive safety training regarding industrial robot inspection-related work defined by each country’s laws and regulations.

– When approaching the robot being stopped, be sure to check the robot operation status.
If the robot is under operation, the robot that has restarted may collide with the worker.
- Check the operation status indicated on the indicator lamp of the controller and on the teach pendant.
- The robot operation status is output by signals (servo operation preparation completion, external mode, operation preparation completion, automatic operation, failure, and emergency stop). Connect to the display equipment, etc.

– Be sure to perform maintenance and inspection periodically.
If maintenance and inspection are neglected, the faulty robot may cause an unpredictable disaster.

⚠ WARNING

– When rotating axis 4 by hand, never insert your hand or finger between the upper or lower mechanical stopper and the robot or under the robot hand.
Otherwise, your hand or finger may get caught in the mechanical stopper and robot hand.
- When axis 4 rotates, the ball screw spline shaft will move up and down.
- The axis-4 motor is not equipped with the brake. If the robot hand is touched by hand when the servo power is off, axis 4 will rotate.

– When releasing the motor brake, be sure to support the ball screw spline shaft.
Otherwise, the shaft will drop by its own weight concurrently with release of the brake.
- When a worker supports the shaft, be sure to prevent hand or finger from getting caught.
- When something strange is felt during work, release the brake release switch immediately. The brake will also be applied by turning off the power to the controller.
- The brake release switch is located on the robot base rear. Do not press the switch by mistake.
- The worker should receive safety training regarding industrial robot inspection-related work defined by each country's laws and restrictions.

⚠ CAUTION

– Be sure to start maintenance when the surface temperature becomes less than 50°C after power off.
If hot motor or periphery of the motor is touched, burn injury may be received.

NOTICE

– Be sure to perform maintenance and inspection periodically.
If maintenance and inspection are neglected, the performance and lifespan of the robot may be reduced and the robot may malfunction.

1.4 Maintenance Tools

We recommend using the following tools for maintaining the robot and controller.

- Screwdrivers (Phillips head screwdrivers, flat head screwdrivers)
- Hexagonal wrench keys, nominal size 2.5 mm to 14 mm • Torque wrenches, nominal size 3 mm to 12 mm
- Torque driver • Wrenches, nominal size 5.5 mm, 7 mm, 46 mm, 50 mm
- Box wrench, nominal size 5.5 mm • Nippers • Needle-nose pliers • Pliers
- Torque wrench extension bar • Plastic hammer • Scale

Recommended provisions other than the tools:

- Alcohol (for cleaning) • Waste cloth • Vise (fixture) • Spatula (for grease application)
- Screw locking adhesive Recommended: Loctite (241 medium strength) (221 low strength)(425)
- Belt tension meter • Anti-corrosive: KLUBER A20
- TSPC, TSAssist : Program creation/teaching support software (option)
- Liquid gasket(THREEBOND1221H)

1.5 Clamping Hexagon Socket Head Cap Screws and Set screws

This robot uses hexagon socket head cap screws at places requiring mechanical strength. At the time of factory-assembly, Loctite is applied and each screw is tightened with the following clamping torque.
When these screws and set screws are tightened again according to the inspection and maintenance as stipulated in this manual, use a torque wrench or torque driver, and Loctite (medium strength) to assure the appropriate clamping torque.
The places which do not fall under the table below are detailed in the paragraph of the replacement procedures.

Table 1.4 Clamping Torque

Hexagon socket head cap screw	-	-	M3	M4	M5	M6	M8	M10	M12
Set screws	M3	M4	M5	M6	M8	M10	M12	M16	M20
Clamping torque	1.5N･m	1.8N･m	2.0 N･m	4.7 N･m	9.0 N･m	15 N･m	37 N･m	75 N･m	128 N･m

For the screws arranged on a circle for mounting the reduction gear, etc., tighten them in the diagonal order, as shown below.

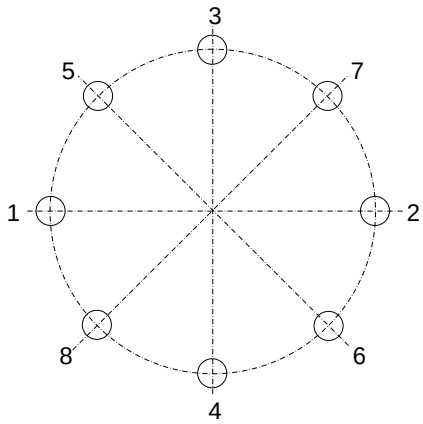


Fig.1.2 Tool set bolts

DO NOT tighten one (1) screw at a time. Tighten each screw in multiple steps, using a hexagonal wrench key, and secure with appropriate clamping torque by means of a torque wrench.

1.6 Maintenance Contract and Repair

1.6.1 Maintenance Contract

For the inspection items for maintenance and inspection to be done semi-annually and after, it is rather difficult for the user to execute, except for greasing and lubrication. We recommend that the user contact SHIBAURA MACHINE Service Department and conclude an after-sale service contract with us.

1.6.2 Repair

If a fault has occurred or if repair is necessary, turn off the controller power and contact the SHIBAURA MACHINE Service Department. At this time, advise us of the details of the fault and the following information stated on the robot and controller. (For the contact information of the SHIBAURA MACHINE Service Department, please refer to Service Network.)

- Robot model
- Manufacture number
- Manufacture number of controller
- Date of manufacture of robot
- Running hours

For the manufacture number and date of manufacture, see the following labels attached to main base of the robot.

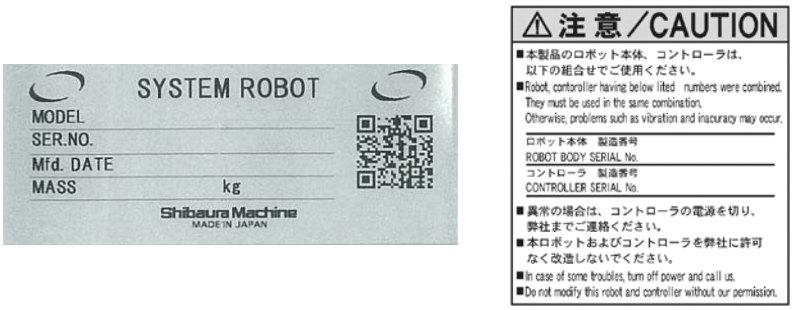


Fig.1.3 Label

1.6.3 Modification

The robot and controller MUST NOT be modified or disassembled without a prior consent from SHIBAURA MACHINE.

⚠DANGER

– Never replace and modify the parts for other than permitted portions.
Otherwise, the faulty robot may cause unpredictable disaster.

NOTICE

– Never replace and modify the parts for other than the permitted portions.
Otherwise, the performance and lifespan of the robot may be reduced, and the robot may malfunction.

2. Maintenance of the Main Robot (THL300, THL400)

2.1 Details of Inspection

2.1.1 Check of Each Bolt (or Screw) for Clamping

⚠ DANGER

– Be sure to perform daily inspection for tool set bolts and robot installation bolts.
Robot hand falls during robot operation. Robot collapses.

NOTICE

– Be sure to perform daily inspection for tool set bolts and robot installation bolts.
If the position is shifted, the robot will bump into peripheral devices.

1) Tool Set Bolts

Using the hexagonal wrench key, make sure that the clamping bolts (M4 x 4 pcs.) of the tool set flange (option), which are clamped to the tool shaft, are tightened completely. If loose, tighten them completely.

Using the hexagonal wrench key, make sure that the mounting bolts (M4 x 4 pcs.), which secure the tool to the tool flange, are tightened completely. If loose, tighten them completely. (Loctite not necessary)

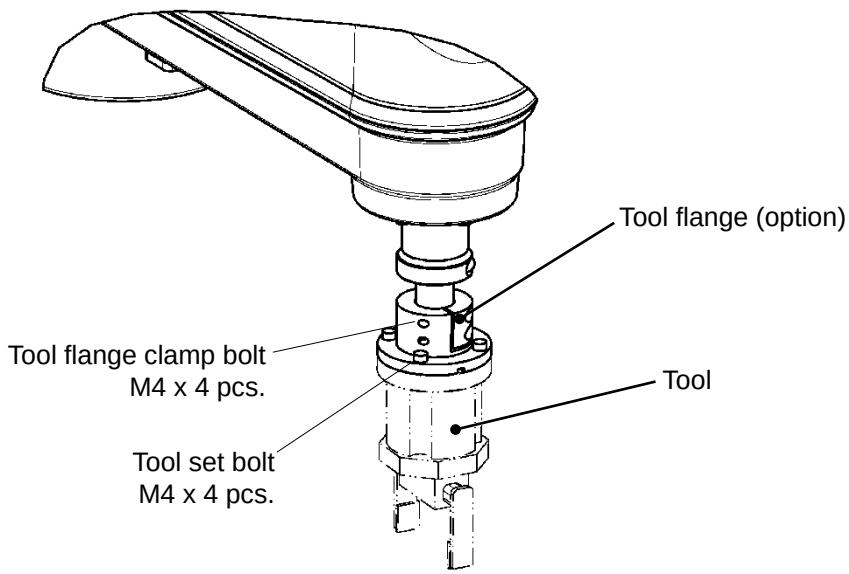


Fig. 2.1 Tool set bolts

2) Robot Installation Bolts

Make sure, using the hexagonal wrench key, that the installation bolts of the main robot base are tightened completely. If loosened, tighten them completely. (Loctite is not necessary)

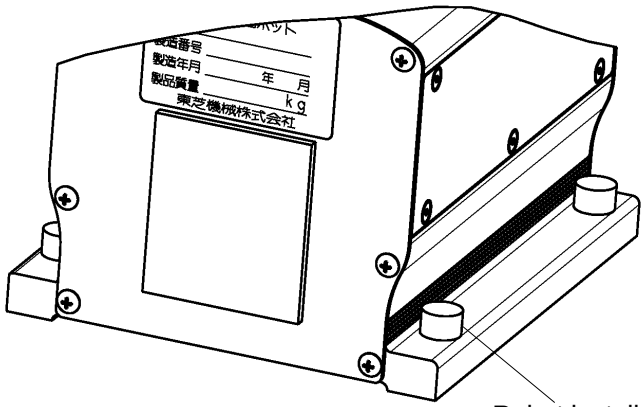


Fig. 2.2 Robot installation bolts (THL300, THL400)

3) Motor Set Bolts

Make sure, using the hexagonal wrench key, that the bolts for securing each axis drive motor are tightened completely. If loosened, tighten them completely. Also make sure that the motor and motor plate bolts are tightened completely. For the places where the following bolts are used, see “Para. 2.4, Replacing Motor.”

Table 2.1 Motor and motor plate set bolt (THL300, THL400)

Location of bolt	Type	Bolts used	Reference Para.
Axis 1 drive motor set bolt	M4 x 14	4 pcs.	2.4.4 Mounting Axis 1 Motor
Axis 2 drive motor set bolt	M4 x 12	4 pcs.	2.4.6 Mounting Axis 2 Motor
Axis 3 drive motor set bolt	M3 x 10	4 pcs.	2.4.8 Mounting Axis 3 Motor
Axis 3 motor plate set bolt	Flange head bolt M4 x 8	4 pcs.	2.4.8 Mounting Axis 3 Motor
Axis 4 drive motor set bolt	M4 x 12	4 pcs.	2.4.10 Mounting Axis 4 Motor
Axis 4 reduction gear set bolt	M5 x 16	4 pcs.	2.7.7 Replacing Axis 4 Reduction Gear
Axis 4 motor plate set bolt	Flange head bolt M4 x 8	4 pcs.	2.5.4 Replacing Axis 4 Timing Belt

* For the recommended clamping torque, see Para. “1.5 Clamping Hexagon Socket Head Cap Screws and Set screws”
* It is not necessary to apply Loctite to the flange head bolts of Axes 3 and 4 motor plate set bolts.

2.1.2 Check of Each Cable and Air Tube for Abrasion

Disassemble the arm 2 cover and the base rear cover, and make sure that each cable is not worn out, broken or cracked. Especially, carefully inspect the vicinity of the cable outlets.

For how to dismount the covers, see “Para. 2.3, Dismounting and Mounting Each Cover.”

The figure below shows an example of air tube installation.

Also inspect the wear and disorder of the air tube as well as the pipe connection. Make sure that the air tube is not worn out and is installed properly, and pipe connection is correct. Otherwise, correct them.

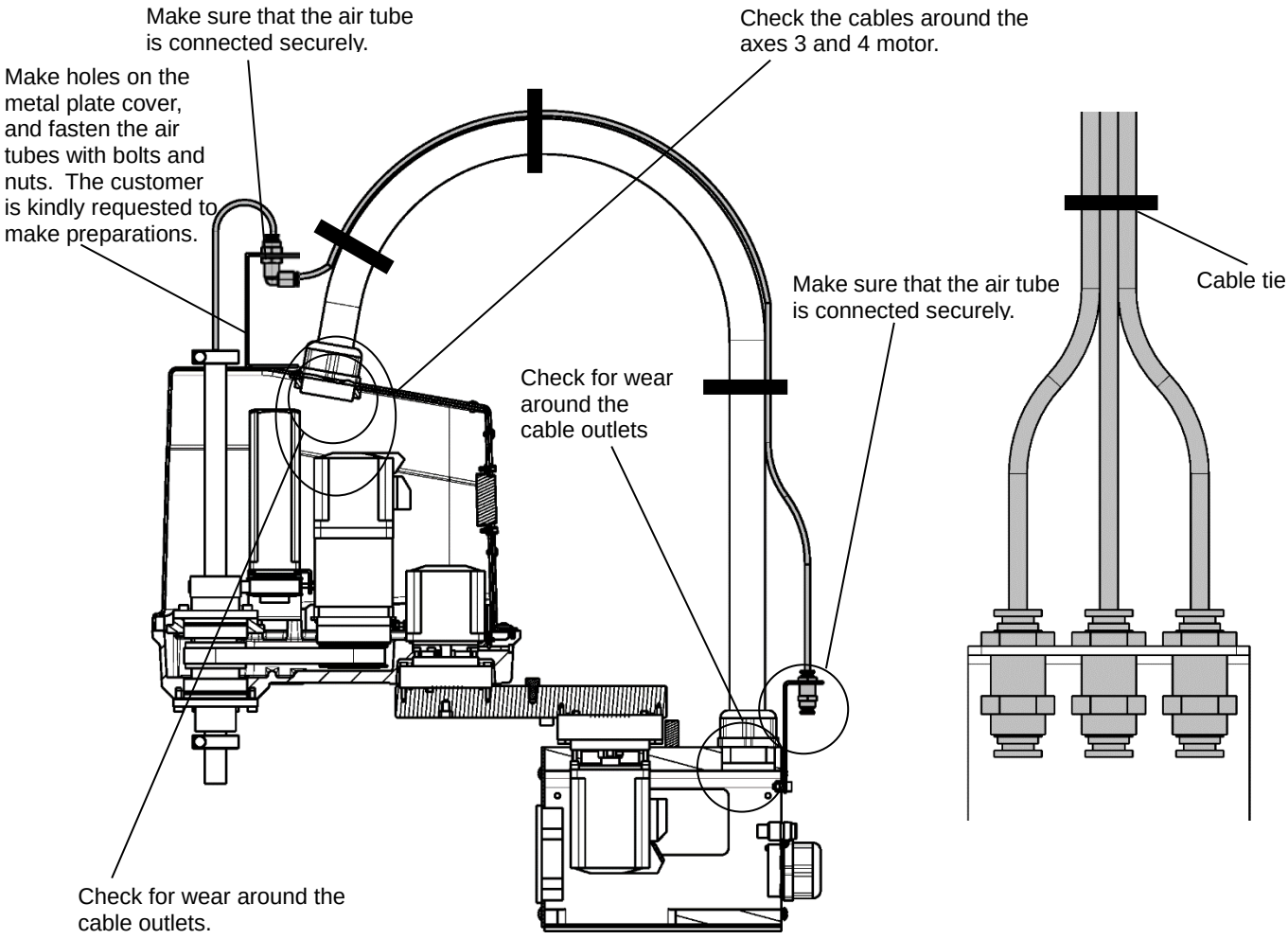


Fig. 2.3 Cable inspection locations (THL300, THL400)

NOTICE

– The air tube is a consumable item. It must be checked periodically, and replaced if broken.
– Fig. 2.3 shows an example of ductwork.
Note that breakage of air tube, etc. is not always under our warranty.

2.1.3 Check of Cable Clamp Tightening

Using a wrench, make sure that the clamps securing the cable are not loose. There are two clamps, one on the arm 2 side and the other on the base side. If they are loose, apply Loctite (low adhesive force) and tighten them. When checking the tightened clamps, check the one on the base side first. By checking the tightness of the clamp on the base side first, the twist of the cable can be adjusted slightly on the arm 2 side. Make sure to tighten the clamps in the posture as shown in the figure below. If the clamps are tightened while the arm is bent, the cable cannot be installed correctly.

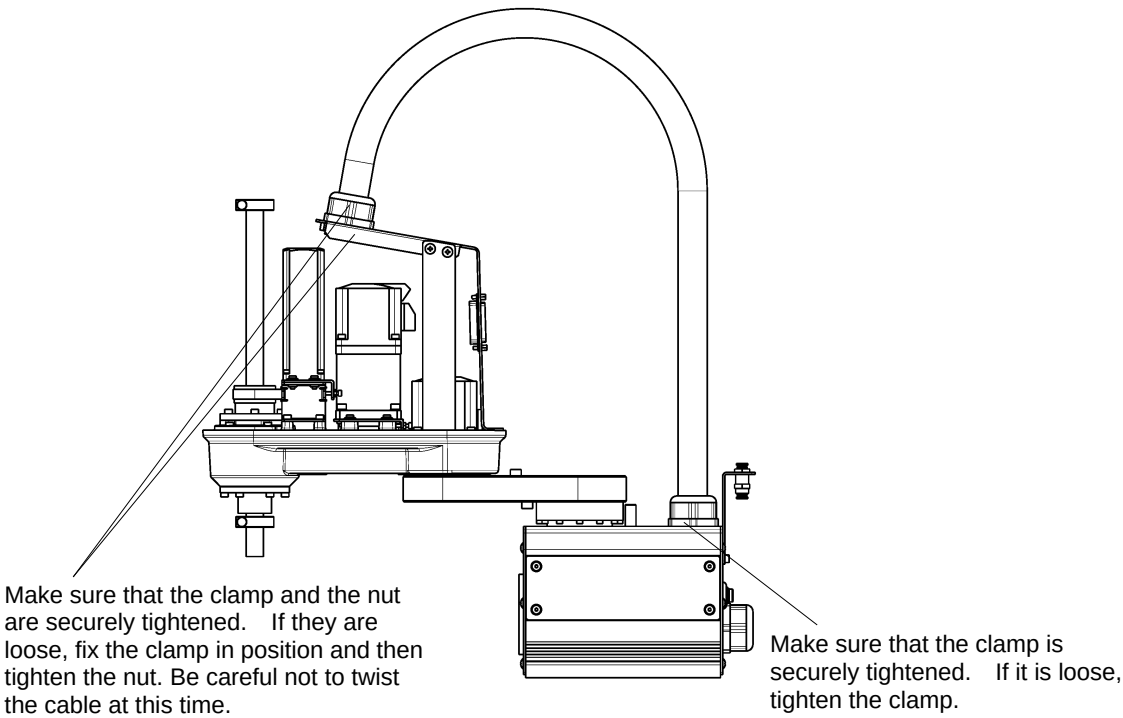


Fig. 2.4 Cable clamp inspection locations (THL300, THL400)

2.1.4 Check of Each Axis for Operation

Connect the power plug of the controller to the power source, then keep the EMERGENCY pushbutton switch in the depressed condition. Move each axis by hand and make sure that it can move smoothly.
For Axis 3, when the brake release switch is pressed, the brake is released. Take careful precautions at this time as the tool shaft may drop according to the weight of the hand and tool.

NOTICE

· When axis 1 and axis 2 are moved near the working envelope limit and a hand is off from them, axis 1 and axis 2 may move due to reaction of the cable.

2.2 Layout of Robot Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 2.5.

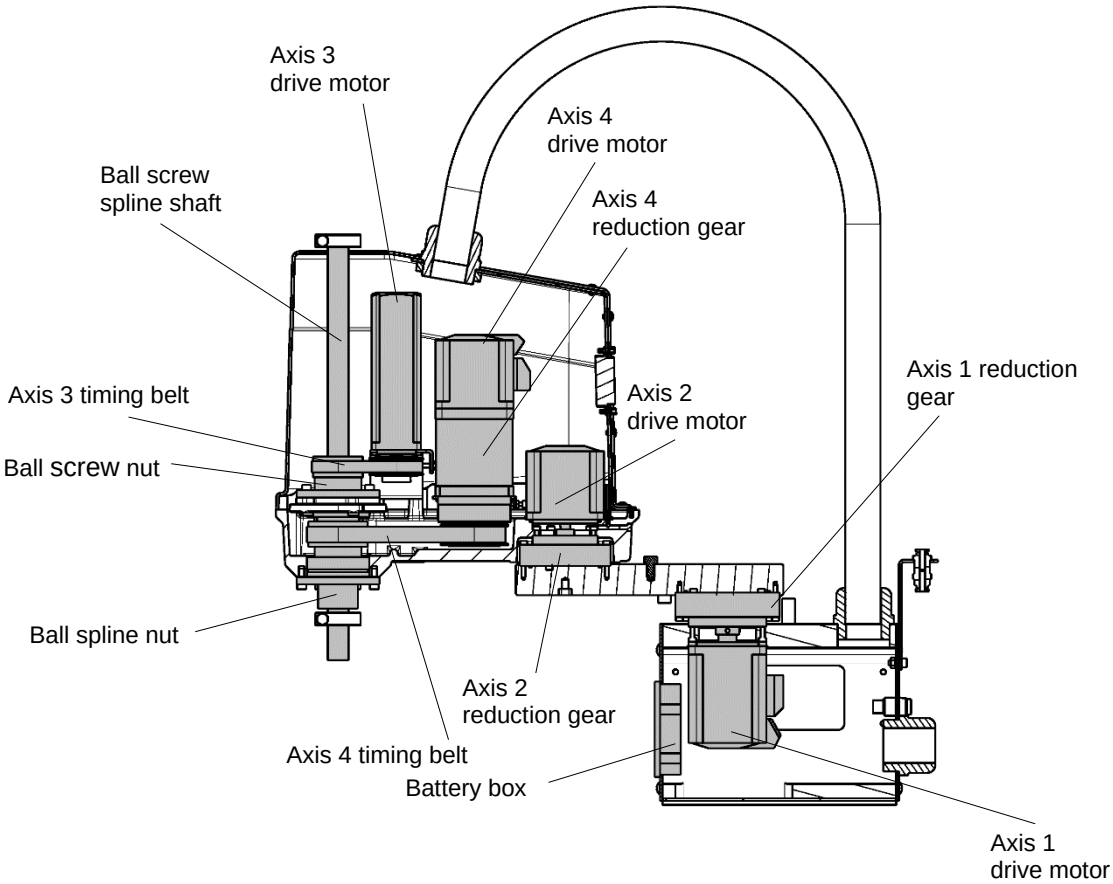


Fig. 2.5 Layout of robot mechanical components (THL300, THL400)

2.3 Dismounting and Mounting Each Cover

This paragraph describes the dismounting and mounting of the covers, which are common to the maintenance and replacement of each unit.

⚠DANGER

– When opening the cover, never allow entry of moisture or contaminant into the robot.
If the power is turned on while moisture or contaminant is left, an electric shock may result.

NOTICE

– When opening the cover, never allow entry of moisture or contaminant into the robot.
If the power is turned on while moisture or contaminant is left, the robot may malfunction.
– When mounting the covers, prevent the cables from being caught.
Otherwise, the cables will be broken.
– Be sure to restore where to route the cables pulled out by removing the cover and where to secure with the cable straps to their original state.
If the cables are pushed by force or where to secure changes, the cables will be worn out or broken.
- Before pulling out the cables, take pictures for restoration

2.3.1 Arm 2 Cover

The arm 2 cover is secured to Arm 2 and the harness guide with 4 hexagon socket head cap screws (M3 x 16) and to cross truss head screws (M3 x 10 x 2 pcs., M3 x 6 x 8 pcs.) (It is not necessary to apply Loctite when installing.)

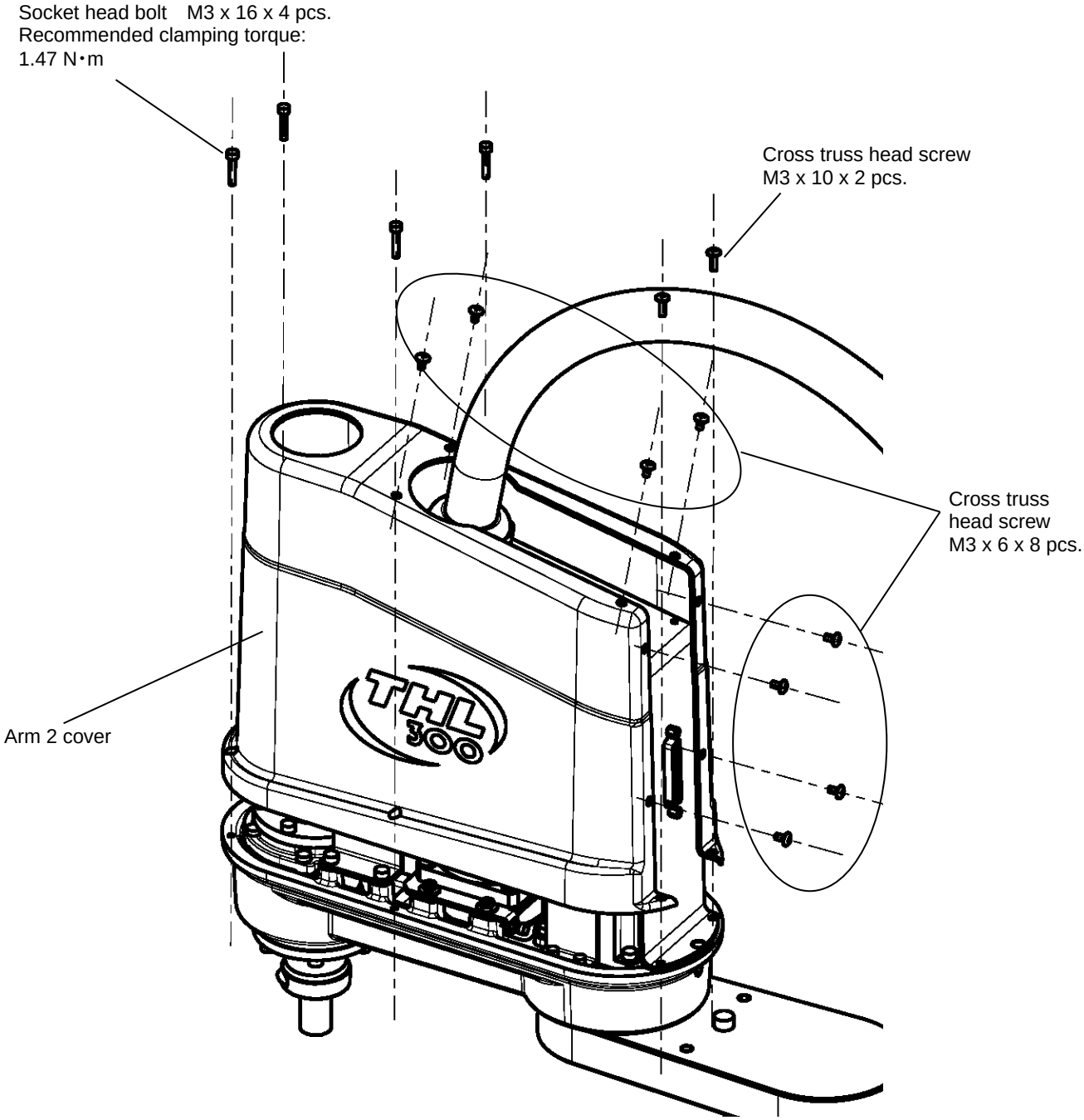


Fig. 2.6 Arm 2 cover (THL300, THL400)

After the cover is mounted, manually move up and down the ball screw spline shaft while pressing the brake release switch, and make sure that the ball screw hole for the arm 2 cover will not interfere with the ball screw stopper.

2.3.2 Base Covers

There are four (4) types of base covers: base front cover, base rear cover, base side cover and base cover. (It is not necessary to apply Loctite to all the covers when installing.)

Each of the base front cover and the base rear cover is secured to the base with six (6) cross truss head screws (M4 x 8). They can be dismounted from the base when these fastening screws are removed, but do not pull them with force as they are connected with the connectors inside.

Each of the left and right base side covers is secured to the base with four (4) hex countersunk screws (M4 x 12).

The base cover is secured to the base with four (4) hex countersunk screws (M4 x 8).

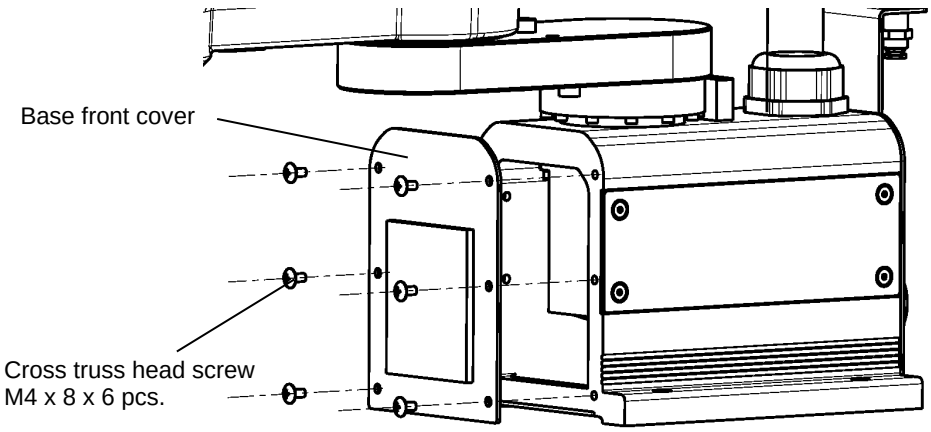


Fig. 2.7 Base front cover (THL300, THL400)

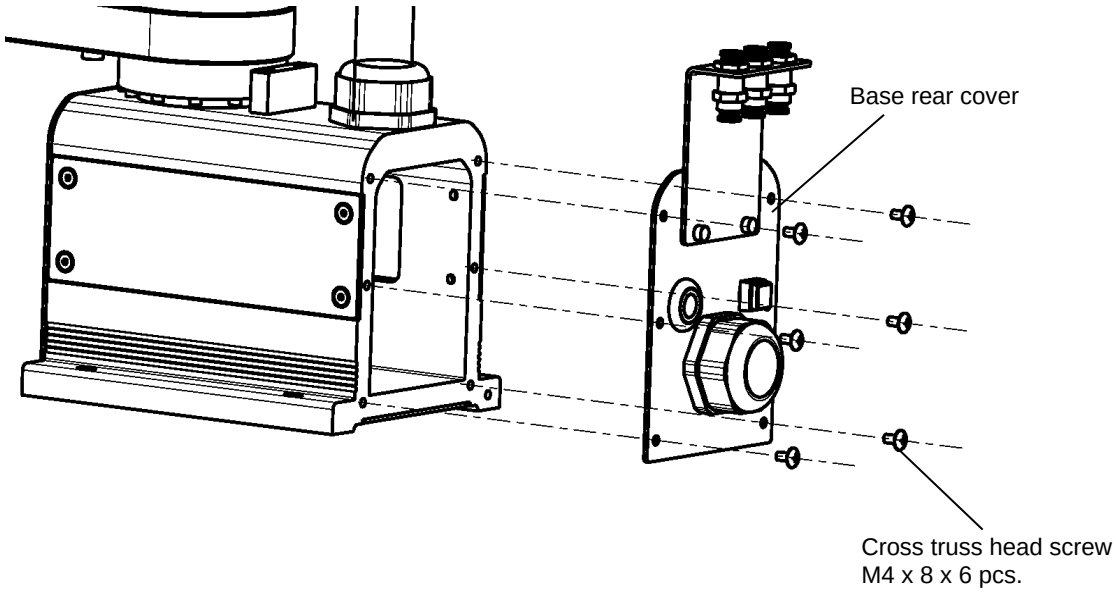


Fig. 2.8 Base rear cover (THL300, THL400)

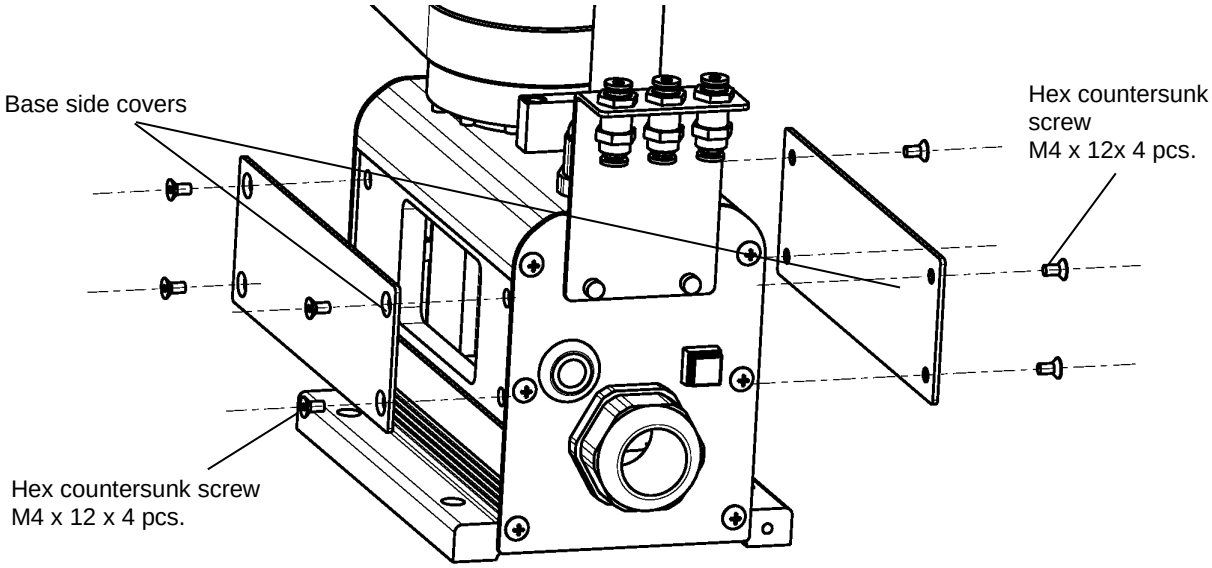


Fig. 2.9 Base side covers (THL300, THL400)

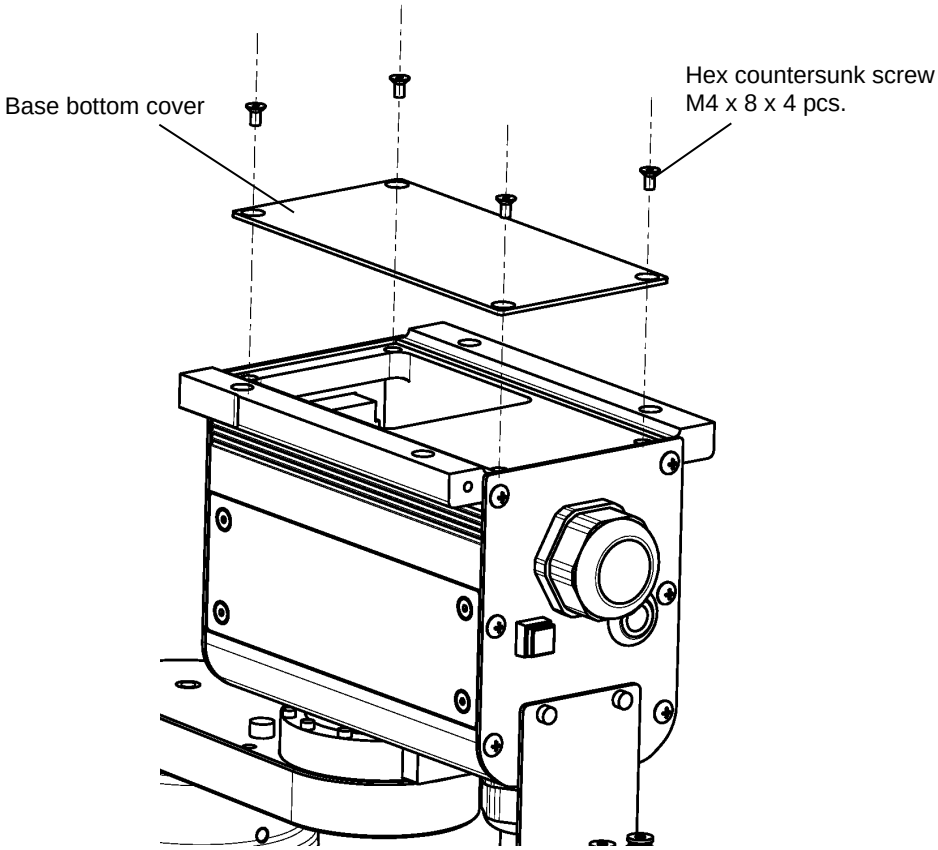


Fig. 2.10 Base bottom cover (THL300, THL400)

2.4 Replacing Motor

The motor is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠ DANGER

· The power plug of the controller must be removed before replacement work.
If replacement work is done with the power turned on, this may result in an electric shock or a malfunction, causing great danger.

NOTICE

– Never cause heavy impact on the motor shaft.
The motor and position detector may malfunction due to impact.
– Never disassemble the motor and position detector.
Otherwise, they may be disabled due to difference in position or intrusion of foreign material.
– When replacing a motor, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– After the motor is replaced, be sure to re-set zero position.
Replacing a motor causes deviation from mechanical zero position; therefore, the robot cannot make normal motion.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error”

2.4.1 Type of Motor

The motors employed in this robot are shown below. When you place an order for a replacement motor, make sure of the robot model (THL300 and THL400), the serial number, the axis name, and our drawing number according to the following table. For the location where the serial number plate is attached, see the “Safety Manual.”

Table 2.2 Type of Motor (THL300, THL400)

Description	Axis name	Type	Our drawing No.	Unit code
AC servo motor	Axis 1		S890967	Y610A3NL0
	Axis 2, 4		S890968	Y610A3NM0
	Axis 3		S777296	Y610A37A0

2.4.2 Motor Locations

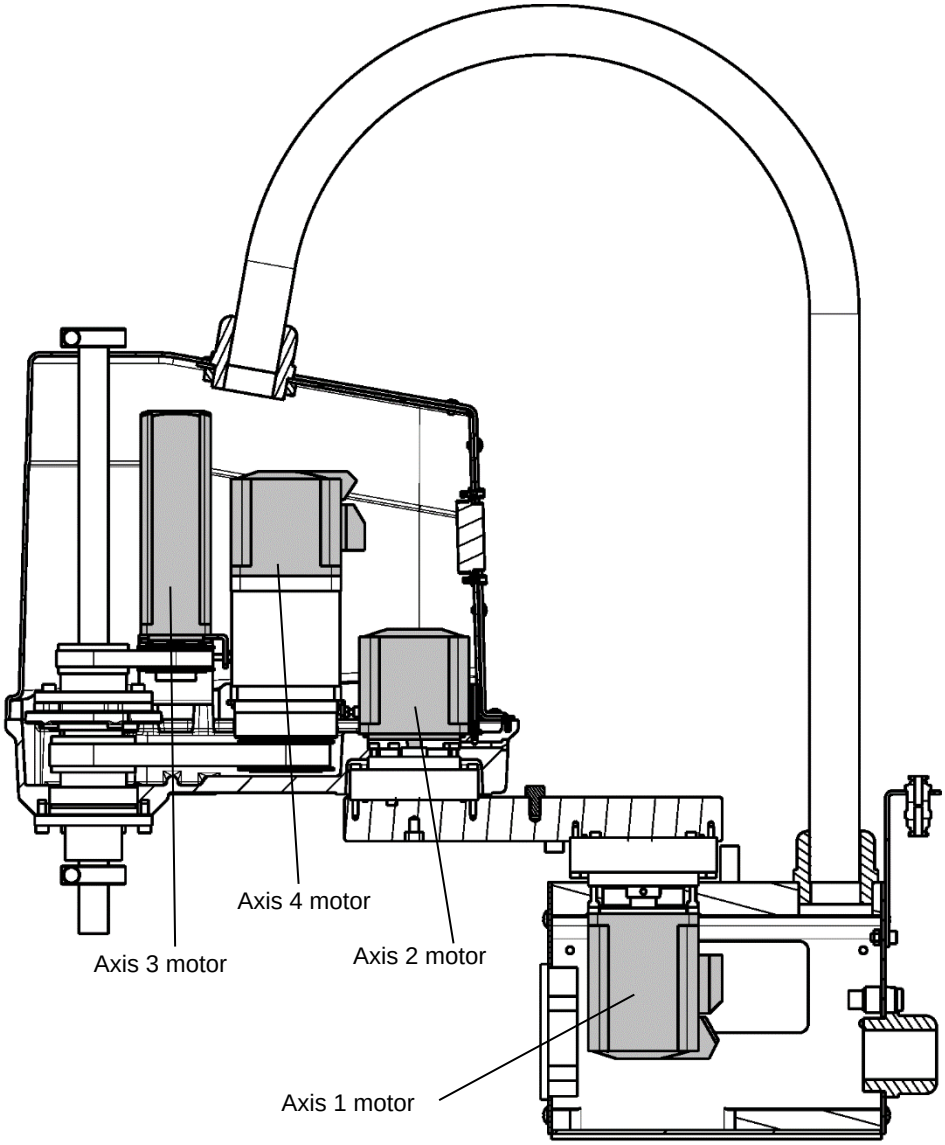


Fig. 2.11 Motor locations (THL300, THL400)

2.4.3 Dismounting Axis 1 Motor

1) Remove the base front cover and both of the base side covers. (See “Para. 2.3.2, Base Covers.”) The battery for position detection is connected to the base front cover, so do not forcibly pull it or unplug the connector.

2) Remove the four hexagon socket head cap screws (M4 x 14 x 4 pcs.) and the washers that secure the axis 1 motor, pull the axis 1 motor assembly in the axis direction and then pull it out. Have a waste cloth handy as grease may drip from the motor mounted section when pulling out the motor assembly. At this time, remove the O-ring (CO0534A) being mounted on the base groove side surface. Also, when pulling out the motor assembly, do not pull it with force as the motor connector is connected.

3) Remove the connectors of the axis 1 motor, i.e., J1AS and J1AP (power drive cable), and J1BS and J1BP (encoder cables).

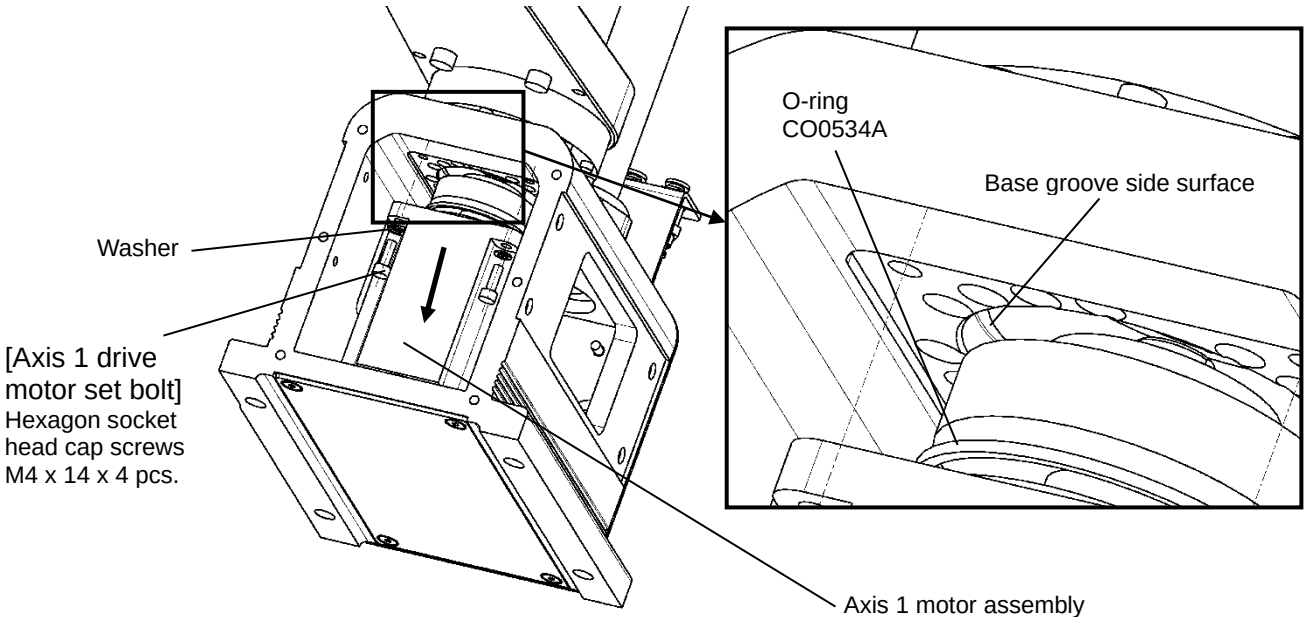


Fig. 2.12 Dismounting axis 1 motor assembly (THL300, THL400)

4) Remove the set screws (2 pcs.) securing the wave generator to the axis 1 motor and pull out the wave generator.

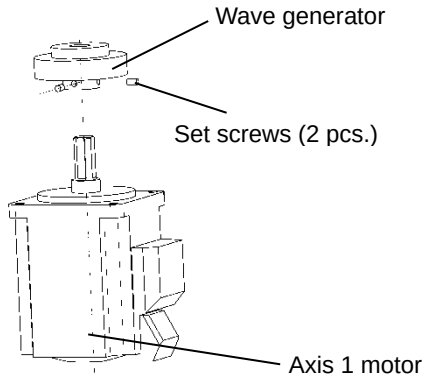


Fig. 2.13 Dismounting axis 1 wave generator (THL300, THL400)

2.4.4 Mounting Axis 1 Motor

1) When inserting the wave generator into the axis 1 motor, be sure to fasten it using the two jigs for fastening Axis 1. If fastened without using these jigs, vibrations may occur when the robot is used. Install the set screws (2 pcs.) to the wave generator. Align the holes of the set screws with the phases of the D cuts of the axis 1 motor and insert the wave generator. Mount the fastening jigs as shown in the figure below. While pushing down each part to make tight contact and paying attention to the D cuts and the phases of the set screws, tighten the set screws to fasten the wave generator. Then, pull out the fastening jigs. In case the fastening jigs come out too easily, remount them in such a way that some force is required to pull them out.

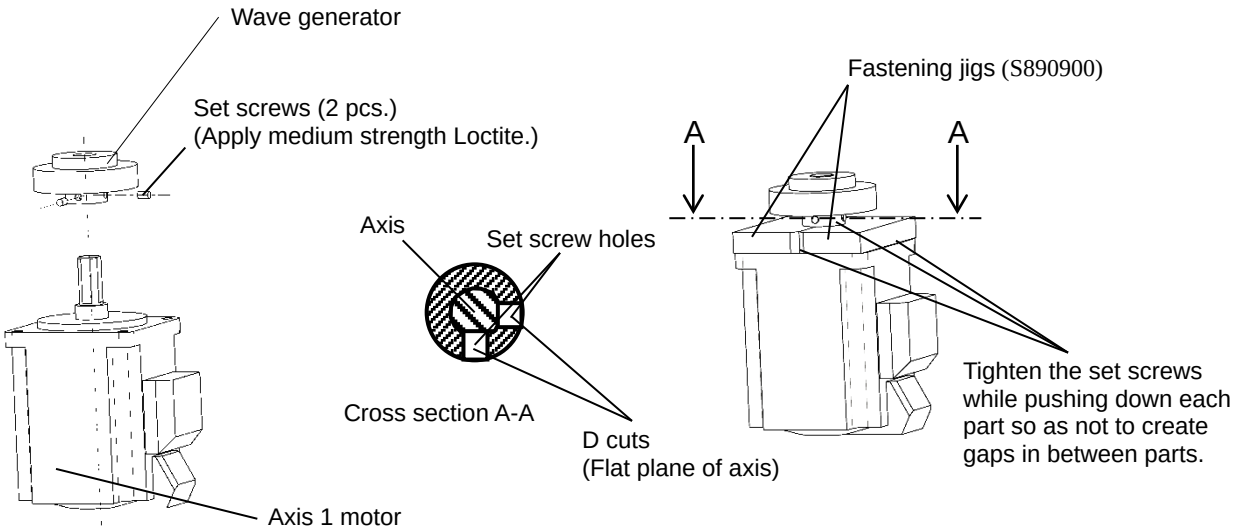


Fig. 2.14 Mounting axis 1 wave generator (THL300, THL400)

NOTICE

– Be sure to use wave generator and reduction gear in the correct combination.
Use of the different combination will result in abnormal noise, reduced service life, and reduced positioning accuracy.

2) Apply grease to the O-ring (CO0534A) and mount it to touch the base groove side surface. Be careful not to drop the O-ring while doing so.

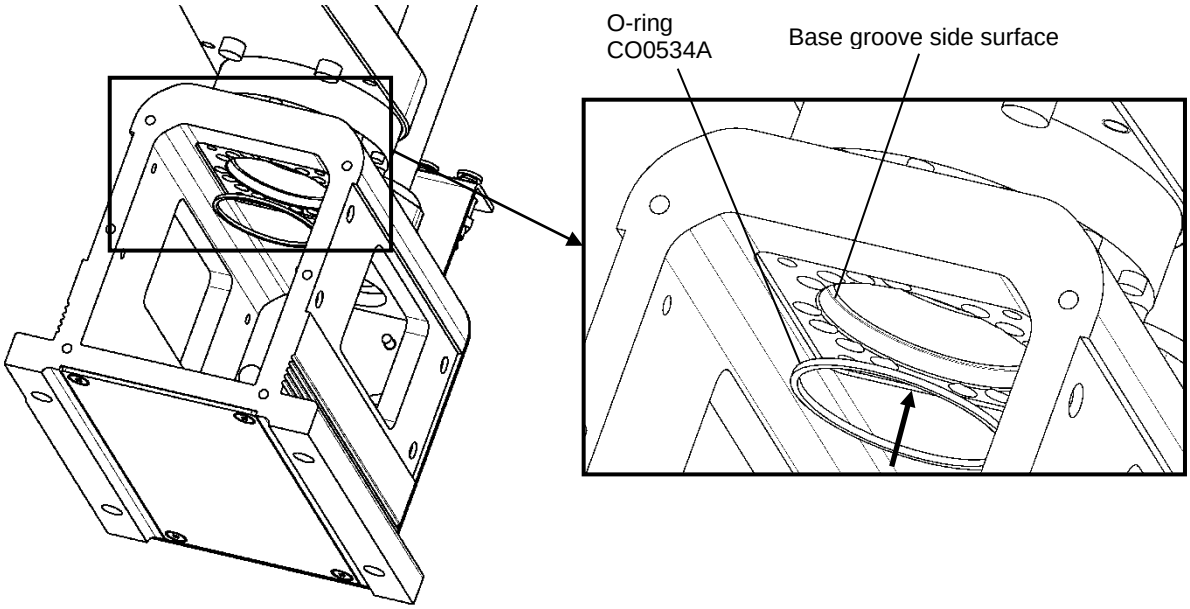


Fig. 2.15 Mounting axis 1 base side O-ring (THL300, THL400)

NOTICE

– Never forget to fit the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.

3) Apply an appropriate amount of grease onto the all surfaces of the wave generator mounted to the axis 1 motor.

Reduction Gear	Recommended grease	Maker
A type	SK-1A	HDS
B ype	SFB No.1	Nidec-Shimpo Corp.

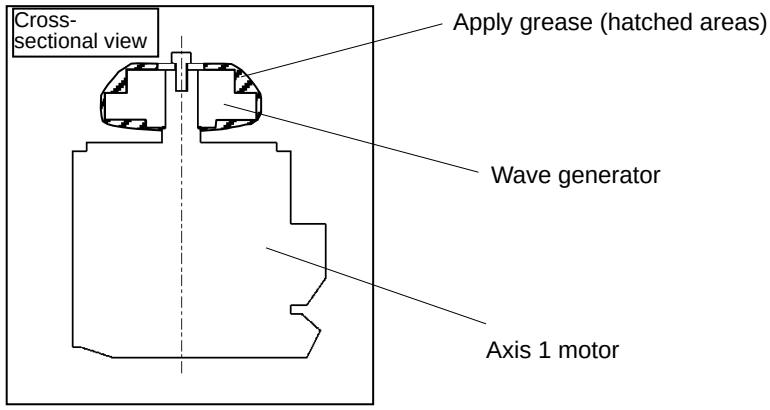


Fig. 2.16 Apply grease onto wave generator (THL300, THL400)

NOTICE

– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

4) Connect the connectors of the axis 1 motor, i.e., J1AS and J1AP (power drive cable), and J1BS and J1BP (encoder cables).

5) Insert the axis 1 motor assembly into the base while paying attention to the motor's mounting phases, the orientations of the reduction gear's main body and wave generator when the motor is inserted (align the long sides of the ovals of the reduction gear and the wave generator), and the position of the O-ring (it should not move). Do not forcibly push in the motor.

6) Secure the axis 1 motor assembly with four (4) hexagon socket head cap screws (M4 x 14) and washers.

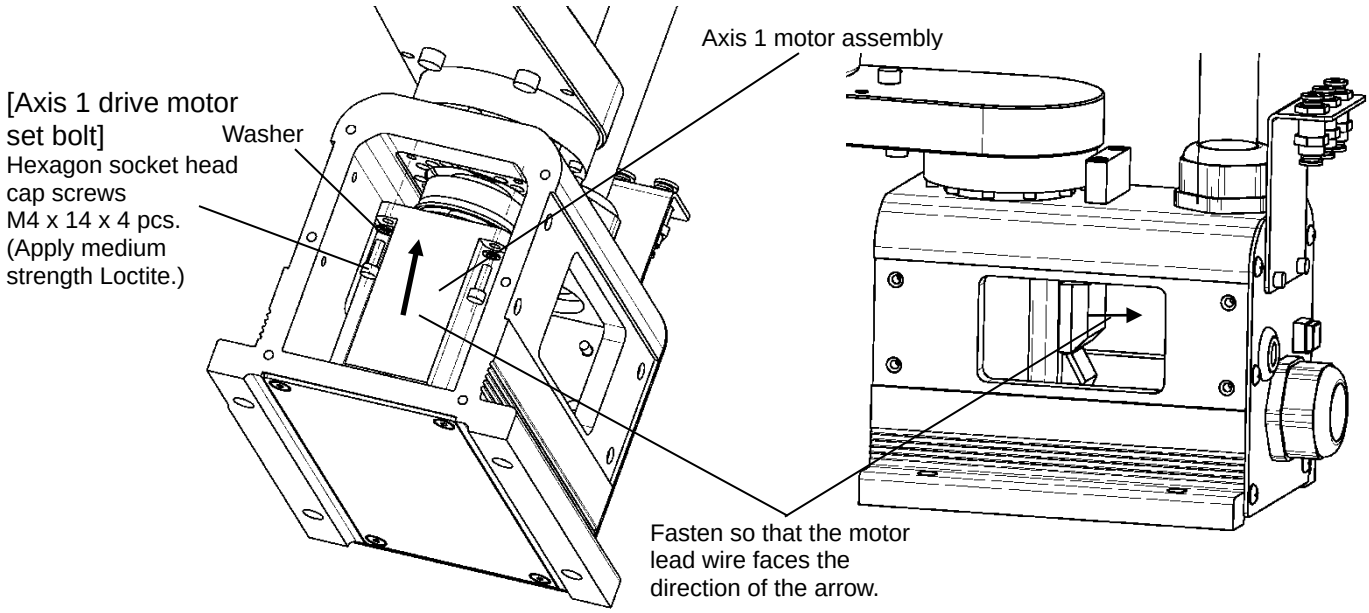


Fig. 2.17 Mounting axis 1 motor assembly (THL300, THL400)

7) After changing Axis 1 motor, move Arm 1 by hand and check that there is no abnormal sound before turning on the power.

8) Mount the base front cover and the base side covers. (See “Para. 2.3.2, Base Covers.”)

9) Turn on the power and set up the axis 1 home position to complete axis 1 motor replacement.
(Check the coordinates of Axes 2 to 4 and set up the home position if necessary. See “Section 8, Robot Home Point and Position Detector Error”.)

2.4.5 Dismounting Axis 2 Motor

1) Disconnect the arm 2 cover. (See “Para. 2.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers or the like, and remove J2AS and J2AP (axis 2 power drive cables), J2BS and J2BP (axis 2 encoder cables). It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Remove the harness guide being secured with one (1) cross countersunk screw (M3 x 6) and the support plate being secured with two (2) hexagon socket head cap screws (M3 x 6).

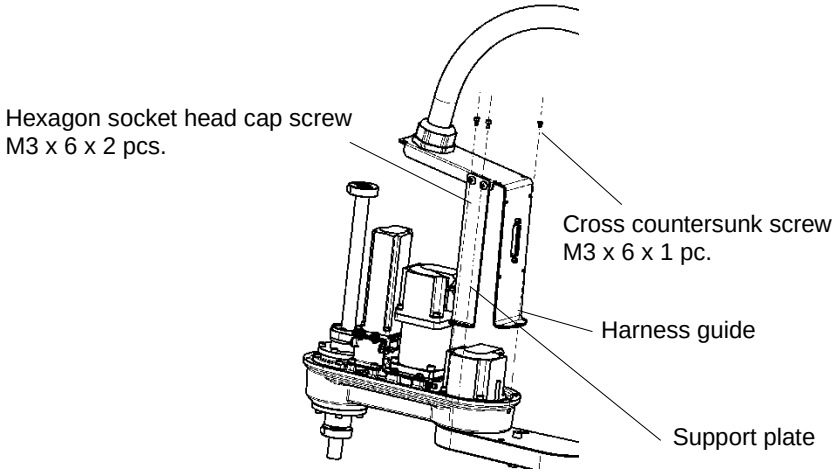


Fig. 2.18 Dismounting harness guide and support plate (THL300, THL400)

4) Remove four (4) hexagon socket head cap screws (M4 x 12) and washers that secure the axis 2 motor. Pull out the axis 2 motor assembly. Also remove the O-ring (CO0538A).

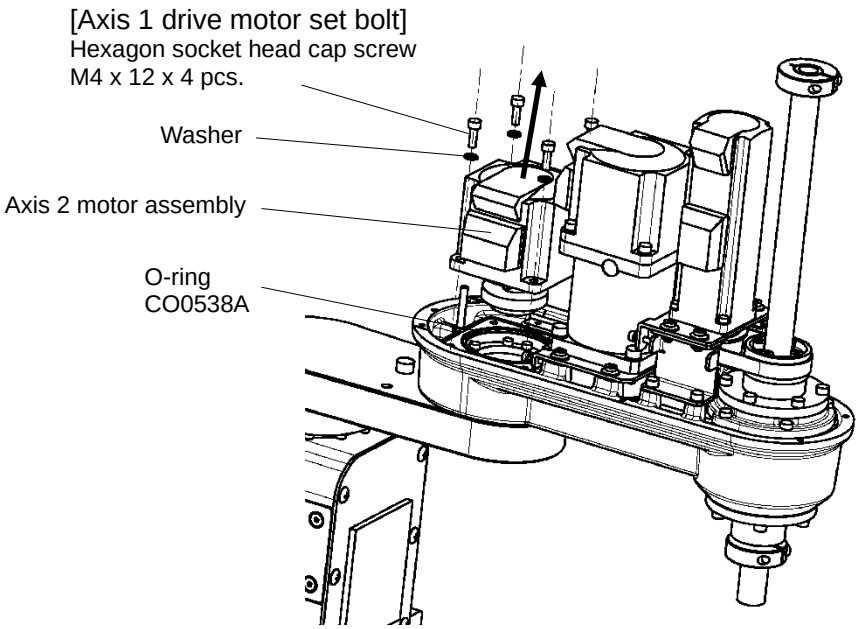
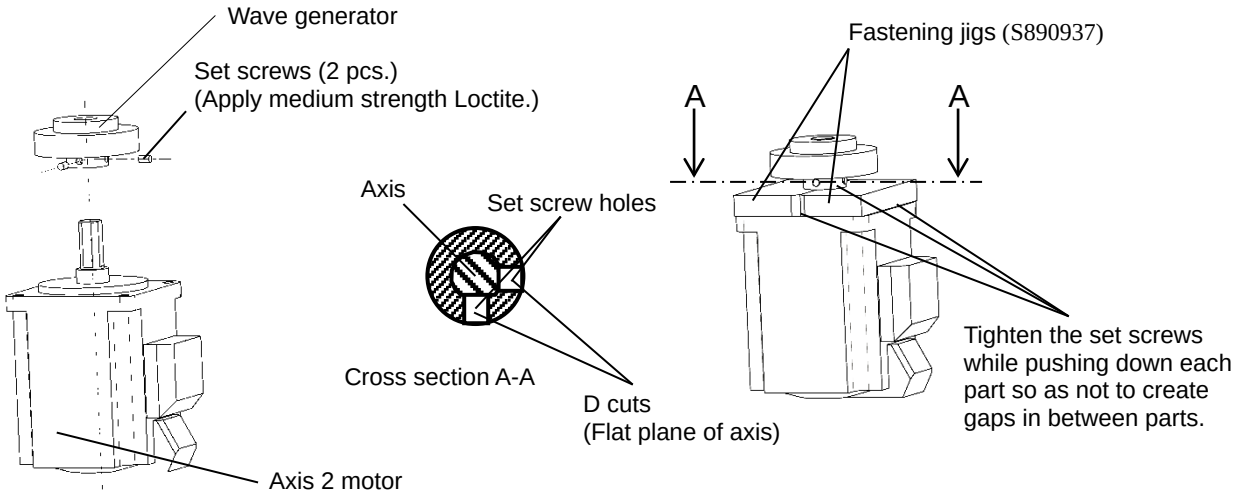


Fig. 2.19 Dismounting axis 2 motor assembly (THL300, THL400)

5) Remove the wave generator from the axis 2 motor assembly. (Same as Step 4 in “Para. 2.4.3, Dismounting Axis 1 Motor.”)

2.4.6 Mounting Axis 2 Motor

1) Mount the wave generator to the axis 2 motor. (Same as Step 1 in “Para. 2.4.4, Mounting Axis 1 Motor.”) When inserting the wave generator into the axis 2 motor, be sure to use the two jigs for fastening Axis 2.



NOTICE

– Be sure to use wave generator and reduction gear in the correct combination.
Use of the different combination will result in abnormal noise, reduced service life, and reduced positioning accuracy.

2) Apply grease to the O-ring (CO0538A) and mount it to the inlaid part of the axis 2 motor.

3) Apply an appropriate amount of grease onto the all surfaces of the wave generator mounted to the axis 2 motor. (Same as Step 3 in “Para. 2.4.4, Mounting Axis 1 Motor.”)

4) Insert the axis 2 motor assembly into Arm 2 while paying attention to the motor's mounting phase, the orientations of the decelerator main body and the wave generator when the motor is inserted (align the long sides of the ovals of the reduction gear and wave generator), and the position of the O-ring (it should not move).

5) Secure the axis 2 motor with four (4) hexagon socket head cap screws (M4 x 12) and washers.

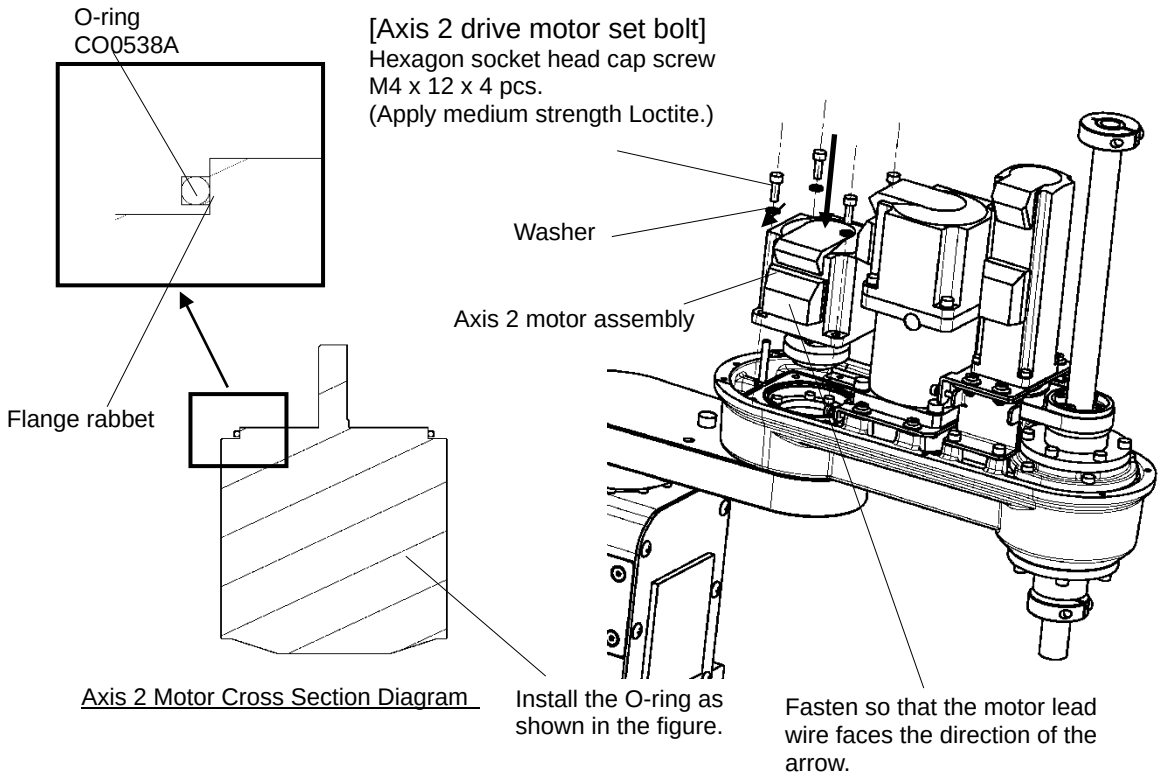


Fig. 2.20 Mounting axis 2 motor assembly (THL300, THL400)

NOTICE

– Never forget to fit the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

6) Secure the harness guide and the support plate with one (1) hex countersunk screw (M3 x 8, application of Loctite not necessary) and two (2) Cross countersunk screw (M3 x 6).

7) After changing the axis 2 motor, move Arm 2 by hand and check that there is no abnormal sound.

8) Connect J2AS and J2AP (axis 2 power drive cables) and J2BS and J2BP (axis 2 encoder cables).

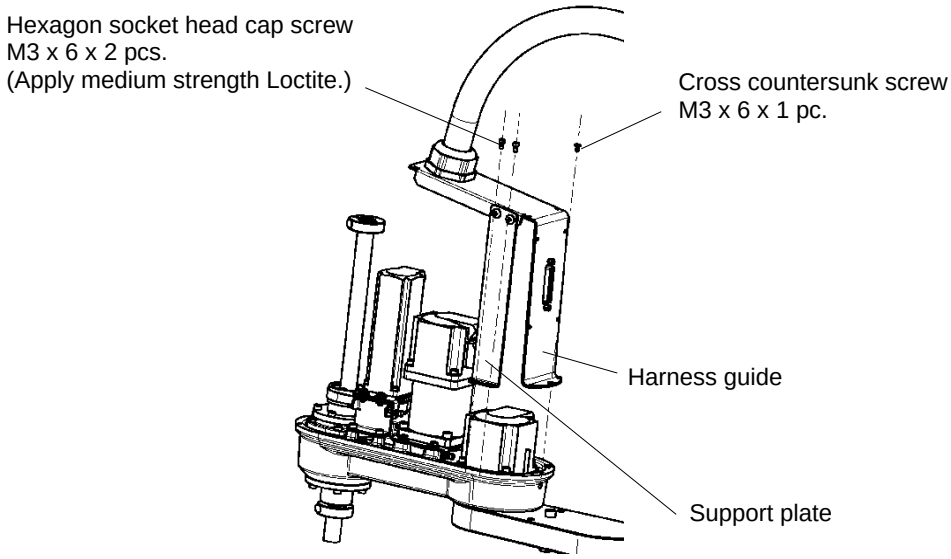


Fig. 2.21 Mounting harness guide and support plate (THL300, THL400)

9) Restore the cables in Arm 2 to the original conditions.

10) Mount arm 2 cover.

11) Turn on the power and set up the home positions of axes 2 to complete axis 2 motor replacement (see “Section 8, Robot Home Point and Position Detector Error”).

2.4.7 Dismounting Axis 3 Motor

⚠ WARNING

– When the axis-3 motor is replaced, be sure to lower the ball screw spline shaft to the lower limit. When the motor or timing belt is removed, the shaft that drops by its own weight could catch your hand or fingers.
- When the ball screw spline or axis-4 timing belt or reduction gear is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

1) Remove the arm 2 cover. (See “Para. 2.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers.
It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Disconnect connectors J3AS and J3AP (power drive cables), connectors J3BS and J3BP (encoder cables) and connectors J3DS and J3DP (brake cables) for Axis 3, which are connected to the connector panel.

4) Loosen the axis 3 tension adjustment bolt (M3 x 12 x 2 and hexagonal nuts) and remove the flange head bolts (M4 x 8 x 4) securing the axis 3 motor plate to cancel the axis 3 timing belt tension. Next, pull out the axis 3 motor assembly upward while making sure that it does not touch the sheet metal and the ball screw spline.

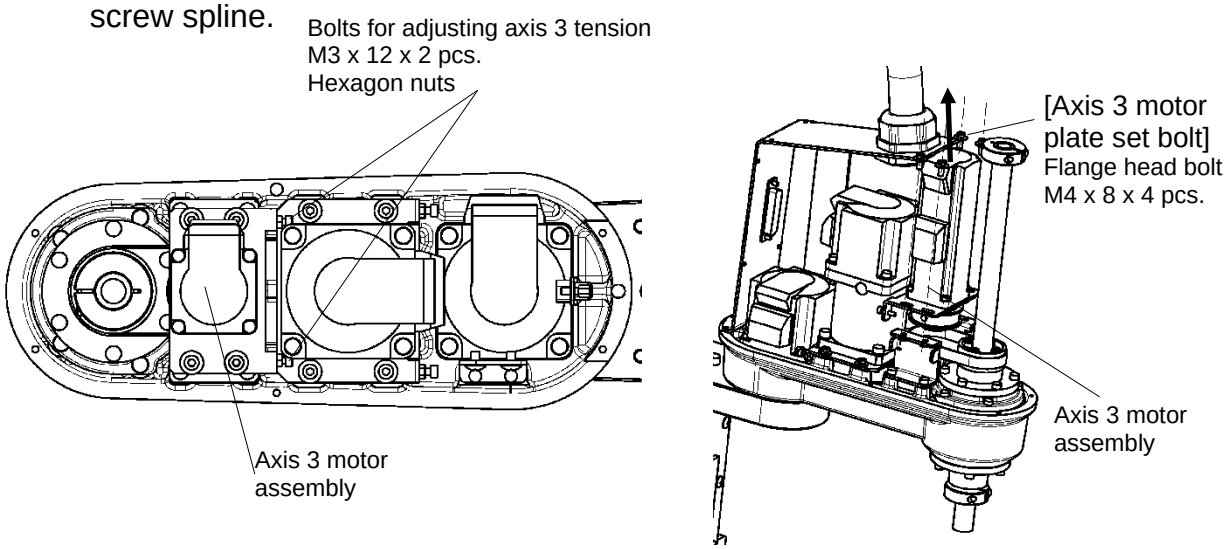


Fig. 2.22 Dismounting axis 3 motor assembly (THL300, THL400)

5) Remove the set screws (4 x 5 x 2 pcs. and 3 x 5 x 2 pcs.). Then, pull out the pulley.

6) Remove four (4) hexagon socket head cap screws (M3 x 10) and washers that secure the axis 3 motor and then disassemble the axis 3 motor plate and the axis 3 motor.

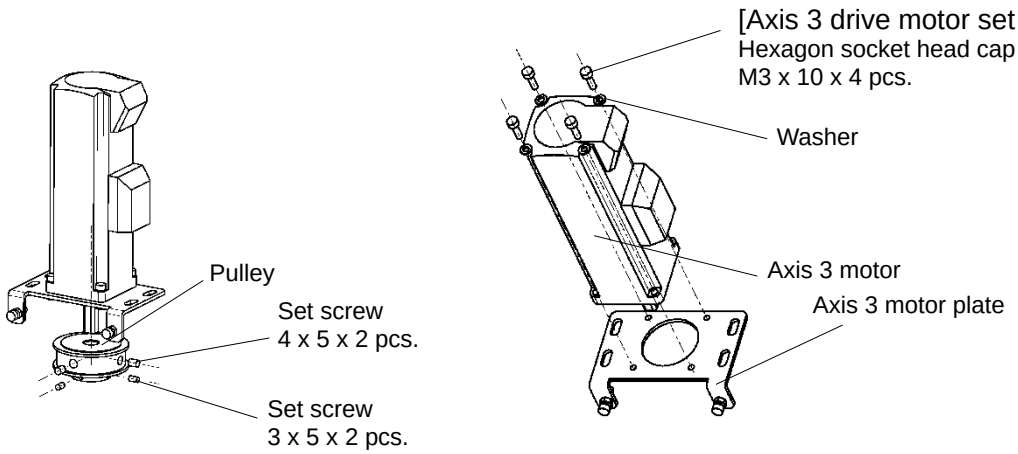


Fig. 2.23 Dismounting axis 3 motor pulley and plate (THL300, THL400)

2.4.8 Mounting Axis 3 Motor

1) Fasten the axis 3 motor to the axis 3 motor plate with the hexagon socket head cap screws (M3 x 10 x 4 pcs.) and the washers. At this time, be careful with the installation phases of the axis 3 motor and axis 3 motor plate.

2) Mount the pulley for the axis 3 motor and fasten it with the set screws (4 x 5 x 2 pcs. and 3 x 5 x 2 pcs.). Fasten the pulley while aligning the holes of the set screws of the pulley with the phases of the D cuts as shown in Cross Section Diagram B-B.

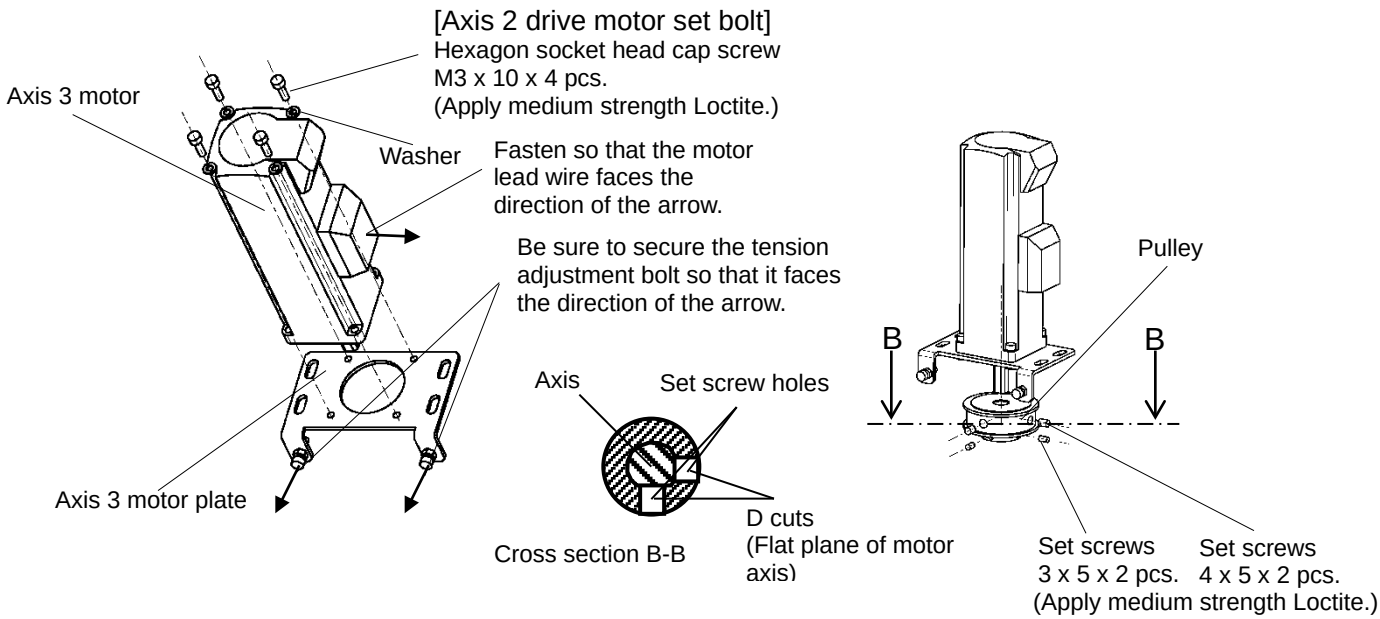


Fig. 2.24 Mounting axis 3 pulley and plate (THL300, THL400)

3) Hang the axis 3 motor timing belt to the arm 3 motor pulley, and temporarily secure the axis 3 motor plate to Arm 2 with four (4) flange head bolts (M4 x 8, application of Loctite not necessary). Be careful with the motor mounting phase at this time. Apply tension using two (2) axis 3 tension adjustment bolts (M3 x 12, application of Loctite not necessary). (For the belt replacement procedures and tension adjustment values, see “Para. 2.5.3, Replacing Axis 3 Timing Belt.”) Then, tighten the temporarily tightened flange head bolts. Next, tighten the tension adjustment bolts and fasten with the hexagon nuts.

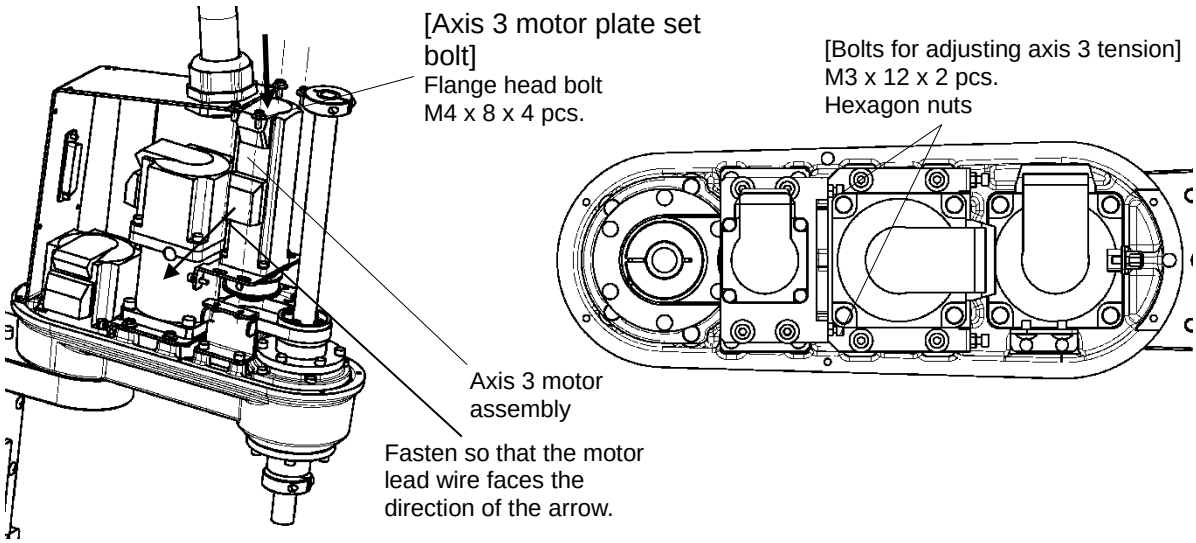


Fig. 2.25 Mounting axis 3 motor assembly (THL300, THL400)

4) Connect the connectors and restore the cables. (See the pictures taken in Section 2.4.7 Dismounting Axis 3 Motor Step 2)

5) Mount the arm 2 cover and perform home setting for Axes 3 and 4. Now, the axis 3 motor replacement is complete. Need perform home setting for Axes 4 at this time.

6） Turn on the power. While pressing the brake release switch in the servo off mode, move the ball screw in the vertical direction. Make sure of the smooth operation of the ball screw.

7) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

2.4.9 Dismounting Axis 4 Motor

1) Remove the arm 2 cover. (See “Para. 2.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers.
It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Disconnect connectors J4AS and J4AP (power drive cables) and connectors J4BS and J4BP (encoder cables) of the axis 4 motor.

4) Remove the cross recessed flat head screw (M3 x 6 x 1 pc.) and the hexagon socket head cap screws (M3 x 6 x 2 pcs.) which are fastening the harness guide and the support plate, and then remove the harness guide and the support plate.

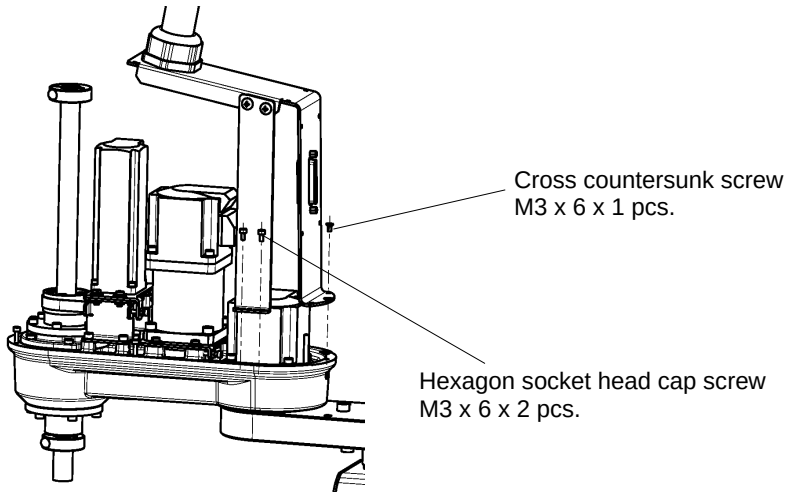


Fig. 2.26 Dismounting harness guide and support plate (THL300, THL400)

5) Remove the cap on the side panel of the axis 4 reduction gear and loosen the bolt (M3) of the coupling which is fastening the input axis of the axis 4 reduction gear. If the phases of the hexagonal holes of the bolt cannot be aligned, manually rotate the ball screw spline shaft and align the phases.

6) Remove four (4) hexagon socket head cap screws (M4 x 12) and washers that secure the axis 4 motor, and then pull the axis 4 motor upward.

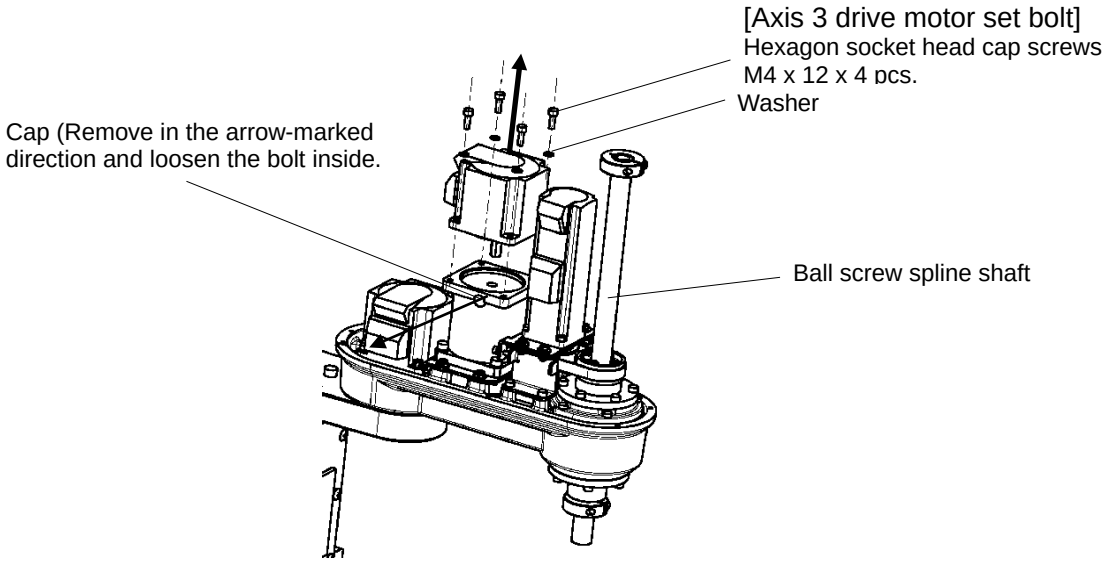


Fig. 2.27 Dismounting axis 4 motor (THL300, THL400)

2.4.10 Mounting Axis 4 Motor

1) Mount the new motor to the reduction gear with four (4) hexagon socket head cap screws (M4 x 12) and washers. Be careful with the phase when mounting the motor. (Recommended clamping torque: 2.3 N·m)

2) Tighten the coupling of the axis 4 reduction gear by means of the attached bolt (M3) to mount the cap. (Recommended clamping torque: 1.8 N·m)

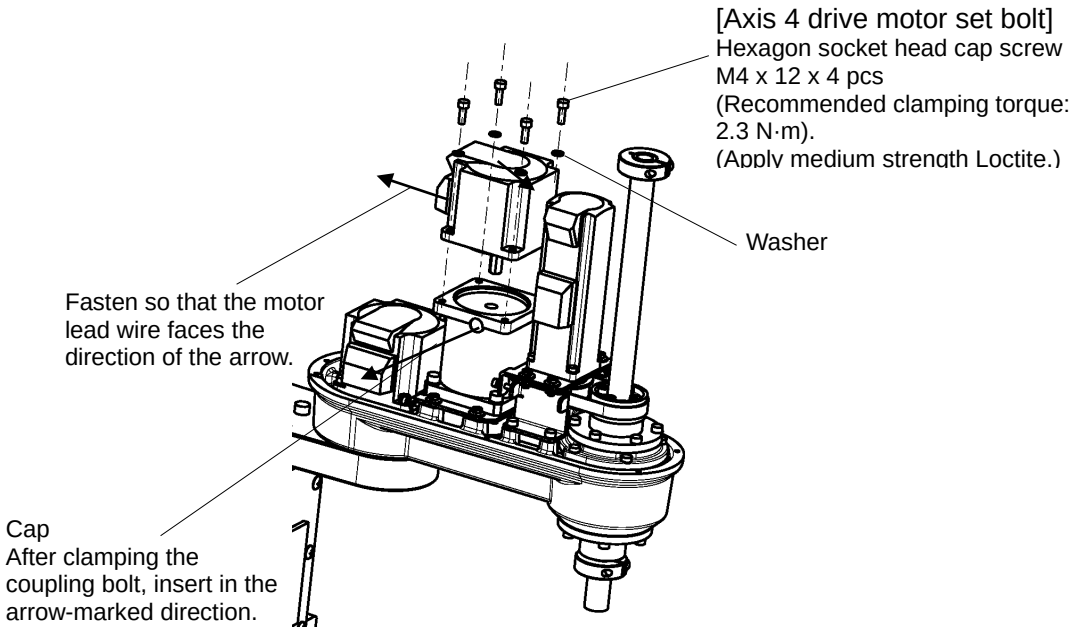


Fig. 2.28 Mounting axis 4 motor (THL300, THL400)

3) Connect the connectors of the axis 4 motor, i.e., J4AS and J4AP (power drive cables), and J4BS and J4BP (encoder cables).

4) Restore the cables to the original conditions and then mount the arm 2 cover.

5) Set up the home positions of Axes 3 and 4 to complete motor replacement. During the replacement of the axis 4 motor, it is necessary to set up the home position of Axis 3.

6) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

2.5 Adjusting and Replacing Timing Belt

The timing belt is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠ DANGER

– When replacing the timing belt, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock.
A worker may be entangled in rotation of the motor that has operated unpredictably.

⚠ WARNING

– When replacing the axis-3 timing belt, be sure to lower the ball screw spline shaft to the lower limit.
When the motor or timing belt is removed, the shaft drops by its own weight.
- When the ball screw spline or axis-4 timing belt is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

NOTICE

– When replacing the timing belt, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– After the timing belt is replaced, be sure to re-set the zero position.
Because a mechanical zero position is shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error”.

2.5.1 Type of Timing Belt

The timing belts used in this robot are shown below.
When you place an order for a replacement belt, specify the robot model (THL300, THL400), the serial number, the axis name, and our drawing number.
For the location where the serial number plate is attached, see the “Safety Manual.”

Table 2.3 Type of timing belt (THL300, THL400)

Description	Axis name	Width	Our drawing No.
Timing belt	Axis 3	9mm	S890882
	Axis 4	12mm	S890883

2.5.2 Timing Belt Locations

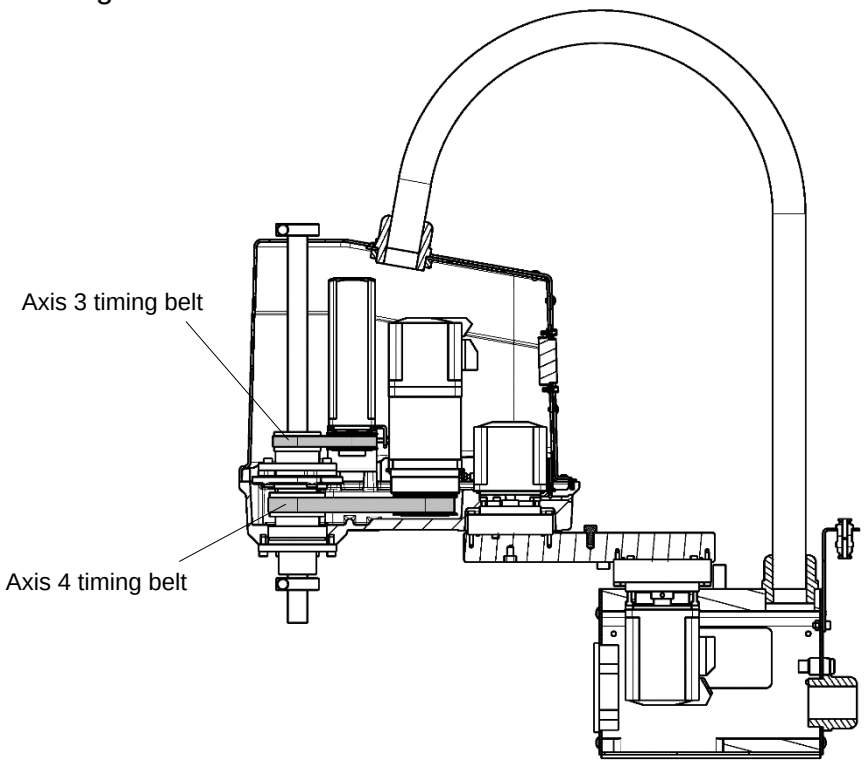


Fig. 2.29 Timing belt locations (THL300, THL400)

2.5.3 Replacing Axis 3 Timing Belt

⚠ WARNING

– When replacing the axis-3 timing belt, be sure to lower the ball screw spline shaft to the lower limit. When the motor or timing belt is removed, the shaft drops by its own weight.
- When the ball screw spline or axis-4 timing belt is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

1) Remove the arm 2 cover. (See “Para. 2.3.1, Arm 2 Cover.”)
2) Cut the cable ties of the cables with nippers. It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.
3) Disconnect the axis 3 motor assembly. For the disconnecting procedures, see Steps 1) through 4) of “Para. 2.4.7, Dismounting Axis 3 Motor.”

4) Disconnect the axis 3 timing belt.

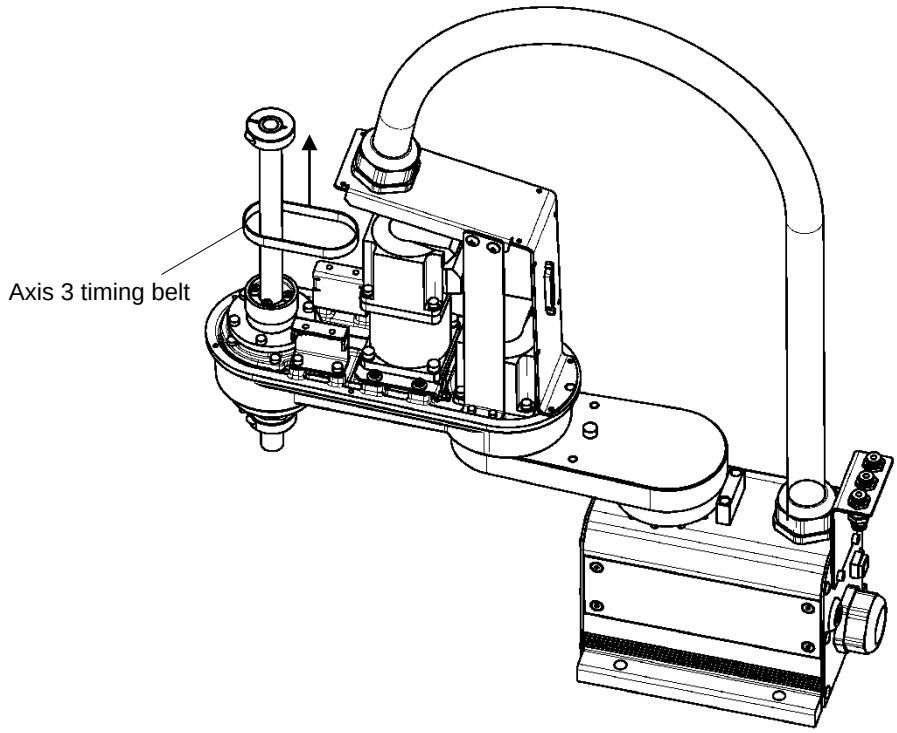


Fig. 2.30 Replacing axis 3 timing belt (THL300, THL400)

5) Mount the new timing belt.

6) Hang the timing belt to the axis 3 motor assembly which was previously removed in 3), and temporarily secure it to Arm 2 with four (4) flange head bolts (M4 x 8, application of Loctite not necessary). Move the ball screw spline unit up and down to make it fit in. Apply tension using two (2) axis 3 tension adjustment bolts (M3 x 12, application of Loctite not necessary).
While pulling the axis 3 motor plate with the axis 3 tension adjustment bolts, measure the tension with a tension meter. Tighten the flange head bolt at the location where the tension reaches a value slightly smaller than 48 N (because the tension gets larger by retightening of the flange head bolt). Adjust so that the tensile value becomes between 48 and 64 N when the flange head bolt is retightened. If the bolt is not tightened any further, the bolt will come out while the robot is in operation. Next, fasten with the hexagon nuts.
The values to be used for the tension meter are as shown in the table below.

Table 2.4 Value of tension of axis 3 timing belt (THL300, THL400)

Value of tension [N]	Unit mass [g/m]	Belt width [mm]	Span [mm]
48 to 64	2.2	9	45

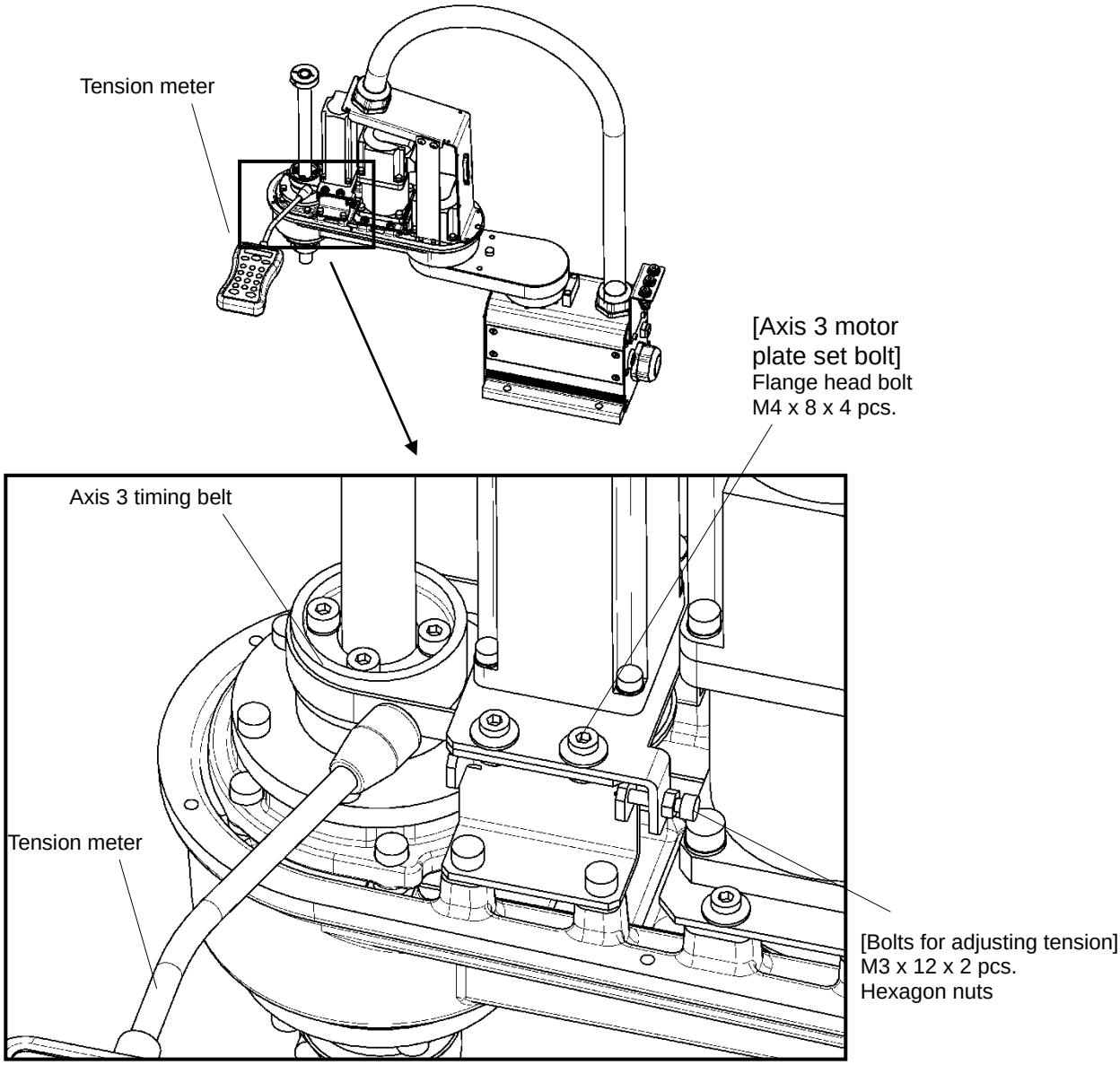


Fig. 2.31 Adjusting axis 3 tension (THL300, THL400)

7) Connect the connectors and return them to their original location, mount the arm 2 cover, and perform home setting for Axes 3 and 4. This completes the replacement of the timing belt. When replacing the 3rd-axis timing belt, be sure to set the 4th-axis home position.

8) Turn on the power. While pressing the brake release switch in the servo off mode, move the ball screw in the vertical direction. Make sure of the smooth operation of the ball screw.

9) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

2.5.4 Replacing Axis 4 Timing Belt

NOTICE

– Never pull out the ball screw nut from the ball screw spline shaft.
Otherwise, the ball screw spline unit from which balls have come out cannot be used.
– After the timing belt is replaced, be sure to re-set the zero position.
Because a mechanical zero position is shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– When axis-4 timing belt is replaced, the axis-3 motor and timing belt must also be removed.
Therefore, the precautions for the replacement of the axis-3 motor and timing belt must be also observed.

1) Remove the arm 2 cover.

2) Cut off the cable ties with nippers. It is recommended to take photos of the routing of the cables and the locations where cable ties are used in advance in case they become necessary for restore.

3) Remove the cross recessed flat head screw (M3 x 6 x 1 pc.) and the hexagon socket head cap screws (M3 x 6 x 2 pcs.), and then remove the harness guide and the support plate.

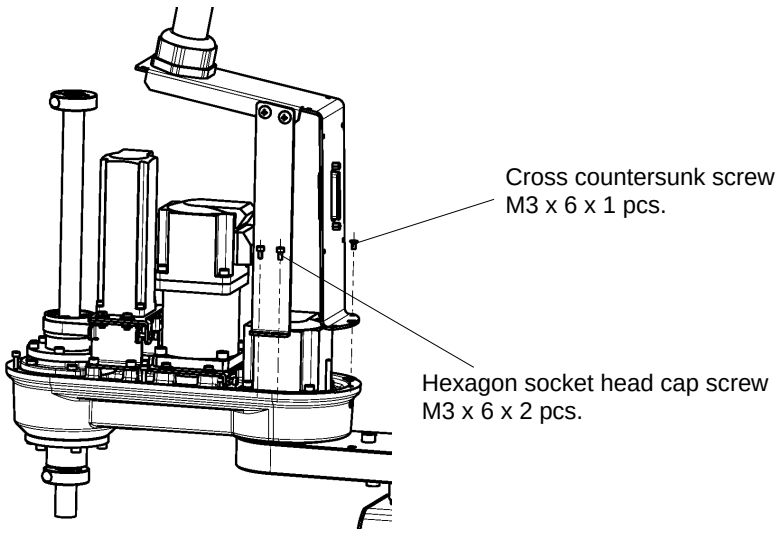


Fig. 2.32 Dismounting harness guide and support plate (THL300, THL400)

4) Remove the axis 3 motor assembly and the axis 3 timing belt. For more information about removal, see items 3) and 4) in "2.5.3, Replacing Axis 3 Timing Belt."

5) Remove the hexagon socket head cap screws (M4 x 8 x 4 pcs.) which are fastening the axis 3 brackets (sheet metal, 2 pcs.), and then remove the axis 3 brackets (sheet metal, 2 pcs.).

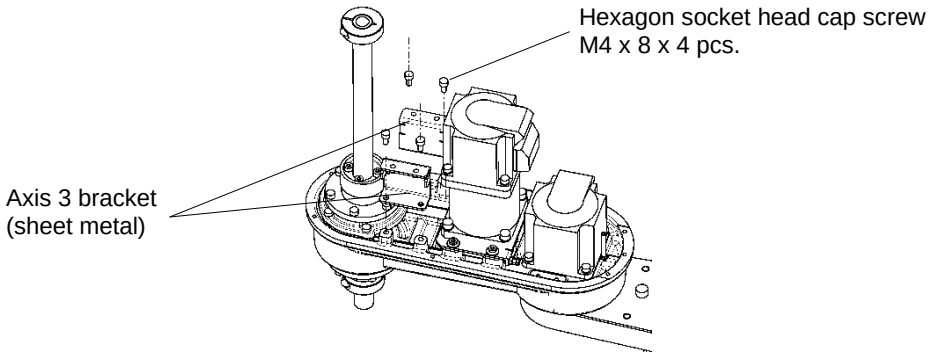


Fig. 2.33 Dismounting axis 3 bracket (sheet metal) (THL300, THL400)

6) Remove the stoppers, the ball screw spline shaft, the ball screw nuts and the axis 3 bracket (casting). For more information about removal, see item 5) in "2.6.4, Dismounting Ball Screw Spline Unit."

7) Remove the flange head bolts (M4 x 8 x 4 pcs.), and then remove the axis 4 motor assembly and the axis 4 timing belt.

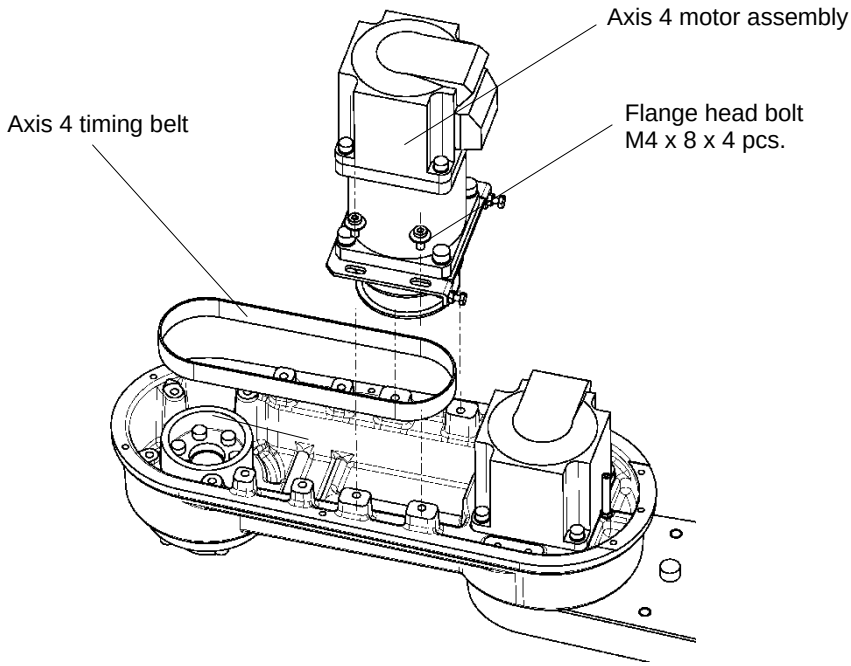


Fig. 2.34 Dismounting axis 4 belt and motor assembly (THL300, THL400)

8) Temporarily fasten the axis 3 bracket (casting) to Arm 2 with the hexagon socket head cap screws (M4 x 16 x 4 pcs.).
At this time, hang a new axis 4 timing belt on the pulley.

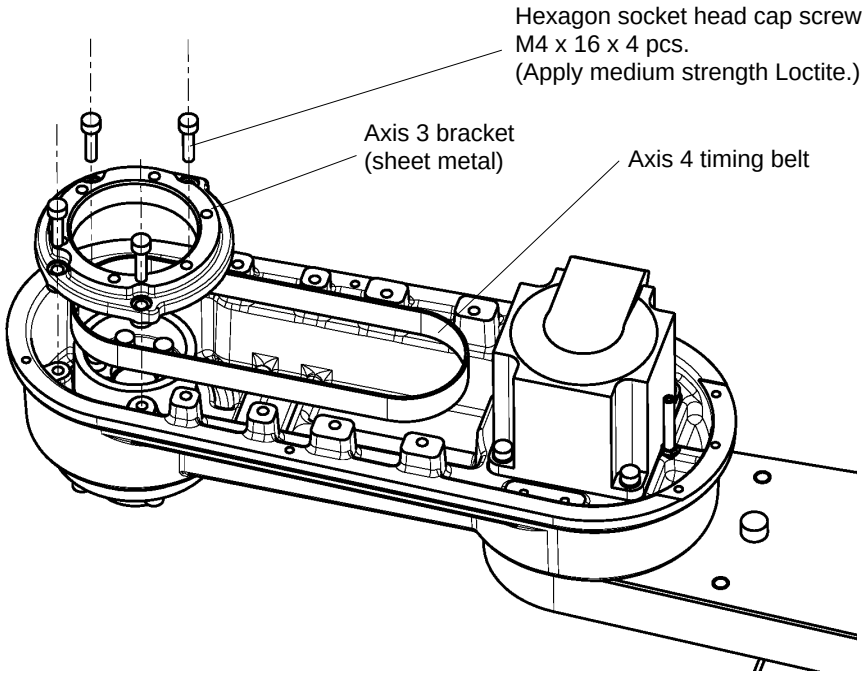


Fig. 2.35 Mounting axis 3 bracket (casting) and axis 4 timing belt (THL300, THL400)

9) Fasten the ball screw nuts, the ball screw spline shaft and the stoppers which were removed in step 6) above to Arm 2. For more information about mounting of the ball screw spline unit and the stoppers, see "2.6.5, Mounting Ball Screw Spline Unit."

10) Temporarily fasten the axis 4 motor assembly to Arm 2 with the flange head bolts (M4 x 8 x 4 pcs.).

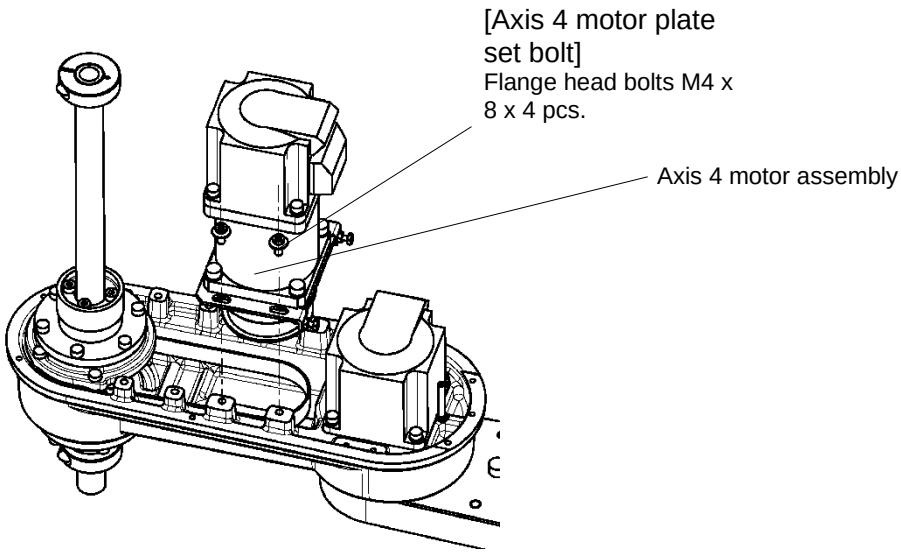


Fig. 2.36 Mounting axis 4 motor assembly (THL300, THL400)

11) While pulling the axis 4 motor plate with the axis 4 tension adjustment bolts, measure the tension with a tension meter. Tighten the flange head bolt at the location where the tension reaches a value slightly smaller than 66 N (because the tension gets larger by retightening of the flange head bolt). Adjust so that the tensile value becomes between 66 and 88 N when the flange head bolt is retightened. Next, tighten the tension adjustment bolts and fasten with the hexagon nuts. If the bolt is not tightened any further, the bolt can come out while the robot is in operation. The values to be used for the tension meter are as shown in the table below.

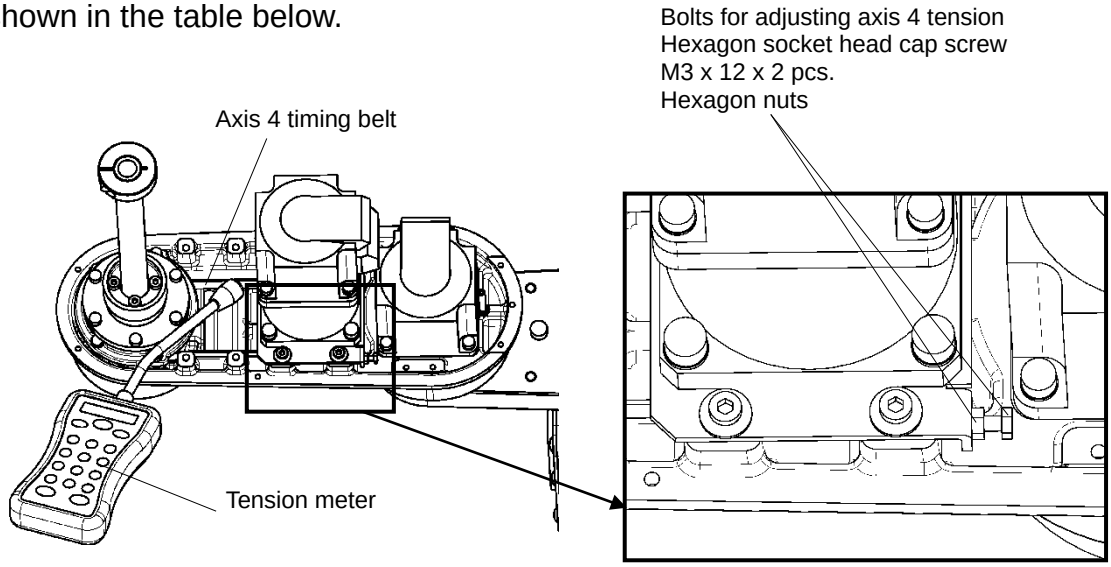


Fig. 2.37 Adjusting axis 4 tension (THL300, THL400)

Table 2.5 Value of tension of axis 4 timing belt (THL300, THL400)

Value of tension [N]	Unit mass [g/m]	Belt width [mm]	Span [mm]
66 to 88	2.2	12	105

12) Mount the axis 3 motor and timing belt. For the mounting procedures, see "Para. 2.5.3, Replacing Axis 3 Timing Belt" above.

13) Arrange the connectors and cables as originally set. (See the pictures you took beforehand.)

14) Perform home setting for Axes 3 and 4. When replacing the 3rd-axis timing belt, be sure to set the 4th-axis home position.

15) Carry out a test operation of Axes 3 and 4 and make sure that the belt tension is appropriate.

16) Attach the arm 2 cover. Now replacement of the axis 4 timing belt completes.

2.5.5 Check Timing Belt Adjustment

Perform inspection and adjustment of each timing belt semi-annually (every six (6) months).

1) Remove the arm 2 cover. (See “Para. 2.3.1, Arm 2 Cover.”)

2) If the belt is worn out, replace it. (See “Para. 2.5.3, Replacing Axis 3 Timing Belt.”) If scratches are found on the belt, please replace it or contact our Service Department.
   If the belt is loosened heavily, adjust its tension, referring the values of tension in Para. 2.5.3, Replacing Axis 3 Timing Belt and 2.5.4, Replacing Axis 4 Timing Belt. Note that the value of tension at replacement of timing belt (i.e., when mounting a new timing belt) is the same as the value of tension at adjustment. If the tension is appropriate, mount the arm 2 cover. Now the inspection is complete.

3) When adjustment of the tension is required, loosen the four (4) flange head bolts (M4 x 8) securing the axis 3 and axis 4 motor plates, respectively. Measure the tension on the tension meter while adjusting the tension adjustment bolt (axis 3: M3 x 12 x 2 bolts, axis 4: M3 x 12 x 2 bolts and hexagonal nut).

4) When the tension has fallen under the appropriate range, tighten the flange head bolts.

5) Tighten the tension adjustment bolts further and fasten with the hexagon nuts. If the bolt is not tightened any further, the bolt can come out while the robot is in operation.

6) Mount the arm 2 cover. Now the adjustment completes.

2.6 Filling Grease to Ball Screw Spline Unit and Replacement

The ball screw spline unit is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠DANGER

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– After the ball screw spline unit is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

2.6.1 Type of Ball Screw Spline Unit

The ball screw spline unit used in this robot is shown below.
When you place an order for the ball screw spline unit for replacement, specify the robot model (THL300, THL400) and our drawing number.

Table 2.6 Ball screw spline unit (THL300, THL400)

Description	Stroke	Our drawing No.	Unit code
Ball screw spline unit	160 mm	H852810	Y610A3NE0

2.6.2 Ball Screw Spline Unit Location

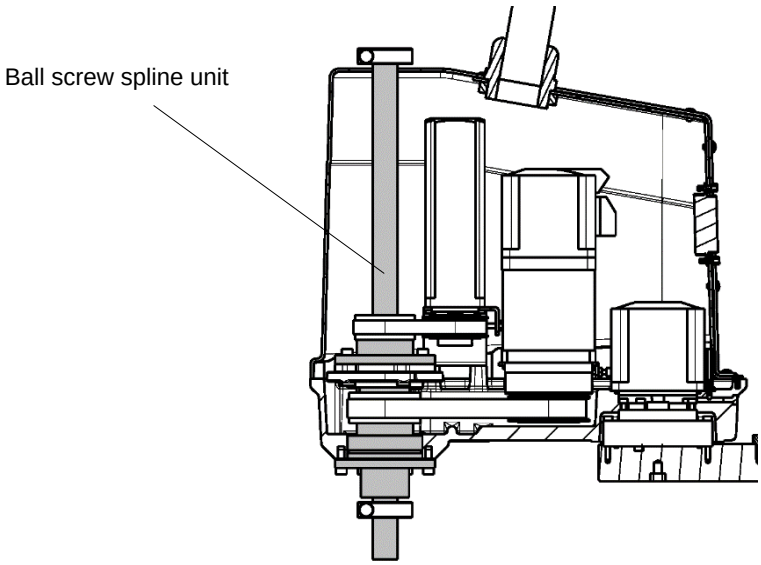


Fig. 2.38 Ball screw spline unit location (THL300, THL400)

2.6.3 Greasing Ball Screw Spline Unit and Applying Anticorrosive

⚠ CAUTION

– When moving the ball screw spline shaft by hand, never insert your hand or finger inside the mechanical stopper or under the robot hand.
Otherwise, your hand or finger may get caught in the mechanical stopper and hand.

NOTICE

– Never touch the ball screw spline unit with bare hands.
Touching it with bare hands may cause the ball screw spline unit to rust.
- Be sure to wear gloves.

– When greasing up the ball screw spline unit and apply corrosion inhibitor, be sure to cover the peripheral equipment, etc.
Grease and corrosion inhibitor may drip on the peripheral equipment.
- Wipe out excess grease and corrosion inhibitor with waste cloth.

– Be sure to check the application state of grease daily.
A shortage of grease and development of rust may cause flaws in the sliding surface, resulting in a reduction in the performance and lifespan of the robot. The robot may malfunction.

– Be sure to use the specified type of grease.
Otherwise, the performance and lifespan of the robot may be reduced. The robot may malfunction.

– Be sure to check the application state of corrosion inhibitor daily.
Otherwise, rust may develop on the ball screw spline unit.
- Corrosion inhibitor must be applied only when it is used up.

Basically, fill the grease to the ball screw spline unit every three (3) months.
If you have verified that there is not much grease, be sure to apply grease.
At daily inspection also, make sure that the ball screw spline unit is filled with a sufficient volume of grease.
Check the condition of anticorrosive in daily inspection. If anticorrosive is dried up, apply it.

Table 2.7 Recommended anticorrosive (THL300, THL400)

Recommended anticorrosive	Maker
KLUBER A20	NOK
WD-40	ST TRADING

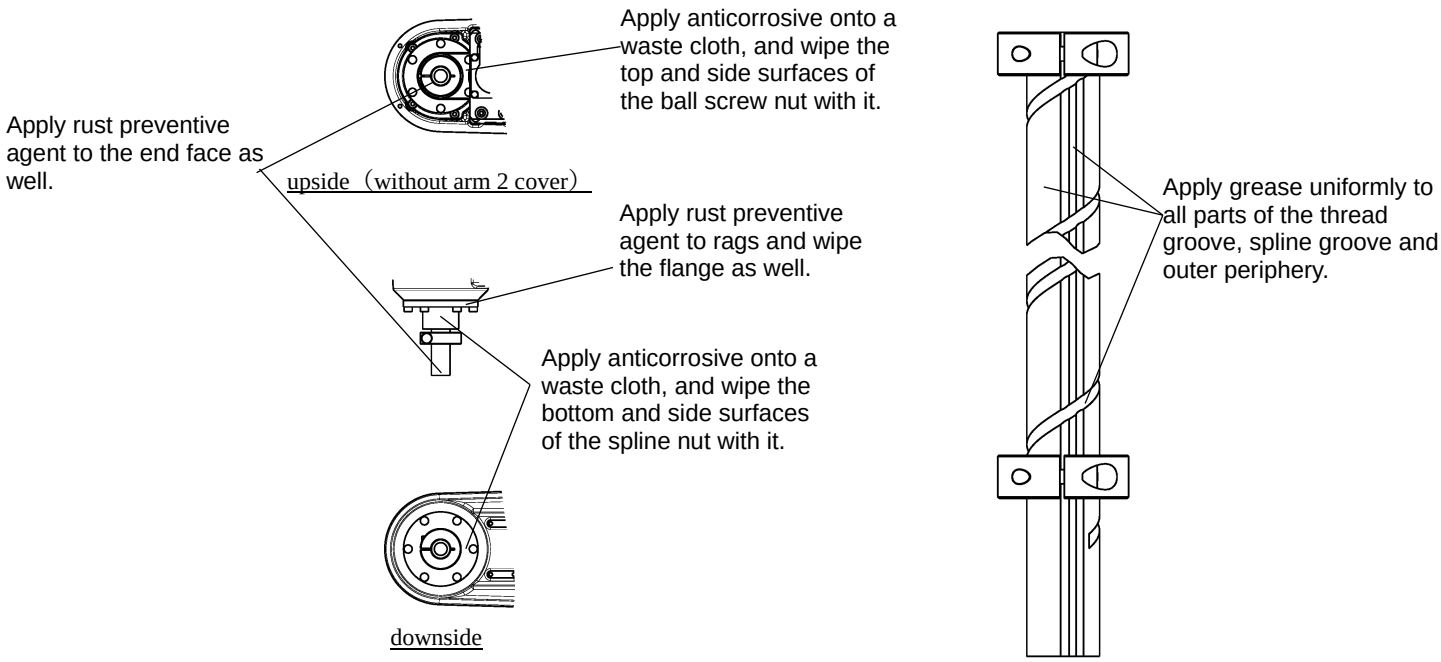


Fig. 2.39 Applying anticorrosive and grease to ball screw spline unit (THL300, THL400)

1) Remove the arm 2 cover (see “Para. 2.3.1, Arm 2 Cover.”)
2) Connect the controller power supply plug and turn off the servo system.
3) Move the arm to a position where Axis 3 can be moved over the full stroke by hand.
4) Push down the ball screw spline shaft to the lower limit while pressing the axis 3 brake release switch.

5) Directly apply the grease to the exposed shaft area by brash.
Apply the grease to such an extent that the shaft groove can be filled.

Table 2.8 Recommended grease (THL300, THL400)

Recommended grease	Maker
AFF grease	THK

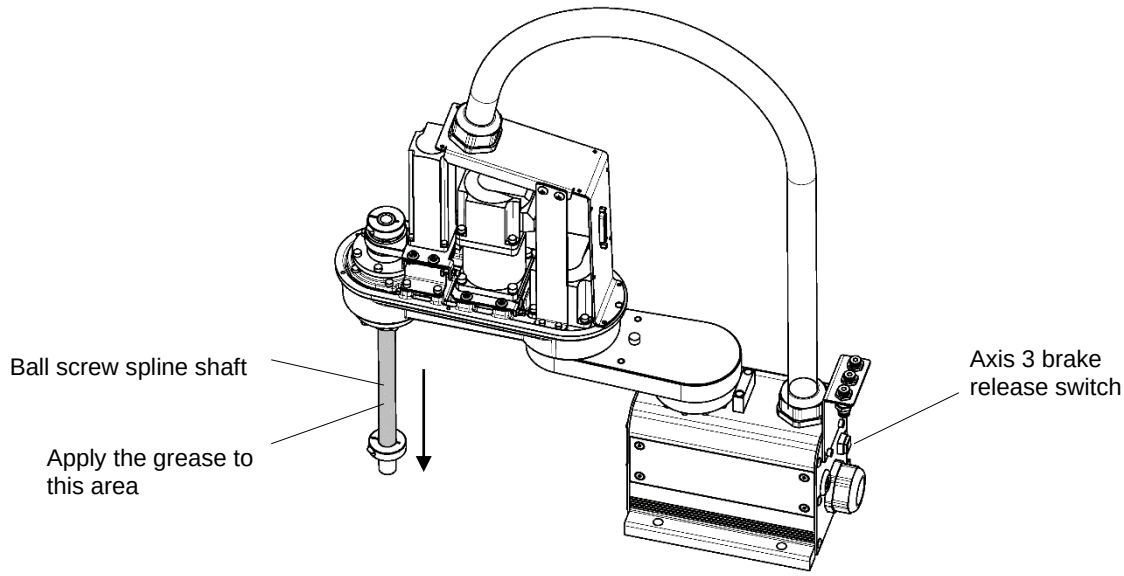


Fig. 2.40 Greasing lower side of ball screw spline unit (THL300, THL400)

6) Push up the shaft up to the upper limit while pressing the axis 3 brake release switch.

7) Directly apply the grease by brash to the shaft area sticking up from the cover. Apply the grease to such an extent that the shaft groove can be filled. For the portions (two positions) to be coated with grease, see Figure 2.39.

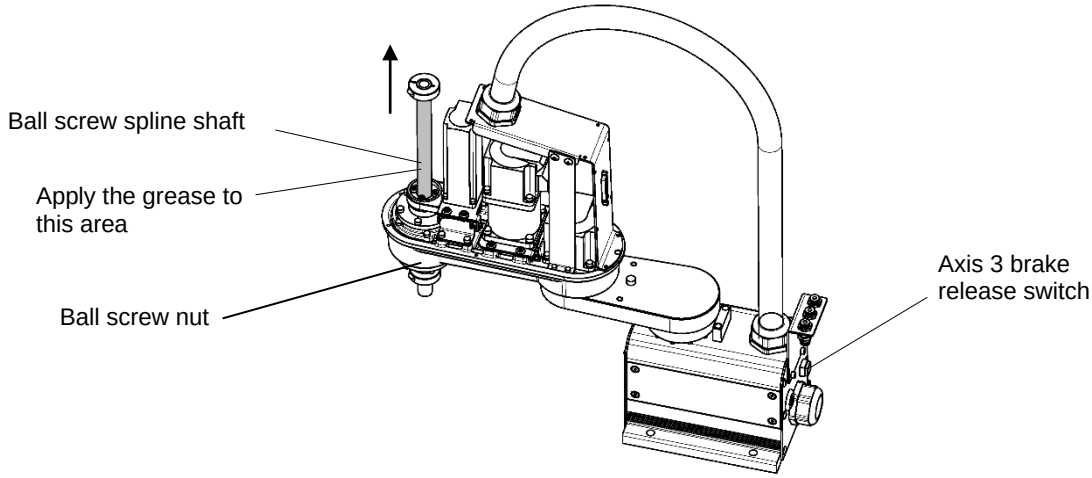


Fig. 2.41 Greasing upper side of ball screw spline unit (THL300, THL400)

8) Move the shaft up and down repeatedly while pressing the axis 3 brake release switch to fill the grease uniformly. Wipe out the surplus grease. Now the work completes.

2.6.4 Dismounting Ball Screw Spline Unit

⚠DANGER

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– Never touch the ball screw spline unit with bare hands.
Touching it with bare hands causes the ball screw spline unit to rust.
- Be sure to wear gloves.
– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The robot may malfunction if you replace it with the power on.
– Be sure to handle the ball screw spline unit carefully.
Drop impact or application of excessive external force could ruin the unit.
– After the ball screw spline unit is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

In the descriptions on replacing the ball screw spline unit, the procedures for changing the axis 3 and 4 motors and timing belts are not included. For details, refer to the descriptions on replacement of each part (see “Para. 2.4.7, Dismounting Axis 3 Motor”, “Para. 2.4.9, Dismounting Axis 4 Motor”, “Para. 2.5.3, Replacing Axis 3 Timing Belt” and “Para. 2.5.4, Replacing Axis 4 Timing Belt.”)

1) Remove the arm 2 cover. (See “Para. 2.3.1, Arm 2 Cover.”)
2) Remove the cross recessed flat head screw (M3 x 6 x 1 pc.) and the hexagon socket head cap screws (M3 x 6 x 2 pcs.), and then remove the harness guide and the support plate.
3) Disconnect the hand, tool, etc. followed by the lower stoppers.
4) Remove the axes 3 motor assemblies as well as the axis 3 timing belt. And, loosen axis 4 belt tension.

5) Remove the hexagon socket head cap screws (M4 x 12 x 6 pcs.), and then pull out the ball screw nuts and the ball screw spline shaft together upward. Be cautious when pulling them out.

NOTICE

– Never pull out the ball screw nut from the ball screw spline shaft.
Otherwise, the ball screw spline unit from which balls have come out cannot be used.

6) Remove the hexagon socket head cap screws (M4 x 16 x 4 pcs.) securing the axis 3 bracket (casting), and then remove the axis 3 bracket from Arm 2.

7) Remove four (4) hexagon socket head cap screws (M4 x 25) and washers, and then remove the axis 3 nut pulley and upper stopper which is secured to the ball screw nut.

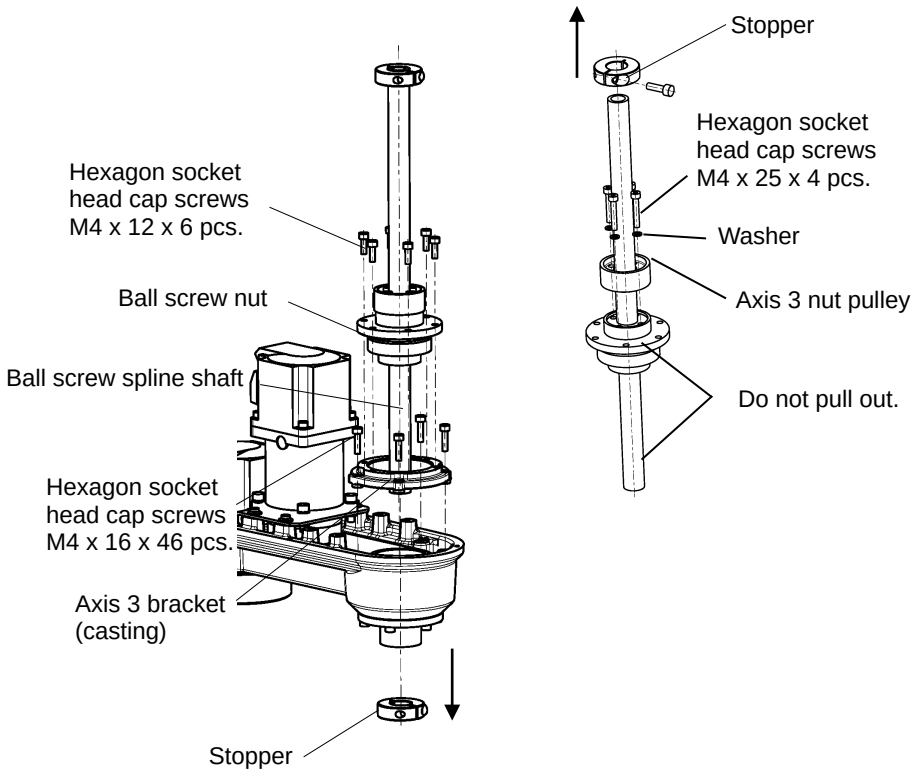


Fig. 2.42 Dismounting ball screw spline unit (THL300, THL400)

8) Remove the hexagon socket head cap screw (M4 x 12 x 6 pcs.) securing the ball spline nut, and pull out the ball spline nut downward.
If it is hard to remove the ball spline nut, lightly tap the pulley with a plastic hammer or similar tool that does not make scratches and then remove the ball spline nut. If the pulley is pounded with a hammer or forcibly pulled out, Arm 2 may be deformed and, as a result, a new ball screw cannot be mounted accurately.

9) Disconnect the axis 4 nut pulley secured to the ball spline nut with the hexagon socket head cap screw (M4 x 10 x 6 pcs.).

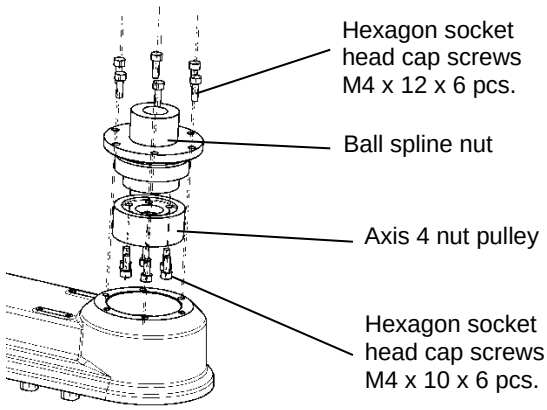


Fig. 2.43 Dismounting ball screw spline nut (THL300, THL400)

10) Insert the disconnected ball spline nut into the ball screw spline shaft.
A marking for phase adjustment is stamped on both the ball screw spline shaft and the ball spline nut. Align the two markings when inserting the ball spline nut.
To prevent the nut from slipping off, wind cable ties around near the top and bottom of the shaft.

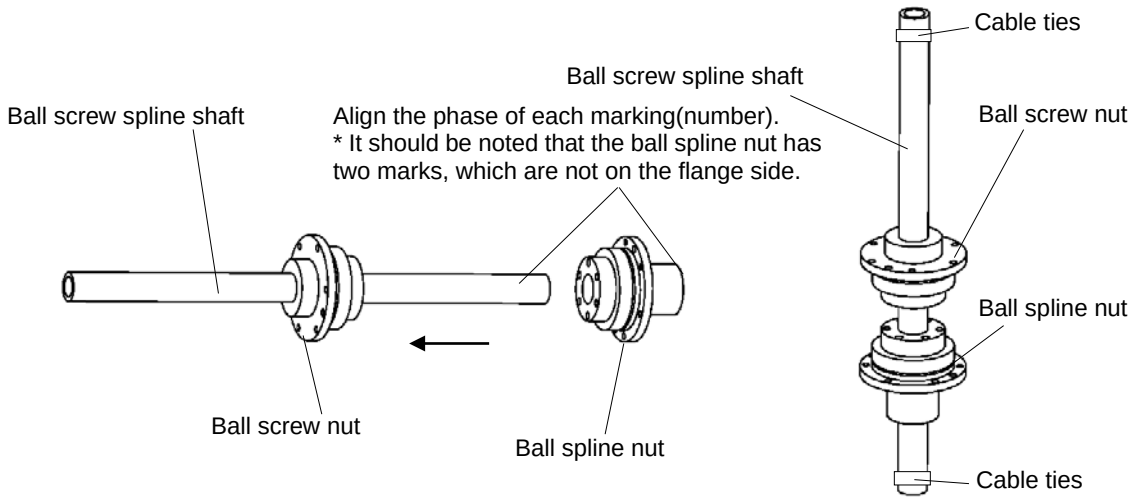


Fig. 2.44 Ball screw spline nut (THL300, THL400)

2.6.5 Mounting Ball Screw Spline Unit

⚠DANGER

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– Never touch the ball screw spline unit with bare hands.
Touching it with bare hands causes the ball screw spline unit to rust.
- Be sure to wear gloves.
– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– Be sure to handle the ball screw spline unit carefully.
Drop impact or application of excessive external force could ruin the unit.
– After the ball screw spline unit is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

1) Disconnect the ball spline nut from the new ball screw spline unit.

2) Mount the axis 4 nut pulley on the ball spline nut with the hexagon socket head cap screws (M4 x 10 x 6 pcs.) and washers.
The bolts are difficult to tighten because the pulley rotates. Thus, tighten them while holding the pulley with pliers or the likes. In doing so, use a waste cloth to protect the pulley from scratches.

3) Mount the ball spline nut on Arm 2 with the hexagon socket head cap screws (M4 x 12 x 6 pcs.) and washers. When doing so, be careful with the orientation of the removal tap.

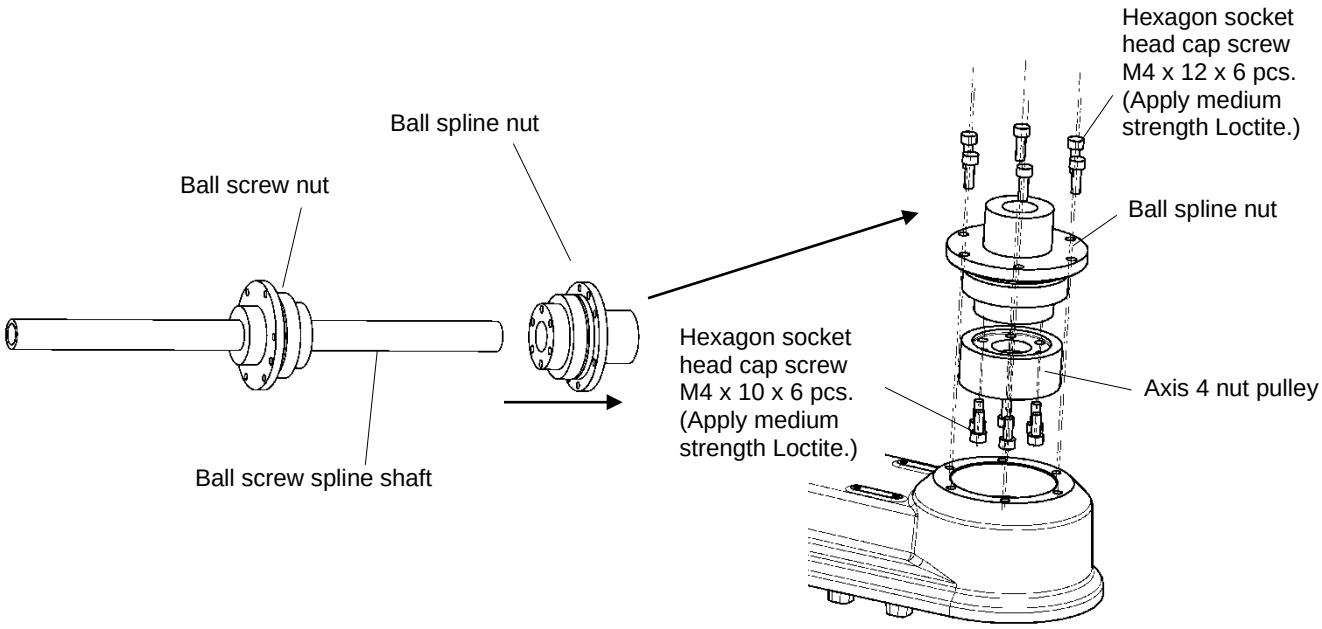


Fig. 2.45 Mounting ball spline nut (THL300, THL400)

NOTICE

– Never pull out the ball screw nut from the ball screw spline shaft.
Otherwise, the ball screw spline unit from which balls have come out cannot be used.

4) Fasten the pulley for the axis 3 nut to the ball screw nut with the hexagon socket head cap screws (M4 x 25 x 4 pcs.) and the washers and fasten upper stopper..

5) Temporarily fasten the axis 3 bracket (casting) to Arm 2 with the hexagon socket head cap screws (M4 x 16 x 4 pcs.). Fasten the ball screw nut to the axis 3 bracket (casting) with the hexagon socket head cap screws (M4 x 12 x 6 pcs.). At this time, hang the axis 4 timing belt on the axis 4 pulley.

6) Mount one stopper at 27 mm from the bottom end of the ball screw spline shaft. Install the stoppers by aligning the notches of the stoppers with the leftmost spline groove.

7) Move the ball screw spline shaft up and down to make it fit. After that, holding the 3rd-axis bracket (casting) so that it does not move, additionally tighten the hexagon socket head bolts (M4 x 16 x 4 bolts) fixing the 3rd-axis bracket (casting).

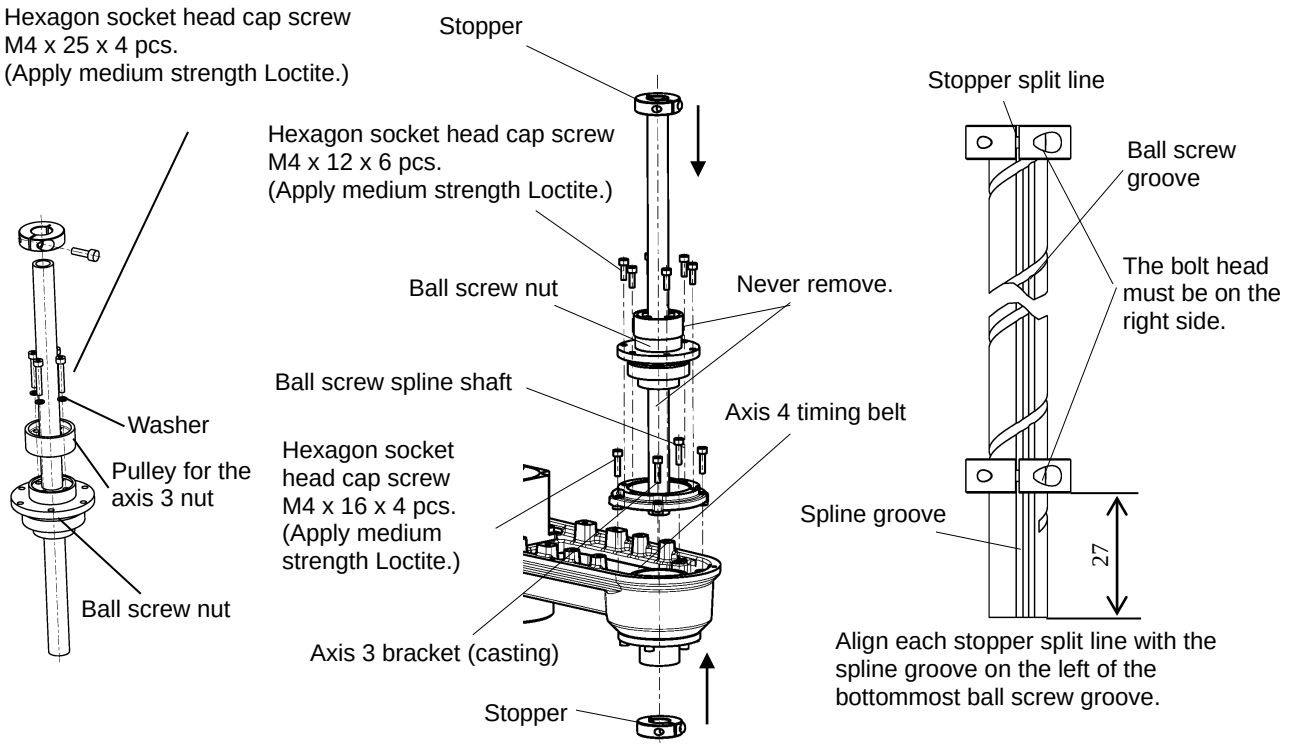


Fig. 2.46 Mounting ball screw spline nut (THL300, THL400)

8) Mount the axis 4 motor assembly and adjust the belt tension. (See "Para. 2.5.4, Replacing Axis 4 Timing Belt.")

9) Mount the axis 3 motor assembly and the axis 3 timing belt, and then adjust the belt tension. (See "Para. 2.5.3, Replacing Axis 3 Timing Belt.")

10) Mount the harness guide and the support plate with the cross recessed flat head screw (M3 x 6 x 1 pc.) and the hexagon socket head cap screws (M3 x 6 x 2 pcs.), and restore the connectors and the cables to the original conditions. (Refer to the photos taken in advance for reference.)

11) Perform home setting for Axes 3 and 4.
When performing home setting for the axis 4, align the stopper split line with the home point match-mark on the ball spline nut.
After home setting, remove the mating mark from the old ball spline nut, and attach it to the new one. For the location of home point match-mark, see Para. 8.4.1, Location of Robot Home Point Match-Marks (THL300, THL400).

12) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

13) Mount the arm 2 cover. Now replacement of the ball screw spline unit completes.

2.7 Reduction Gear Replacement

The reduction gear is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠DANGER

– When replacing the reduction gear, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– When replacing the reduction gear, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.
– After the reduction gear is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”

2.7.1 Type of Reduction Gear

The reduction gears used in this robot are shown below.
When you place an order for the reduction gear for replacement, specify the robot model (THL300, THL400), axis name, type and our drawing number.

Table 2.9 Type of reduction gear (THL300, THL400)

Description	Axis name	Our drawing No.	Unit code
Reduction gear (A type)	Axis 1	S890907	Y610A3NG0
	Axis 2	S890906	Y610A3NH0
Reduction gear (B type)	Axis 1	S958148	Y610D06N0
	Axis 2	S958147	Y610D06M0
Reduction gear	Axis 4	S890969	Y610A3NJ0

2.7.2 Reduction Gear Locations

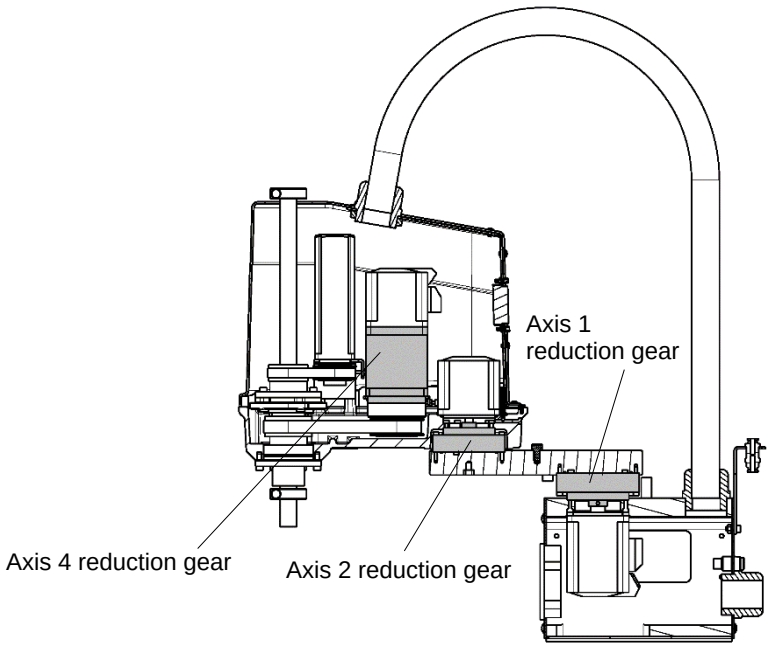


Fig. 2.47 Reduction gear locations (THL300, THL400)

2.7.3 Dismounting Axis 1 Reduction Gear

1) Remove the base front cover and the base side covers. (See “Para. 2.3.2, Base Covers.”)

2) Remove four (4) hexagon socket head cap screws (M4 x 14) and the washers that secure the axis 1 motor assembly, pull out the axis 1 motor assembly, and then remove the O-ring (CO0534A).
(See “Para. 2.4.3, Dismounting Axis 1 Motor.”)

3) Remove sixteen (16) hexagon socket head cap screws (M3 x 20) that secure the base, and then remove Arm 1.

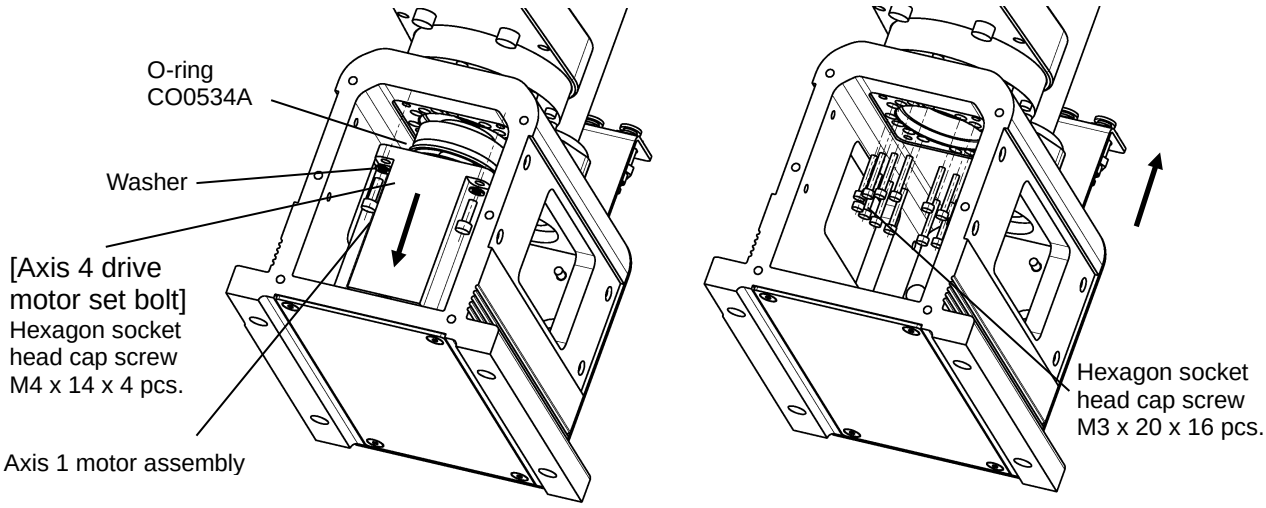


Fig. 2.48 Dismounting axis 1 motor assembly and Arm 1 (THL300, THL400)

⚠ CAUTION

– When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

– Be sure to handle the arm carefully.
Drop impact or application of excessive external force could ruin the arm.

4) Remove twelve (12) hexagon socket head cap screws (M3 x 25) that secure the axis 1 reduction gear to Arm 1, and then remove the axis 1 reduction gear. Also remove the O-ring mounted in the groove of Arm 1 at this time. Also, remove the liquid gasket attached to the base mounting surface of the axis 1 reduction gear with the back of a cutter or a similar tool without damaging.

5) Remove the wave generator from the axis 1 motor. (See “Para. 2.4.3, Dismounting Axis 1 Motor.”)

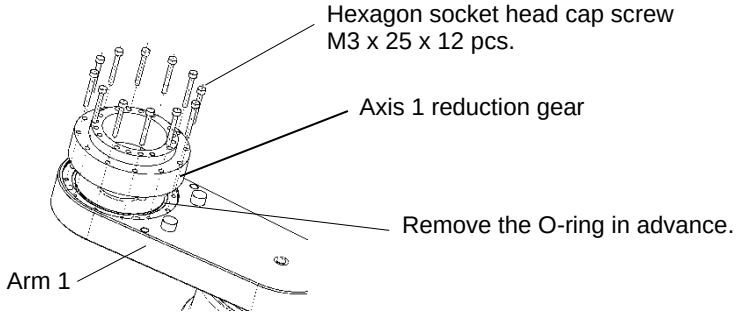


Fig. 2.49 Dismounting axis 1 reduction gear (THL300, THL400)

2.7.4 Mounting Axis 1 Reduction Gear

NOTICE

– Be sure to use wave generator provided for new reduction gear to be replaced.
The combination of the reduction gear and wave generator is determined. If they are used in a wrong combination, abnormal sound will occur. The positioning accuracy and lifespan will be reduced.
– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.

1) Mount a new wave generator provided as an accessory with a new reduction gear. For the mounting procedure, see item 1) in “Para. 2.4.4, Mounting Axis 1 Motor.”)

2) Cleanly wipe off dusts and stains on the axis 1 reduction gear of Arm 1. Apply grease to an O-ring attached to the new reduction gear, and then mount it in the O-ring groove of Arm 1.

3) Secure the axis 1 reduction gear with twelve (12) hexagon socket head cap screws (M3 x 25).

Reduction Gear	Recommended anticorrosive	Maker
A type	SK-Ⅱ	HDS
B type	SFB No.1	Nidec-Shimpo Corp.

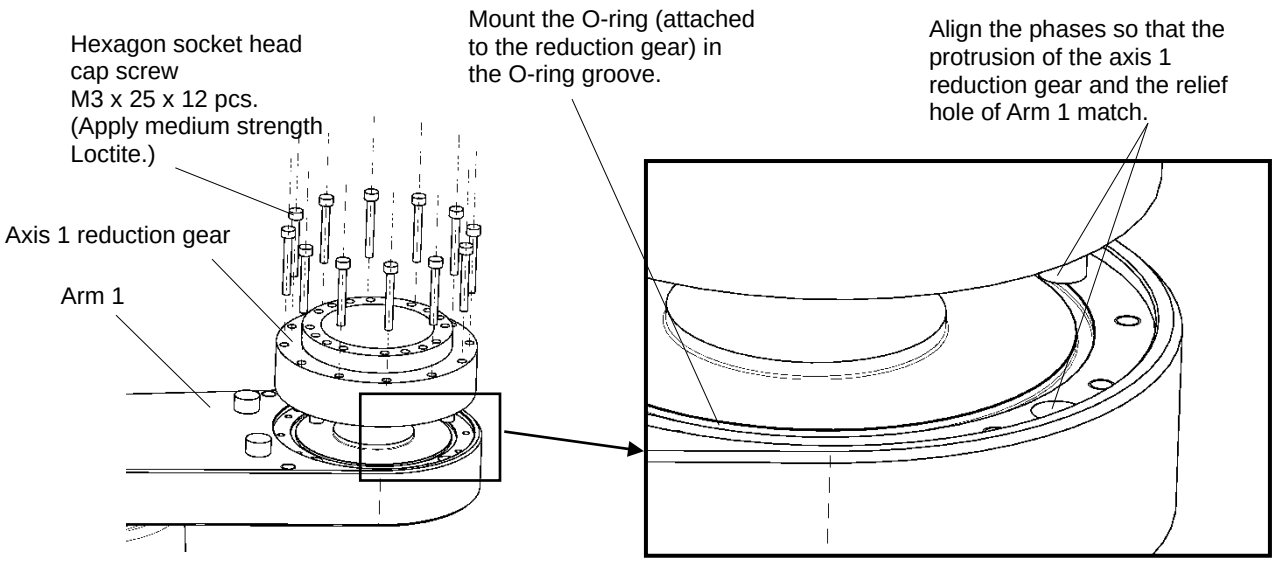


Fig. 2.50 Mounting axis 1 reduction gear (THL300, THL400)

4) Apply grease at a thickness of the mark onto the inside of the axis 1 reduction gear's main body. The mark is to gear inside wall from the protrusion on Arm 1. Apply an appropriate amount of grease to the wave generator also. Apply liquid gasket onto the base mounting surface of the axis 1 reduction gear.

Reduction gear	Amount of grease filled in Axis 1
A type	13 g
B type	25 g

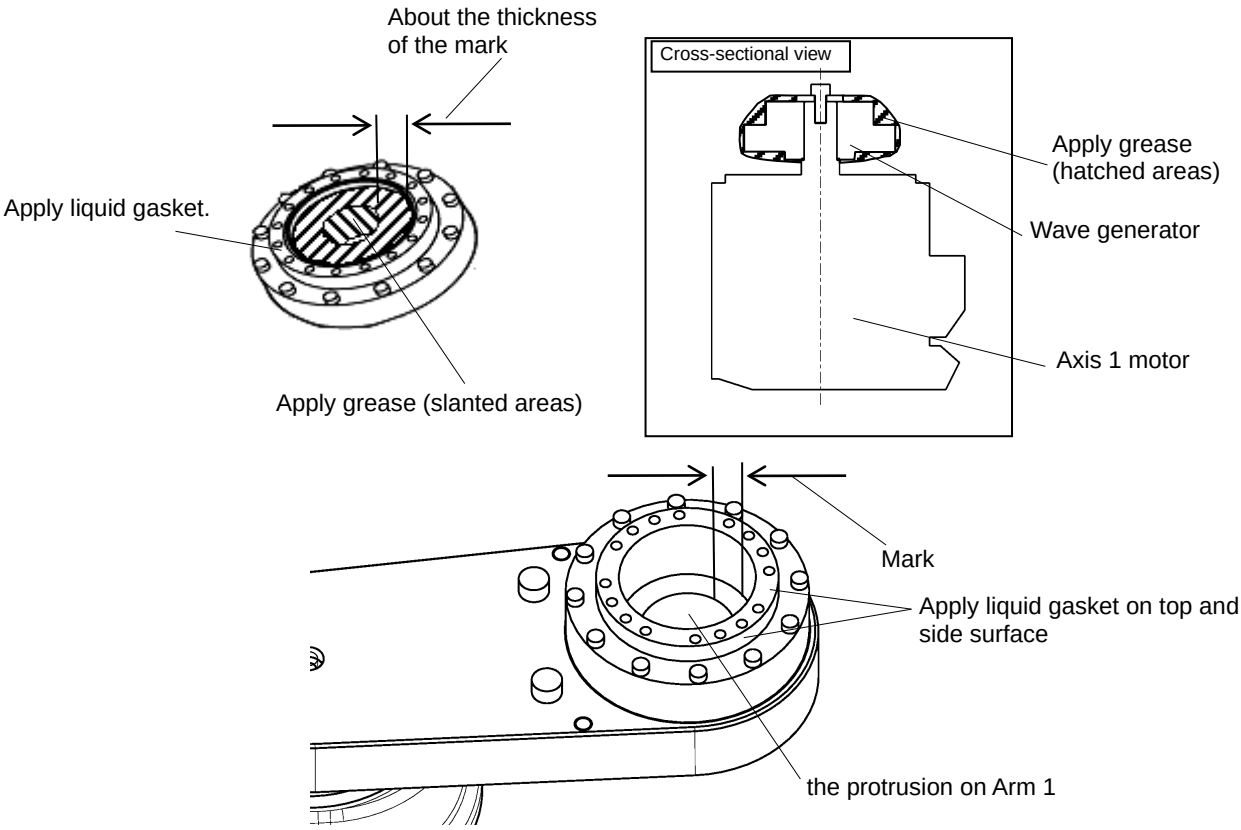


Fig. 2.51 Application of grease to axis 1 reduction gear (THL300, THL400)

NOTICE

– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

– Be sure to observe the filling amount of grease.
If the filling amount is tool large, the starting torque increases and the internal seal is broken due to rise in the internal pressure in the reduction gear. If the filling amount is too small, malfunction is caused by poor lubrication.

5) Secure Arm 1 to the base with sixteen (16) hexagon socket head cap screws (M3 x 20). Apply grease to the O-ring (CO0534A), and then mount it to the base groove side surface.

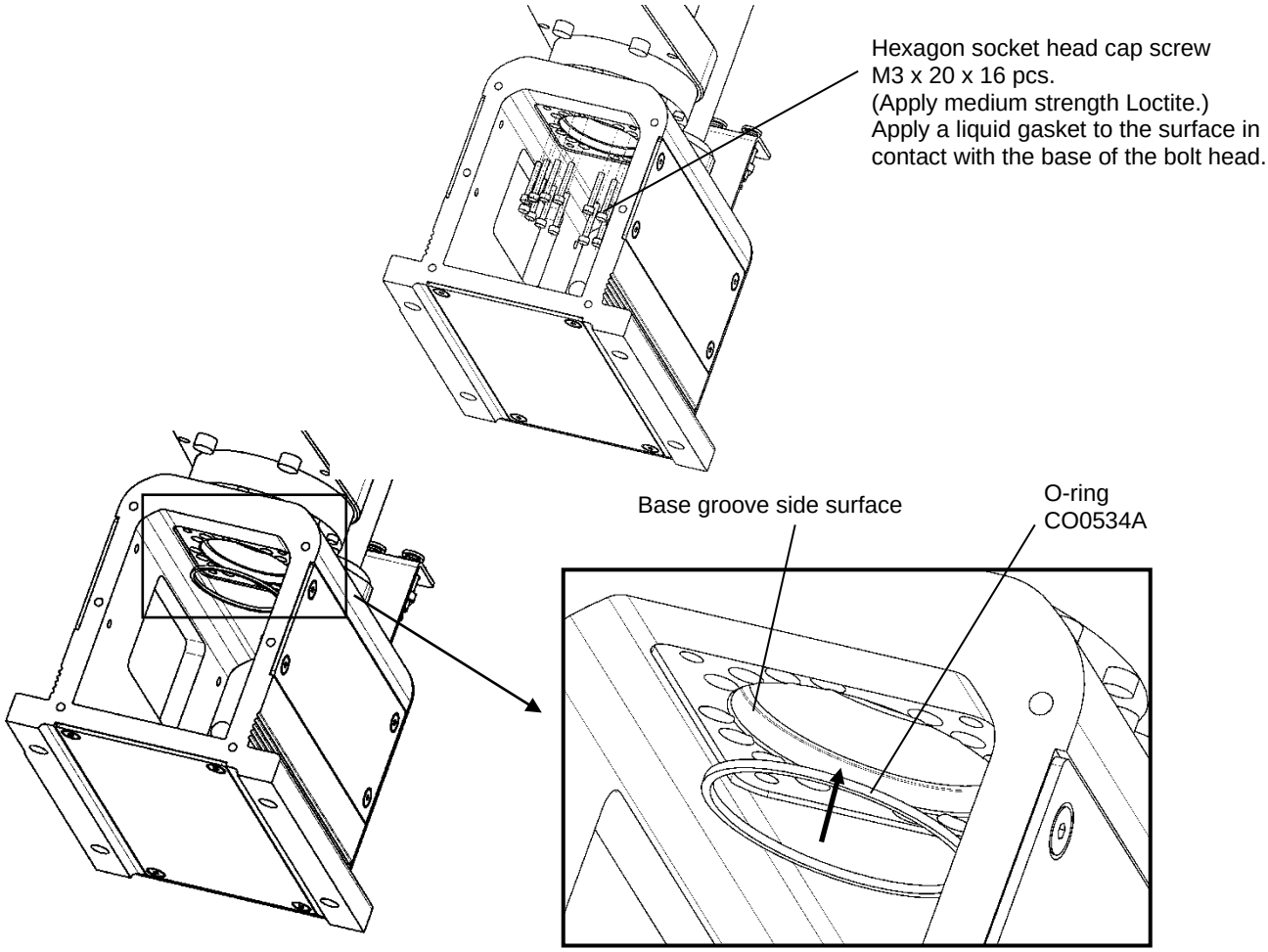


Fig. 2.52 Mounting arm 1 and O-ring (THL300, THL400)

NOTICE

– Never forget fitting the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
- Use the new O ring provided for the reduction gear.

6) Connect the connectors (J1AS and J1BS) of the cables.

7) Mount the axis 1 motor assembly. (See “Para. 2.4.4, Mounting Axis 1 Motor.”)

8) Manually move the 1st arm and make sure of its smooth movement.

9) Mount the base front cover and the base side covers. (See “Para. 2.3.2, Base Covers.”)

10) Set up the axis 1 home position to complete axis 1 reduction gear replacement.

11) Carry out a test operation of Axis 1 to make sure that each part operates properly.

2.7.5 Dismounting Axis 2 Reduction Gear

1) Remove Arm 2 cover. (See “Para. 2.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers. Disconnect connectors J2AS and J2AP (power drive cables), connectors J2BS and J2BP (encoder cables).
It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Remove the harness guide and the support plate, and then remove the axis 2 motor assembly. Also remove the O-ring (CO0538A) of the O-ring groove. (See “Para. 2.4.5, Dismounting Axis 2 Motor.”)

4) Remove eight (8) hexagon socket head cap screws (M3 x 16) that secure Arm 2 and the reduction gear.

⚠ CAUTION

– When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

– Be sure to handle the arm carefully.
Drop impact or application of excessive external force could ruin the arm.

5) Remove eight (8) hexagon socket head cap screws (M3 x 25) that secure the axis 2 reduction gear mounted to Arm 1, and then remove the axis 2 reduction gear. At this time, remove the O-ring mounted in the O-ring groove of Arm 1. At this time, remove the liquid gasket attached to the base mounting surface of the axis 1 reduction gear without damaging. Also, remove the O-ring (that comes with the reduction gear) in the O-ring groove.

6) Remove the wave generator from the axis 2 motor. (See “Para. 2.4.5, Dismounting Axis 2 Motor.”)

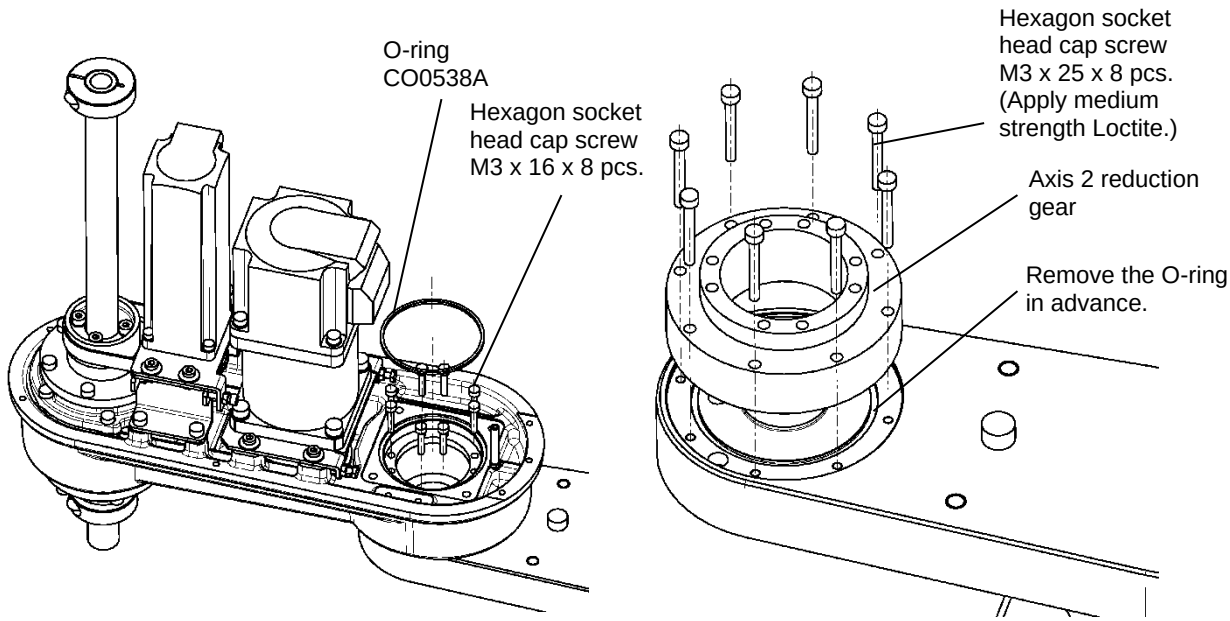


Fig. 2.53 Dismounting axis 2 reduction gear (THL300, THL400)

2.7.6 Mounting Axis 2 Reduction Gear

NOTICE

– Be sure to use wave generator provided for new reduction gear to be replaced.
The combination of the reduction gear and wave generator is determined. If they are used in a wrong combination, abnormal sound will occur. The positioning accuracy and lifespan will be reduced.
– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.

1) Mount the wave generator attached to the new reduction gear to the axis 2 motor. At this time, apply grease to the O-ring (CO0538A) and then mount it to the axis 2 motor flange rabbet. (See “Para. 2.4.6, Mounting Axis 2 Motor.”)

2) Cleanly wipe off dusts and stains on the reduction gear of Arm 1.
Apply grease to an O-ring attached to the new reduction gear, and then mount it in the O-ring groove of Arm 1.

3) Secure the axis 2 reduction gear to Arm 1 with eight (8) hexagon socket head cap screws (M3 x 25).

4) Apply grease to the axis 2 reduction gear. (See “Para. 2.7.4, Mounting Axis 1 Reduction Gear.”) However, the amount of grease filled is different from that for the axis 1 reduction gear. Regarding the filled amount for the axis 2 reduction gear, see the following.)

Reduction gear	Amount of grease filled in Axis 2
A type	8.9 g
B type	20 g

5) Apply liquid gasket onto the arm 2 mounting surface of the new reduction gear.

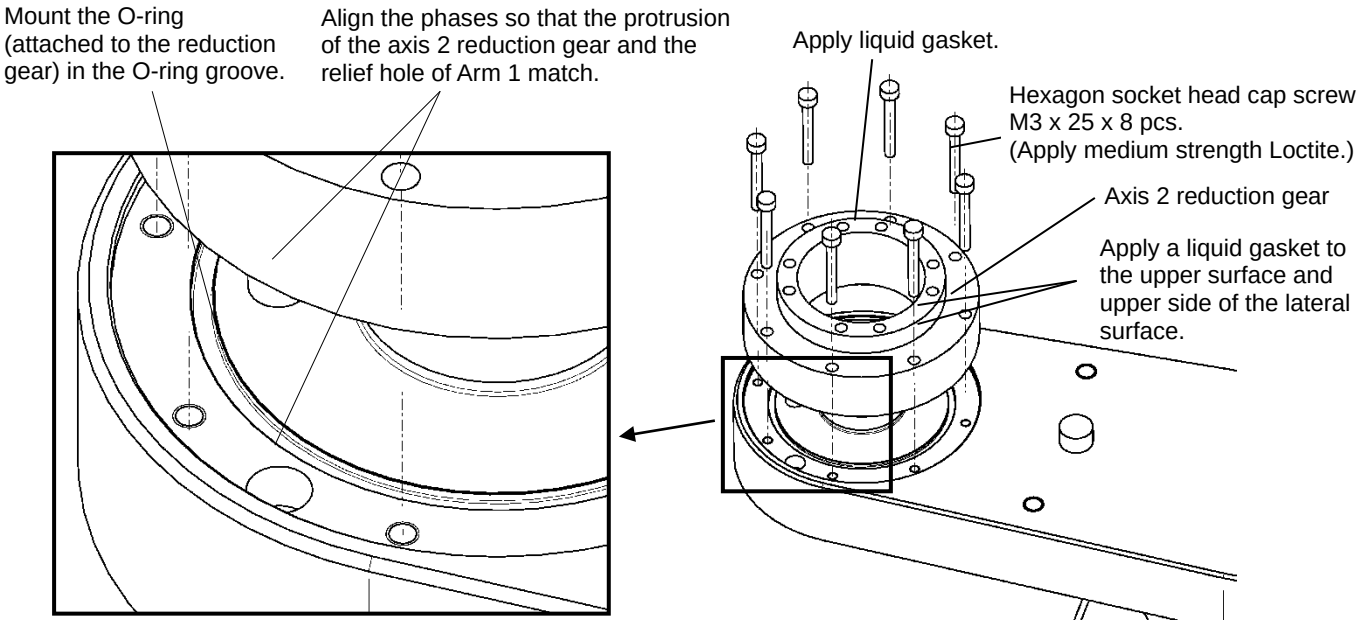


Fig. 2.54 Mounting axis 2 reduction gear (THL300, THL400)

NOTICE

– Never forget fitting the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
- Use the new O ring provided for the reduction gear.

– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

– Be sure to observe the filling amount of grease.
If the filling amount is tool large, the starting torque increases and the internal seal is broken due to rise in the internal pressure in the reduction gear. If the filling amount is too small, malfunction is caused by poor lubrication.

6) Secure Arm 2 to the new reduction gear with eight (8) hexagon socket head cap screws (M3 x 16).

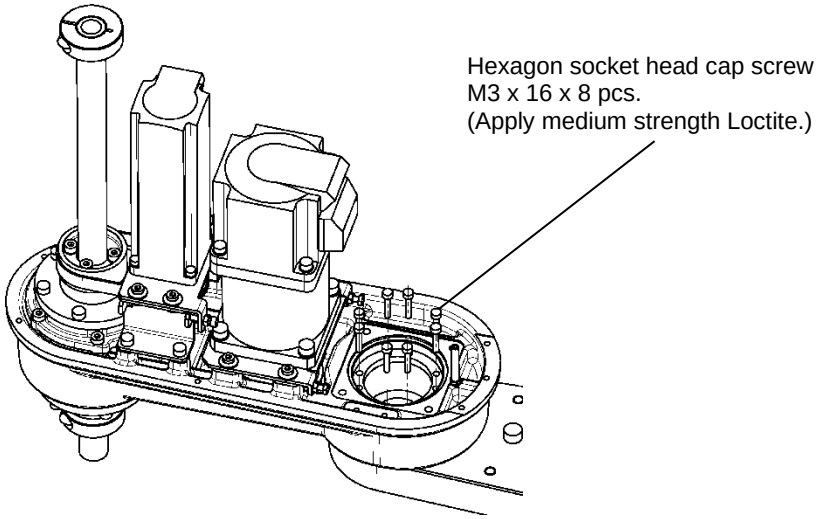


Fig. 2.55 Mounting Arm 2 (THL300, THL400)

⚠ CAUTION

– When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

– Be sure to handle the arm carefully.
Drop impact or application of excessive external force could ruin the arm.

7) Mount the axis 2 motor assembly, the harness guide and the support plate. Put back the cables to the original locations, and then mount Arm 2 cover. For the installation procedure, see "2.4.6 Installation of the 2nd-axis motor"

8) Manually move the 2nd arm and make sure of its smooth movement.

9) Set up the home positions of axes 2 to complete axis 4 reduction gear replacement.

2.7.7 Replacing Axis 4 Reduction Gear

NOTICE

– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.
– Be careful not to forget to loosen the coupling that fastens the reduction gear and motor shaft when dismounting reduction gear.
– Be careful not to forget tightening the coupling that fastens the reduction gear and motor shaft when dismounting reduction gear.

1) Remove Arm 2 cover. (See “Para. 2.3.1, Arm 2 Cover.”)
2) Cut off the cable tie with a nipper, and remove the J4AS and J4AP (2nd axis power cable) and the J4BS and J4BP (2nd axis encoder cable). It is recommended to take a photograph beforehand to make sure that the cable routing position and cable tie position can be easily identified at the time of re-assembling.
3) Referring to "2.5.4 Replacement of 4th-axis timing belt", remove only the 3rd-axis motor assembly and 4th-axis motor assembly from the 2nd arm. Other parts need not be removed.
4) Remove the cap attached to the lateral side of the axis 4 reduction gear, then loosen the coupling bolt (M3) securing the axis 4 motor shaft and the input shaft of the axis 4 reduction gear.
5) Remove the hexagon socket head cap screws (M4 x 12 x 4 pcs.) and washers securing the motor, and draw out the axis 4 motor upward.
6) While holding the axis 4 motor pulley with pliers or the likes, remove the hexagon socket head cap screw (M4 x 12 x 6 pcs.) to remove the pulley. In doing so, use a waste cloth (or the like) as a cushion to protect the pulley from scratches.
7) Remove the hexagon socket head cap screws (M5 x 16 x 4 pcs.) and washers securing the axis 4 motor plate and reduction gear to the motor base, then dismount the reduction gear.

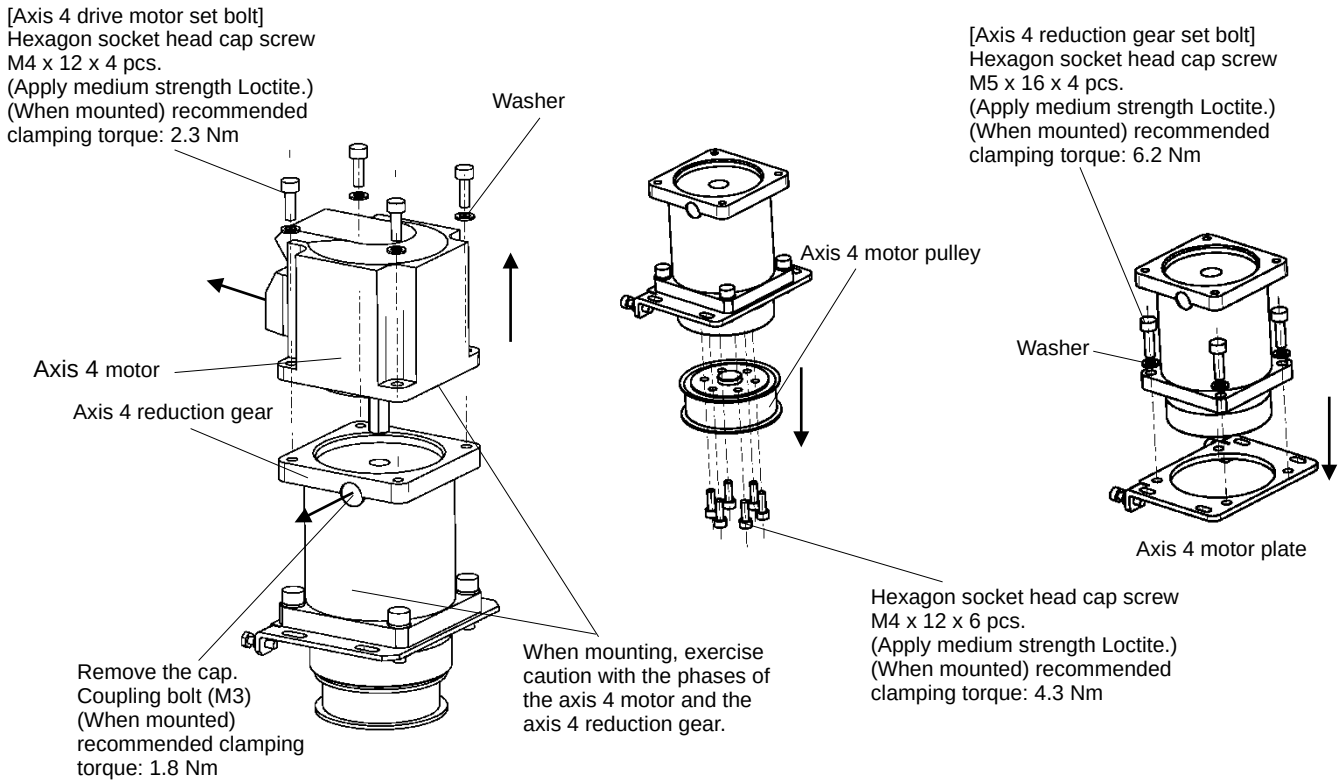


Fig. 2.56 Replacing axis 4 reduction gear (THL300, THL400)

8) Assemble the new reduction gear, motor and pulley in the reverse order of Steps 4) to 7) above. Be careful with the phases of the axis 4 motor, the axis 4 reduction gear and the axis 4 motor plate at this time.

9) Assemble the axis 4 motor assembly and timing belt according to Steps 11) through 15) of “Para. 2.5.4, Replacing Axis 4 Timing Belt” above, then mount Arm 2 cover.

10) Mount the arm 2 cover. Now replacement of Axis 4 Reduction Gear completes.

3. Maintenance of the Main Robot (THL500, THL600, THL700)

3.1 Details of Inspection

3.1.1 Check of Each Bolt (or Screw) for Clamping

⚠ DANGER

– Be sure to perform daily inspection for tool set bolts and robot installation bolts.
Robot hand falls during robot operation. Robot collapses.

NOTICE

– Be sure to perform daily inspection for tool set bolts and robot installation bolts.
If the position is shifted, the robot will bump into peripheral devices.

1) Tool Set Bolts

Using the hexagonal wrench key, make sure that the clamping bolts (M4 x 4 pcs.) of the tool set flange (option), which are clamped to the tool shaft, are tightened completely. If loose, tighten them completely.

Using the hexagonal wrench key, make sure that the mounting bolts (M4 x 4 pcs.), which secure the tool to the tool flange, are tightened completely. If loose, tighten them completely. (Loctite not necessary)

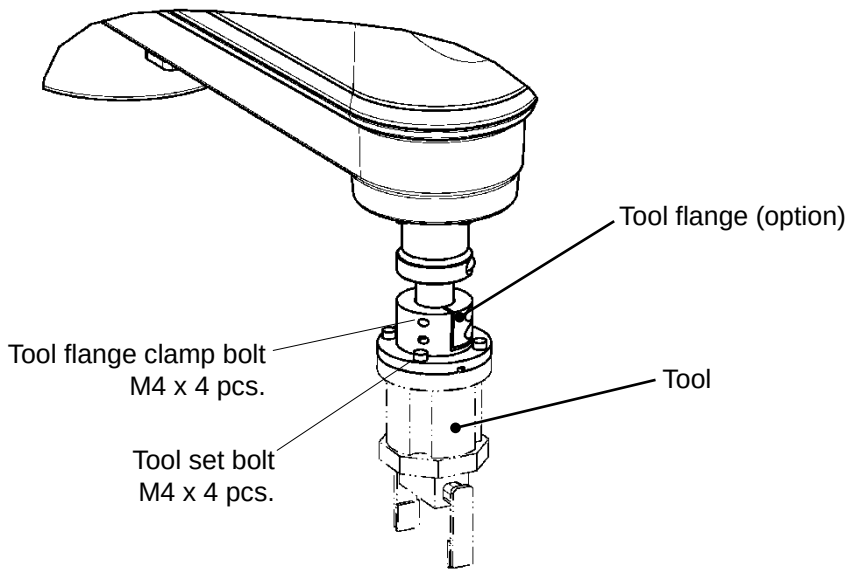


Fig. 3.1 Tool set bolts

2) Robot Installation Bolts

Make sure, using the hexagonal wrench key, that the installation bolts of the main robot base are tightened completely. If loosened, tighten them completely. (Loctite is not necessary)

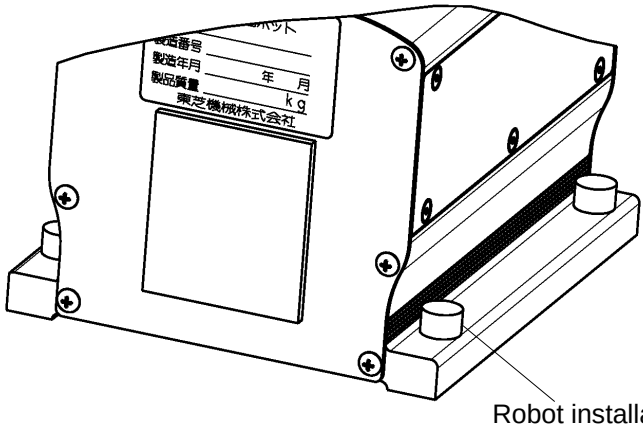


Fig. 3.2 Robot installation bolts (THL500, THL600, THL700)

3) Motor Set Bolts

Make sure, using the hexagonal wrench key, that the bolts for securing each axis drive motor are tightened completely. If loosened, tighten them completely. Also make sure that the motor and motor plate bolts are tightened completely. For the places where the following bolts are used, see “Para. 3.4, Replacing Motor.”

Table 3.1 Motor and motor plate set bolt (THL500, THL600, THL700)

Location of bolt	Type	Bolts used	Reference Para.
Axis 1 drive motor set bolt	M5 x 16	4 pcs.	3.4.4 Mounting Axis 1 Motor
Axis 2 drive motor set bolt	M4 x 14	4 pcs.	3.4.6 Mounting Axis 2 Motor
Axis 3 drive motor set bolt	M4 x 12	4 pcs.	3.4.8 Mounting Axis 3 Motor
Axis 3 motor plate set bolt	Flange head bolt M4 x 10	4 pcs.	3.4.8 Mounting Axis 3 Motor
Axis 4 drive motor set bolt	M4 x 12	4 pcs.	3.4.10 Mounting Axis 4 Motor
Axis 4 reduction gear set bolt	M5 x 16	4 pcs.	3.7.7 Replacing Axis 4 Reduction Gear
Axis 4 motor plate set bolt	Flange head bolt M4 x 10	4 pcs.	3.5.4 Replacing Axis 4 Timing Belt

* For the recommended clamping torque, see Para. 1.5 Clamping Hexagon Socket Head Cap Screws and Set screws
* It is not necessary to apply Loctite to the flange head bolts of Axes 3 and 4 motor plate set bolts.

3.1.2 Check of Each Cable and Air Tube for Abrasion

Disassemble the arm 2 cover and the base rear cover, and make sure that each cable is not worn out, broken or cracked. Especially, carefully inspect the vicinity of the cable outlets.

For how to dismount the covers, see “Para. 3.3, Dismounting and Mounting Each Cover.”

The figure below shows an example of air tube installation.

Also inspect the wear and disorder of the air tube as well as the pipe connection.

Make sure that the air tube is not worn out and is installed properly, and pipe connection is correct. Otherwise, correct them.

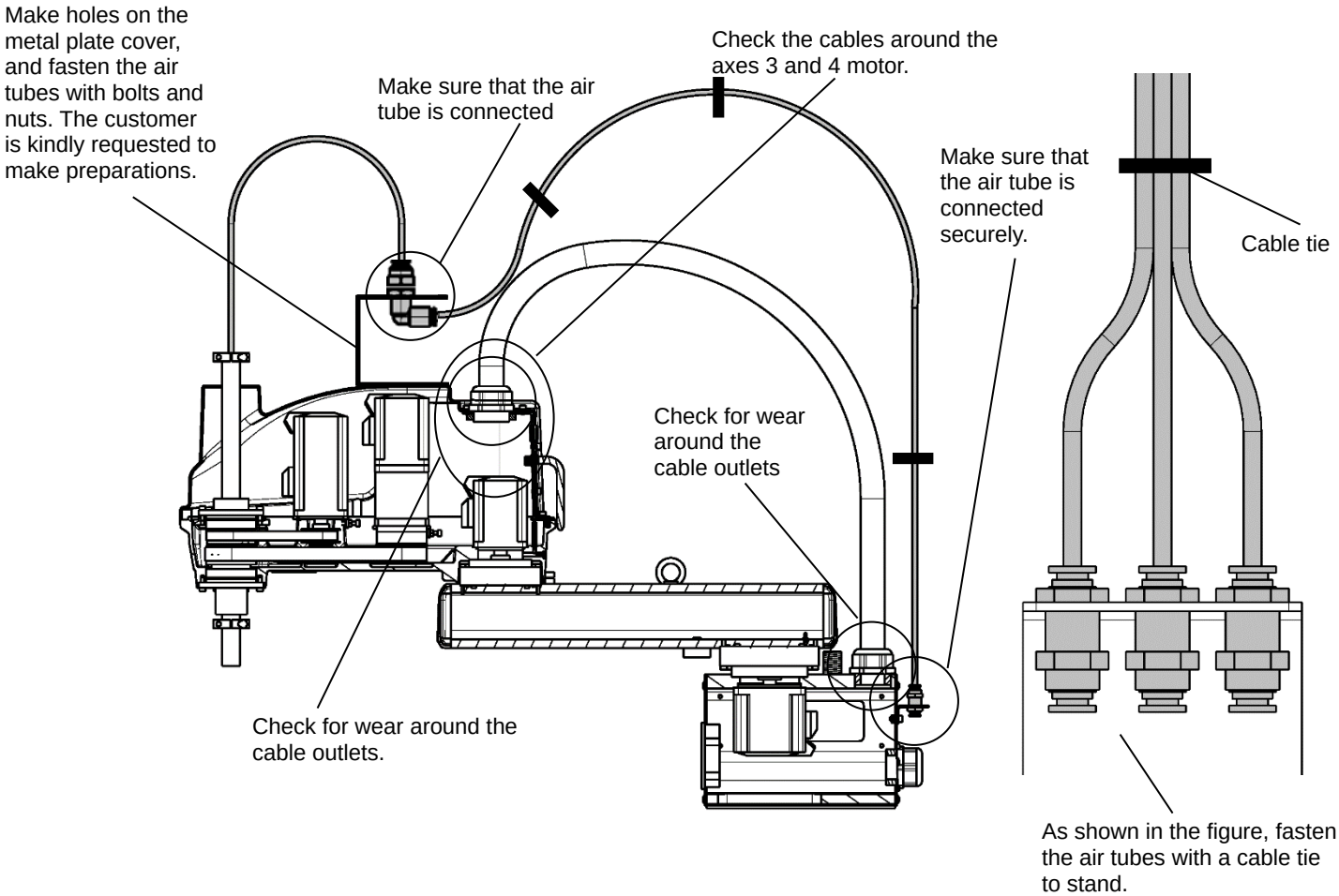


Fig. 3.3 Cable inspection locations (THL500, THL600, THL700)

NOTICE

– The air tube is a consumable item. It must be checked periodically, and replaced if broken.

– Fig. 3.3 shows an example of ductwork.
Note that breakage of air tube, etc. is not always under our warranty.

3.1.3 Check of Cable Clamp Tightening

Using a wrench, make sure that the clamps securing the cable are not loose. There are two clamps, one on the arm 2 side and the other on the base side. If they are loose, apply Loctite (low adhesive force) and tighten them. When checking the tightened clamps, check the one on the base side first. By checking the tightness of the clamp on the base side first, the twist of the cable can be adjusted slightly on the arm 2 side. Make sure to tighten the clamps in the posture as shown in the figure below. If the clamps are tightened while the arm is bent, the cable cannot be installed correctly.

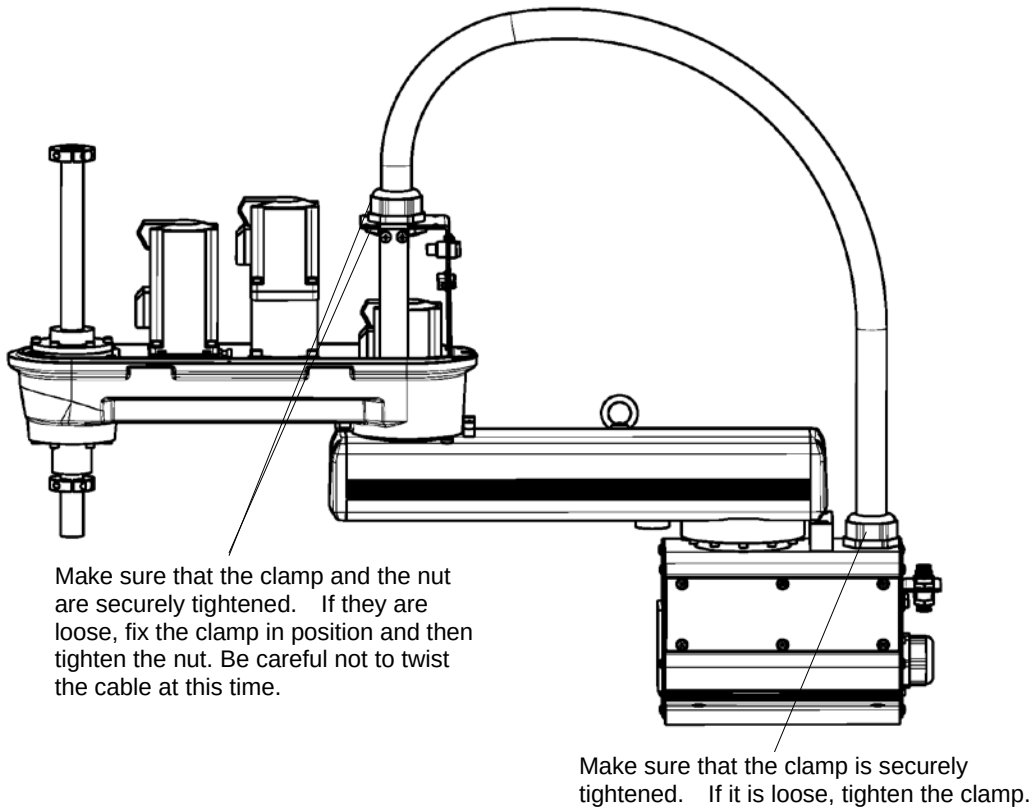


Fig. 3.4 Cable clamp inspection locations (THL500, THL600, THL700)

3.1.4 Check of Each Axis for Operation

Connect the power plug of the controller to the power source, then keep the EMERGENCY pushbutton switch in the depressed condition. Move each axis by hand and make sure that it can move smoothly.
For Axis 3, when the brake release switch is pressed, the brake is released. Take careful precautions at this time as the tool shaft may drop according to the weight of the hand and tool.

NOTICE

· When axis 1 and axis 2 are moved near the working envelope limit and a hand is off from them, axis 1 and axis 2 may move due to reaction of the cable.

3.2 Layout of Robot Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 3.5.

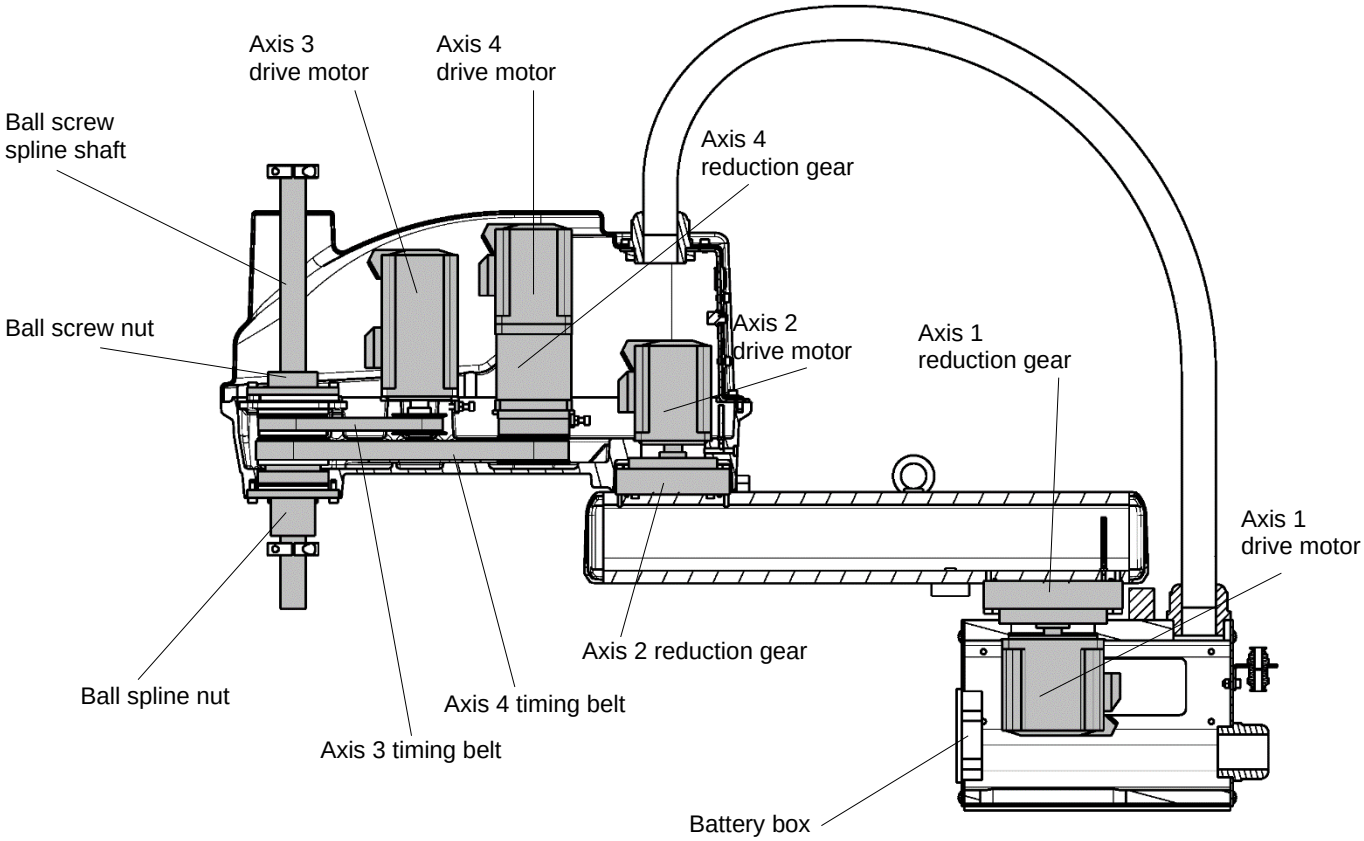


Fig. 3.5 Layout of robot mechanical components (THL500, THL600, THL700)

3.3 Dismounting and Mounting Each Cover

This paragraph describes the dismounting and mounting of the covers, which are common to the maintenance and replacement of each unit.

⚠DANGER

– When opening the cover, never allow entry of moisture or contaminant into the robot.
If the power is turned on while moisture or contaminant is left, an electric shock may result.

NOTICE

– When opening the cover, never allow entry of moisture or contaminant into the robot.
If the power is turned on while moisture or contaminant is left, the robot may malfunction.
– When mounting the covers, prevent the cables from being caught.
Otherwise, the cables will be broken.
– Be sure to restore where to route the cables pulled out by removing the cover and where to secure with the cable straps to their original state.
If the cables are pushed by force or where to secure changes, the cables will be worn out or broken.
- Before pulling out the cables, take pictures for restoration

3.3.1 Arm 2 Cover

The arm 2 cover is secured to Arm 2 and the harness guide with 14 hexagon socket head cap screws (M3 x 16 x 4 pcs., M4 x 6 x 8 pcs., M4 x 10 x 2 pcs.) and nylon washers. (It is not necessary to apply Loctite when installing.)

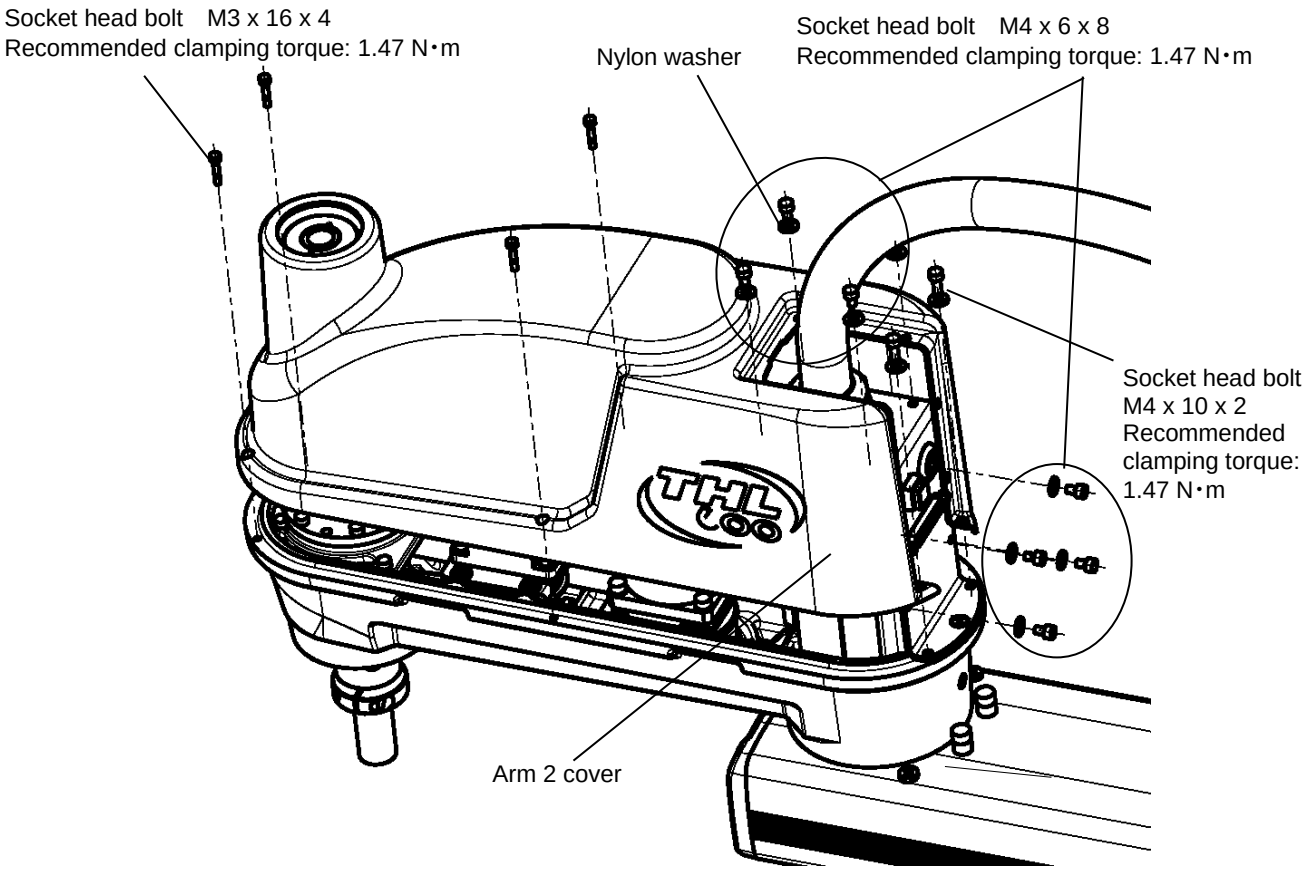


Fig. 3.6 Arm 2 cover (THL500, THL600, THL700)

After the cover is mounted, manually move up and down the ball screw spline shaft while pressing the brake release switch, and make sure that the ball screw hole for the arm 2 cover will not interfere with the ball screw stopper.

3.3.2 Base Covers

There are four (4) types of base covers: base front cover, base rear cover, base side cover and base cover. (It is not necessary to apply Loctite to all the covers when installing.)

Each of the base front cover and the base rear cover is secured to the base with six (6) cross truss head screws (M4 x 6). They can be dismounted from the base when these fastening screws are removed, but do not pull them with force as they are connected with the connectors inside.
Each of the left and right base side covers is secured to the base with six (6) hex countersunk screws (M4 x 12).
The base cover is secured to the base with six (6) hex countersunk screws (M4 x 8).

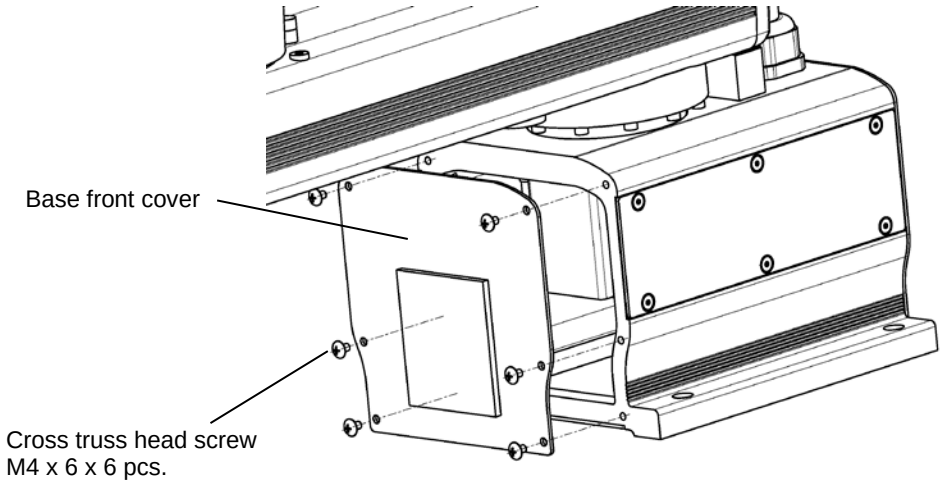


Fig. 3.7 Base front cover (THL500, THL600, THL700)

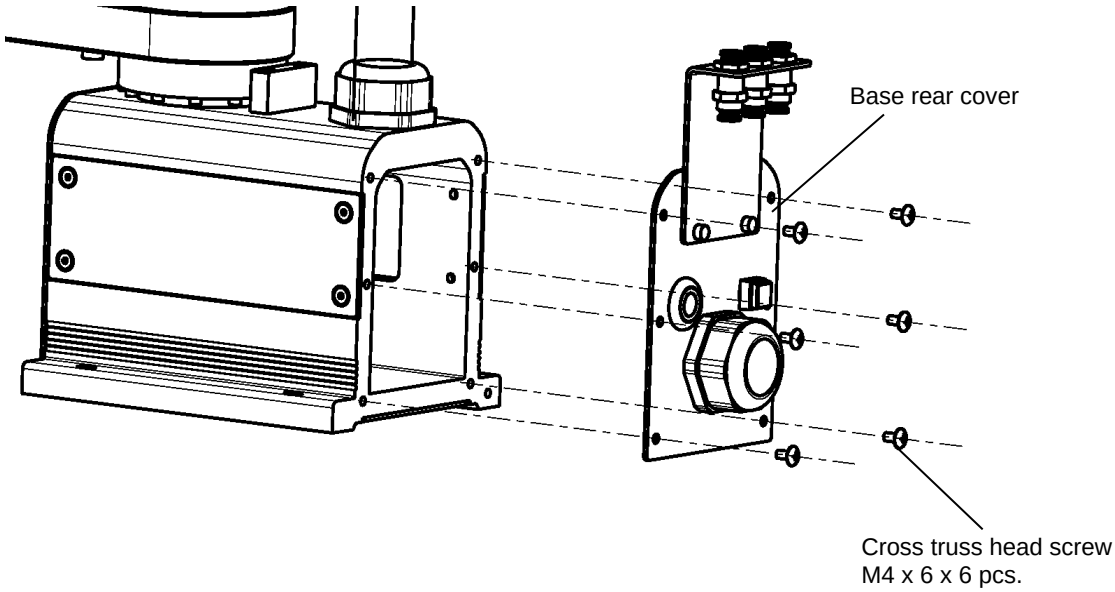


Fig. 3.8 Base rear cover (THL500, THL600, THL700)

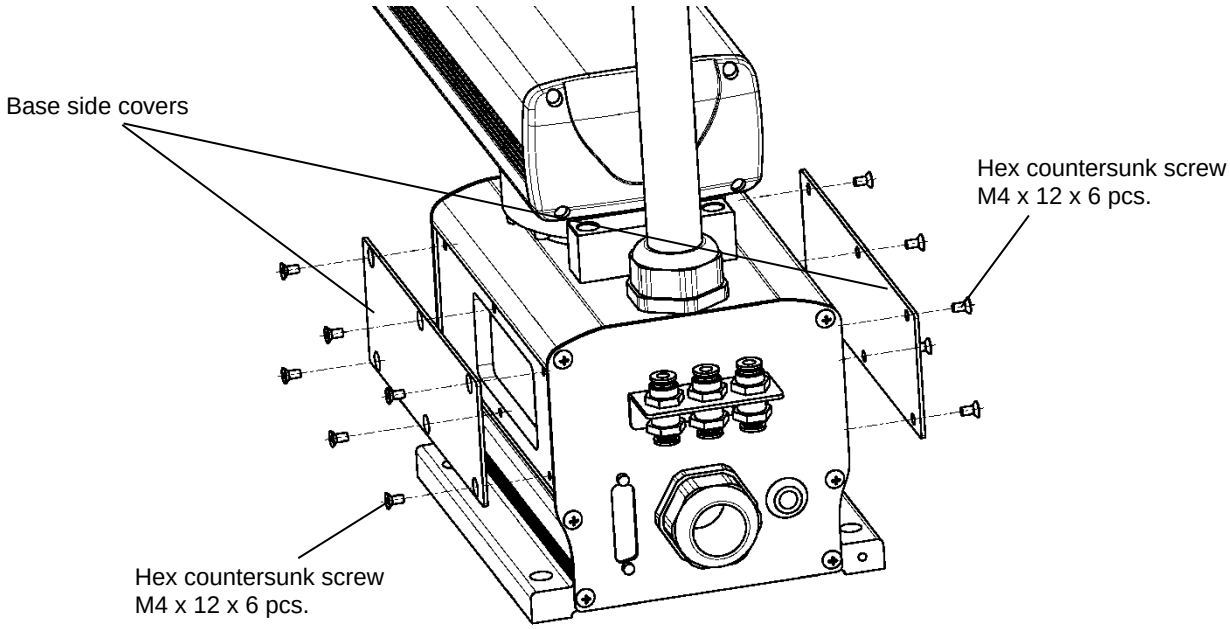


Fig. 3.9 Base side covers (THL500, THL600, THL700)

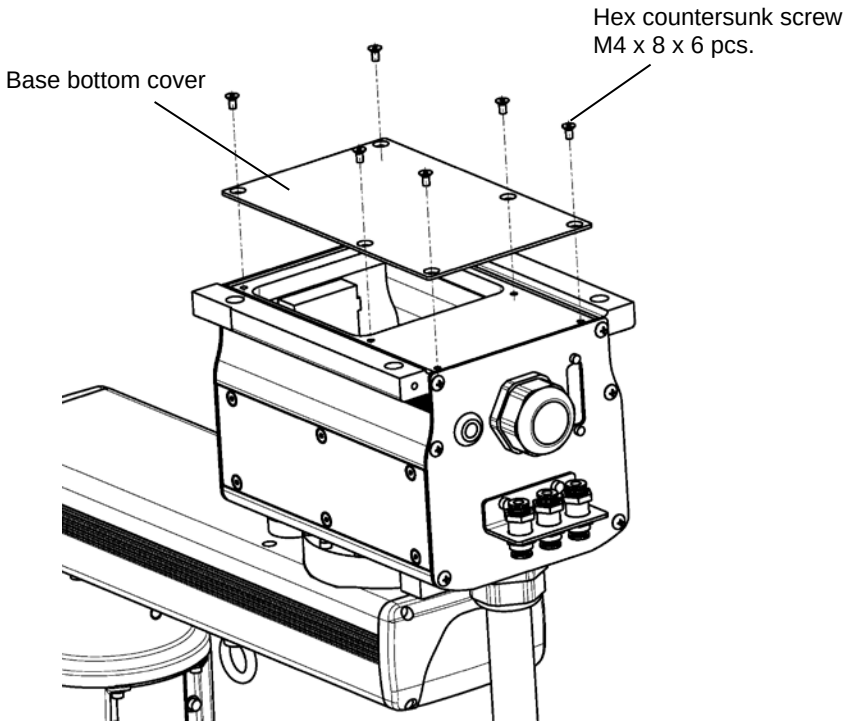


Fig. 3.10 Base bottom cover (THL500, THL600, THL700)

3.4 Replacing Motor

The motor is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠ DANGER

· The power plug of the controller must be removed before replacement work.
If replacement work is done with the power turned on, this may result in an electric shock or a malfunction, causing great danger.

NOTICE

– Never cause heavy impact on the motor shaft.
The motor and position detector may malfunction due to impact.
– Never disassemble the motor and position detector.
Otherwise, they may be disabled due to difference in position or intrusion of foreign material.
– When replacing a motor, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– After the motor is replaced, be sure to re-set zero position.
Replacing a motor causes deviation from mechanical zero position; therefore, the robot cannot make normal motion.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”

3.4.1 Type of Motor

The motors employed in this robot are shown below. When you place an order for a replacement motor, make sure of the robot model (THL500, THL600, THL700), the serial number, the axis name, and our drawing number according to the following table.
For the location where the serial number plate is attached, see the “Safety Manual.”
The motor and pulley for Axis 3 are connected using a key. Thus, key alignment is necessary during motor replacement.
For the type of the key, also see the table below.

Table 3.2 Type of Motor (THL500, THL600, THL700)

Description	Axis name	Type	Our drawing No.	Unit code
AC servo motor	Axis 1		S875289	Y610A3M20
	Axis 2		S875290	Y610A3M30
	Axis 3		S746337	Y610A3430
	Axis 4		S875291	Y610A3M40
Key	Axis 3	Square key 4 x 4 x 16		

* The key to be used with Axis 2 is attached to the motor.

3.4.2 Motor Locations

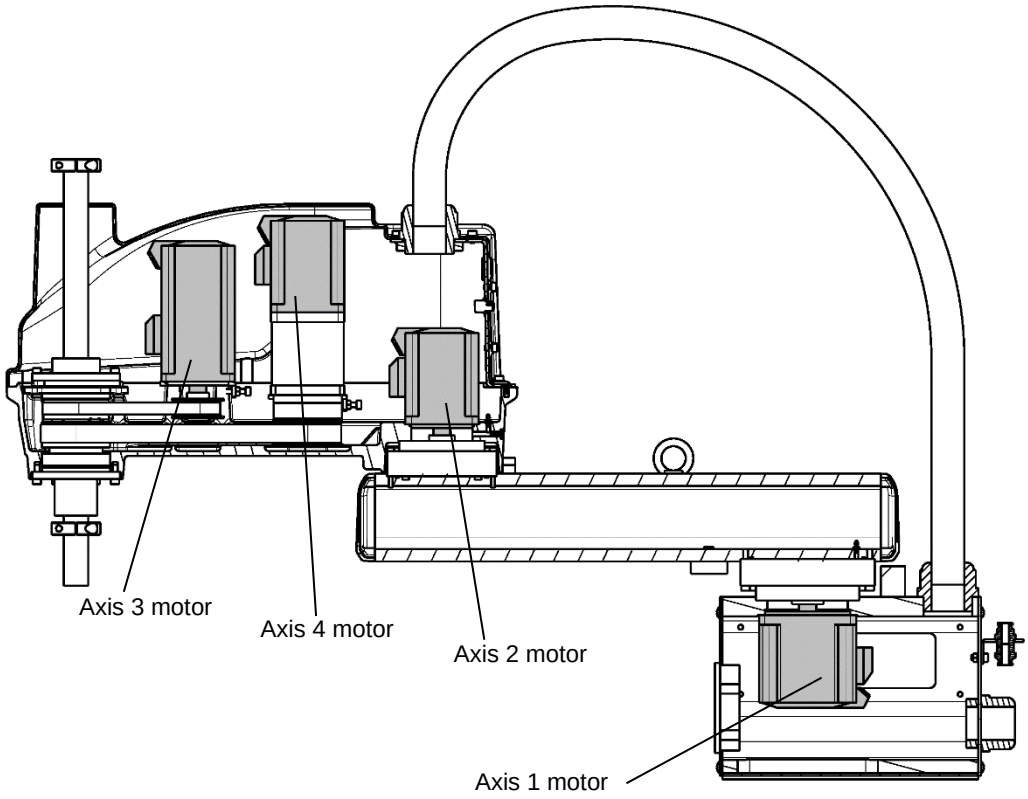


Fig. 3.11 Motor locations (THL500, THL600, THL700)

3.4.3 Dismounting Axis 1 Motor

1) Remove the base front cover and both of the base side covers. (See “Para. 3.3.2, Base Covers.”) The battery for position detection is being connected to the base front cover, so do not forcibly pull it or unplug the connector.

2) Remove the four hexagon socket head cap screws (M5 x 16 x 4 pcs.) and the washers that secure the axis 1 motor, pull the axis 1 motor assembly in the axis direction and then pull it out. Have a waste cloth handy as grease may drip from the motor mounted section when pulling out the motor assembly. At this time, remove the O-ring (CO0545A) being mounted on the base groove side surface. Also, when pulling out the motor assembly, do not pull it with force as the motor connector is connected.

3) Remove the connectors of the axis 1 motor, i.e., J1AS and J1AP (power drive cable), and J1BS and J1BP (encoder cables).

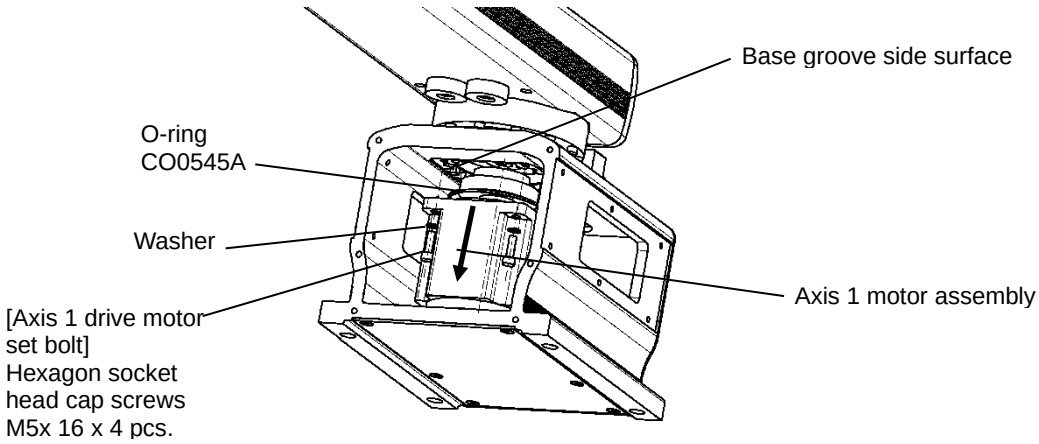


Fig. 3.12 Dismounting axis 1 motor assembly (THL500, THL600, THL700)

4) Remove the hexagon socket head cap screw (M4 x 10 x 1 pc.) from the end of the axis while securing the protrusion on the wave generator with pliers or the like. Use a cushioning material such as a waste cloth to protect the wave generator from being scratched when securing it. After removing the hexagon socket head cap screw, remove the holding plate and the wave generator.

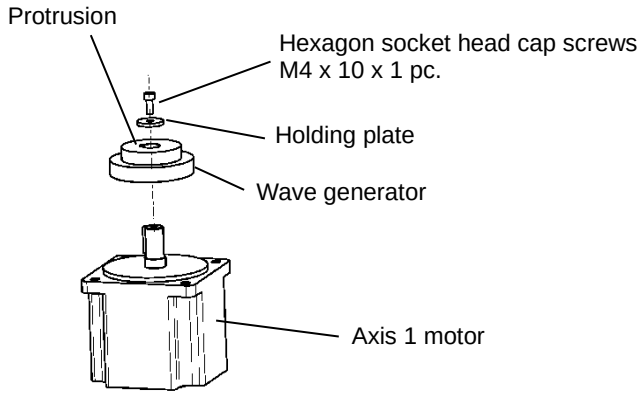


Fig. 3.13 Dismounting axis 1 wave generator (THL500, THL600, THL700)

3.4.4 Mounting Axis 1 Motor

1) Mount the wave generator, and secure it with the hexagon socket head cap screw (M4 x 10 x 1 pc.) and the holding plate. When securing the wave generator, hold the protrusion on the wave generator with pliers or the like, and then tighten the hexagon socket head cap screw (M4 x 10 x 1 pc.) at the end of the axis. Use a cushioning material such as a waste cloth to protect the wave generator from being scratched when securing it.

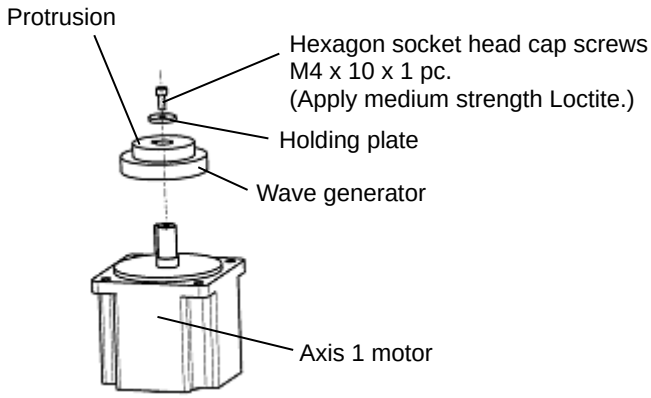


Fig. 3.14 Mounting axis 1 wave generator (THL500, THL600, THL700)

NOTICE

– Be sure to use wave generator and reduction gear in the correct combination.
Use of the different combination will result in abnormal noise, reduced service life, and reduced positioning accuracy.

2) Apply grease to the O-ring (CO0545A) and mount it to touch the base groove side surface. Be careful not to drop the O-ring while doing so.

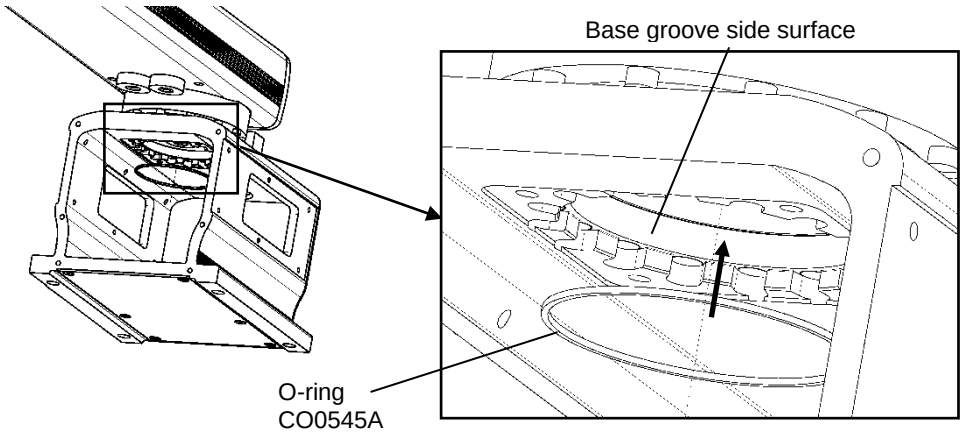


Fig. 3.15 Mounting axis 1 base side O-ring (THL500, THL600, THL700)

NOTICE

– Never forget to fit the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.

3) Apply an appropriate amount of grease onto the all surfaces of the wave generator mounted to the axis 1 motor.

Reduction Gear	Recommended grease	Maker
A type	SK-1A	HDS
B ype	SFB No.1	Nidec-Shimpo Corp.

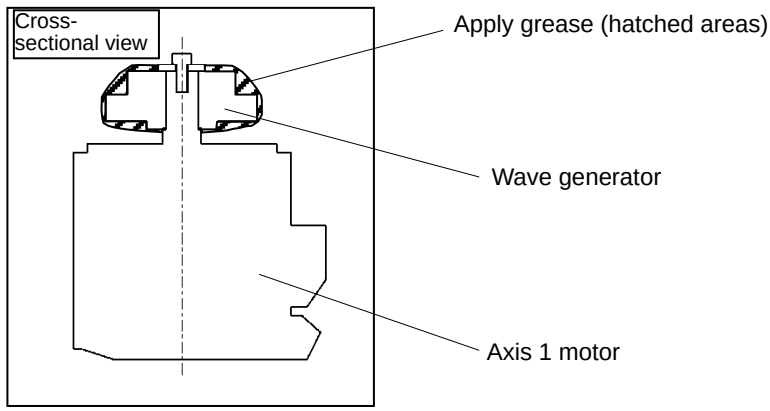


Fig. 3.16 Apply grease onto wave generator (THL500, THL600, THL700)

NOTICE

– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

4) Connect the connectors of the axis 1 motor, i.e., J1AS and J1AP (power drive cable), and J1BS and J1BP (encoder cables).

5) Insert the axis 1 motor assembly into the base while paying attention to the motor's mounting phase, the orientations of the reduction gear main body and the wave generator when the motor is inserted, and the position of the O-ring (it should not move).

6) Secure the axis 1 motor assembly with four (4) hexagon socket head cap screws (M5 x 16) and washers.

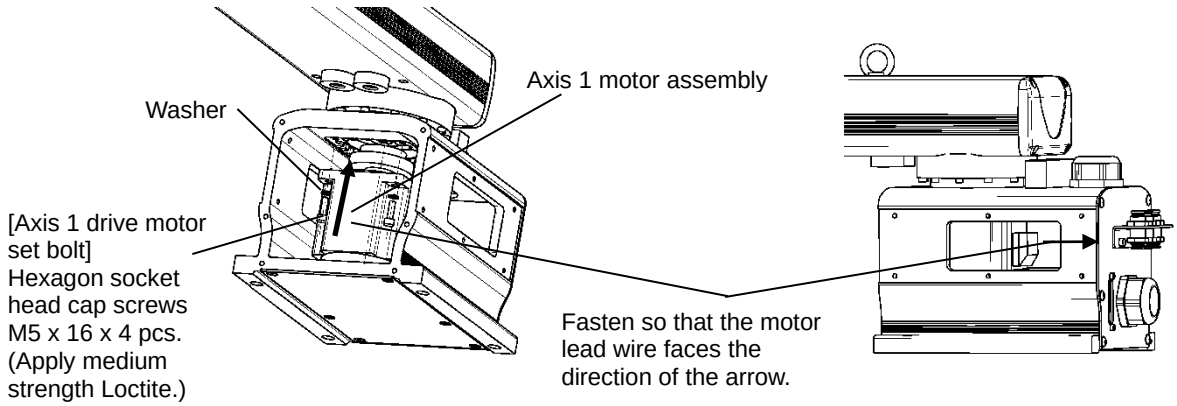


Fig. 3.17 Mounting axis 1 motor assembly (THL500, THL600, THL700)

7) After changing Axis 1 motor, move Arm 1 by hand and check that there is no abnormal sound before turning on the power.

8) Mount the base front cover and the base side covers. (See “Para. 3.3.2, Base Covers.”)

9) Turn on the power and set up the axis 1 home position to complete axis 1 motor replacement.
(Check the coordinates of Axes 2 to 4 and set up the home position if necessary. See “Section 7, Robot Home Point and Position Detector Error”.)

3.4.5 Dismounting Axis 2 Motor

1) Disconnect the arm 2 cover. (See “Para. 3.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers or the like, and remove J2AS and J2AP (axis 2 power drive cables), J2BS and J2BP (axis 2 encoder cables). It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Remove the harness guide being secured with one (1) hex countersunk screw (M4 x 8) and the support plate being secured with two (2) hexagon socket head cap screws (M4 x 8).

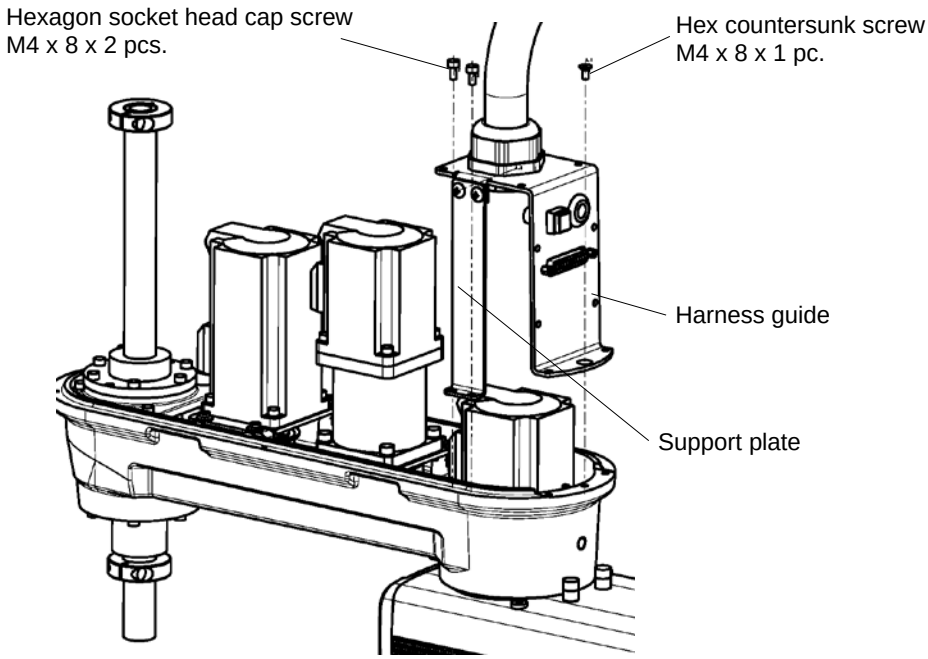


Fig. 3.18 Dismounting harness guide and support plate (THL500, THL600, THL700)

4) Remove four (4) hexagon socket head cap screws (M4 x 16) and washers securing the axis 2 motor. Also remove the O-ring (CO0538A).

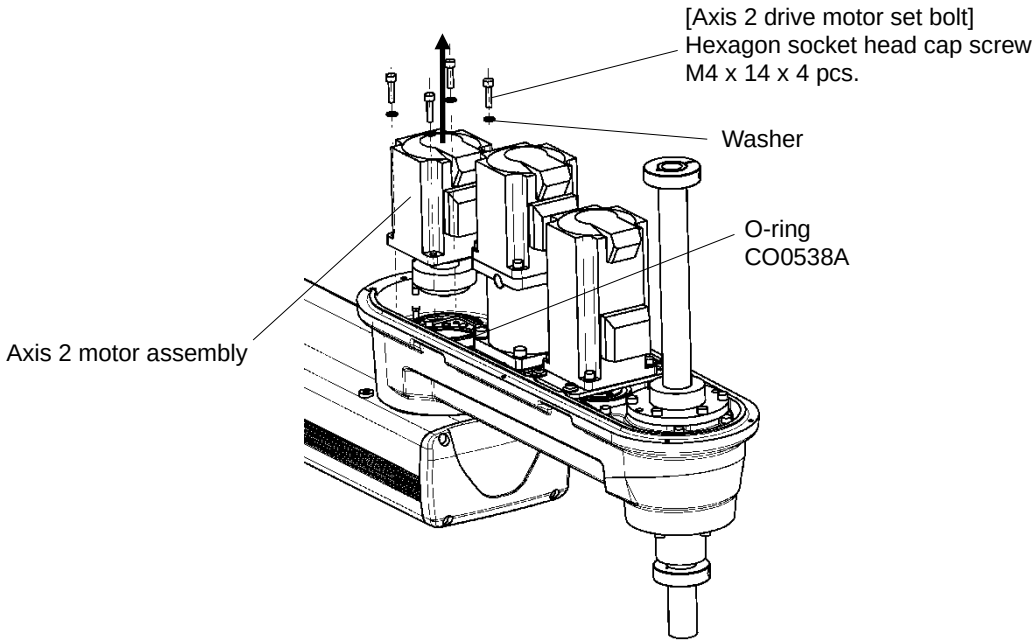


Fig. 3.19 Dismounting axis 2 motor assembly (THL500, THL600, THL700)

5) Remove the wave generator from the axis 2 motor assembly. (Same as Step 4 in “Para. 3.4.3, Dismounting Axis 1 Motor.”)

3.4.6 Mounting Axis 2 Motor

1) Apply grease to the O-ring (CO0538A) and mount it to the inlaid part of the axis 2 motor.

2) Mount the wave generator to the axis 2 motor. (Same as Step 1 in “Para. 3.4.4, Mounting Axis 1 Motor.”)

3) Apply an appropriate amount of grease onto the all surfaces of the wave generator mounted to the axis 2 motor. (Same as Step 3 in “Para. 3.4.4, Mounting Axis 1 Motor.”)

4) Insert the axis 2 motor assembly into Arm 2 while paying attention to the motor's mounting phase, the orientations of the decelerator main body and the wave generator when the motor is inserted (align the long sides of the ovals of the reduction gear and wave generator), and the position of the O-ring (it should not move).

5) Secure the axis 2 motor with four (4) hexagon socket head cap screws (M4 x 16) and washers.

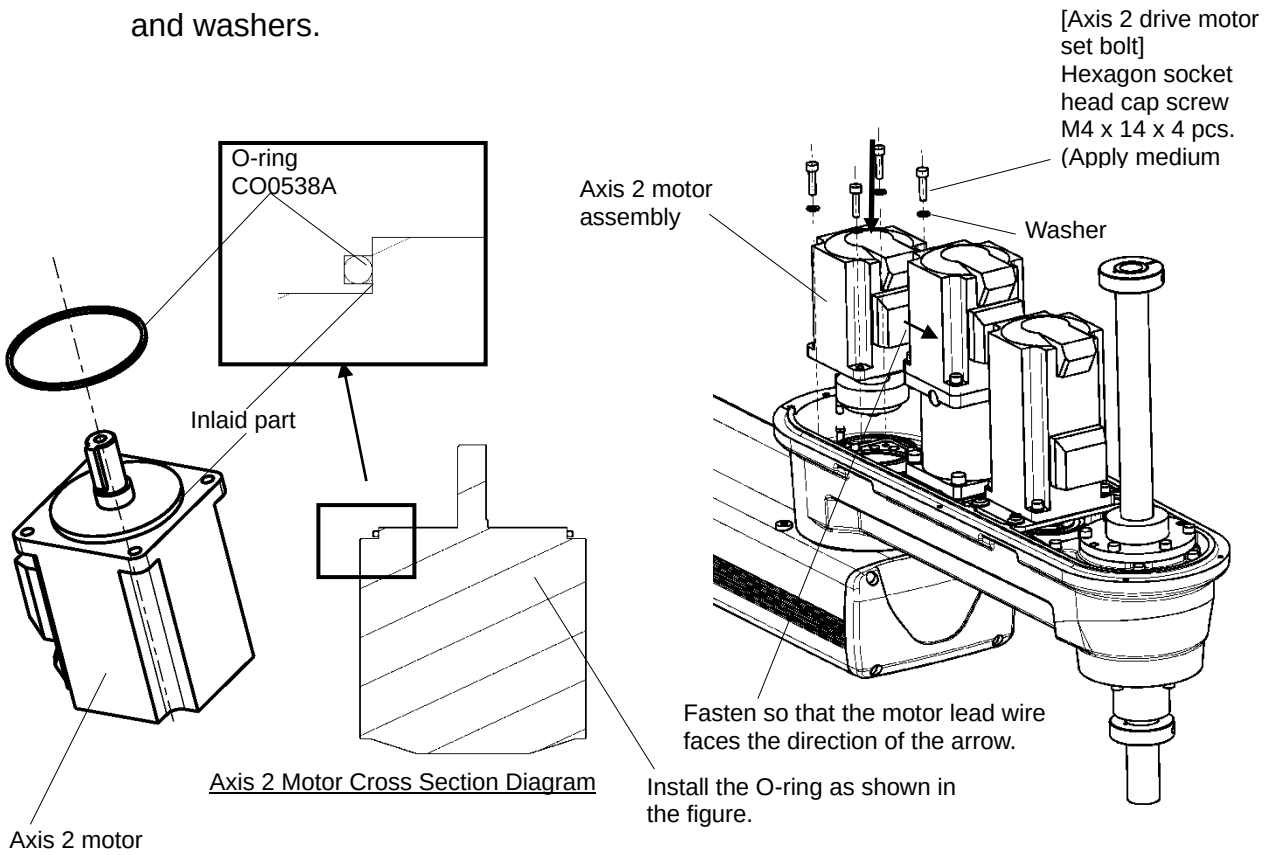


Fig. 3.20 Mounting axis 2 motor assembly (THL500, THL600, THL700)

NOTICE

– Never forget to fit the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
– Be sure to use wave generator and reduction gear in the correct combination.
Use of the different combination will result in abnormal noise, reduced service life, and reduced positioning accuracy.
– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

6) Secure the harness guide and the support plate with one (1) hex countersunk screw (M4 x 8, application of Loctite not necessary) and two (2) hexagon socket head cap screws (4 x 8).

7) Move Arm 2 by hand and check that there is no abnormal sound before turning on the power.

8) Connect J2AS and J2AP (axis 2 power drive cables), J2BS and J2BP (axis 2 encoder cables).

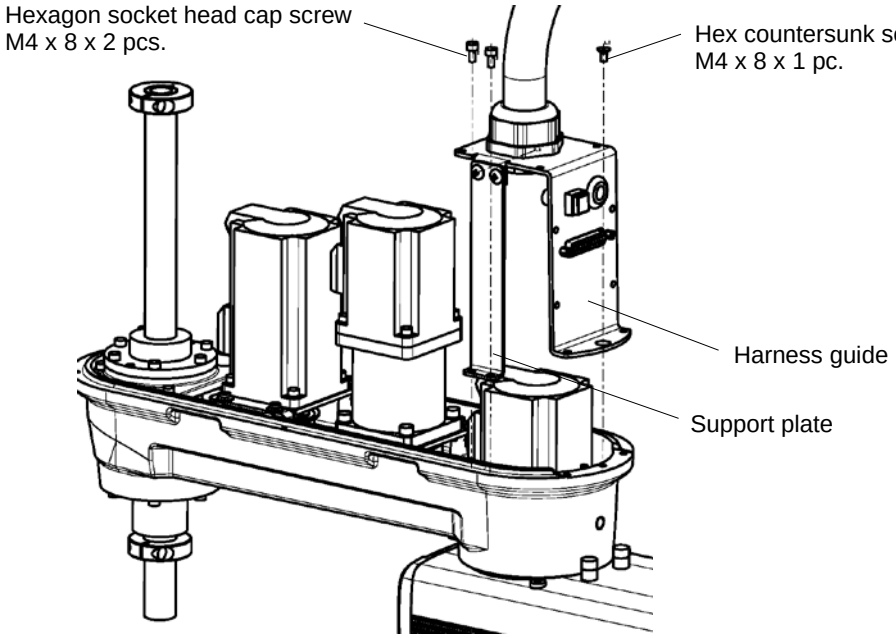


Fig. 3.21 Mounting harness guide and support plate (THL500, THL600, THL700)

9) Restore the cables in A 2 to the original conditions.

10) mount Arm 2 cover.

11) Turn on the power and set up the home positions of axes 2 to complete axis 2 motor replacement (see “Section 7, Robot Home Point and Position Detector Error”).

3.4.7 Dismounting Axis 3 Motor

⚠ WARNING

– When the axis-3 motor is replaced, be sure to lower the ball screw spline shaft to the lower limit.
When the motor or timing belt is removed, the shaft that drops by its own weight could catch your hand or fingers.
- When the ball screw spline or axis-4 timing belt or reduction gear is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

1) Remove the arm 2 cover. (See “Para. 3.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers.
It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Disconnect connectors J3AS and J3AP (power drive cables), connectors J3BS and J3BP (encoder cables) and connectors J3DS and J3DP (brake cables) for the axis 3, which are connected to the connector panel.

4) Loosen the axis 3 tension adjustment bolt (M4 x 16 x 2 and hexagonal nuts) and remove the flange head bolts (M4 x 10 x 4) securing the axis 3 motor plate to cancel the axis 3 timing belt tension. Then draw out axis 3 the motor assembly upward.

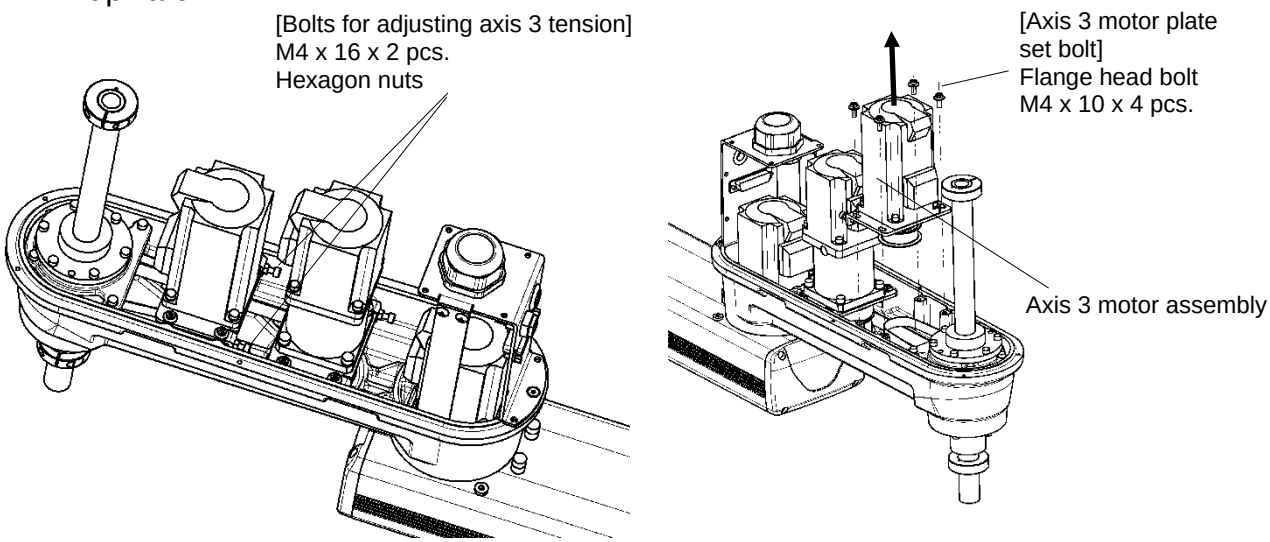


Fig. 3.22 Dismounting axis 3 motor assembly (THL500, THL600, THL700)

5) Remove one (1) hexagon socket head cap screw at the end of the axis (M3 x 8). (Same as Step 4 in “Para. 3.4.3, Dismounting Axis 1 Motor.”) Next, pull out the washer, holding plate, pulley and key.

6) Remove four (4) hexagon socket head cap screws (M4 x 12) and washers that secure the axis 3 motor and then disassemble the axis 3 motor plate and the axis 3 motor.

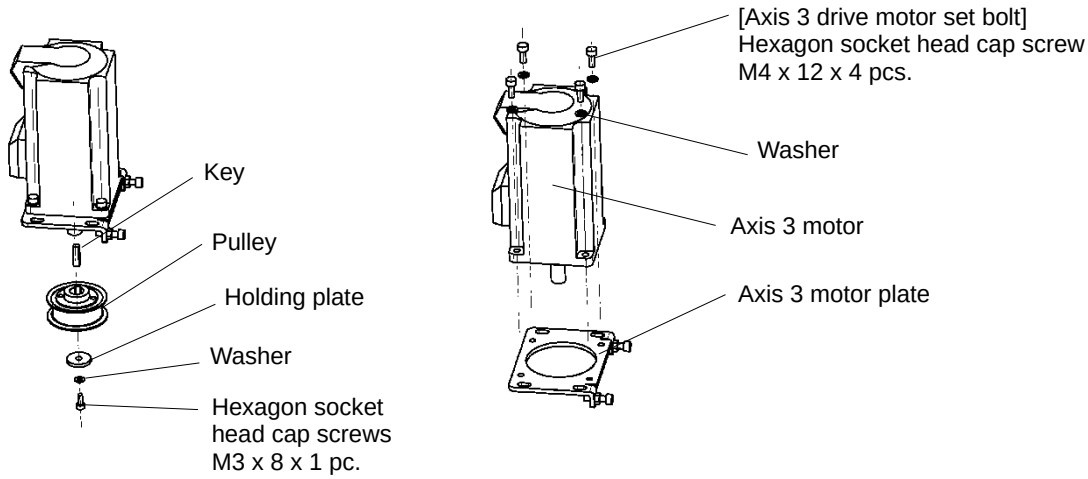


Fig. 3.23 Dismounting axis 3 motor pulley and plate (THL500, THL600, THL700)

3.4.8 Mounting Axis 3 Motor

1) Perform key alignment, using the new axis 3 motor and pulley. At this time, no clearance should exist between the motor shaft and pulley.

2) Secure the axis 3 motor to the axis 3 motor plate with four (4) hexagon socket head cap screws (M4 x 12) and washers. Be careful with the mounting phases of the axis 3 motor and the axis 3 motor plate at this time.

3) Mount the key and the axis 3 motor pulley, and then secure them with the holding plate, washers and one (1) hexagon socket head cap screw (M3 x 8).

NOTICE

– Perform key alignment very carefully. If there is a clearance between the motor shaft and pulley, positioning accuracy will drop and the life of the parts will shorten.

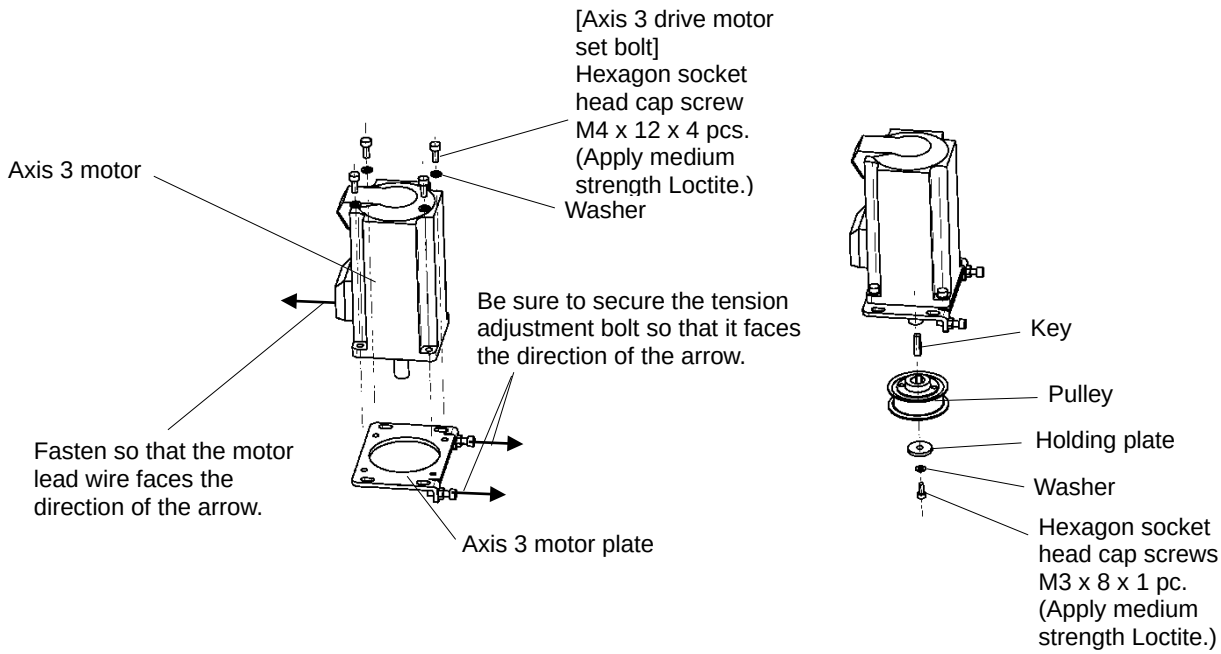


Fig. 3.24 Mounting axis 3 pulley and plate (THL500, THL600, THL700)

4) Hang the axis 3 motor timing belt to the arm 3 motor pulley, and temporarily secure the axis 3 motor plate to Arm 2 with four (4) flange head bolts (M4 x 10, application of Loctite not necessary). Be careful with the motor mounting phase at this time. Apply tension using two (2) axis 3 tension adjustment bolts (M4 x 16, application of Loctite not necessary). (For the belt replacement procedures and tension adjustment values, see “Para. 3.5.3, Replacing Axis 3 Timing Belt.”) Then, tighten and secure the temporarily tightened flange head bolts with hex nuts.

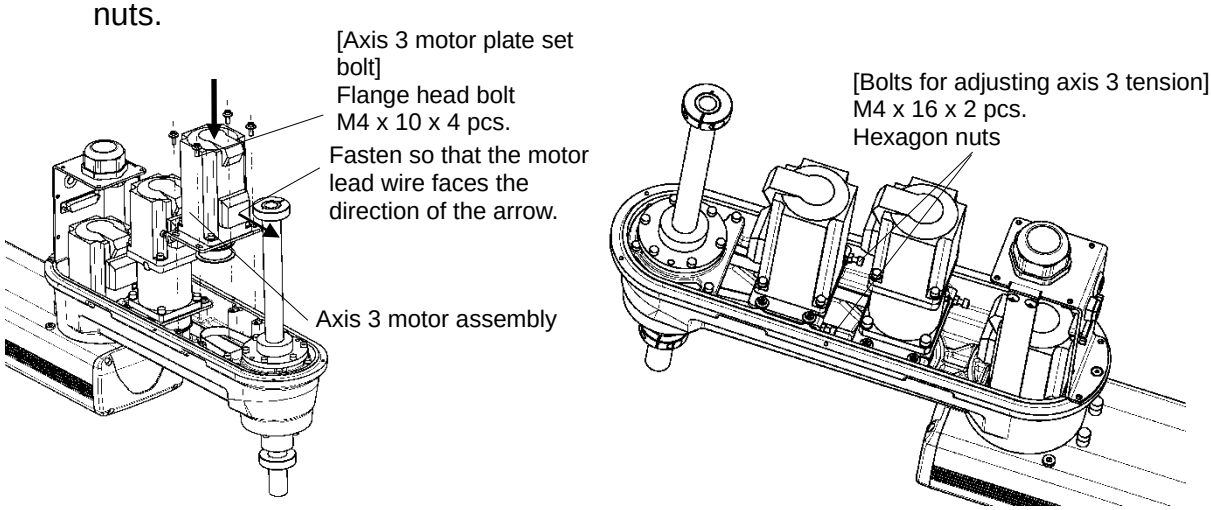


Fig. 3.25 Mounting axis 3 motor assembly (THL500, THL600, THL700)

5) Connect the connectors. (See the pictures taken in Para. 3.4.7, Step 2.)

6) Mount the arm 2 cover and perform home setting for Axes 3 and 4. Now, the axis 3 motor replacement is complete. Need perform home setting for Axes 4 at this time.

7) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

3.4.9 Dismounting Axis 4 Motor

1) Remove the arm 2 cover. (See “Para. 3.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers.
It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Disconnect connectors J4AS and J4AP (power drive cables) and connectors J4BS and J4BP (encoder cables) of the axis 4 motor.

4) Remove the cap attached to the lateral side of the axis 4 reduction gear and loosen the coupling bolt (M4) securing the axis 4 motor shaft and input shaft of the axis 4 reduction gear.
If the hexagonal hole of the bolt is out of phase, adjust the phase by turning the ball screw spline shaft by hand.

5) Remove four (4) hexagon socket head cap screws (M4 x 12) and washers that secure the axis 4 motor, and then pull the axis 4 motor upward.

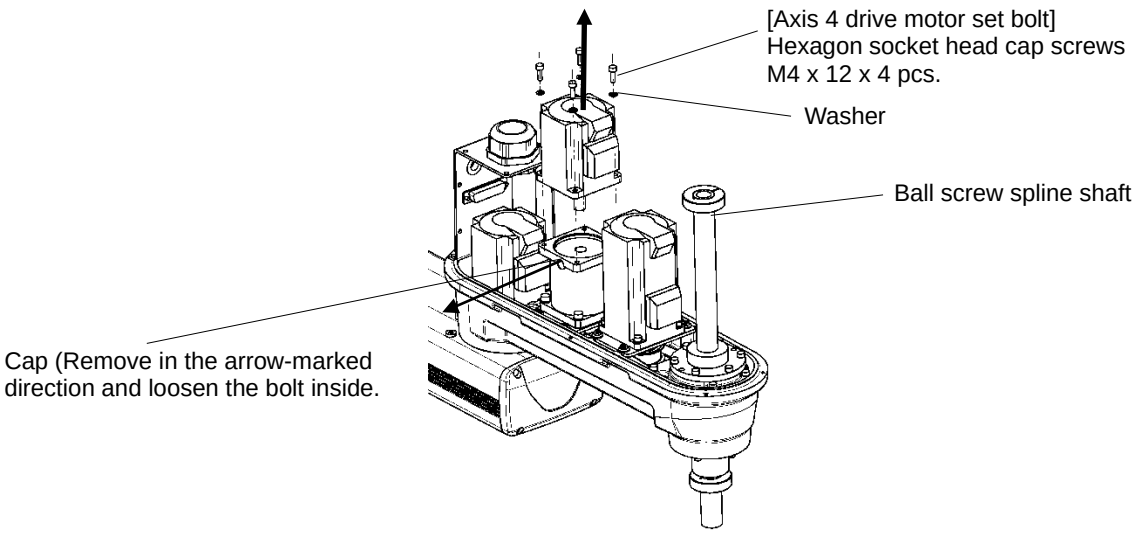


Fig. 3.26 Dismounting axis 4 motor (THL500, THL600, THL700)

3.4.10 Mounting Axis 4 Motor

1) Mount the new motor to the reduction gear with four (4) hexagon socket head cap screws (M4 x 12) and washers. Be careful with the phase when mounting the motor. (Recommended clamping torque: 3.3 N·m)

2) Tighten the coupling of the axis 4 reduction gear by means of the attached bolt (M4) to mount the cap. (Recommended clamping torque: 4.3 N·m)

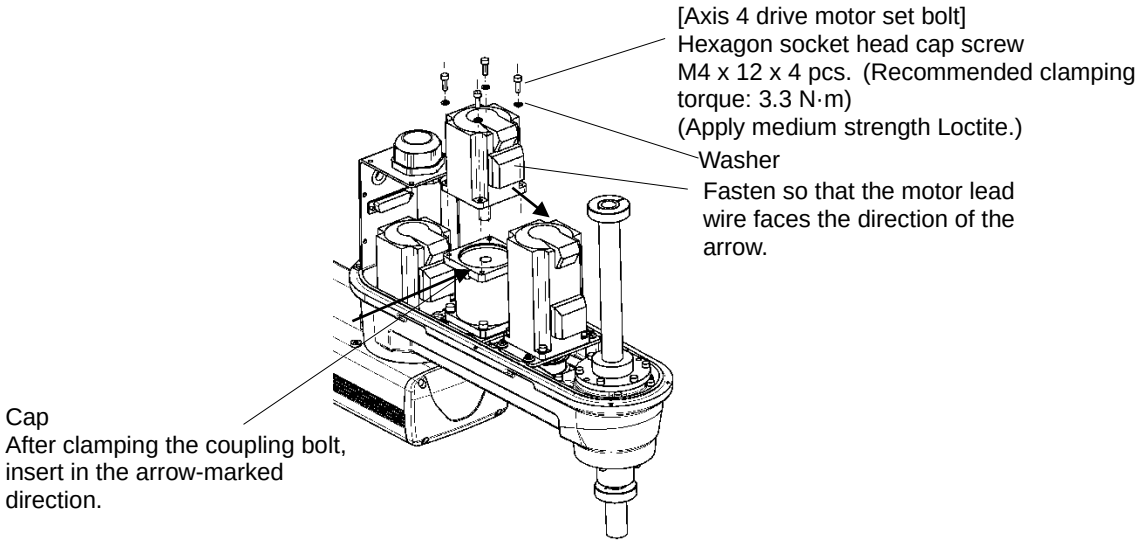


Fig. 3.27 Mounting axis 4 motor (THL500, THL600, THL700)

3) Connect the connectors of the axis 4 motor, i.e., J4AS and J4AP (power drive cables), and J4BS and J4BP (encoder cables).

4) Restore the cables to the original conditions and then mount the arm 2 cover.

5) Set up the home positions of Axes 3 and 4 to complete motor replacement. During the replacement of the axis 4 motor, it is necessary to set up the home position of Axis 3.

6) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

3.5 Adjusting and Replacing Timing Belt

The timing belt is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠ DANGER

– When replacing the timing belt, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock.
A worker may be entangled in rotation of the motor that has operated unpredictably.

⚠ WARNING

– When replacing the axis-3 timing belt, be sure to lower the ball screw spline shaft to the lower limit.
When the motor or timing belt is removed, the shaft drops by its own weight.
- When the ball screw spline or axis-4 timing belt is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

NOTICE

– When replacing the timing belt, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– After the timing belt is replaced, be sure to re-set the zero position.
Because a mechanical zero position is shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error”

3.5.1 Type of Timing Belt

The timing belts used in this robot are shown below.
When you place an order for a replacement belt, specify the robot model (THL500, THL600, THL700), the serial number, the axis name, and our drawing number.
For the location where the serial number plate is attached, see the “Safety Manual.”

Table 3.3 Type of timing belt (THL500, THL600, THL700)

Description	Axis name	Width	Our drawing No.
Timing belt	Axis 3	10 mm	S875174
	Axis 4	15 mm	S875175

3.5.2 Timing Belt Locations

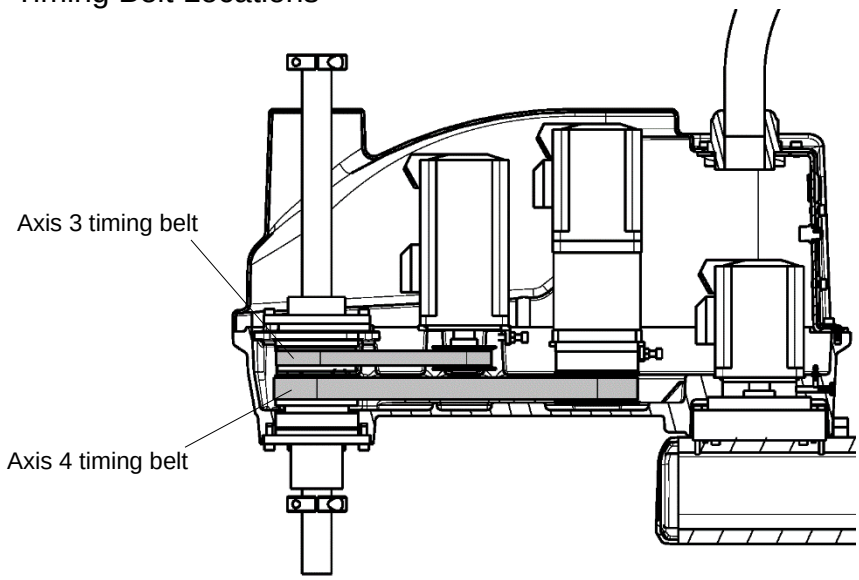


Fig. 3.28 Timing belt locations (THL500, THL600, THL700)

3.5.3 Replacing Axis 3 Timing Belt

⚠ WARNING

– When replacing the axis-3 timing belt, be sure to lower the ball screw spline shaft to the lower limit. When the motor or timing belt is removed, the shaft drops by its own weight.
- When the ball screw spline or axis-4 timing belt is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

1) Remove the arm 2 cover. (See “Para. 3.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers. It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Disconnect the axis 3 motor assembly. For the disconnecting procedures, see Steps 1) through 4) of “Para. 3.4.7, Dismounting Axis 3 Motor.”

4) Remove the four (4) hexagon socket head cap screws (M4 x 16) and the bottom stopper that secure the axis 4 bracket (remove the hand and other parts). Pull out the ball screw nut together with the axis 4 bracket, the ball screw shaft and the top stopper from Arm 2. Caution must be exercised when pulling them out. Note that if the ball screw nut is pulled out of the shaft, the ball falls off from the ball screw nut and, consequently, the ball screw spline unit can no longer be used.

5) Disconnect the axis 3 timing belt.

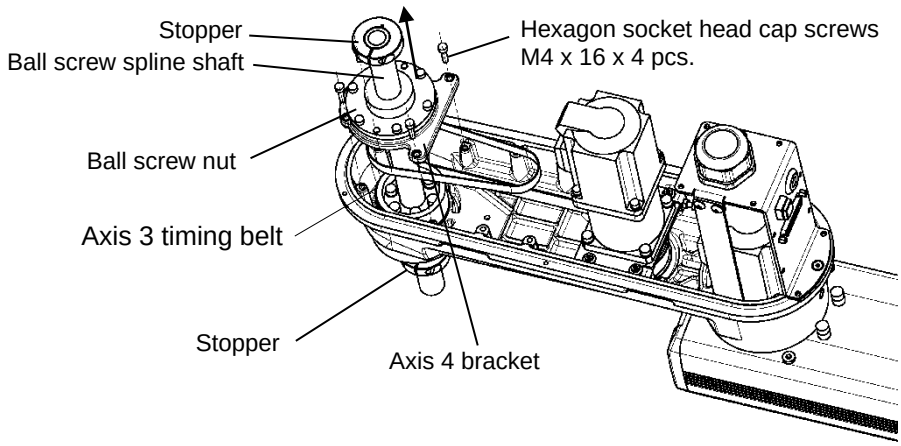


Fig. 3.29 Replacing axis 3 timing belt (THL500, THL600, THL700)

6) Mount the new timing belt. Insert the ball screw spline shaft into the ball spline nut, and mount the stoppers. (For the insertion of the ball screw spline shaft, see “Para. 3.6.5, Mounting Ball Screw Spline Unit.”)

7) Hang the timing belt to the axis 3 motor assembly which was previously removed in 3), and temporarily secure it to Arm 2 with four (4) flange head bolts (M4 x 10, application of Loctite not necessary). Move the ball screw spline unit up and down to make it fit in. Apply tension using two (2) axis 3 tension adjustment bolts (M4 x 16, application of Loctite not necessary).
While pulling the axis 3 motor plate with the axis 3 tension adjustment bolts, measure the tension with a tension meter. Tighten the flange head bolt at the location where the tension reaches a value slightly smaller than 49 N (because the tension gets larger by retightening of the flange head bolt). Adjust so that the tensile value becomes between 49 and 57 N when the flange head bolt is retightened. Then, secure with the hex nut.
The values to be used for the tension meter are as shown in the table below.

Table 3.4 Value of tension of axis 3 timing belt (THL500, THL600, THL700)

Value of tension [N]	Unit mass [g/m]	Belt width [mm]	Span [mm]
49 to 57	2.2	10	100

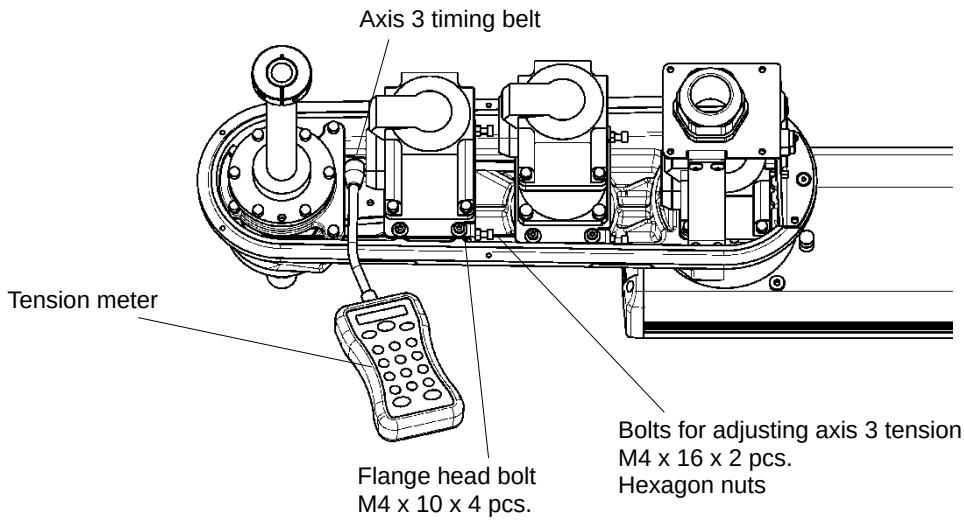


Fig. 3.30 Adjusting axis 3 tension (THL500, THL600, THL700)

8) Connect the connectors and return them to their original location, mount Arm 2 cover, and perform home setting for Axes 3 and 4. This completes the replacement of the timing belt.

9) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

3.5.4 Replacing Axis 4 Timing Belt

NOTICE

– Never pull out the ball screw nut from the ball screw spline shaft.
Otherwise, the ball screw spline unit from which balls have come out cannot be used.
– After the timing belt is replaced, be sure to re-set the zero position.
Because a mechanical zero position is shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– When axis-4 timing belt is replaced, the axis-3 motor and timing belt must also be removed.
Therefore, the precautions for the replacement of the axis-3 motor and timing belt must be also observed.

1) Remove the arm 2 cover, the axis 3 motor assembly, the axis 3 timing belt, the ball screw nut and the ball screw spline shaft.
For the disconnecting procedures, see Steps 1) through 5) of “Para. 3.5.3, Replacing Axis 3 Timing Belt.”

2) Loosen two (2) tension adjustment bolts (M4 x 16) for the axis 4, remove four (4) flange head bolts (M4 x 10) securing the axis 4 motor plate, and then pull out the axis 4 motor assembly. Pull out the axis 4 timing belt upward and then install a new belt.

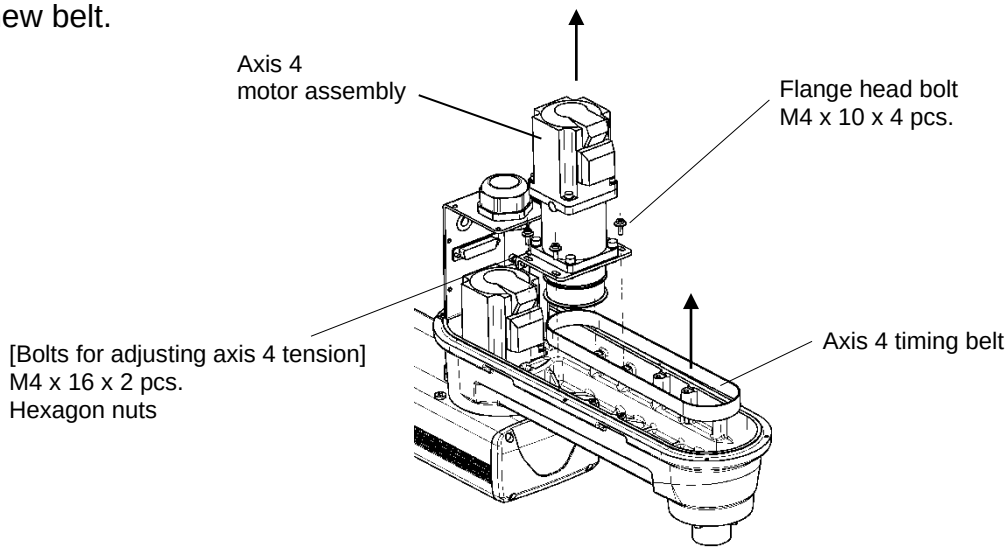


Fig. 3.31 Dismounting harness guide and support plate (THL500, THL600, THL700)

3) Temporarily secure the ball screw nut and the ball screw spline shaft which were removed in Step 1) above and the axis 4 bracket above to Arm 2 using four (4) hexagon socket head cap screws (M4 x 16). In doing so, mount the axis 3 timing belt and the axis 4 timing belt through the ball screw spline shaft first. For the mounting of the ball screw spline unit and the stoppers, see “Para. 3.6.5, Mounting Ball Screw Spline Unit.”

4) Temporarily set the axis 4 motor assembly which was disconnected in Step 2) above to Arm 2 with the four (4) flange head bolts (M4 x 10 x 4 Loctite is not necessary) after setting the axis 4 timing belt to the axis 4 motor pulley. Be careful with the motor assembly mounting phase at this time.

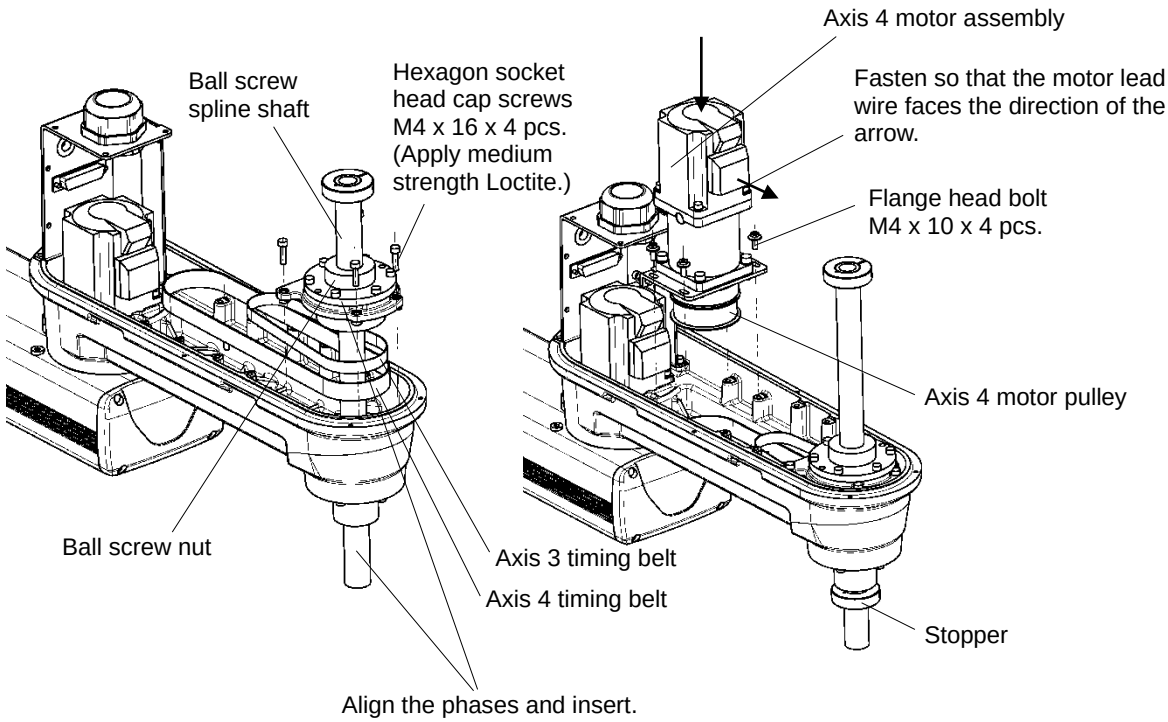


Fig. 3.32 Ball screw and axis 4 motor assembly (THL500, THL600, THL700)

5) While pulling the axis 4 motor plate with the axis 4 tension adjustment bolts, measure the tension with a tension meter. Tighten the flange head bolt at the location where the tension reaches a value slightly smaller than 150 N (because the tension gets larger by retightening of the flange head bolt). Adjust so that the tensile value becomes between 150 and 180 N when the flange head bolt is retightened. After that, gradually tighten the tension adjusting bolt, and fix it with a nut. If the tension adjusting bolt is not tightened in this step, the tension adjusting bolt may be disconnected during the robot operation. The values to be used for the tension meter are as shown in the table below.

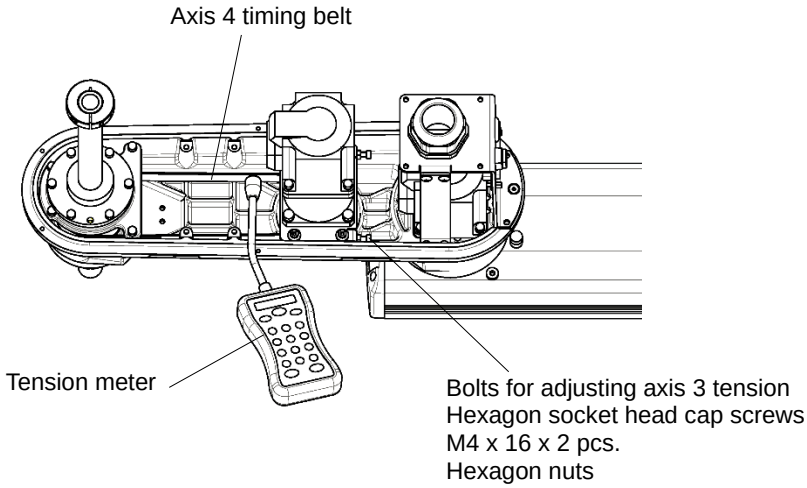


Fig. 3.33 Ball screw and axis 4 motor assembly (THL500, THL600, THL700)

Table 3.5 Value of tension of axis 4 timing belt (THL500, THL600, THL700)

Value of tension [N]	Unit mass [g/m]	Belt width [mm]	Span [mm]
150 to 180	4.1	15	190

6) Mount the axis 3 motor and timing belt. For the mounting procedures, see “Para. 3.4.8, Mounting Axis 3 Motor” and “Para. 3.5.3, Replacing Axis 3 Timing Belt” above.

7) Arrange the connectors and cables as originally set. (See the pictures you took beforehand.)

8) Perform home setting for Axes 3 and 4.

9) Carry out a test operation of Axes 3 and 4 and make sure that the belt tension is appropriate.

10) Attach the arm 2 cover. Now replacement of the axis 4 timing belt completes.

3.5.5 Check Timing Belt Adjustment

Perform inspection and adjustment of each timing belt semi-annually (every six (6) months).

1) Remove the arm 2 cover. (See “Para. 3.3.1, Arm 2 Cover.”)
2) If the belt is worn out, replace it. If cracks are observed or there is a serious abrasion (see "3.5.3 Replacement of the 3rd-axis timing belt"), replace the belt. If the belt is scratched, replace the belt or contact our service office. If the belt is seriously loosened, adjust the tension in conformity to "3.5.3 Replacement of the 3rd-axis timing belt" and 3.5.4 Replacement of the 4th-axis timing belt". If the belt is loosened heavily, adjust its tension, referring the following table. Note that the value of tension at replacement of timing belt (i.e., when mounting a new timing belt) is the same as the value of tension at adjustment If the tension is appropriate, mount the arm 2 cover. Now the inspection is complete.
3) When adjustment of the tension is required, loosen the four (4) flange head bolts (M4 x 10) securing the axis 3 and axis 4 motor plates, respectively. Measure the tension on the tension meter while adjusting the tension adjustment bolt (axis 3: M4 x 16 x 2 bolts, axis 4: M4 x 16 x 2 bolts and hexagonal nut).
4) When the tension has fallen under the appropriate range, tighten the flange head bolts.
5) Tighten the tension adjustment bolts further and fasten with the hexagon nuts. If the bolt is not tightened any further, the bolt can come out while the robot is in operation.
6) Mount the arm 2 cover. Now the adjustment completes.

3.6 Filling Grease to Ball Screw Spline Unit and Replacement

The ball screw spline unit is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠DANGER

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– After the ball screw spline unit is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

3.6.1 Type of Ball Screw Spline Unit

The ball screw spline unit used in this robot is shown below.
When you place an order for the ball screw spline unit for replacement, specify the robot model (THL500, THL600, THL700) and our drawing number.

Table 3.6 Ball screw spline unit (THL500, THL600, THL700)

Description	Stroke	Our drawing No.	Unit code
Ball screw spline unit	150mm	H852666	Y610A3LT0
	300mm	H852702	Y610A3ME0

* Stroke 300 mm is optional.

3.6.2 Ball Screw Spline Unit Location

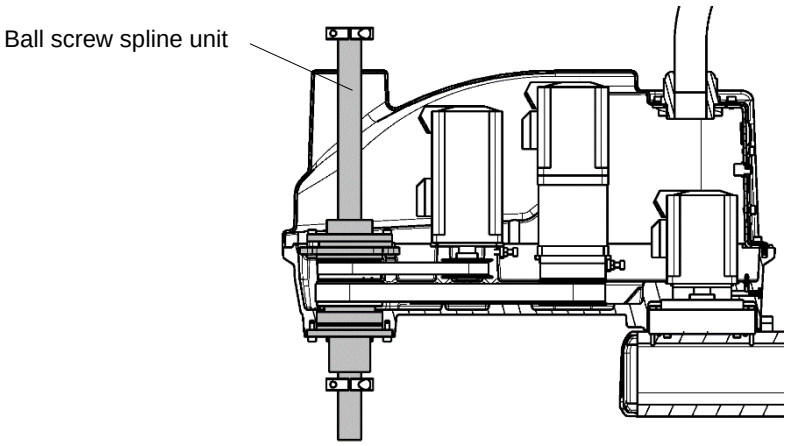


Fig. 3.34 Ball screw spline unit location (THL500, THL600, THL700)

3.6.3 Greasing Ball Screw Spline Unit and Applying Anticorrosive

⚠ CAUTION

– When moving the ball screw spline shaft by hand, never insert your hand or finger inside the mechanical stopper or under the robot hand.
Otherwise, your hand or finger may get caught in the mechanical stopper and hand.

NOTICE

– Never touch the ball screw spline unit with bare hands.
Touching it with bare hands may cause the ball screw spline unit to rust.
- Be sure to wear gloves.
– When greasing up the ball screw spline unit and apply corrosion inhibitor, be sure to cover the peripheral equipment, etc.
Grease and corrosion inhibitor may drip on the peripheral equipment.
- Wipe out excess grease and corrosion inhibitor with waste cloth.
– Be sure to check the application state of grease daily.
A shortage of grease and development of rust may cause flaws in the sliding surface, resulting in a reduction in the performance and lifespan of the robot. The robot may malfunction.
– Be sure to use the specified type of grease.
Otherwise, the performance and lifespan of the robot may be reduced. The robot may malfunction.
– Be sure to check the application state of corrosion inhibitor daily.
Otherwise, rust may develop on the ball screw spline unit.
- Corrosion inhibitor must be applied only when it is used up.

Basically, fill the grease to the ball screw spline unit every three (3) months.
If you have verified that there is not much grease, be sure to apply grease.
At daily inspection also, make sure that the ball screw spline unit is filled with a sufficient volume of grease.
Check the condition of anticorrosive in daily inspection. If anticorrosive is dried up, apply it.

Table 3.7 Recommended anticorrosive (THL500, THL600, THL700)

Recommended anticorrosive	Maker
KLUBER A20	NOK
WD-40	ST TRADING

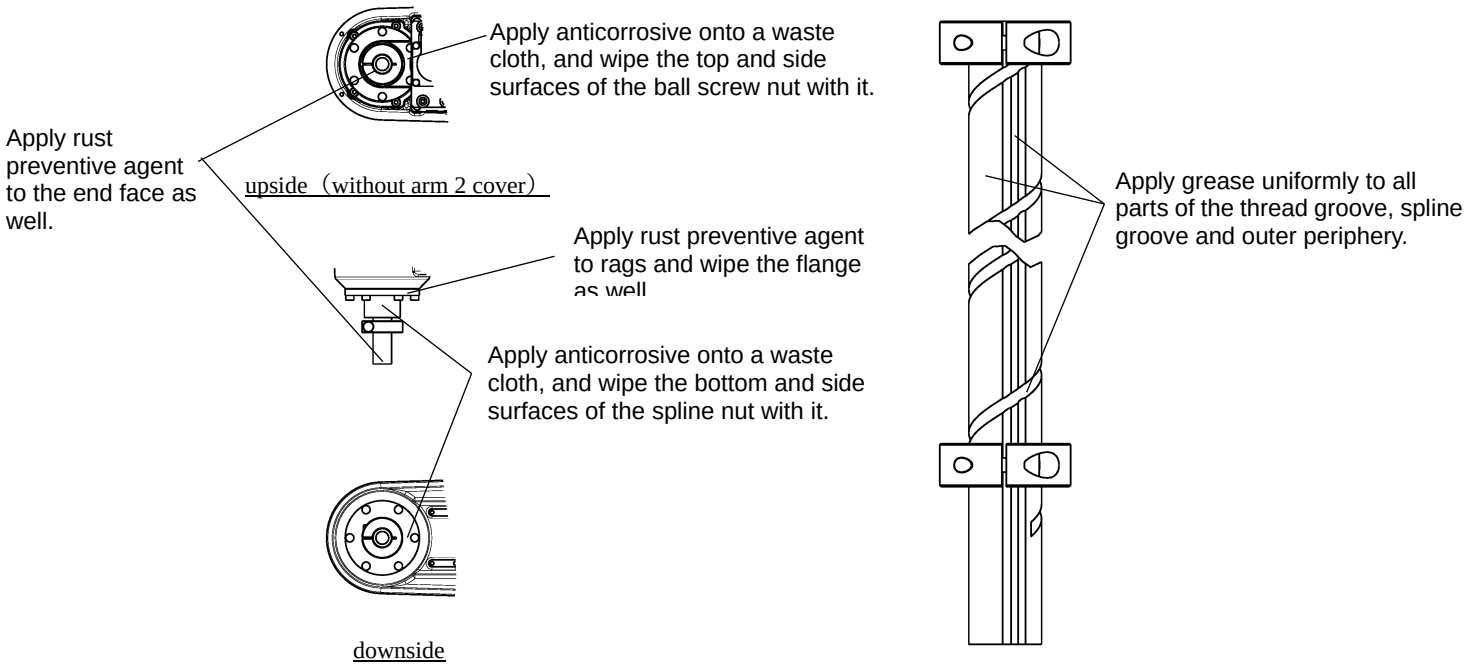


Fig. 3.35 Applying anticorrosive and grease to ball screw spline unit (THL500, THL600, THL700)

1) Remove the arm 2 cover (see “Para. 3.3.1, Arm 2 Cover.”)
2) Connect the controller power supply plug and turn off the servo system.
3) Move the arm to a position where Axis 3 can be moved over the full stroke by hand.
4) Push down the ball screw spline shaft to the lower limit while pressing the axis 3 brake release switch.

5) Directly apply the grease to the exposed shaft area by brash.
Apply the grease to such an extent that the shaft groove can be filled. Wipe out the surplus grease.

Table 3.8 Recommended grease (THL500, THL600, THL700)

Recommended grease	Maker
AFF grease	THK

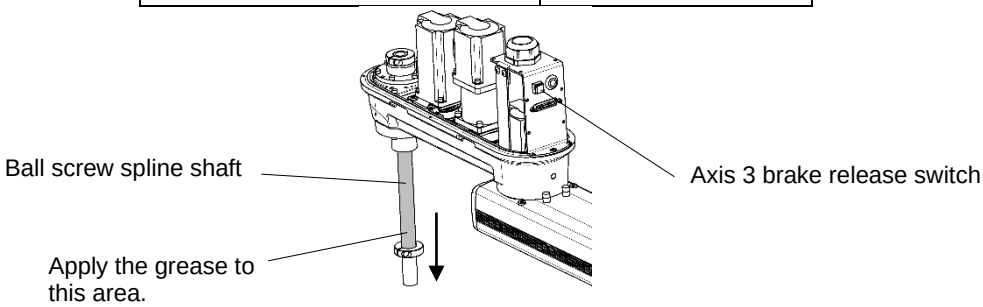


Fig. 3.36 Greasing lower side of ball screw spline unit (THL500, THL600, THL700)

6) Push up the shaft up to the upper limit while pressing the axis 3 brake release switch.

7) Directly apply the grease by brash to the shaft area sticking up from the cover.
Apply the grease to such an extent that the shaft groove can be filled. Wipe out the surplus grease.

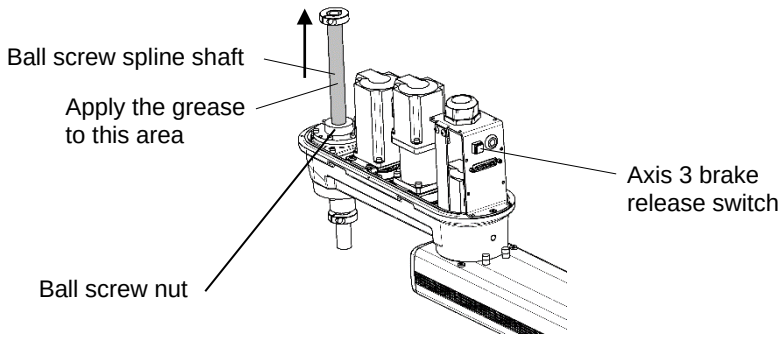


Fig. 3.37 Greasing upper side of ball screw spline unit (THL500, THL600, THL700)

8) Move the shaft up and down repeatedly while pressing the axis 3 brake release switch to fill the grease uniformly. Wipe out the surplus grease. Now the work completes.

3.6.4 Dismounting Ball Screw Spline Unit

⚠DANGER

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– Never touch the ball screw spline unit with bare hands.
Touching it with bare hands causes the ball screw spline unit to rust.
- Be sure to wear gloves.
– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The robot may malfunction if you replace it with the power on.
– Be sure to handle the ball screw spline unit carefully.
Drop impact or application of excessive external force could ruin the unit.
– After the ball screw spline unit is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

In the descriptions on replacing the ball screw spline unit, the procedures for changing the axis 3 and 4 motors and timing belts are not included. For details, refer to the descriptions on replacement of each part (see “Para. 3.4.7, Dismounting Axis 3 Motor”, “Para. 3.4.9, Dismounting Axis 4 Motor”, “Para. 3.5.3, Replacing Axis 3 Timing Belt” and “Para. 3.5.4, Replacing Axis 4 Timing Belt.”)

1) Remove the arm 2 cover. (See “Para. 3.5.1, Arm 2 Cover.”)
2) Disconnect the hand, tool, etc. followed by the upper and lower stoppers.
3) Disconnect the axis 3 and 4 motor assemblies.

4) Remove four (4) hexagon socket head cap screws (M4 x 16) securing the axis 4 bracket, and then pull the ball screw nut, the ball screw spline shaft and the axis 4 bracket upward together. Caution must be exercised when pulling them out. A

NOTICE

– Never pull out the ball screw nut from the ball screw spline shaft.
Otherwise, the ball screw spline unit from which balls have come out cannot be used.

5) Remove the timing belts from Axes 3 and 4.

6) Remove six (6) hexagon socket head cap screws (M4 x 12) securing the ball screw nut, and then pull it out of the axis 4 bracket. If it cannot be pulled out by hand, use a removal tap. Remove the stopper on the upper side as well.

7) Remove four (4) hexagon socket head cap screws (M5 x 20) and washers, and then remove the axis 3 nut pulley which is secured to the ball screw nut.

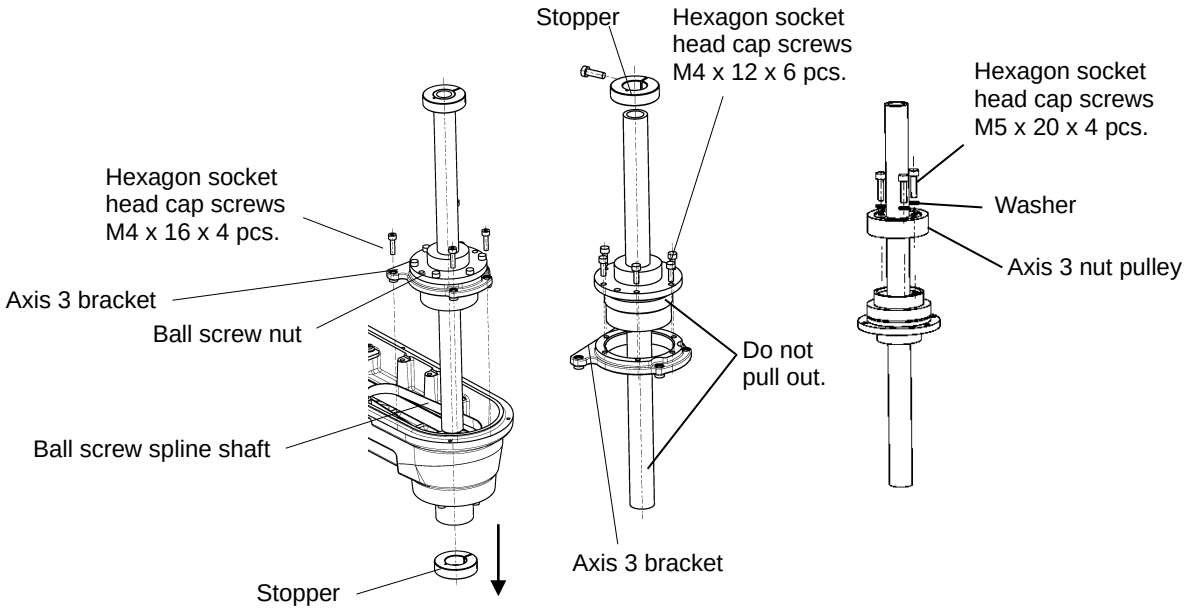


Fig. 3.38 Dismounting ball screw spline unit (THL500, THL600, THL700)

8) Remove the hexagon socket head cap screw (M4 x 12 x 6 pcs.) securing the ball spline nut, and pull out the ball spline nut downward.
If the ball spline nut is hard to remove, screw a bolt into the ball spline nut tap (M4) to remove it.

9) Disconnect the axis 4 nut pulley secured to the ball spline nut with the hexagon socket head cap screw (M5 x 12 x 6 pcs.). When it is hard to disconnect, screw the bolt into the tap for disassembly (M4) machined on the pulley and disconnect.

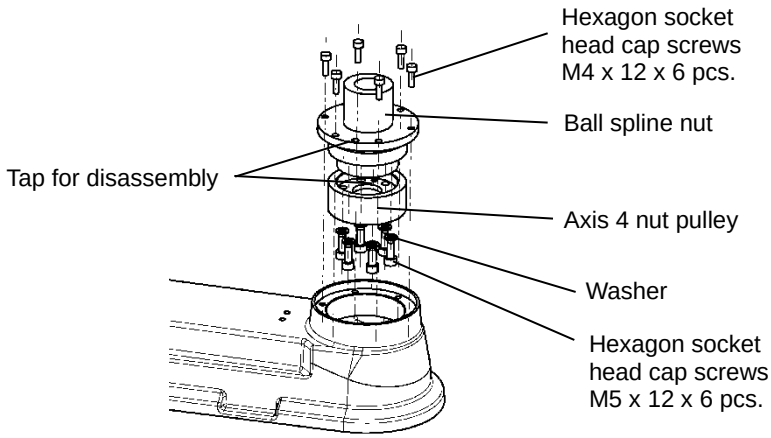


Fig. 3.39 Dismounting ball screw spline nut (THL500, THL600, THL700)

10) Insert the disconnected ball spline nut into the ball screw spline shaft.
A marking for phase adjustment is stamped on both the ball screw spline shaft and the ball spline nut. Align the two markings when inserting the ball spline nut.
To prevent the nut from slipping off, wind cable ties around near the top and bottom of the shaft.

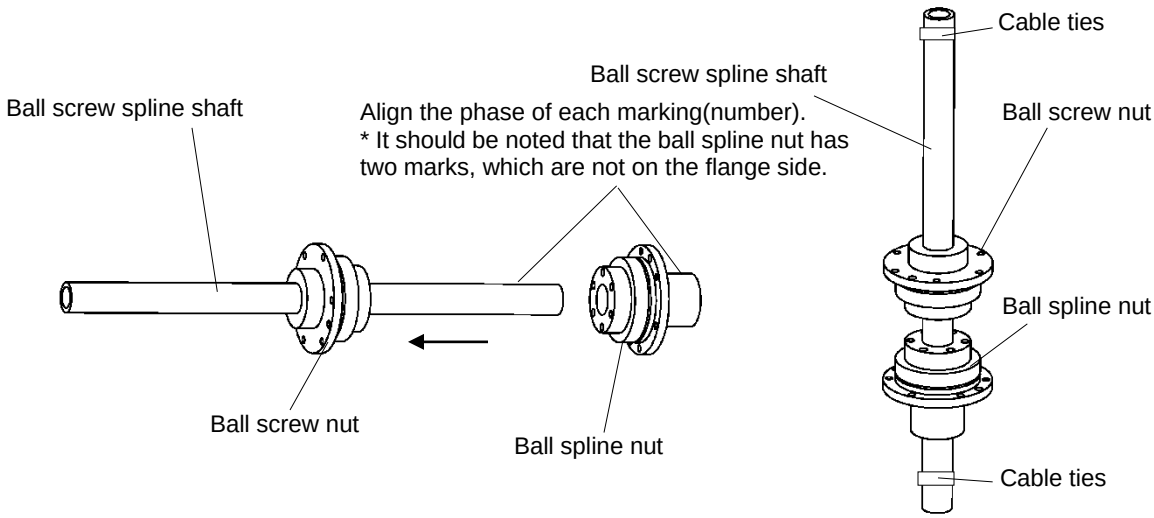


Fig. 3.40 Ball screw spline nut (THL500, THL600, THL700)

3.6.5 Mounting Ball Screw Spline Unit

⚠DANGER

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– Never touch the ball screw spline unit with bare hands.
Touching it with bare hands causes the ball screw spline unit to rust.
- Be sure to wear gloves.
– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– Be sure to handle the ball screw spline unit carefully.
Drop impact or application of excessive external force could ruin the unit.
– After the ball screw spline unit is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

1) Disconnect the ball spline nut from the new ball screw spline unit.

2) Mount the axis 4 nut pulley on the ball spline nut with the hexagon socket head cap screws (M5 x 12 x 6 pcs.) and washers.
The bolts are difficult to tighten because the pulley rotates. Thus, tighten them while holding the pulley with pliers or the likes. In doing so, use a waste cloth to protect the pulley from scratches.

3) Mount the ball spline nut on Arm 2 with the hexagon socket head cap screws (M4 x 12 x 6 pcs.) and washers. When doing so, be careful with the orientation of the removal tap.

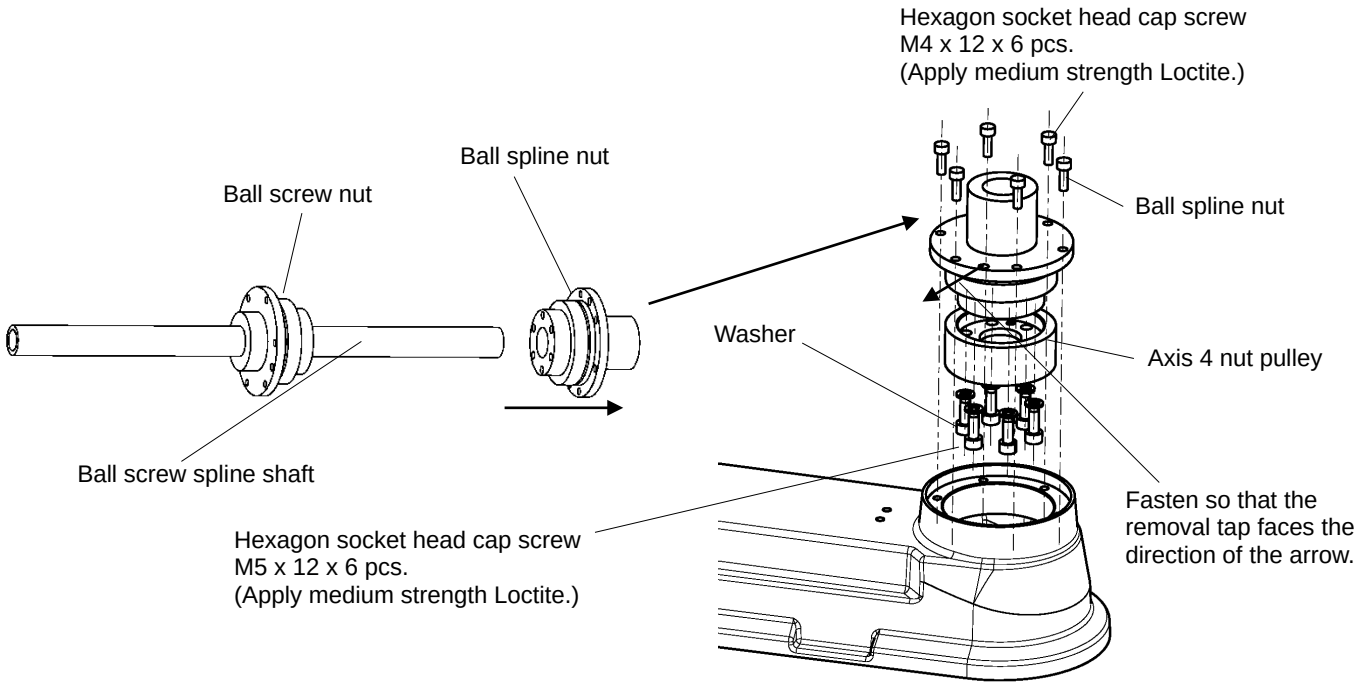


Fig. 3.41 Mounting ball spline nut (THL500, THL600, THL700)

4) Mount the axis 4 timing belt and axis 4 motor assembly, then adjust the belt tension. (See “Para. 3.5.4, Replacing Axis 4 Timing Belt.”)

5) Temporarily tighten the axis 4 bracket to Arm 2 with four (4) hexagon socket head cap screws (M4 x 16).

6) Secure the axis 3 nut pulley to the ball screw nut with four (4) hexagon socket head cap screws (M5 x 20) and washers.

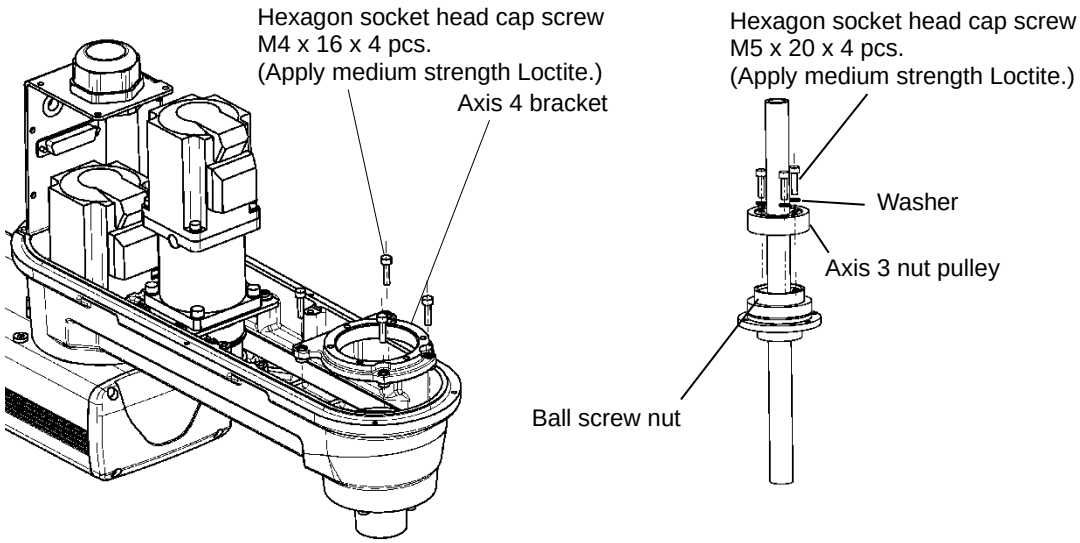


Fig. 3.42 Axis 3 bracket and pulley (THL500, THL600, THL700)

NOTICE

– Never pull out the ball screw nut from the ball screw spline shaft.
Otherwise, the ball screw spline unit from which balls have come out cannot be used.

7) Mount the ball screw nut and the ball screw spline shaft together through the ball spline nut, and then mount it to the axis 3 bracket with six (6) hexagon socket head cap screws (M4 x 12). At this time, align the phases of both markings put on the tip of the ball spline nut and the ball screw spline shaft.

8) Mount a stopper at a position 42 mm from both the top and the bottom of the ball screw spline shaft.
When mounting each stopper, align its split line with the left end of the spline groove.

9) Manually move the shaft in the vertical direction. Make sure of its smooth operation. After that, hold the bracket so that it does not move, and tighten the bolt temporarily locking the 3rd-axis bracket.

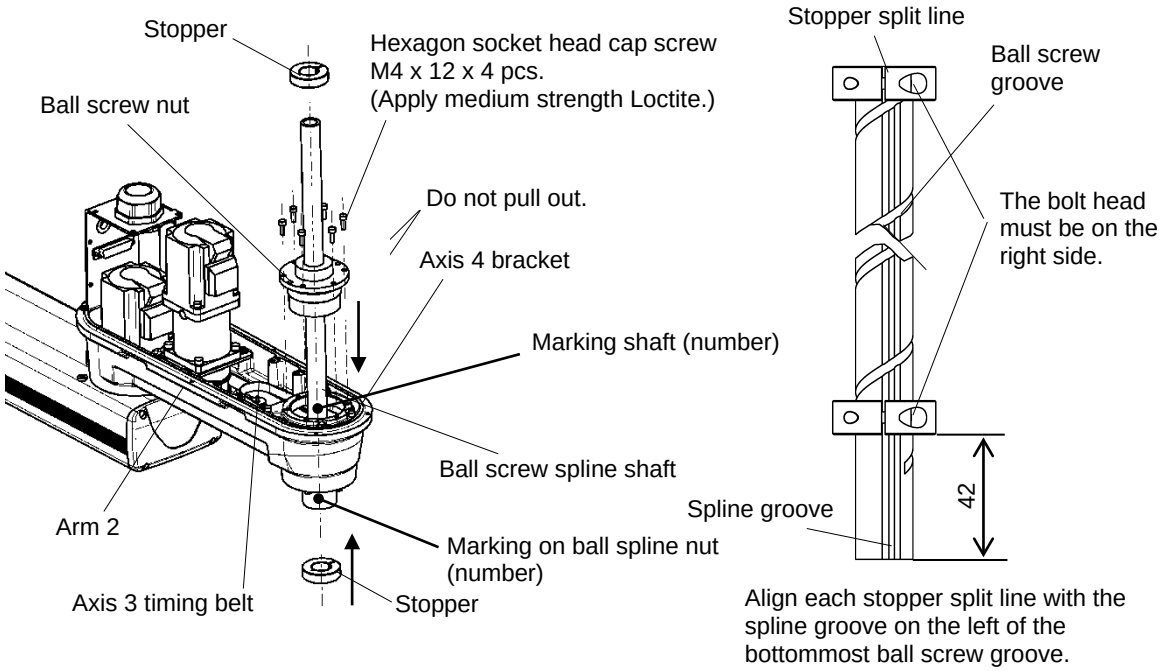


Fig. 3.43 Mounting ball screw spline nut (THL500, THL600, THL700)

10) Mount the axis 3 motor assembly, then adjust the belt tension. (See “Para. 3.5.3, Replacing Axis 3 Timing Belt.”)

11) Arrange the connectors and cables as originally set. (See the pictures you took beforehand.)

12) Perform home setting for Axes 3 and 4.
When performing home setting for Axis 4, align the stopper split line with the mating mark on the ball spline nut.
After home setting, remove the mating mark from the old ball spline nut, and attach it to the new one.

13) Turn on the power. Press the brake release switch in the servo off mode, and manually move the ball screw in the vertical direction. Make sure of its smooth operation.

14) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

15) Mount the arm 2 cover. Now replacement of the ball screw spline unit completes.

3.7 Reduction Gear Replacement

The reduction gear is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠DANGER

– When replacing the reduction gear, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– When replacing the reduction gear, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.
– After the reduction gear is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”

3.7.1 Type of Reduction Gear

The reduction gears used in this robot are shown below.
When you place an order for the reduction gear for replacement, specify the robot model (THL500, THL600, THL700), axis name, type and our drawing number.

Table 3.9 Type of reduction gear (THL500, THL600, THL700)

Description	Axis name	Our drawing No.	Unit code
Reduction gear (A type)	Axis 1	S875237	Y610A3LW0
	Axis 2	S875238	Y610A3LX0
Reduction gear (B type)	Axis 1	S994595	Y610D0FW0
	Axis 2	S994596	Y610D0FX0
Reduction gear	Axis 4	S875239	Y610A3LY0

3.7.2 Reduction Gear Locations

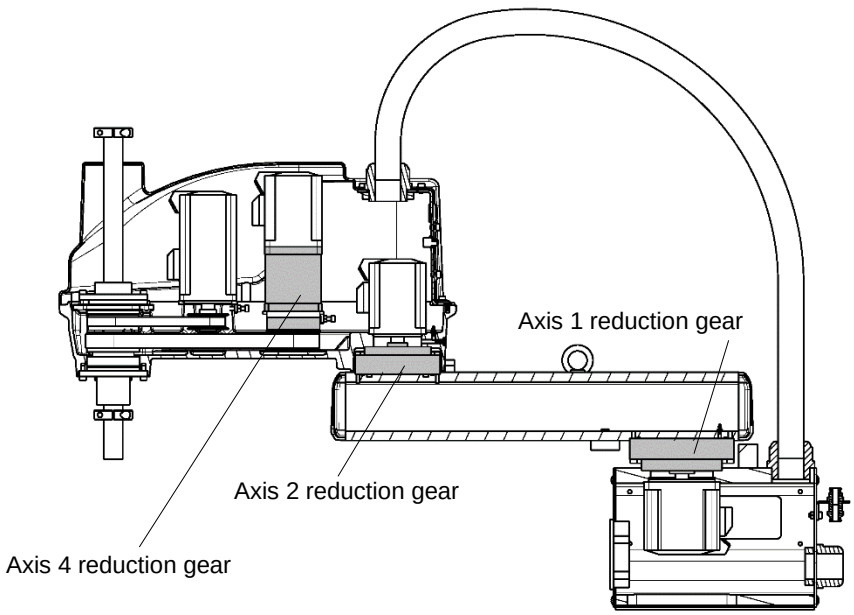


Fig. 3.44 Reduction gear locations (THL500, THL600, THL700)

3.7.3 Dismounting Axis 1 Reduction Gear

1) Remove the base front cover and the base side covers. (See “Para. 3.3.2, Base Covers.”)

2) Remove four (4) hexagon socket head cap screws (M5 x 16) and the washers that secure the axis 1 motor assembly, pull out the axis 1 motor assembly, and then remove the O-ring (CO0545).
(See “Para. 3.4.3, Dismounting Axis 1 Motor.”)

3) Remove sixteen (16) hexagon socket head cap screws (M4 x 20) that secure the base, and then remove Arm 1.

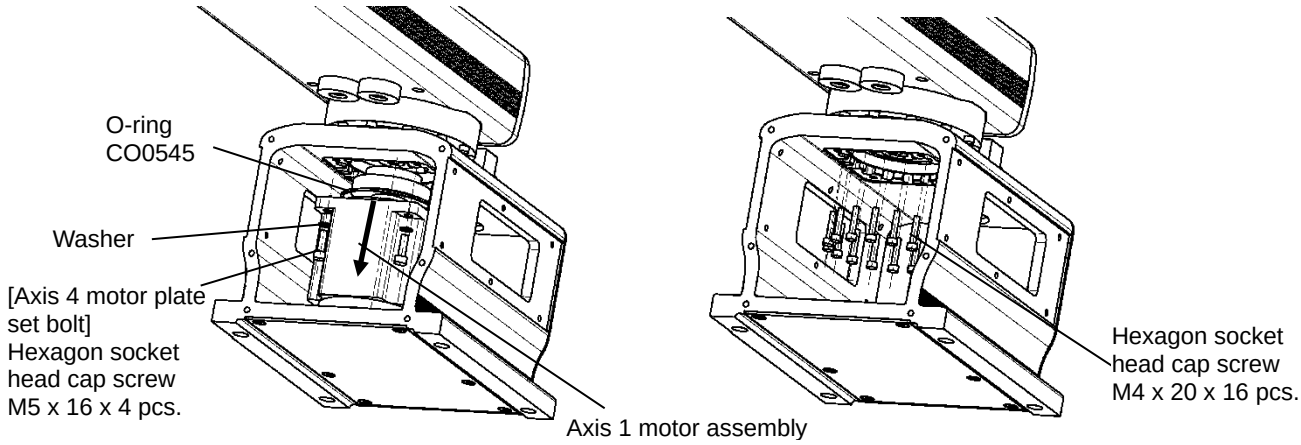


Fig. 3.45 Dismounting axis 1 motor assembly and Arm 1 (THL500, THL600, THL700)

⚠ CAUTION

– When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

– Be sure to handle the arm carefully.
Drop impact or application of excessive external force could ruin the arm.

4) Remove twelve (12) hexagon socket head cap screws (M4 x 30) that secure the axis 1 reduction gear to Arm 1, and then remove the axis 1 reduction gear. Also remove the O-ring mounted in the groove of Arm 1 at this time. Also, remove the liquid gasket attached to the base mounting surface of the axis 1 reduction gear with the back of a cutter or a similar tool without damaging.

5) Remove the wave generator from the axis 1 motor. (See “Para. 3.4.3, Dismounting Axis 1 Motor.”)

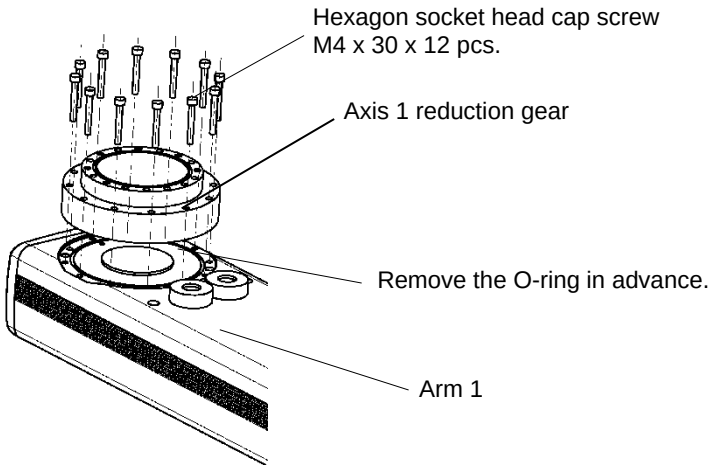


Fig. 3.46 Dismounting axis 1 reduction gear (THL500, THL600, THL700)

3.7.4 Mounting Axis 1 Reduction Gear

NOTICE

– Be sure to use wave generator provided for new reduction gear to be replaced.
The combination of the reduction gear and wave generator is determined. If they are used in a wrong combination, abnormal sound will occur. The positioning accuracy and lifespan will be reduced.
– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.
– Never forget fitting the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
- Use the new O ring provided for the reduction gear.

1) Mount a new wave generator provided as an accessory with a new reduction gear. For the mounting procedure, see “Para. 3.4.4, Mounting Axis 1 Motor.”)

2) Cleanly wipe off dusts and stains on the axis 1 reduction gear of Arm 1. Apply grease to an O-ring attached to the new reduction gear, and then mount it in the O-ring groove of Arm 1.

3) Secure the axis 1 reduction gear with twelve (12) hexagon socket head cap screws (M4 x 30).

Reduction Gear	Recommended anticorrosive	Maker
A type	SK-1A	HDS
B type	SFB No.1	Nidec-Shimpo Corp.

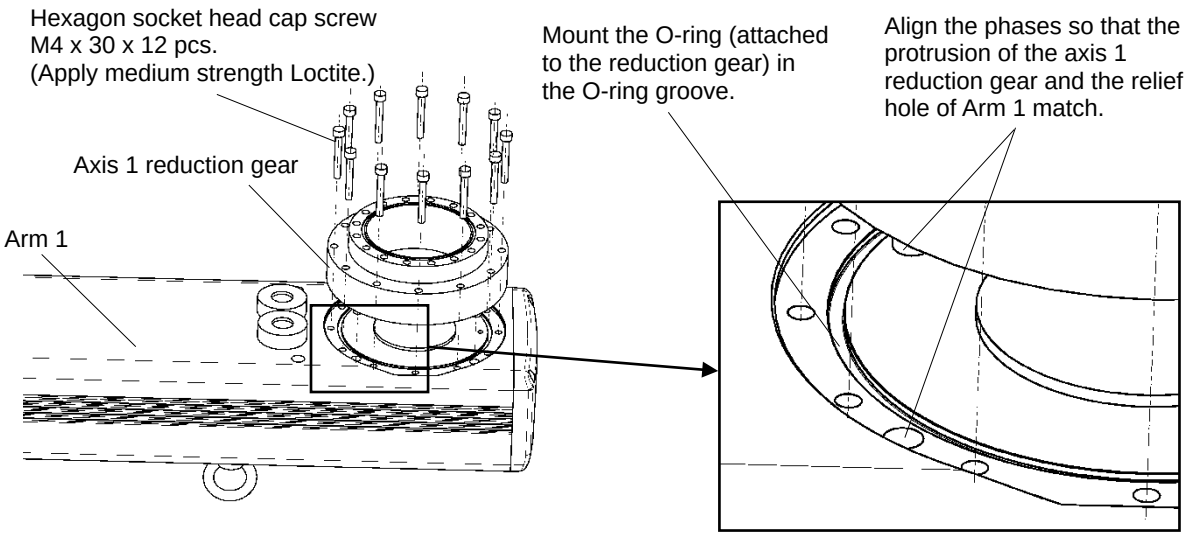


Fig. 3.47 Mounting axis 1 reduction gear (THL500, THL600, THL700)

4) Apply grease inside the 1st-axis reduction gear proper as shown on the diagram. Apply an adequate volume of grease on the wave generator side as well. Further, apply grease to the O-ring attached to the reducing gear, and insert the O-ring into the O-ring groove of the reducing gear.

Reduction ear	Amount of grease filled in Axis 1
A type	30 g
B type	56 g

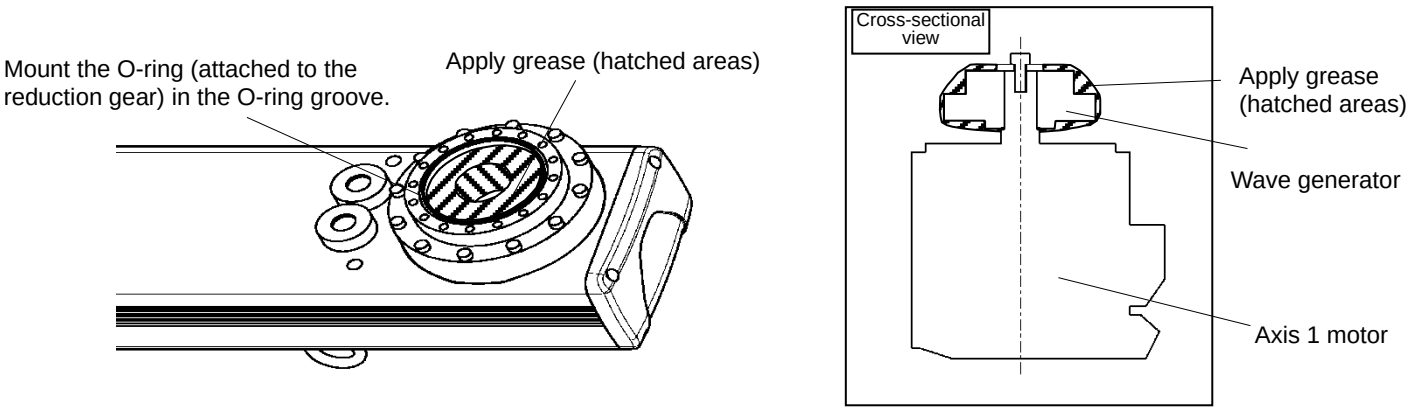


Fig. 3.48 Application of grease to axis 1 reduction gear (THL500, THL600, THL700)

NOTICE

– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.
– Be sure to observe the filling amount of grease.
If the filling amount is tool large, the starting torque increases and the internal seal is broken due to rise in the internal pressure in the reduction gear. If the filling amount is too small, malfunction is caused by poor lubrication.

5) Secure Arm 1 to the base with sixteen (16) hexagon socket head cap screws (M4 x 20). Apply grease to the O-ring (CO0545), and then mount it to the base groove side surface.

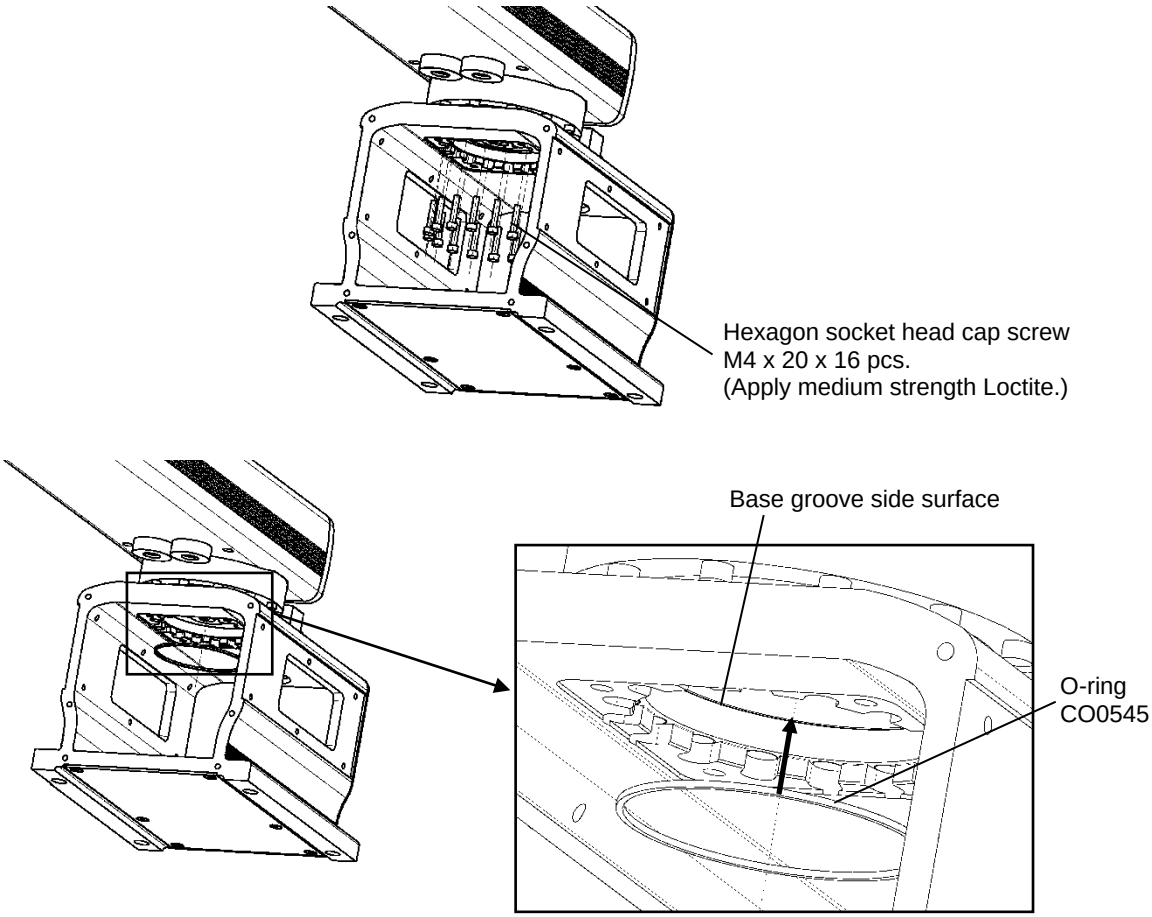


Fig. 3.49 Mounting arm 1 and O-ring (THL500, THL600, THL700)

NOTICE

– Never forget fitting the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
- Use the new O ring provided for the reduction gear.

6) Connect the connectors (J1AS and J1BS) of the cables.

7) Mount the axis 1 motor assembly. (See “Para. 3.4.4, Mounting Axis 1 Motor.”)

8) After changing Axis 1 motor, move Arm 1 by hand and check that there is no abnormal sound .

9) Mount the base front cover and the base side covers. (See “Para. 3.3.2, Base Covers.”)

10) Set up the axis 1 home position to complete axis 1 reduction gear replacement. Because the reduction gear is changed, home position settings of Axes 2 through 4 are required.

11) Carry out a test operation of Axis 1 to make sure that each part operates properly.

3.7.5 Dismounting Axis 2 Reduction Gear

1) Remove the arm 2 cover. (See “Para. 3.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers. It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Remove J2AS and J2AP (axis 2 power drive cables), J2BS and J2BP (axis 2 encoder cables).

4) Remove the harness guide and the support plate, and then remove the axis 2 motor assembly. Also remove the O-ring (CO0583A) of the O-ring groove. (See “Para. 3.4.5, Dismounting Axis 2 Motor.”)

5) Remove twelve (12) hexagon socket head cap screws (M3 x 20) that secure Arm 2 and the reduction gear.

⚠ CAUTION

– When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

– Be sure to handle the arm carefully.
Drop impact or application of excessive external force could ruin the arm.

6) Remove twelve (12) hexagon socket head cap screws (M3 x 30) that secure the axis 2 reduction gear mounted to Arm 1, and then remove the axis 2 reduction gear. At this time, remove the O-ring mounted in the O-ring groove of Arm 1.

7) Remove the wave generator from the axis 2 motor. (See “Para. 3.4.5, Dismounting Axis 2 Motor.”)

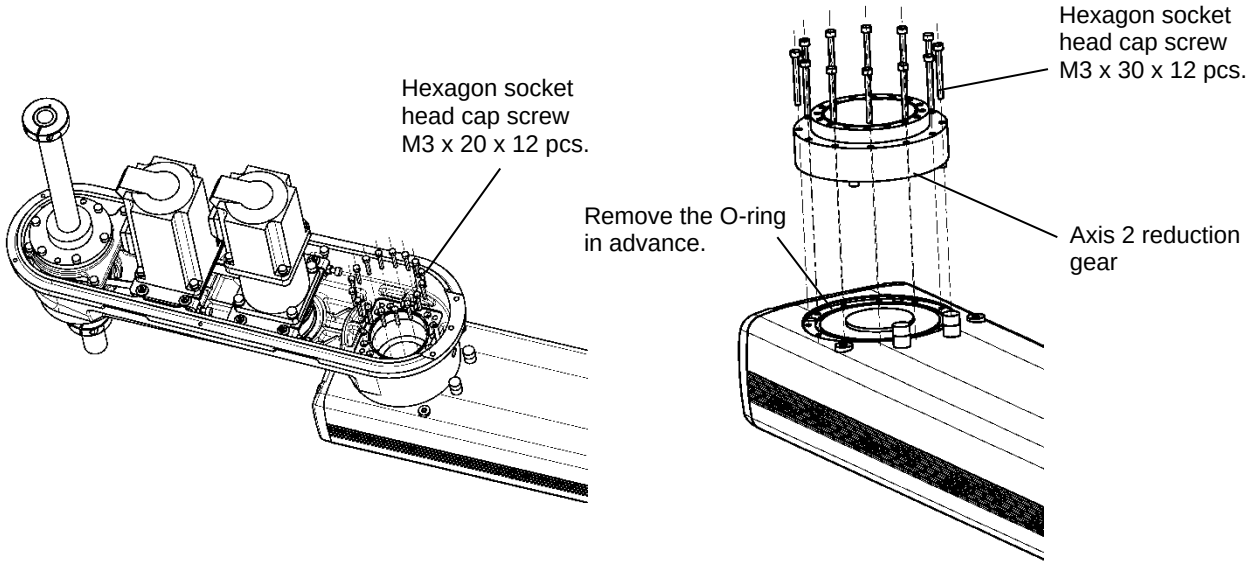


Fig. 3.50 Dismounting axis 2 reduction gear (THL500, THL600, THL700)

3.7.6 Mounting Axis 2 Reduction Gear

NOTICE

– Be sure to use wave generator provided for new reduction gear to be replaced.
The combination of the reduction gear and wave generator is determined. If they are used in a wrong combination, abnormal sound will occur. The positioning accuracy and lifespan will be reduced.
– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.
– Never forget fitting the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
- Use the new O ring provided for the reduction gear.

1) Mount the wave generator attached to the new reduction gear to the axis 2 motor. (See “Para. 3.4.6, Mounting Axis 2 Motor.”)

2) Cleanly wipe off dusts and stains on the reduction gear of Arm 1.
Apply grease to an O-ring attached to the new reduction gear, and then mount it in the O-ring groove of Arm 1.

3) Secure the axis 2 reduction gear to Arm 1 with twelve (12) hexagon socket head cap screws (M3 x 30).

4) Apply grease to the axis 2 reduction gear. (See “Para. 3.7.4, Mounting Axis 1 Reduction Gear.” However, the amount of grease filled is different from that for the axis 1 reduction gear. Regarding the filled amount for the axis 2 reduction gear, see the following.)

Reduction gear	Amount of grease filled in Axis 2
A type	15 g
B type	35 g

5) Apply grease to the other O-ring attached to the new reduction gear, and then mount it in the O-ring groove of the axis 2 reduction gear.

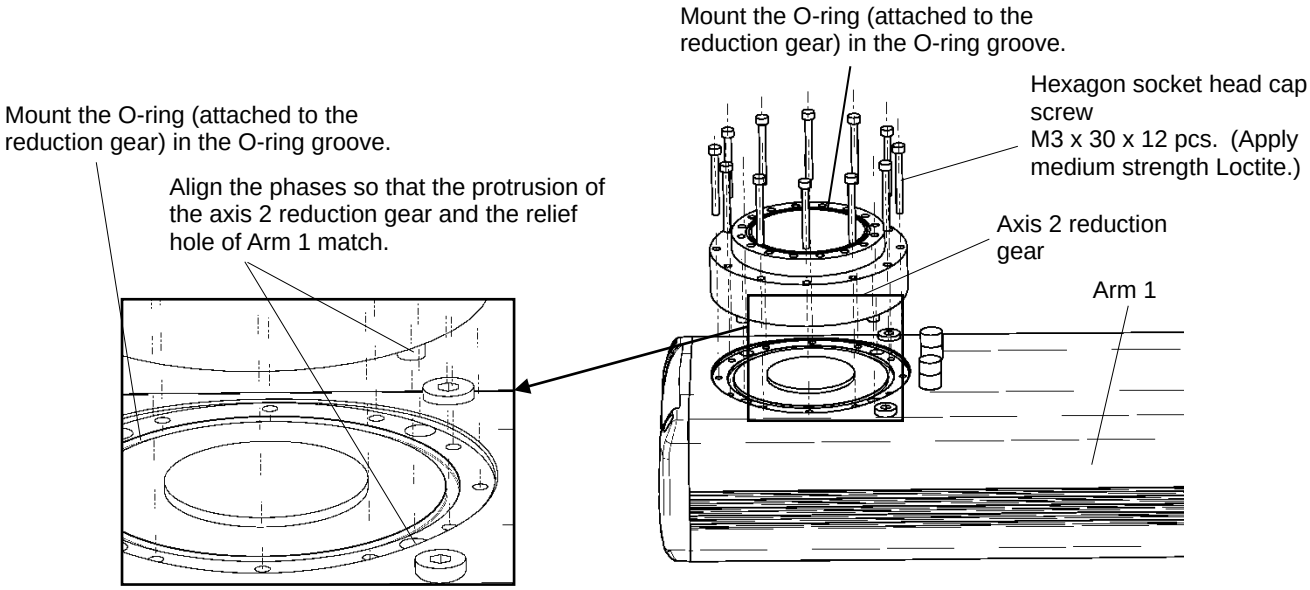


Fig. 3.51 Mounting axis 2 reduction gear (THL500, THL600, THL700)

NOTICE

– Never forget fitting the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
- Use the new O ring provided for the reduction gear.

– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

– Be sure to observe the filling amount of grease.
If the filling amount is tool large, the starting torque increases and the internal seal is broken due to rise in the internal pressure in the reduction gear. If the filling amount is too small, malfunction is caused by poor lubrication.

6) Secure Arm 2 to the new reduction gear with twelve (12) hexagon socket head cap screws (M3 x 20).

7) Apply grease to the O-ring (CO0538A), and then mount it in the O-ring groove of Arm 2.

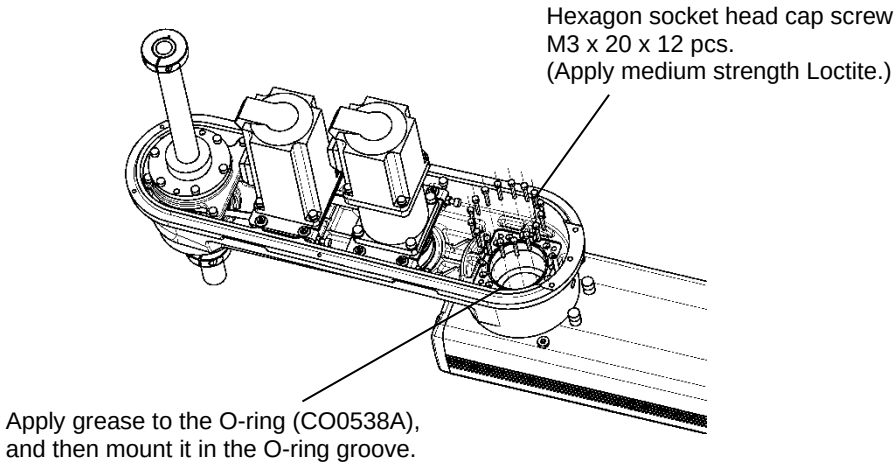


Fig. 3.52 Mounting arm 2 (THL500, THL600, THL700)

⚠ CAUTION

– When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

– Be sure to handle the arm carefully.
Drop impact or application of excessive external force could ruin the arm.

8) Mount the axis 2 motor assembly, the harness guide and the support plate. Put back the cables to the original locations, and then mount the arm 2 cover.

9) Move Arm 2 by hand and check that there is no abnormal sound before turning on the power.

10) Set up the home positions of axes 2 to complete axis 2 reduction gear replacement.

3.7.7 Replacing Axis 4 Reduction Gear

NOTICE

– Be sure to handle the reduction gear carefully.
  Drop impact or application of excessive external force could ruin the reduction gear.
– Be careful not to forget to loosen the coupling that fastens the reduction gear and motor shaft when dismounting reduction gear.
– Be careful not to forget tightening the coupling that fastens the reduction gear and motor shaft when dismounting reduction gear.

1) Remove the arm 2 cover. (See “Para. 3.3.1, Arm 2 Cover.”)
2) Cut the cable ties of the cables with nippers or the like, and remove J4AS and J4AP (axis 4 power drive cables), J4BS and J4BP (axis 4 encoder cables). It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.
3) Disconnect the axis 3 motor assembly and axis 4 motor assembly only from Arm 2. Other parts need not be removed.
4) Remove the cap attached to the lateral side of the axis 4 reduction gear, then loosen the coupling bolt (M4) securing the axis 4 motor shaft and the input shaft of the axis 4 reduction gear.
5) Remove the hexagon socket head cap screws (M4 x 12 x 4 pcs.) and washers securing the motor, and draw out the axis 4 motor upward.
6) While holding the axis 4 motor pulley with pliers or the likes, remove the hexagon socket head cap screw (M4 x 12 x 6 pcs.) to remove the pulley. In doing so, use a waste cloth (or the like) as a cushion to protect the pulley from scratches.
7) Remove the hexagon socket head cap screws (M5 x 16 x 4 pcs.) and washers securing the axis 4 motor plate and reduction gear to the motor base, then dismount the reduction gear.

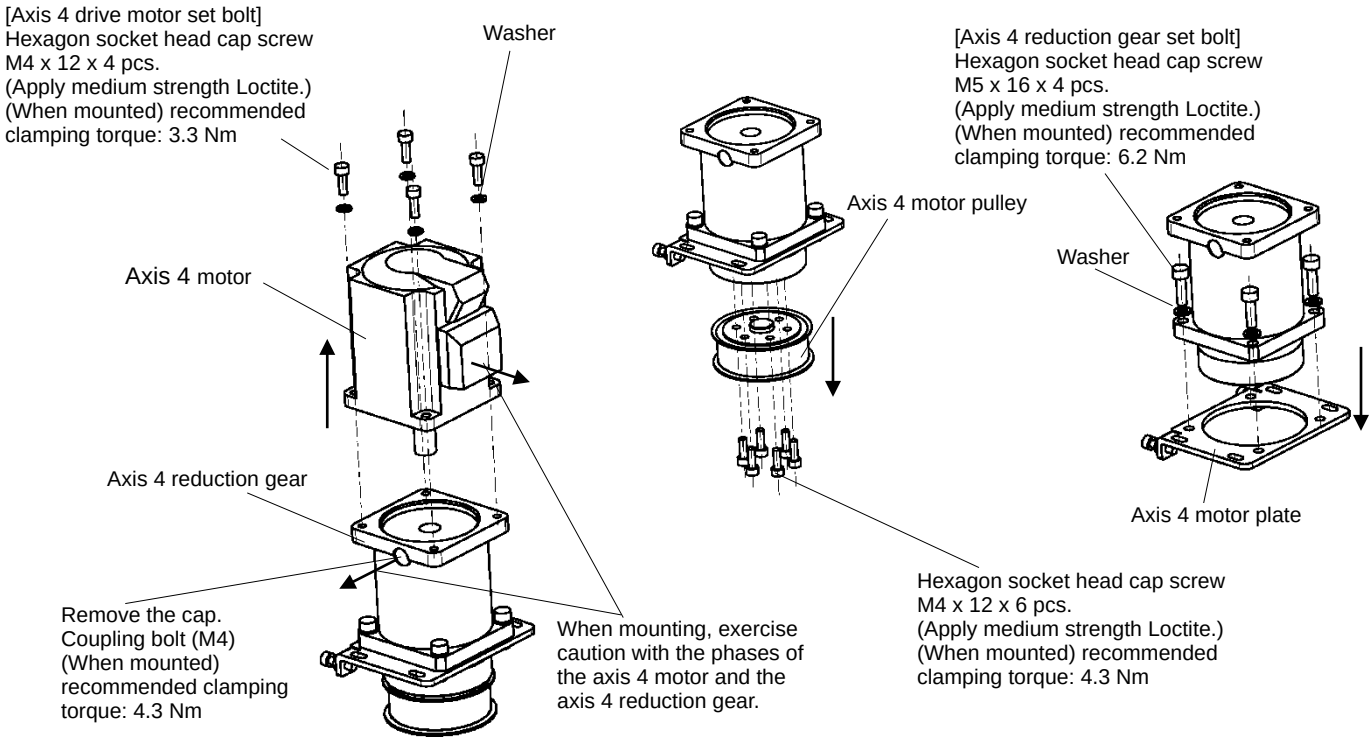


Fig. 3.53 Replacing axis 4 reduction gear (THL500, THL600, THL700)

8) Assemble the new reduction gear, motor and pulley in the reverse order of Steps 3) to 6) above. Be careful with the phases of the axis 4 motor, the axis 4 reduction gear and the axis 4 motor plate at this time.

9) Assemble the axis 4 motor assembly and timing belt according to Steps 11) through 15) of “Para. 3.5.4, Replacing Axis 4 Timing Belt” above.

10) Mount the arm 2 cover. Now replacement of the axis 4 reduction gear completes.

4. Maintenance of the Main Robot (THL800, THL900, THL1000,THL1200)

4.1 Details of Inspection

4.1.1 Check of Each Bolt (or Screw) for Clamping

⚠ DANGER

– Be sure to perform daily inspection for tool set bolts and robot installation bolts.
Robot hand falls during robot operation. Robot collapses.

NOTICE

– Be sure to perform daily inspection for tool set bolts and robot installation bolts.
If the position is shifted, the robot will bump into peripheral devices.

1) Tool Set Bolts

Using the hexagonal wrench key, make sure that the clamping bolts (M4 x 4 pcs.) of the tool set flange (option), which are clamped to the tool shaft, are tightened completely. If loose, tighten them completely.

Using the hexagonal wrench key, make sure that the mounting bolts (M4 x 4 pcs.), which secure the tool to the tool flange, are tightened completely. If loose, tighten them completely. (Loctite not necessary)

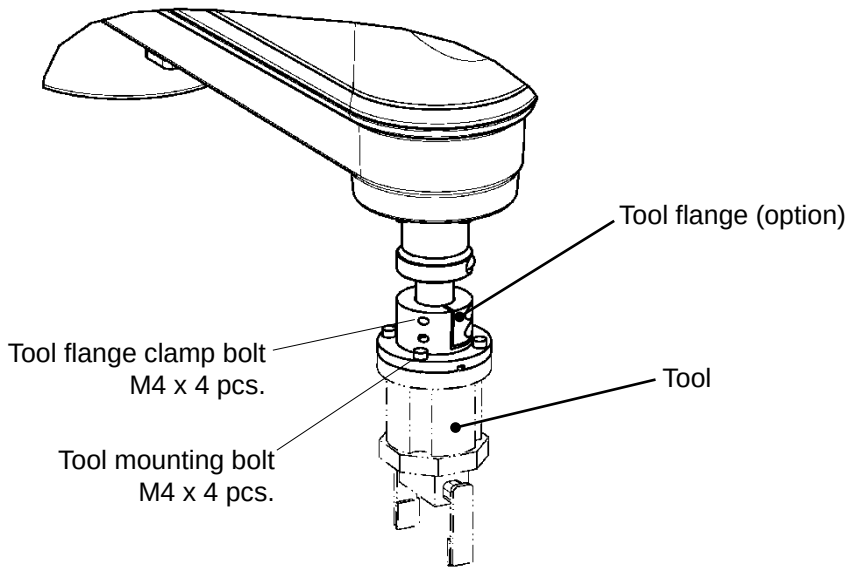


Fig. 4.1 Tool mounting bolts

2) Robot Installation Bolts

Make sure, using the hexagonal wrench key, that the installation bolts of the main robot base are tightened completely. If loosened, tighten them completely. (Loctite is not necessary)

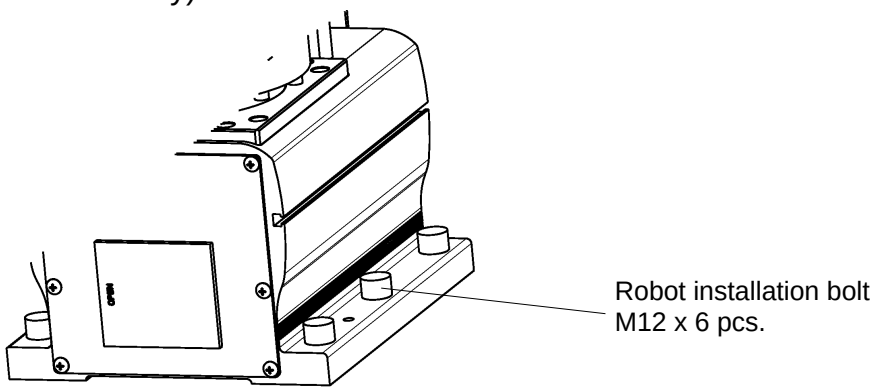


Fig. 4.2 Robot installation bolts (THL800, THL900, THL1000, THL1200)

3) Motor Set Bolts

Make sure, using the hexagonal wrench key, that the bolts for securing each axis drive motor are tightened completely. If loosened, tighten them completely. Also make sure that the motor and motor plate bolts are tightened completely. For the places where the following bolts are used, see “Para. 4.4, Replacing Motor.”

Table 4.1 Motor and motor plate mounting bolt (THL800, THL900, THL1000, THL1200)

Location of bolt	Type	Bolts used	Reference Para.
Axis 1 drive motor mounting bolt	M4 x 14	4 pcs.	4.4.4 Mounting Axis 1 Motor
Axis 2 drive motor mounting bolt	M4 x 12	4 pcs.	4.4.6 Mounting Axis 2 Motor
Axis 3 drive motor mounting bolt	M4 x 12	4 pcs.	4.4.8 Mounting Axis 3 Motor
Axis 3 motor plate mounting bolt	Flange head bolt M4 x 10	4 pcs.	4.4.8 Mounting Axis 3 Motor
Axis 4 drive motor mounting bolt	M4 x 12	4 pcs.	4.4.10 Mounting Axis 4 Motor
Axis 4 reduction gear mounting bolt	M5 x 16	4 pcs.	4.7.7 Replacing Axis 4 Reduction Gear
Axis 4 motor plate mounting bolt	Flange head bolt M4 x 10	4 pcs.	4.5.4 Replacing Axis 4 Timing Belt

* For the recommended clamping torque, see Para. 1.5 Clamping Hexagon Socket Head Cap Screws and Set screws
* It is not necessary to apply Loctite to the flange head bolts of Axes 3 and 4 motor plate mounting bolts.

4.1.2 Check of Each Cable and Air Tube for Abrasion

Disassemble the arm 2 cover and the base rear cover, and make sure that each cable is not worn out, broken or cracked. Especially, carefully inspect the vicinity of the cable outlets.

For how to dismount the covers, see “Para. 4.3, Dismounting and Mounting Each Cover.”

The figure below shows an example of air tube installation.

Also inspect the wear and disorder of the air tube as well as the pipe connection. Make sure that the air tube is not worn out and is installed properly, and pipe connection is correct. Otherwise, correct them.

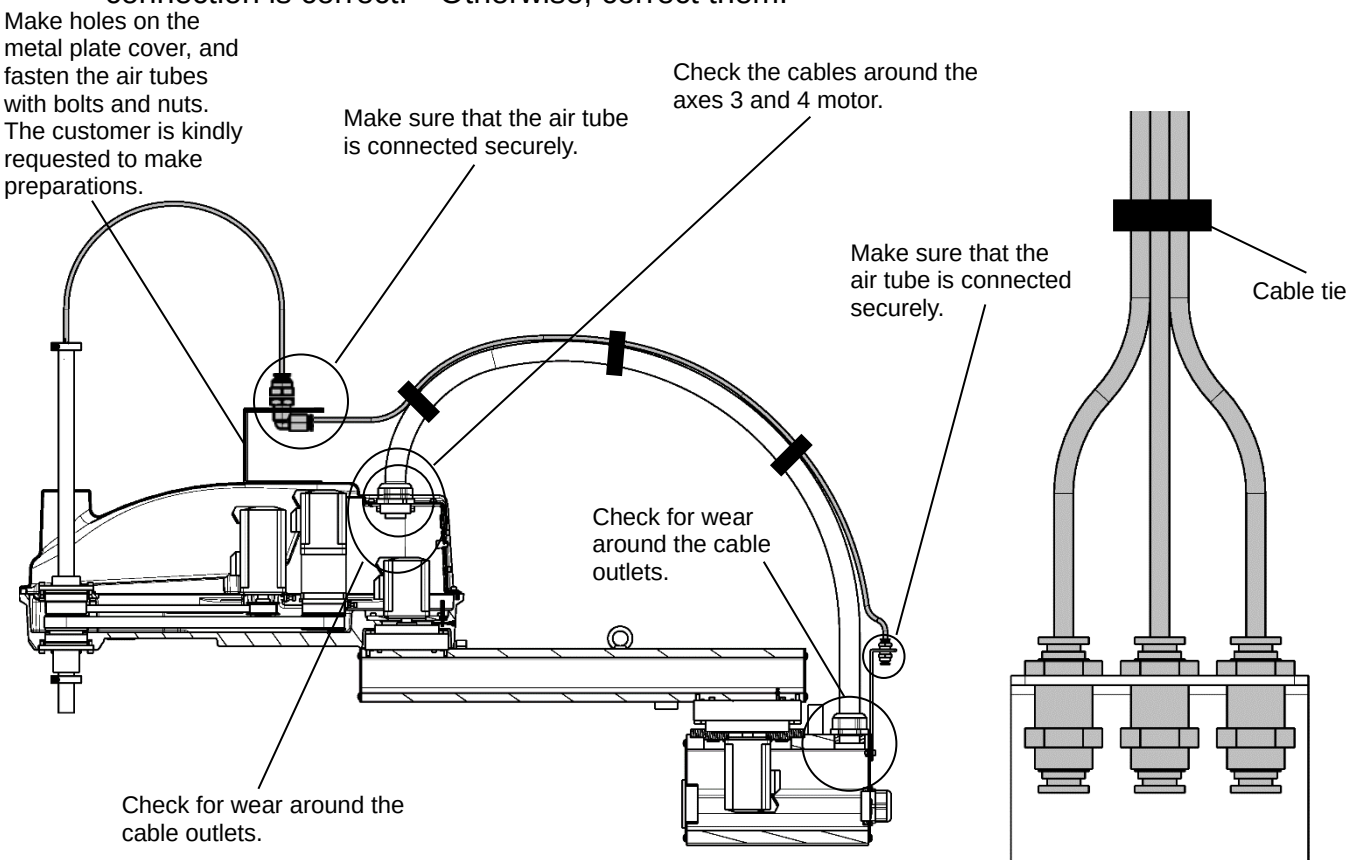


Fig. 4.3 Cable inspection locations (THL800, THL900, THL1000)

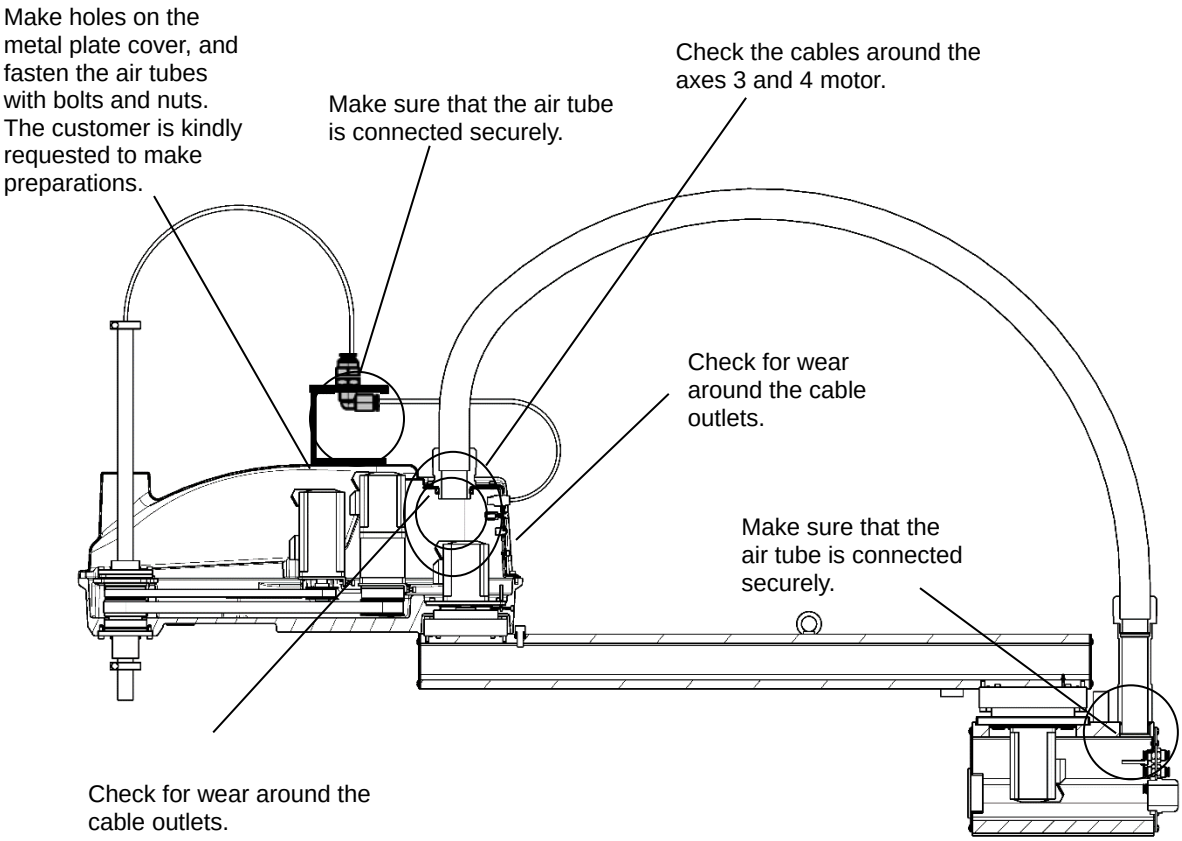


Fig. 4.4 Cable inspection locations (THL1200)

NOTICE

– The air tube is a consumable item. It must be checked periodically, and replaced if broken.
– Fig. 4.3 and Fig. 4.4 shows an example of ductwork.
Note that breakage of air tube, etc. is not always under our warranty.

4.1.3 Check of Cable Clamp Tightening

Using a wrench, make sure that the clamps securing the cable are not loose. There are two clamps, one on the arm 2 side and the other on the base side. If they are loose, apply Loctite (low adhesive force) and tighten them. When checking the tightened clamps, check the one on the base side first. By checking the tightness of the clamp on the base side first, the twist of the cable can be adjusted slightly on the arm 2 side. Make sure to tighten the clamps in the posture as shown in the figure below. If the clamps are tightened while the arm is bent, the cable cannot be installed correctly.

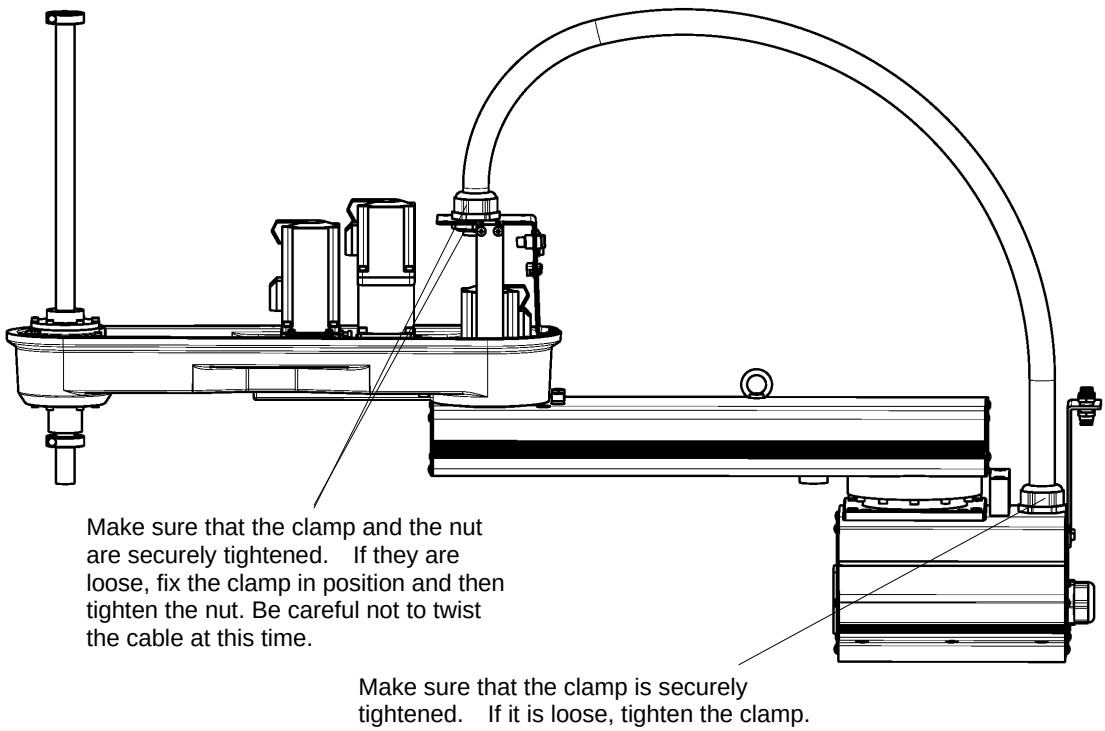


Fig. 4.5 Cable clamp inspection locations (THL800, THL900, THL1000)

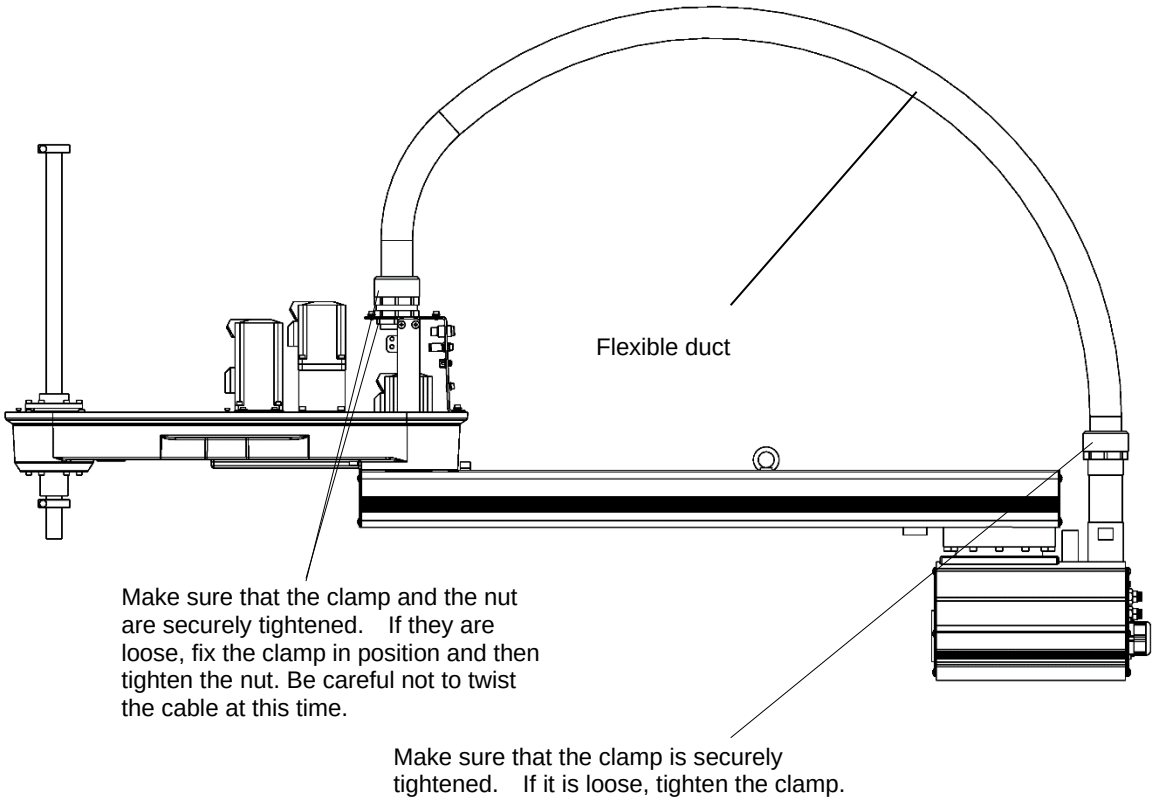


Fig. 4.6 Cable clamp inspection locations (THL1200)

4.1.4 Check of Each Axis for Operation

Connect the power plug of the controller to the power source, then keep the EMERGENCY pushbutton switch in the depressed condition. Move each axis by hand and make sure that it can move smoothly.

For Axis 3, when the brake release switch is pressed, the brake is released. Take careful precautions at this time as the tool shaft may drop according to the weight of the hand and tool.

NOTICE

· When axis 1 and axis 2 are moved near the working envelope limit and a hand is off from them, axis 1 and axis 2 may move due to reaction of the cable.(Except THL1200)

4.2 Layout of Robot Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 4.5.

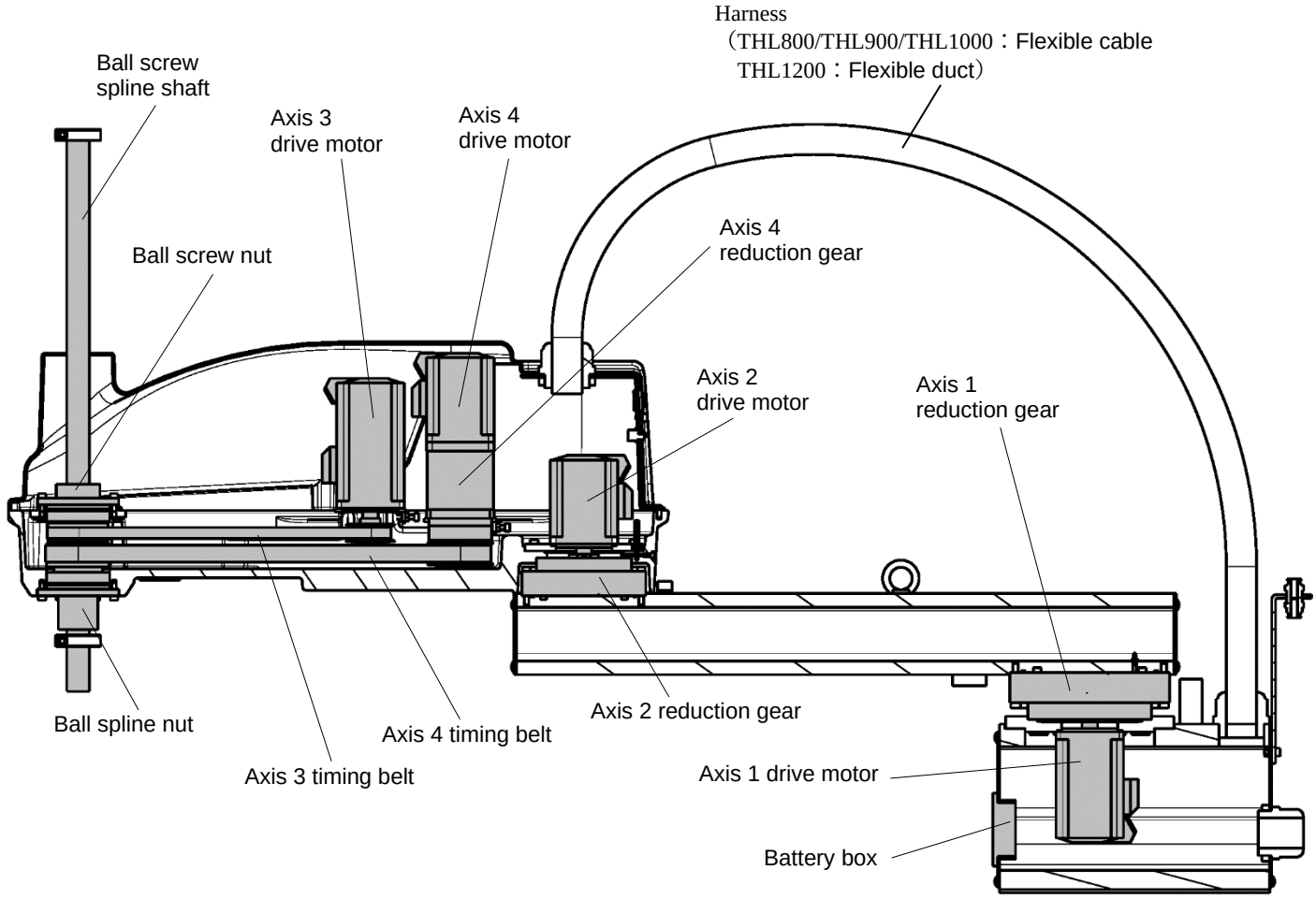


Fig. 4.7 Layout of robot mechanical components (THL800, THL900, THL1000,THL1200)

4.3 Dismounting and Mounting Each Cover

This paragraph describes the dismounting and mounting of the covers, which are common to the maintenance and replacement of each unit.

⚠DANGER

– When opening the cover, never allow entry of moisture or contaminant into the robot.
If the power is turned on while moisture or contaminant is left, an electric shock may result.

NOTICE

– When opening the cover, never allow entry of moisture or contaminant into the robot.
If the power is turned on while moisture or contaminant is left, the robot may malfunction.
– When mounting the covers, prevent the cables from being caught.
Otherwise, the cables will be broken.
– Be sure to restore where to route the cables pulled out by removing the cover and where to secure with the cable straps to their original state.
If the cables are pushed by force or where to secure changes, the cables will be worn out or broken.
- Before pulling out the cables, take pictures for restoration

4.3.1 Arm 2 Cover

The arm 2 cover is secured to Arm 2 and the harness guide with 14 hexagon socket head bolts (M3 x 6 x 4 pcs., M4 x 6 x 8 pcs., M4 x 10 x 2 pcs.) and nylon washers. (It is not necessary to apply Loctite when installing.)

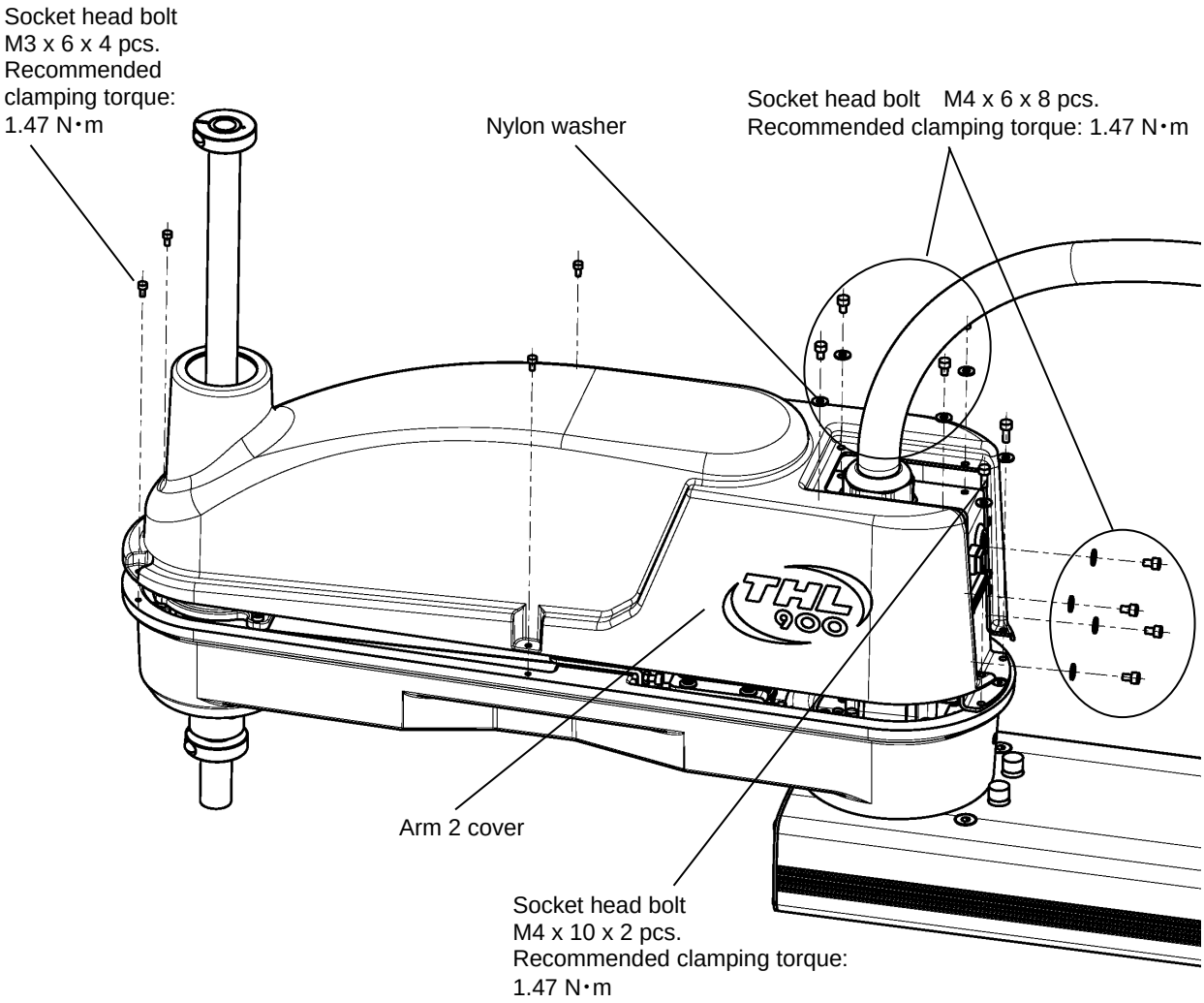


Fig. 4.8 Arm 2 cover (THL800, THL900, THL1000, THL1200)

After the cover is mounted, manually move up and down the ball screw spline shaft while pressing the brake release switch, and make sure that the ball screw hole for the arm 2 cover will not interfere with the ball screw stopper.

4.3.2 Base Covers

There are two (2) types of base covers: base front cover and base rear cover. (It is not necessary to apply Loctite to all the covers when installing.)
The base front cover is secured to the base with six (6) cross truss head screws (M4 x 8). It can be dismounted from the base when these fastening screws are removed, but do not pull them with force as they are connected with the connectors inside. In the case of THL 800, THL900 and THL1000, the base rear cover is secured to the base with six (6) cross truss head screws (M4 x 8). It can be dismounted from the base when these fastening screws are removed, but do not pull them with force as they are connected with the connectors inside. In the case of THL 1200, the base rear cover is composed of two types of base covers, the base rear cover 1 and the base rear cover 2, tightened together by six (6) cross truss head screws (M4 x 10). A cable connected to the Arm 2 is fixed to the base rear cover 1, and another cable is connected inside the base rear cover 2, so be sure not to jerk any of them with force. For the cable fixed to the base rear cover 1, pull out the pins after removing the binding bands. We recommend taking photos at this stage so that the binding bands can be restored to the original state.

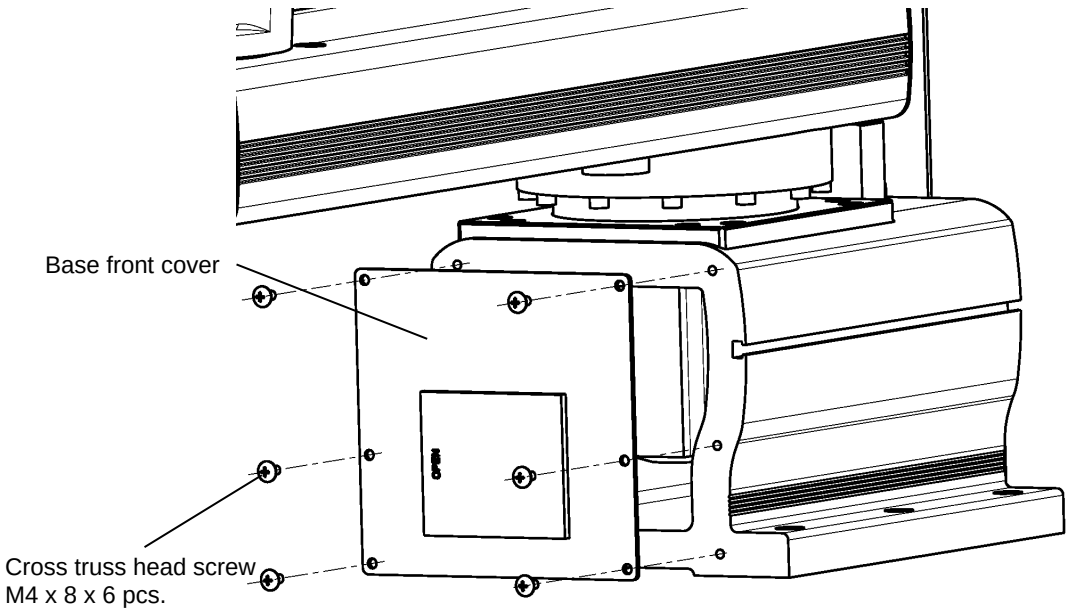


Fig. 4.9 Base front cover (THL800, THL900, THL1000, THL1200)

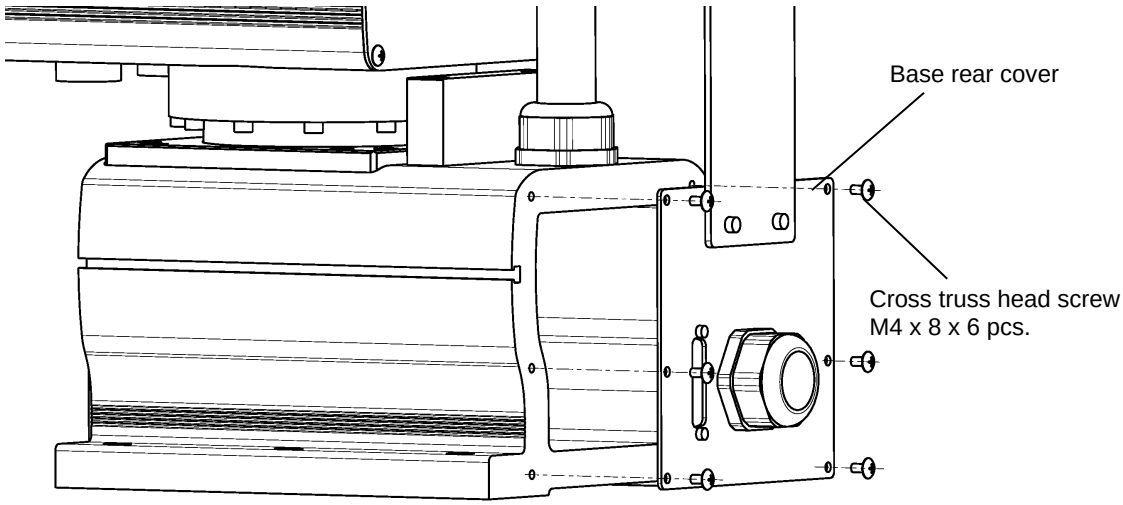


Fig. 4.10 Base rear cover (THL800, THL900, THL1000)

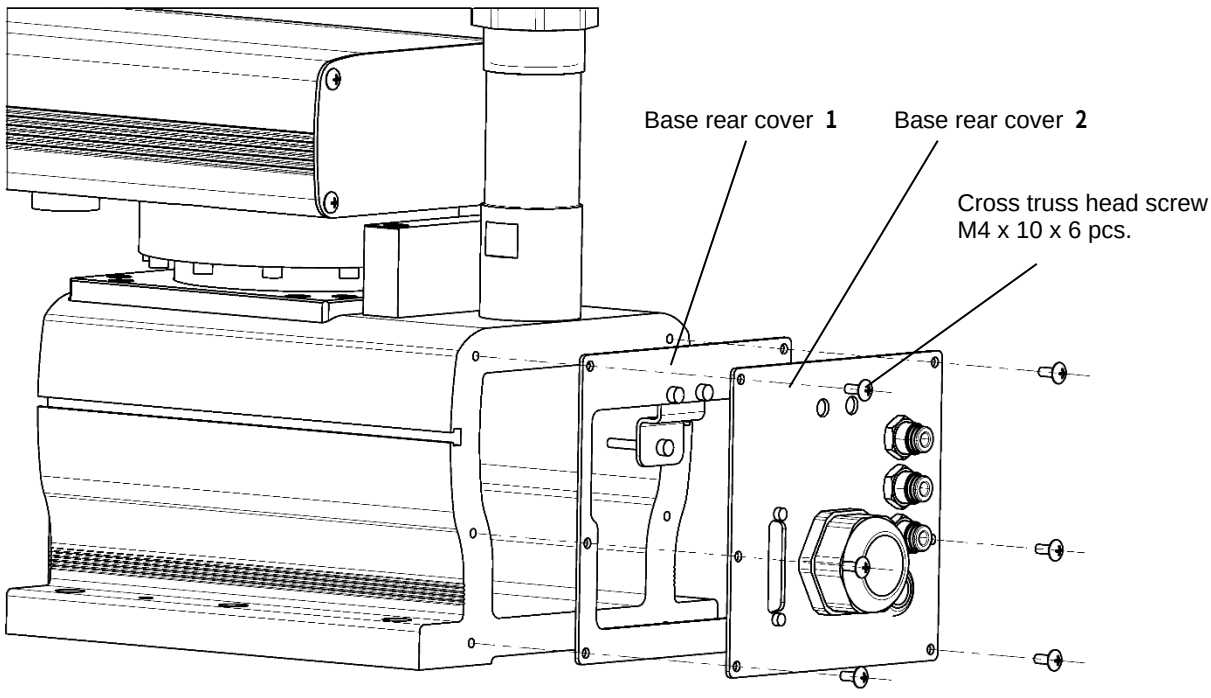


Fig. 4.11 Base rear cover (THL1200)

4.4 Replacing Motor

The motor is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠ DANGER

· The power plug of the controller must be removed before replacement work.
If replacement work is done with the power turned on, this may result in an electric shock or a malfunction, causing great danger.

NOTICE

– Never cause heavy impact on the motor shaft.
The motor and position detector may malfunction due to impact.
– Never disassemble the motor and position detector.
Otherwise, they may be disabled due to difference in position or intrusion of foreign material.
– When replacing a motor, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– After the motor is replaced, be sure to re-set zero position.
Replacing a motor causes deviation from mechanical zero position; therefore, the robot cannot make normal motion.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”

4.4.1 Type of Motor

The motors employed in this robot are shown below. When you place an order for a replacement motor, make sure of the robot model (THL800, THL900, THL1000, THL1200), the serial number, the axis name, and our drawing number according to the following table.
For the location where the serial number plate is attached, see the “Safety Manual.”
The motor and pulley for Axis 3 are connected using a key. Thus, key alignment is necessary during motor replacement.
For the type of the key, also see the table below.

Table 4.2 Type of Motor (THL800, THL900, THL1000, THL1200)

Description	Axis name	Type	Our drawing No.	Unit code
AC servo motor	Axis 1		S746328	Y610A34C0
	Axis 2		S875290	Y610A3M30
	Axis 3		S746337	Y610A3430
	Axis 4		S875291	Y610A3M40
Key	Axis 1	Square key 5 x 5 x 16		
	Axis 3	Square key 4 x 4 x 16		

* The key to be used with Axis 2 is attached to the motor.

4.4.2 Motor Locations

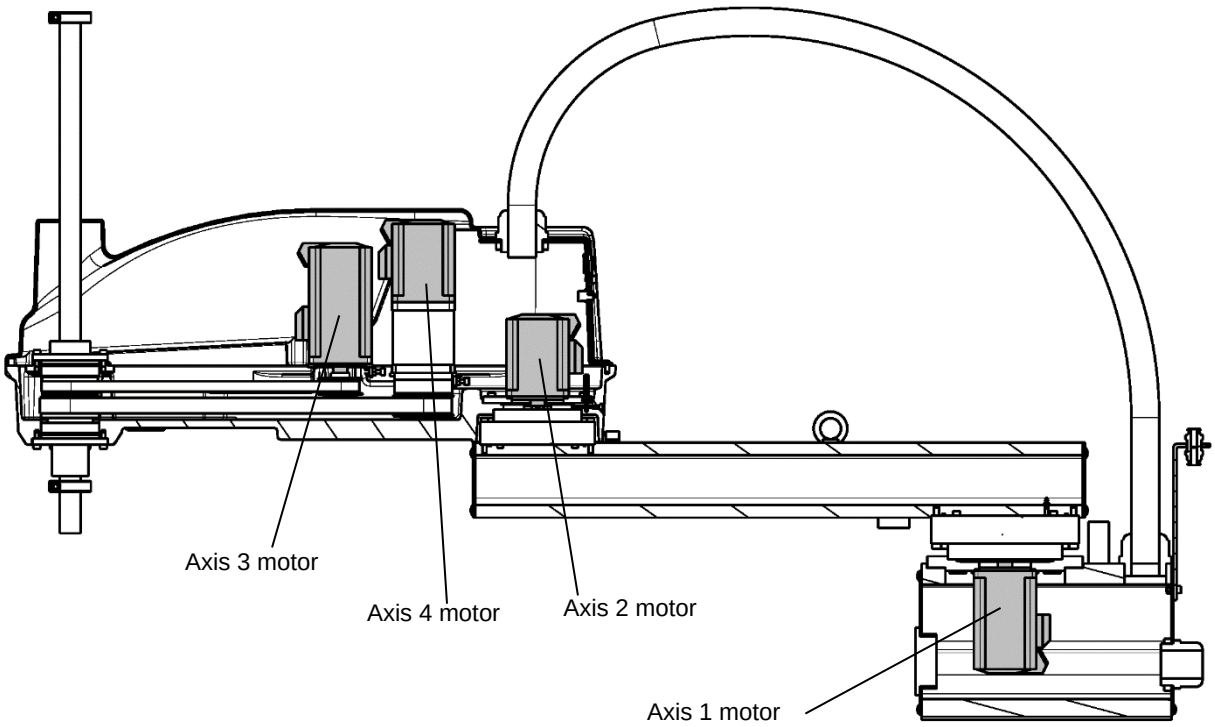


Fig. 4.12 Motor locations (THL800, THL900, THL1000, THL1200)

4.4.3 Dismounting Axis 1 Motor

1) Remove the hexagon socket head bolts (M6 x 16 x 8 pcs.) securing the axis 1 motor base while changing the position of the arm 1 position. Since the arms 1 and 2 are heavy, two persons must cooperate with each other for this work. Further, the motor cable is connected, so do not pull it forcibly. The axis 1 motor base is connected by a pin (MS5-15). Do not lose this pin. The pin need not be removed.

2) Remove the connector from the motor. Lay the arms 1 and 2 down so that undue force will not be applied to the harness of the main body.

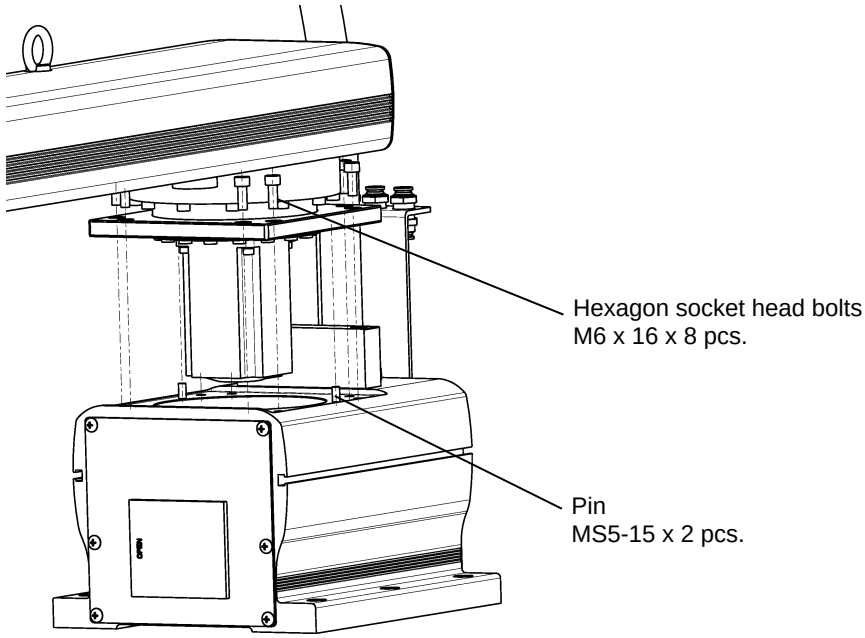


Fig. 4.13 Removal of motor mount (THL800, THL900, THL1000, THL1200)

3) Remove hexagon socket head bolts (M5 x 30 x 16 pcs.) and washers securing the axis 1 motor assembly, and pull out the axis 1 motor assembly. At the same time remove the O-ring (CO0551A) from the axis 1 motor base as well.

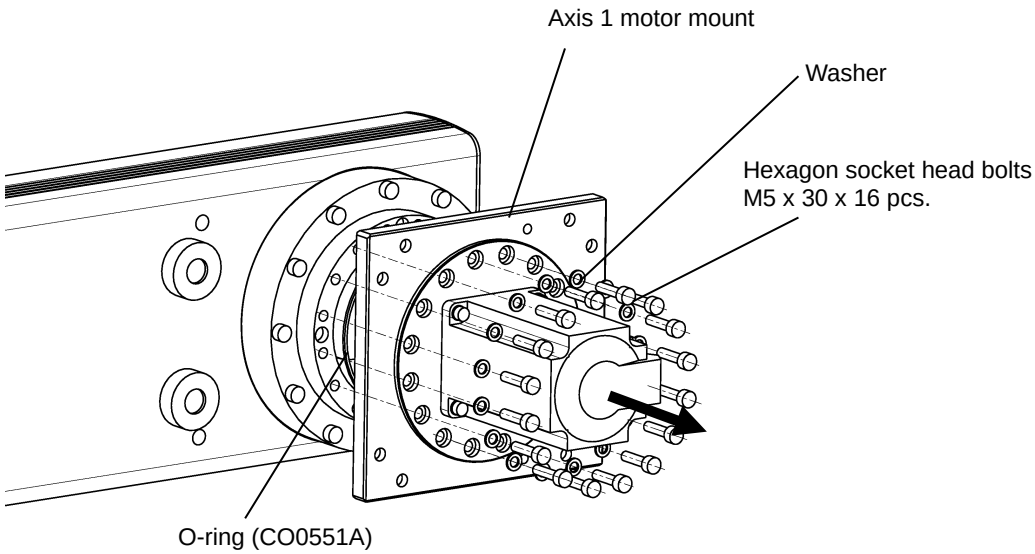


Fig. 4.14 Dismounting axis 1 motor assembly (THL800, THL900, THL1000, THL1200)

4) Remove a hexagon socket head bolt (M4 x 12 x 1 pc.) and washer securing the wave generator, and remove the wave generator.

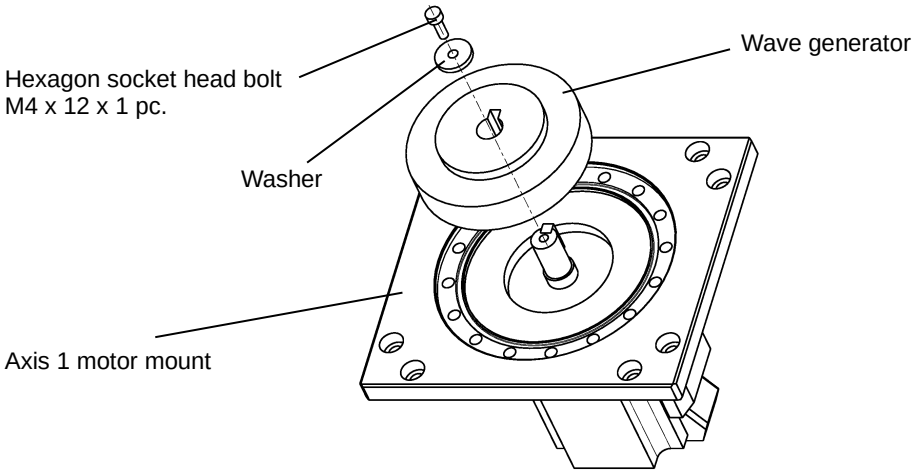


Fig. 4.15 Dismounting axis 1 wave generator (THL800, THL900, THL1000, THL1200)

5) Remove hexagon socket head bolts (M4 x14 x 4 pcs.) and washers securing the axis 1 motor. At the same time, remove the O-ring (CO0540A) from the axis 1 motor base as well.

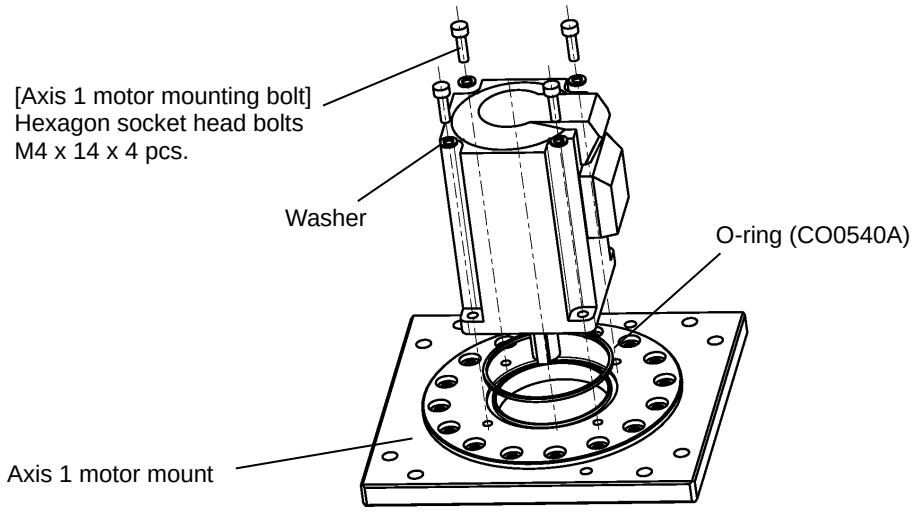


Fig. 4.16 Dismounting axis 1 motor (THL800, THL900, THL1000, THL1200)

4.4.4 Mounting Axis 1 Motor

1) Using a new axis 1 motor and key, perform key adjustment. At the time of mounting, apply SK-1A grease to the key and motor shaft end (to ensure easier insertion of the wave generator).

2) Apply grease to the O-ring (CO0540A) and set the O-ring into the O-ring groove of the axis 1 motor mount.
Use hexagon socket head bolts (M4 x 14 x 4 pcs.) and washers to mount the axis 1 motor on the axis 1 motor mount. In this case, take care of the phases of the 1 axis motor mount and motor.

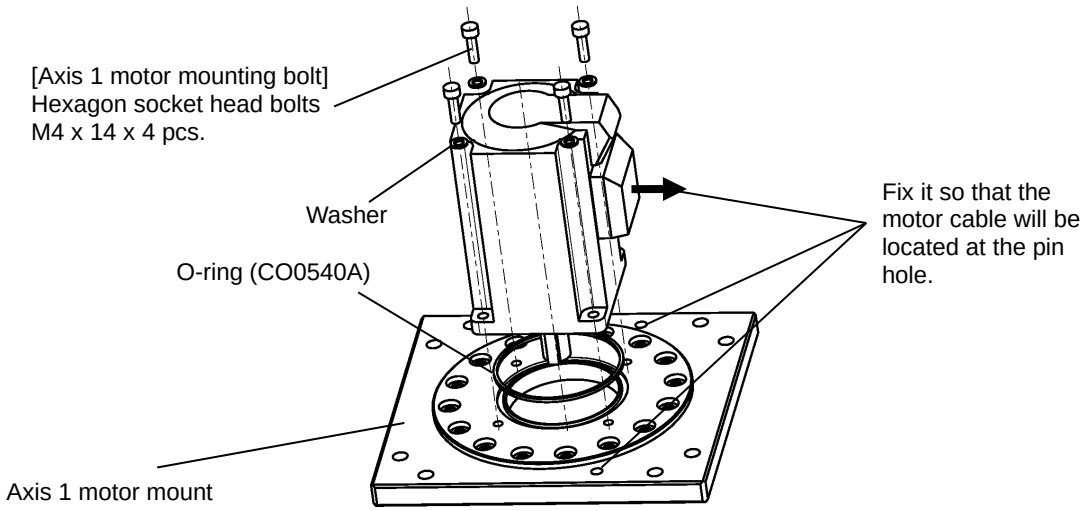


Fig. 4.17 Mounting axis 1 motor (THL800, THL900, THL1000, THL1200)

NOTICE

– Never forget to fit the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.

3) Fix the wave generator using a hexagon socket head bolt (M12 x 4 x 1 pc.) and washer. Apply grease to the O-ring (CO0551A) and set the O-ring into the O-ring groove of the axis 1 motor mount.

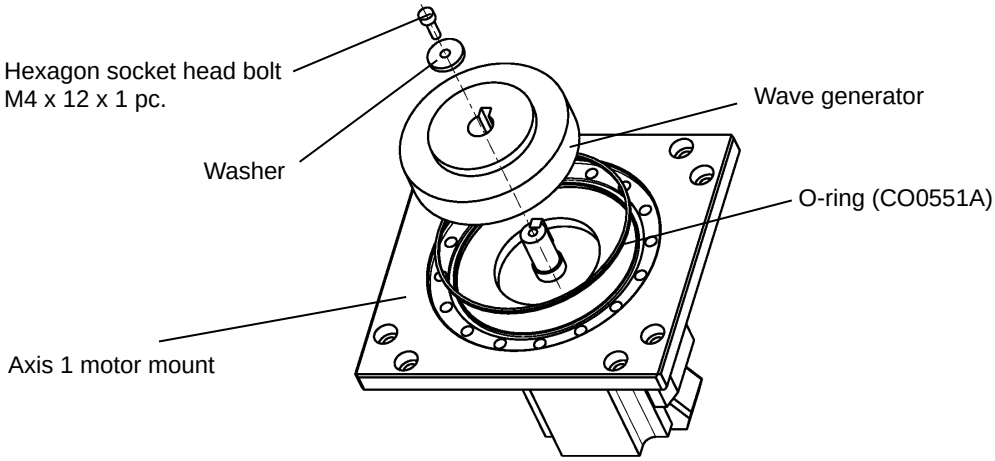


Fig. 4.18 Mounting axis 1 wave generator (THL800, THL900, THL1000, THL1200)

NOTICE

– Never forget to fit the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
– Be sure to use wave generator and reduction gear in the correct combination.
Use of the different combination will result in abnormal noise, reduced service life, and reduced positioning accuracy.

4) Apply an appropriate amount of grease onto the top and bottom surfaces of the wave generator mounted to the axis 1 motor.

Recommended grease	Maker
SK-1A	HDS

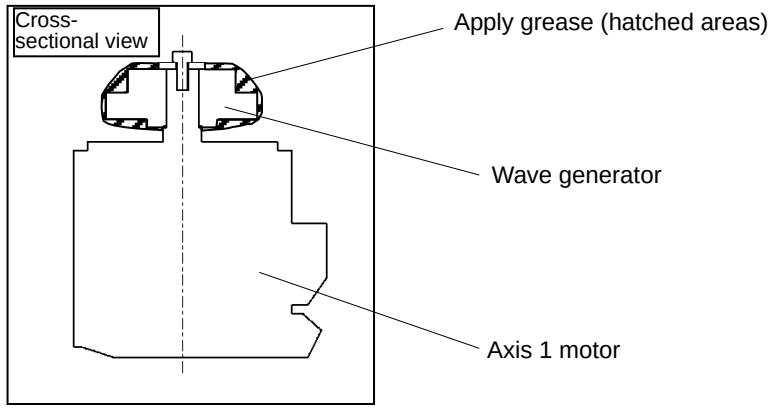


Fig. 4.19 Apply grease onto wave generator (THL800, THL900, THL1000, THL1200)

NOTICE

– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

5) Take care of the motor mounting phase, and directions of the reduction gear and wave generator at the time of insertion of the motor (elliptical portions of the reduction gear proper and wave generator should be adjusted). Make sure that the O-ring is not misaligned. Then insert the axis 1 motor assembly into the reduction gear proper.

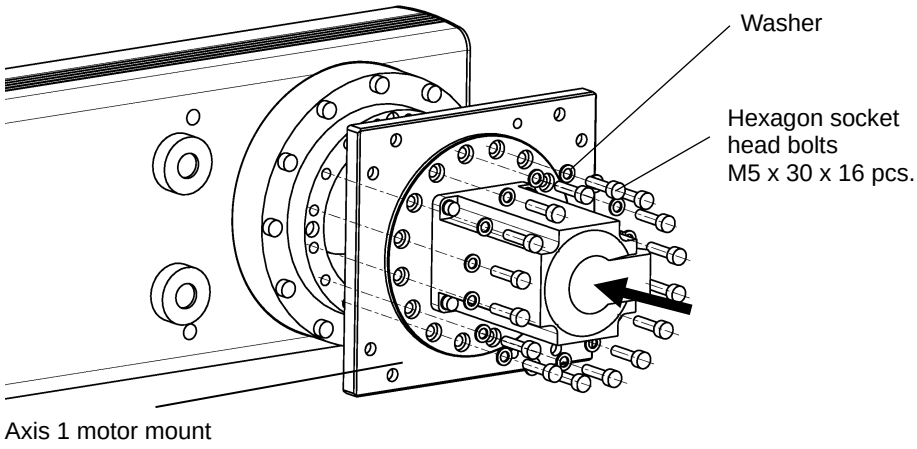


Fig. 4.20 Mounting axis 1 motor assembly (THL800, THL900, THL1000, THL1200)

6) Connect the connectors of the axis 1 motor, i.e., J1AS and J1AP (power drive cable), and J1BS and J1BP (encoder cables).

7) Taking care of the pin (MS5-15), fix the motor mount on the base using hexagon socket head bolts (M6 x 16 x 8 pcs.).

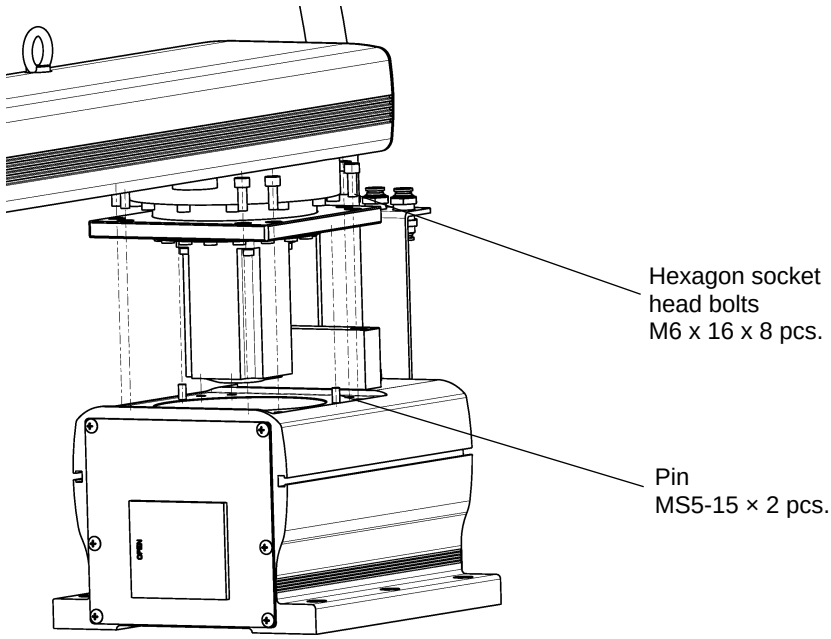


Fig. 4.21 Mounting motor mount (THL800, THL900, THL1000, THL1200)

8) Before turning on power subsequent to replacement of the axis 1 motor, move the arm 1 by hand and make sure that there is no abnormal noise.

9) Turn on the power and set up the axis 1 home position to complete axis 1 motor replacement.
(Check the coordinates of Axes 2 to 4 and set up the home position if necessary. See “Section 8, Robot Home Point and Position Detector Error”.)

4.4.5 Dismounting Axis 2 Motor

1) Disconnect the arm 2 cover. (See “Para. 4.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers or the like, and remove J2AS and J2AP (axis 2 power drive cables), J2BS and J2BP (axis 2 encoder cables). It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Remove the harness guide being secured with one (1) hex countersunk screw (M4 x 8) and the support plate being secured with two (2) hexagon socket head bolts (M4 x 6).

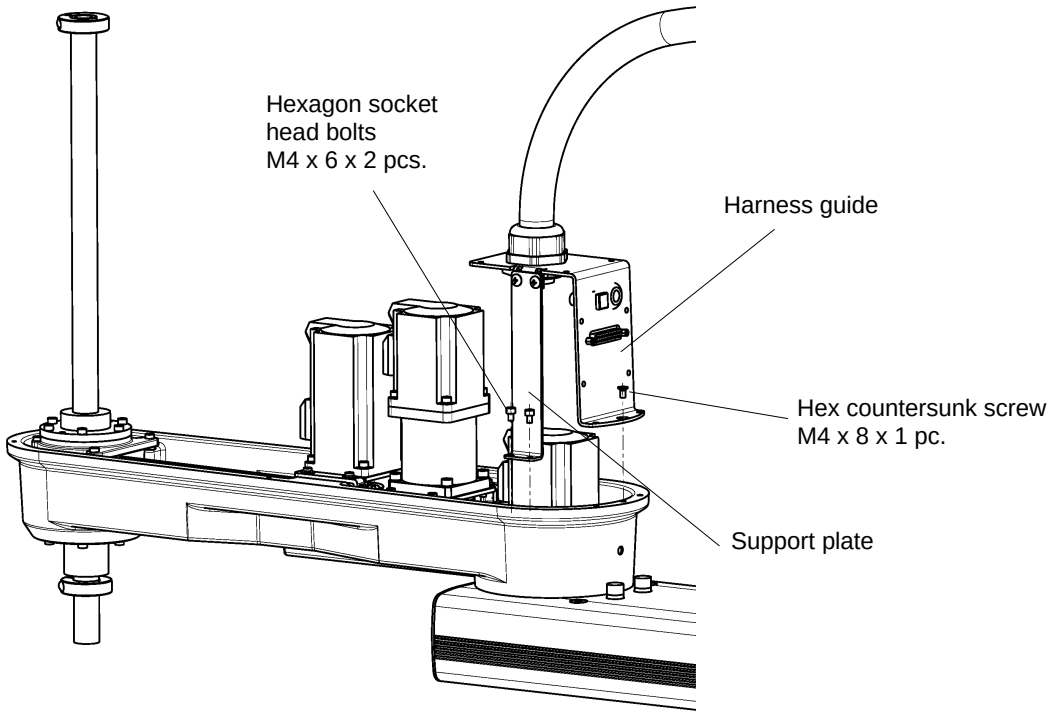


Fig. 4.22 Dismounting harness guide and support plate
(THL800, THL900, THL1000, THL1200)

4) Remove eight (8) hexagon socket head bolts (M4 x 12), two (2) hexagon socket head bolts (M4 x 22), four (4) hexagon socket head bolts (M4 x 30) and washers securing the axis 2 motor. Also remove the O-ring (CO0545A).

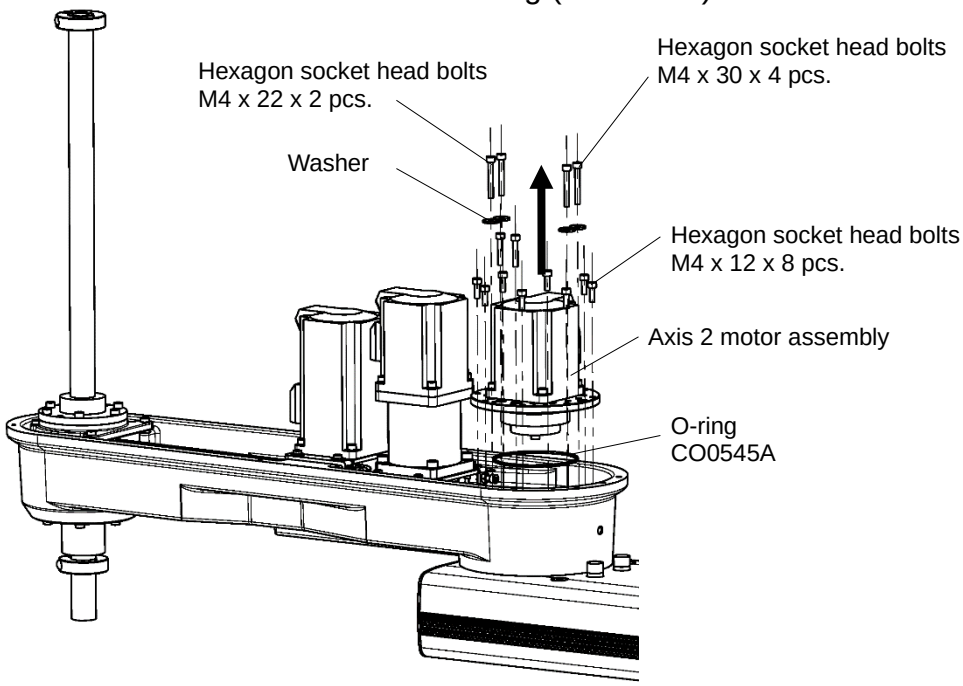


Fig. 4.23 Dismounting axis 2 motor assembly (THL800, THL900, THL1000, THL1200)

5) Remove a hexagon socket head bolt (M4 x 10 x 1 pc.) and washer securing the wave generator, and remove the wave generator from the axis 1 motor assembly.

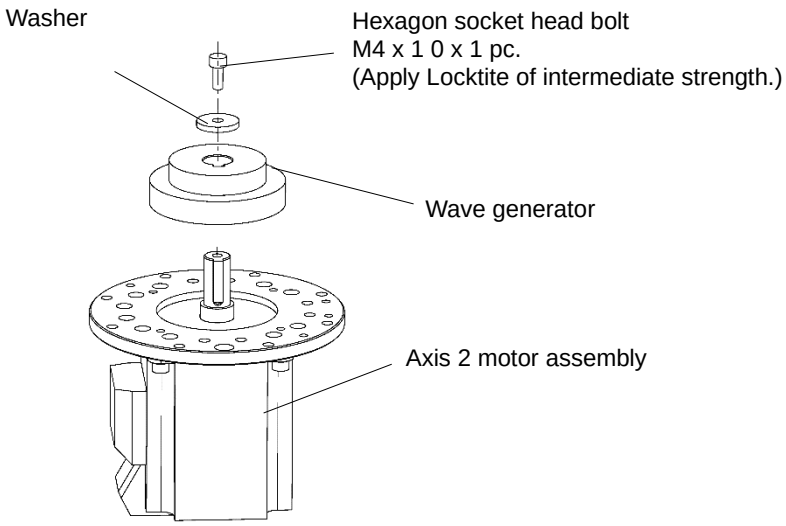


Fig. 4.24 Removal of axis 2 wave generator (THL800, THL900, THL1000, THL1200)

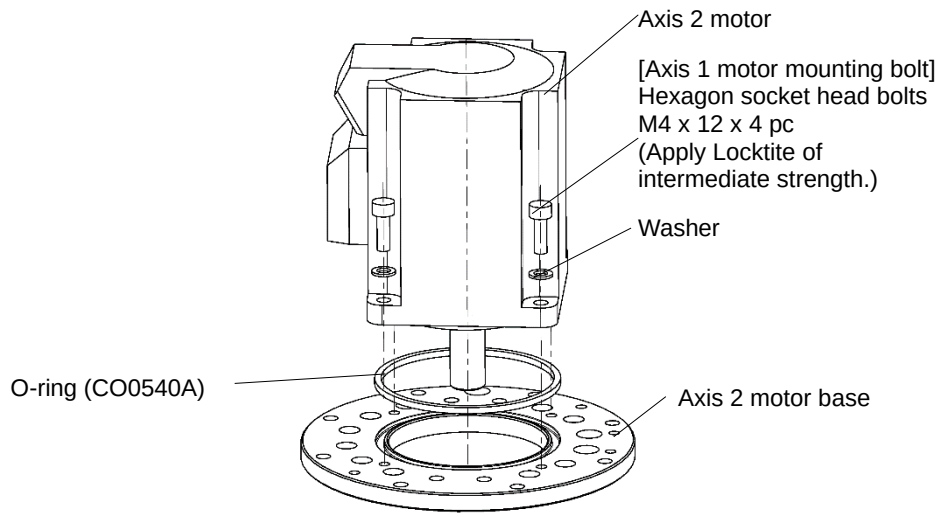


Fig. 4.25 Removal of axis 2 motor (THL800, THL900, THL1000, THL1200)

4.4.6 Mounting Axis 2 Motor

1) Apply grease to the O-ring (CO0540A) and set the O-ring into the O-ring groove of the axis 2 motor mount.

2) Use hexagon socket head bolts (M4 x 12 x 4 pcs.) and washers to mount the axis 2 motor on the axis 2 motor base. In this case, take care of the phases of the 2 axis motor and 2 axis motor base.

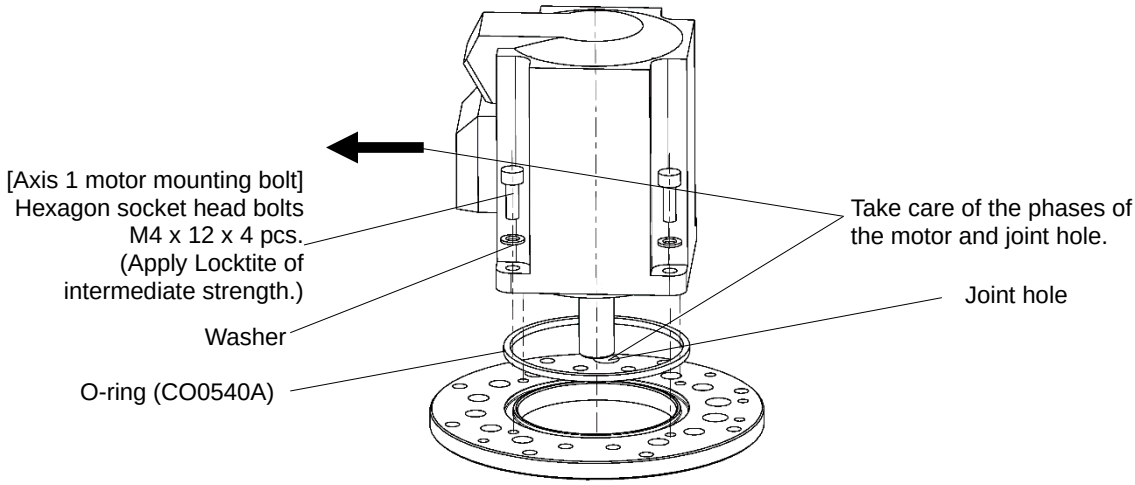


Fig. 4.26 Mounting axis 2 motor (THL800, THL900, THL1000, THL1200)

NOTICE

– Never forget to fit the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.

3) Fix the wave generator to the axis 2 motor assembly using a hexagon socket head bolt (M4 x 10 x 1 pc.) and washer. In this case, apply grease SK-1A to the key.

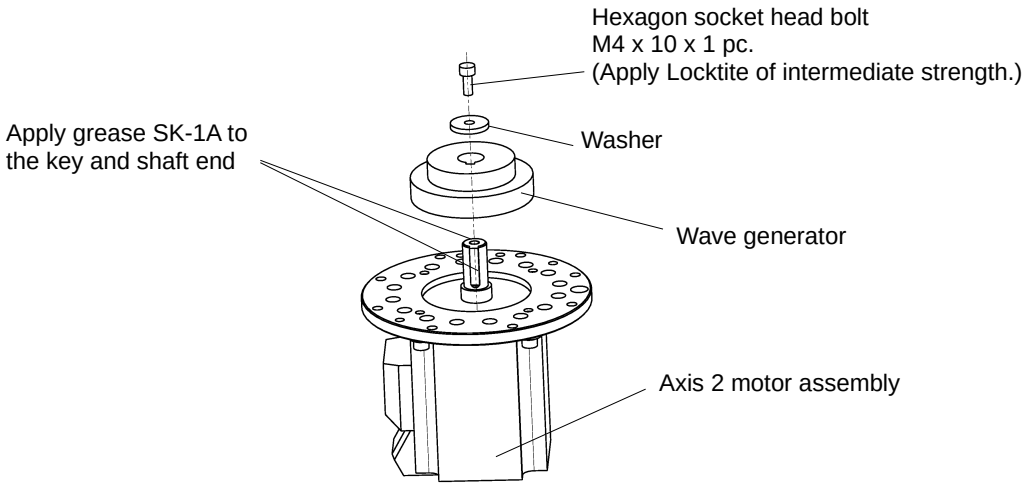


Fig. 4.27 Mounting axis 2 wave generator (THL800, THL900, THL1000, THL1200)

NOTICE

– Be sure to use wave generator and reduction gear in the correct combination.
Use of the different combination will result in abnormal noise, reduced service life, and reduced positioning accuracy.
– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

4) Apply grease to the O-ring (CO0545A) and set the O-ring into the O-ring groove of the axis 2 arm.

5) Insert the axis 2 motor assembly into Arm 2 while paying attention to the motor's mounting phase, the orientations of the decelerator main body and the wave generator when the motor is inserted (align the long sides of the ovals of the reduction gear and wave generator), and the position of the O-ring (it should not move).

6) Secure the axis 2 motor assembly with eight (8) hexagon socket head bolts (M4 x 12) , two (2) hexagon socket head bolts (M4 x 22), four (4) hexagon socket head bolts (M4 x 30) and washers.

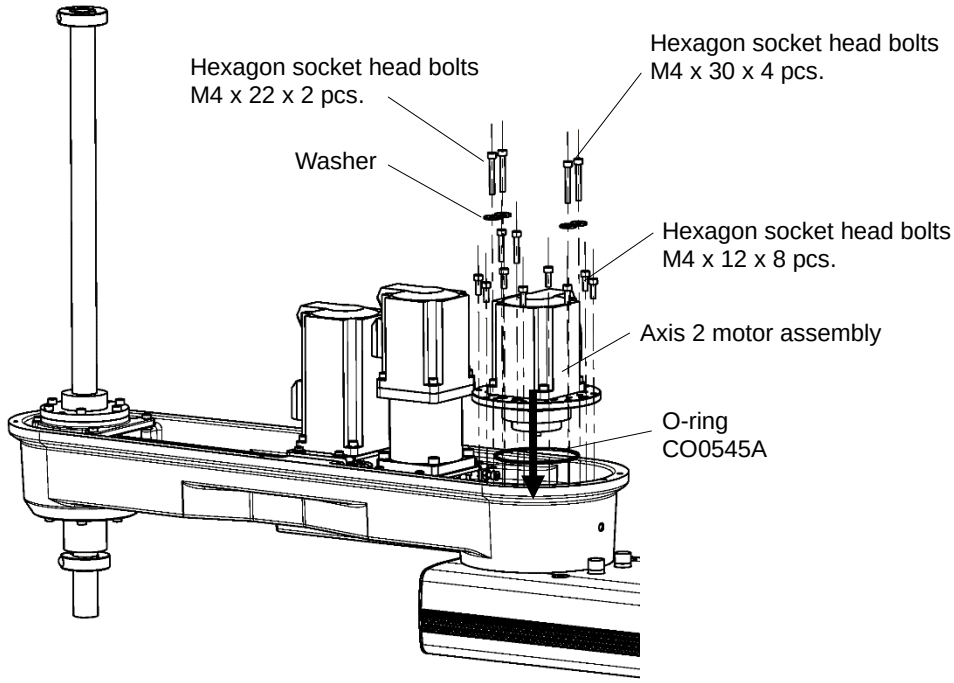


Fig. 4.28 Mounting axis 2 motor assembly (THL800, THL900, THL1000, THL1200)

7) Secure the harness guide and the support plate with one (1) hex countersunk screw (M4 x 8) and two (2) hexagon socket head bolts (M4 x 6).

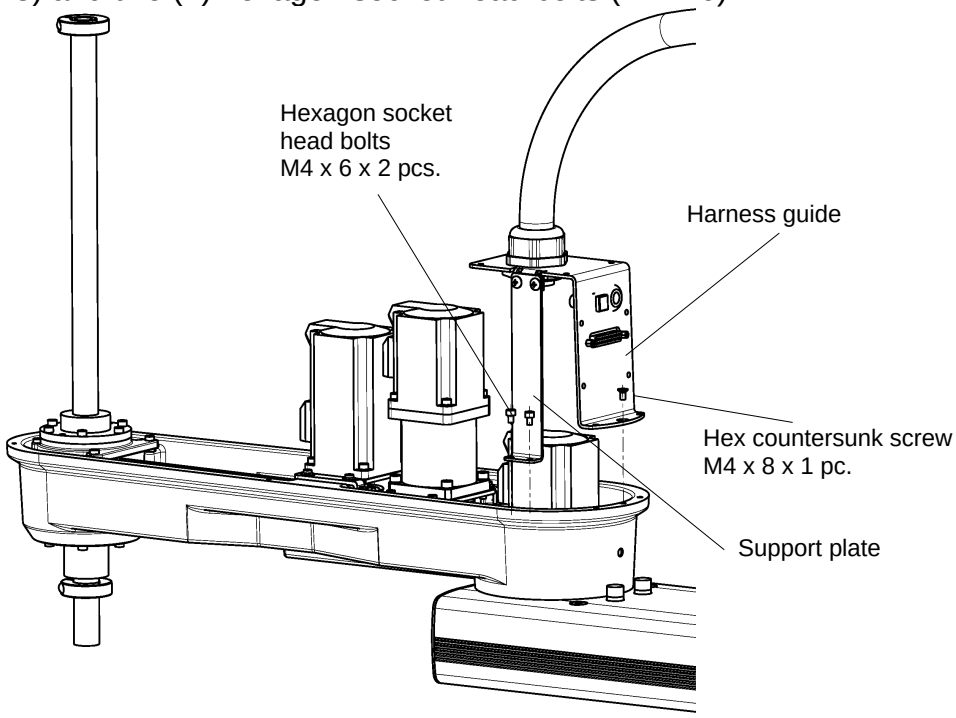


Fig. 4.29 Mounting harness guide and support plate
(THL800, THL900, THL1000, THL1200)

8) Before turning on power subsequent to replacement of the axis 2 motor, move the arm 2 by hand and make sure that there is no abnormal noise.

9) Mount the axis 2 arm cover after the cables and others in the axis 2 arm have been placed back to the original position.

10) Turn on the power and set up the home positions of axes 2 to complete axis 2 motor replacement (see “Section 8, Robot Home Point and Position Detector Error”).

4.4.7 Dismounting Axis 3 Motor

⚠ WARNING

– When the axis-3 motor is replaced, be sure to lower the ball screw spline shaft to the lower limit. When the motor or timing belt is removed, the shaft that drops by its own weight could catch your hand or fingers.
- When the ball screw spline or axis-4 timing belt or reduction gear is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

1) Remove the arm 2 cover. (See “Para. 4.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers.
It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Disconnect connectors J3AS and J3AP (power drive cables), connectors J3BS and J3BP (encoder cables) and connectors J3DS and J3DP (brake cables) for the axis 3, which are connected to the connector panel.

4) Loosen the axis 3 tension adjustment bolt (M4 x 16 x 2 pcs. and hexagonal nuts) and remove the flange head bolts (M4 x 10 x 4 pcs.) securing the axis 3 motor plate to cancel the axis 3 timing belt tension. Then draw out axis 3 the motor assembly upward.

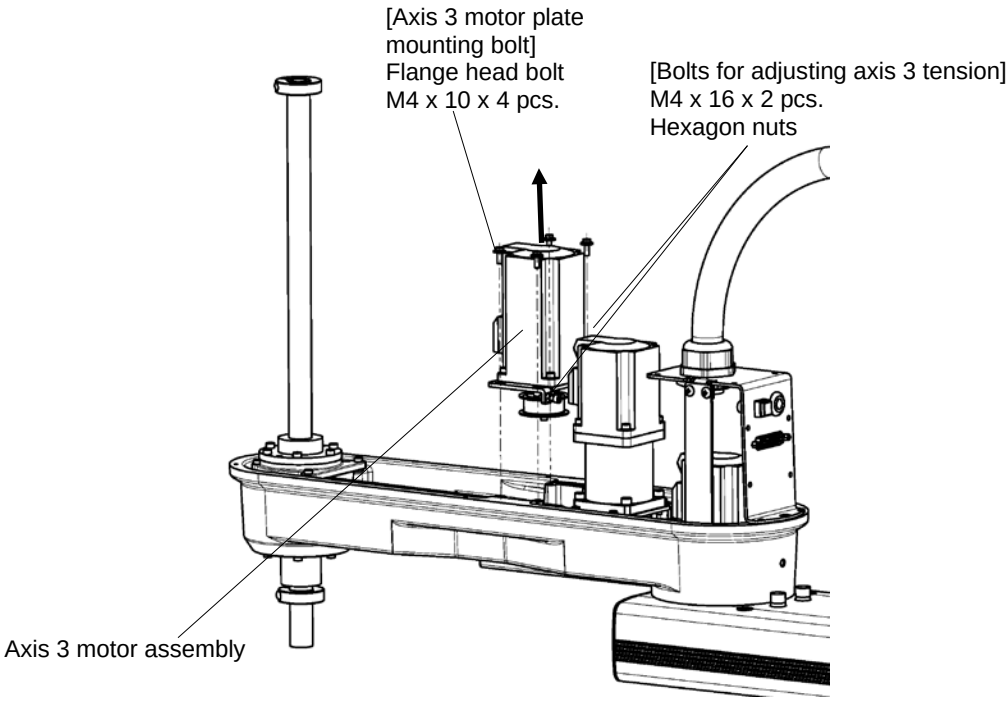


Fig. 4.30 Dismounting axis 3 motor assembly (THL800, THL900, THL1000, THL1200)

5) Remove one (1) hexagon socket head bolt at the end of the axis (M3 x 8). (Same as Step 4 in “Para. 4.4.3, Dismounting Axis 1 Motor 4.”) Next, pull out the washer, holding plate, pulley and key.

6) Remove four (4) hexagon socket head bolts (M4 x 10) and washers that secure the axis 3 motor and then disassemble the axis 3 motor plate and the axis 3 motor.

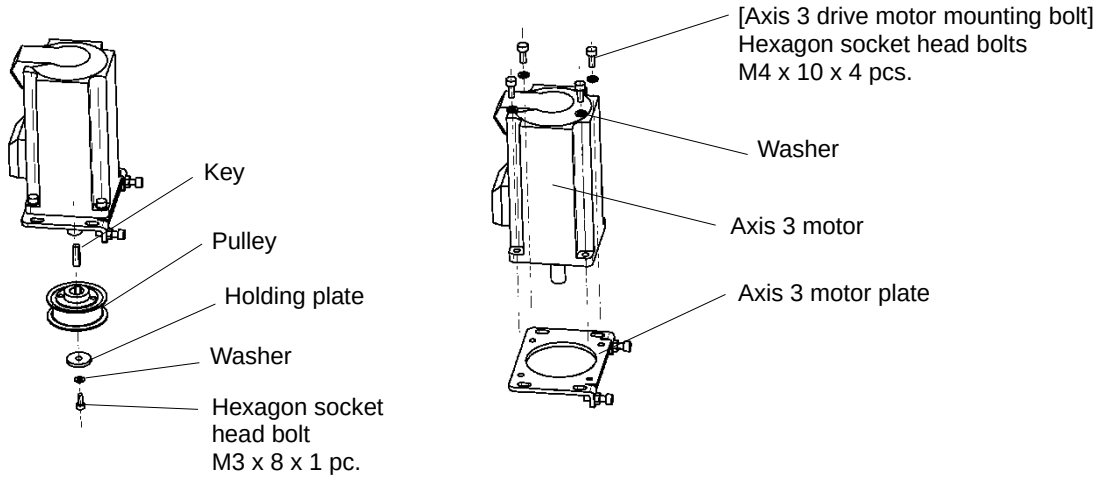


Fig. 4.31 Dismounting axis 3 motor pulley and plate (THL800, THL900, THL1000, THL1200)

4.4.8 Mounting Axis 3 Motor

1) Perform key alignment, using the new axis 3 motor and pulley. At this time, no clearance should exist between the motor shaft and pulley.

2) Secure the axis 3 motor to the axis 3 motor plate with four (4) hexagon socket head bolts (M4 x 10) and washers. Be careful with the mounting phases of the axis 3 motor and the axis 3 motor plate at this time.

3) Mount the key and the axis 3 motor pulley, and then secure them with the holding plate, washers and one (1) hexagon socket head bolt (M3 x 8).

NOTICE

– Perform key alignment very carefully. If there is a clearance between the motor shaft and pulley, positioning accuracy will drop and the life of the parts will shorten.

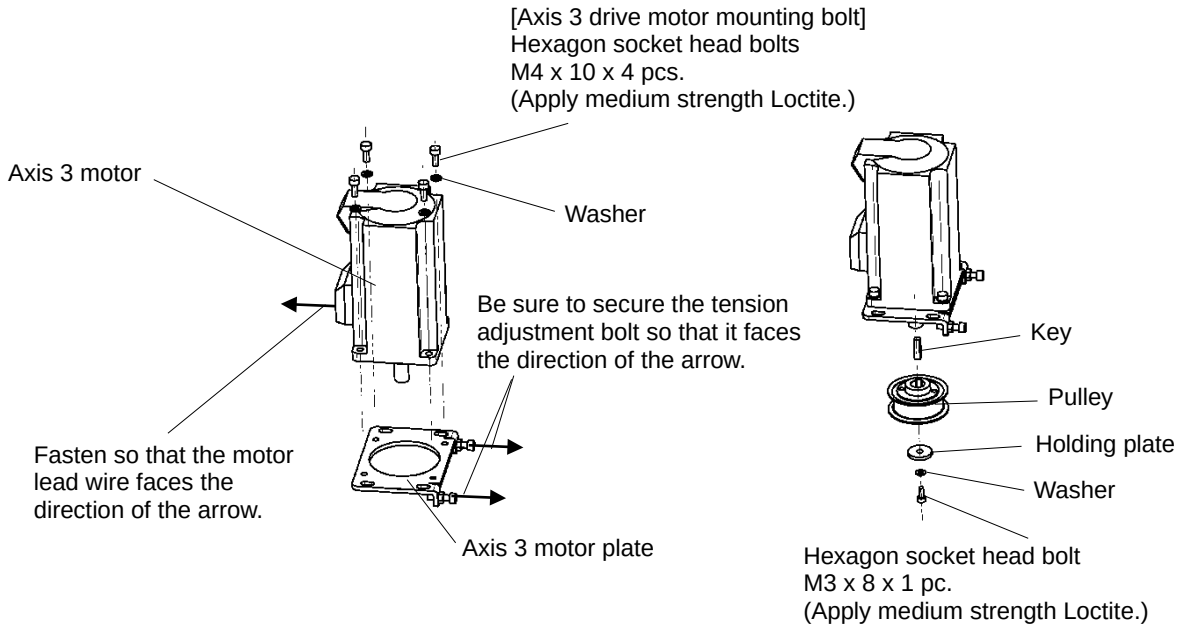


Fig. 4.32 Mounting axis 3 pulley and plate (THL800, THL900, THL1000, THL1200)

4) Hang the axis 3 motor timing belt to the arm 3 motor pulley, and temporarily secure the axis 3 motor plate to Arm 2 with four (4) flange head bolts (M4 x 10, application of Loctite not necessary). Be careful with the motor mounting phase at this time. Apply tension using two (2) axis 3 tension adjustment bolts (M4 x 16, application of Loctite not necessary). (For the belt replacement procedures and tension adjustment values, see “Para. 4.5.3, Replacing Axis 3 Timing Belt.”) Then, tighten and secure the temporarily tightened flange head bolts with hex nuts.

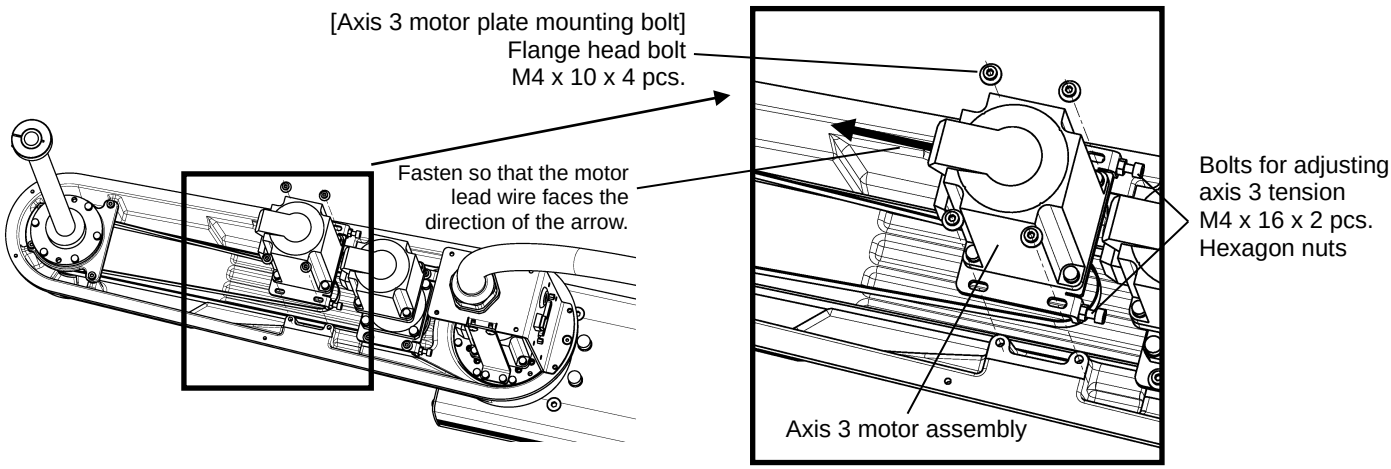


Fig. 4.33 Mounting axis 3 motor assembly (THL800, THL900, THL1000, THL1200)

5) Connect the connectors and set the cables back to the original state. (See the pictures taken in Para. 4.4.7, Step 2.)

6) Mount the arm 2 cover and perform home setting for Axes 3 and 4. Now, the axis 3 motor replacement is complete.

7) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

4.4.9 Dismounting Axis 4 Motor

1) Remove the arm 2 cover. (See “Para. 4.3.1, Arm 2 Cover.”)

2) Cut the cable ties of the cables with nippers.
It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.

3) Disconnect connectors J4AS and J4AP (power drive cables) and connectors J4BS and J4BP (encoder cables) of the axis 4 motor.

4) Remove the cap attached to the lateral side of the axis 4 reduction gear and loosen the coupling bolt (M4) securing the axis 4 motor shaft and input shaft of the axis 4 reduction gear.
If the hexagonal hole of the bolt is out of phase, adjust the phase by turning the ball screw spline shaft by hand.

5) Remove four (4) hexagon socket head bolts (M4 x 12) and washers that secure the axis 4 motor, and then pull the axis 4 motor upward.

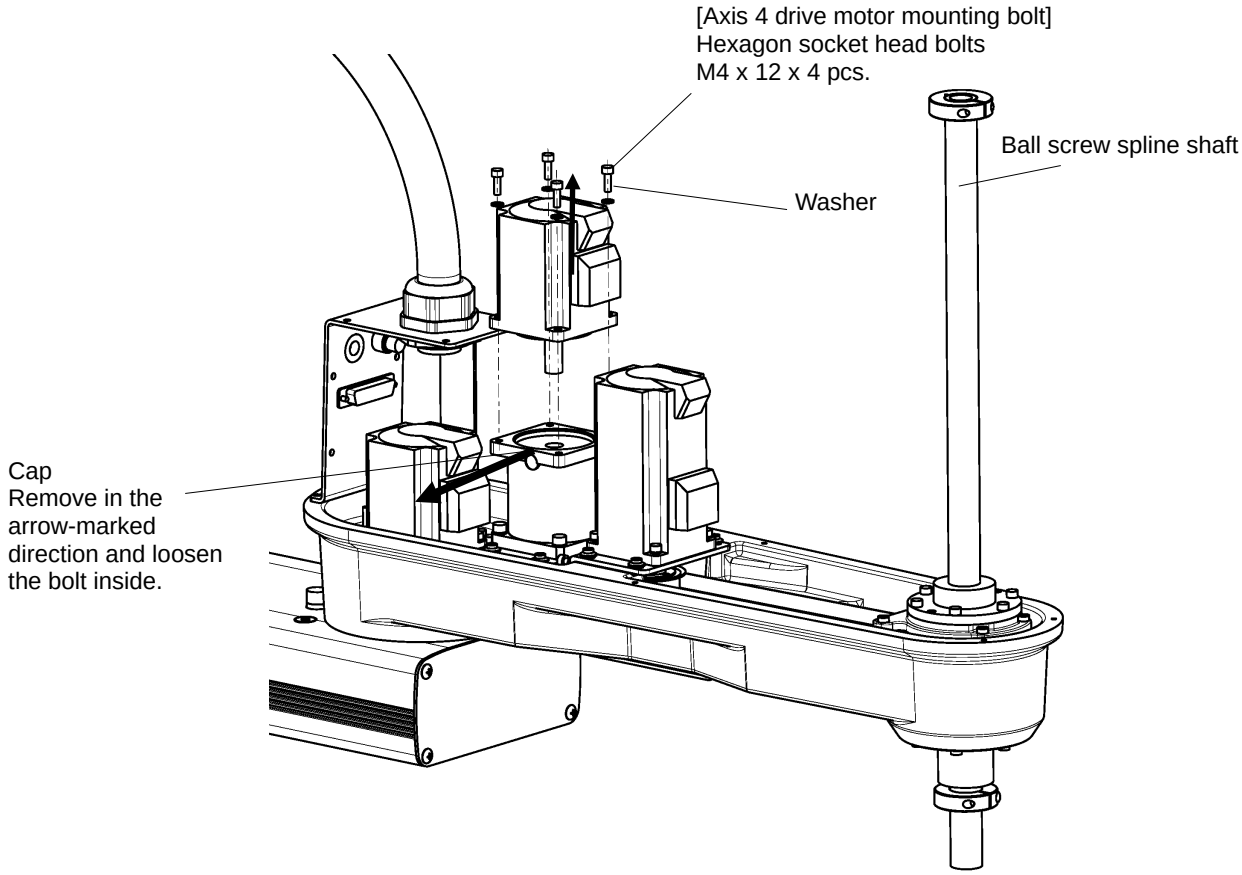


Fig. 4.34 Dismounting axis 4 motor (THL800, THL900, THL1000, THL1200)

4.4.10 Mounting Axis 4 Motor

1) Mount the new motor to the reduction gear with four (4) hexagon socket head bolts (M4 x 12) and washers. Be careful with the phase when mounting the motor. (Recommended clamping torque: 3.3 N·m)

2) Tighten the coupling of the axis 4 reduction gear by means of the attached bolt (M4) to mount the cap. (Recommended clamping torque: 4.3 N·m)

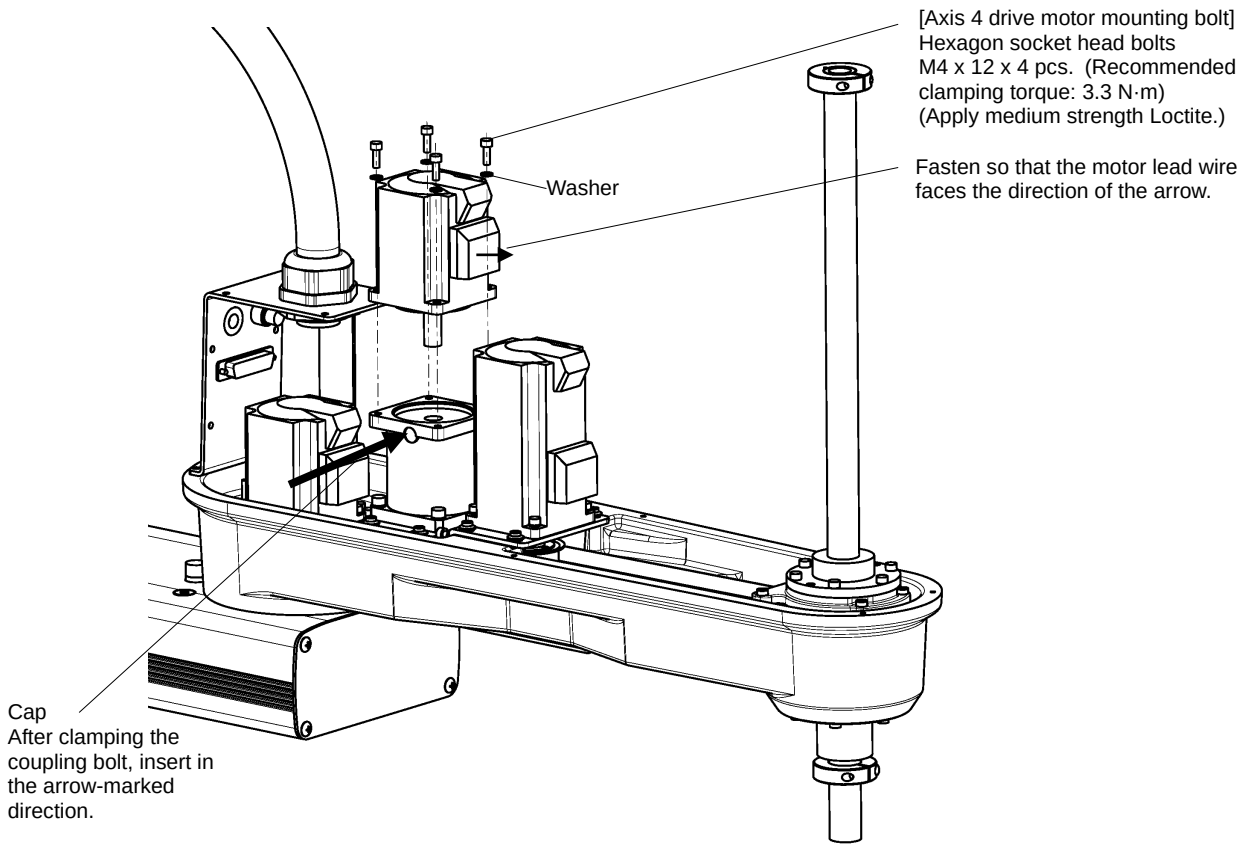


Fig. 4.35 Mounting axis 4 motor (THL800, THL900, THL1000, THL1200)

3) Connect the connectors of the axis 4 motor, i.e., J4AS and J4AP (power drive cables), and J4BS and J4BP (encoder cables).

4) Restore the cables to the original conditions and then mount the arm 2 cover.

5) Set up the home positions of Axes 3 and 4 to complete motor replacement. During the replacement of the axis 4 motor, it is necessary to set up the home position of Axis 3.

6) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

4.5 Adjusting and Replacing Timing Belt

The timing belt is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠ DANGER

– When replacing the timing belt, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock.
A worker may be entangled in rotation of the motor that has operated unpredictably.

⚠ WARNING

– When replacing the axis-3 timing belt, be sure to lower the ball screw spline shaft to the lower limit.
When the motor or timing belt is removed, the shaft drops by its own weight.
- When the ball screw spline or axis-4 timing belt is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

NOTICE

– When replacing the timing belt, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– After the timing belt is replaced, be sure to re-set the zero position.
Because a mechanical zero position is shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”

4.5.1 Type of Timing Belt

The timing belts used in this robot are shown below.
When you place an order for a replacement belt, specify the robot model (THL800, THL900, THL1000, THL1200), the serial number, the axis name, and our drawing number.
For the location where the serial number plate is attached, see the “Safety Manual.”

Table 4.3 Type of timing belt (THL800, THL900, THL1000, THL1200)

Description	Axis name	Width	Our drawing No.
Timing belt	Axis 3	10 mm	S891108
	Axis 4	15 mm	S891109

4.5.2 Timing Belt Locations

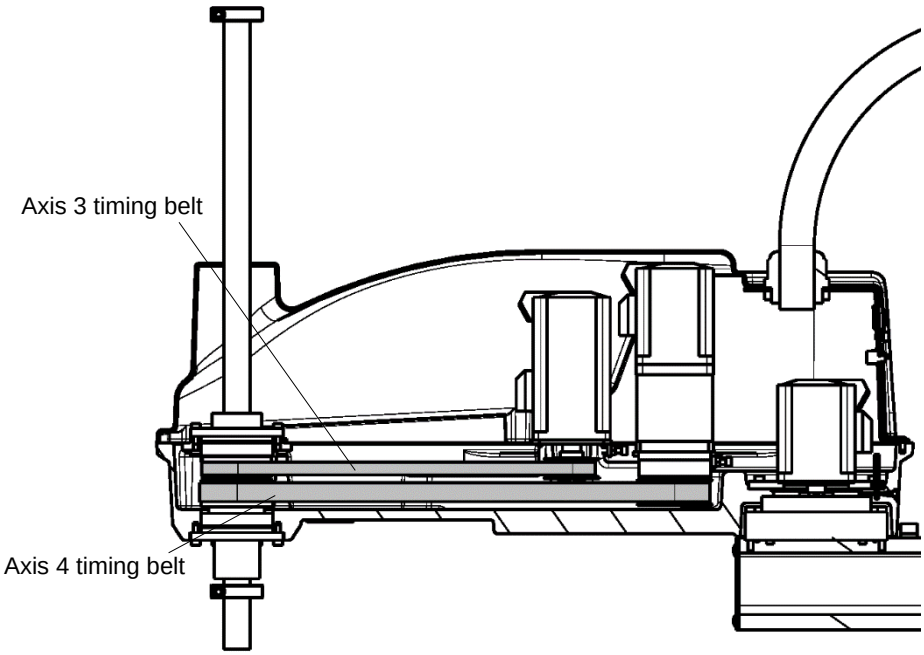


Fig. 4.36 Timing belt locations (THL800, THL900, THL1000, THL1200)

4.5.3 Replacing Axis 3 Timing Belt

⚠ WARNING

– When replacing the axis-3 timing belt, be sure to lower the ball screw spline shaft to the lower limit. When the motor or timing belt is removed, the shaft drops by its own weight.
- When the ball screw spline or axis-4 timing belt is replaced, the axis-3 motor and timing belt must also be removed. Follow the same instruction.

1) Remove the arm 2 cover. (See “Para. 4.3.1, Arm 2 Cover.”)
2) Cut the cable ties of the cables with nippers. It is recommended to take photos of the cable installation positions and the locations where cable ties are used so that these locations can be restored when needed.
3) Disconnect the axis 3 motor assembly. For the disconnecting procedures, see Steps 3) through 4) of “Para. 4.4.7, Dismounting Axis 3 Motor.”

4) Remove six (6) hexagon socket head bolts (M4 x 12) securing the ball screw nut, and remove the ball screw nut from the axis 3 bracket. (Keep the hand wheel removed.)
   Remove the four (4) hexagon socket head bolts (M4 x 16) securing the axis 3 bracket.

5) Hold the ball screw nut and the axis 3 bracket upward, and remove the axis 3 timing belt.

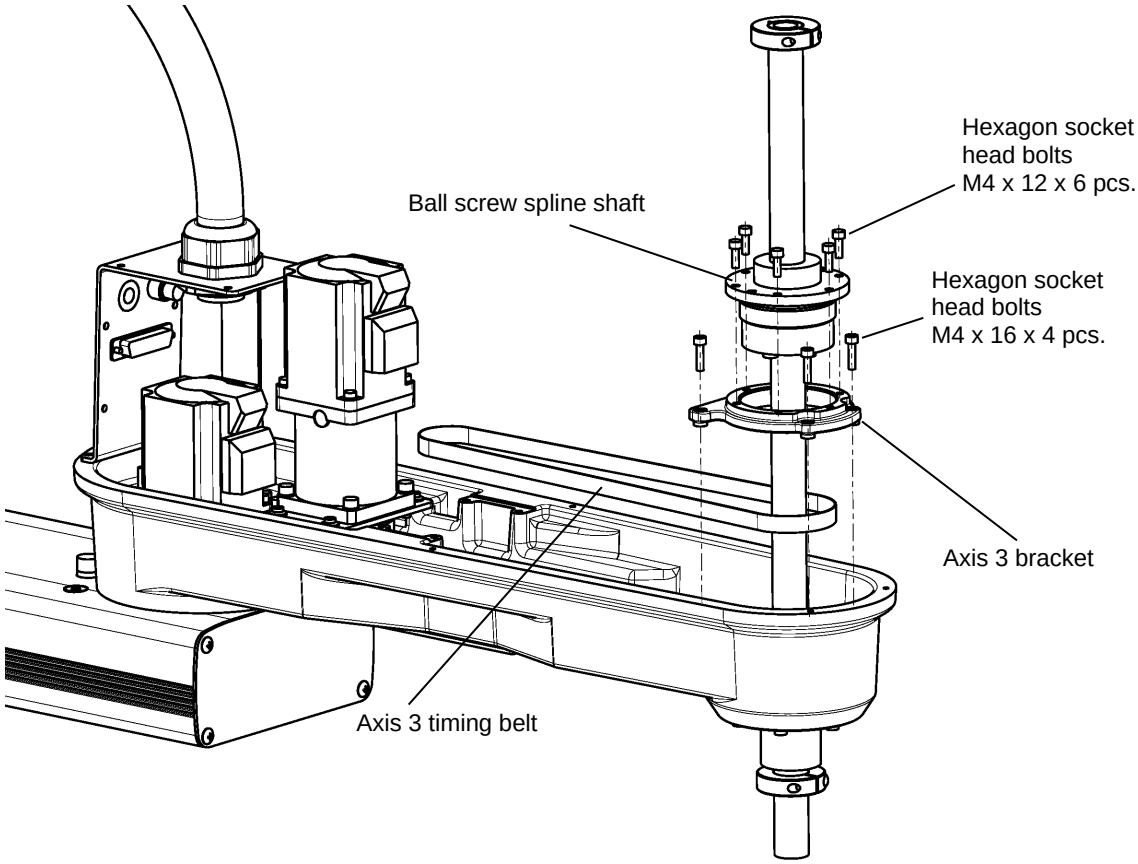


Fig. 4.37 Replacing axis 3 timing belt (THL800, THL900, THL1000, THL1200)

6) Hold the ball screw nut and the axis 3 bracket upward, and mount a new timing belt.
7) Temporarily clamp the axis 3 bracket on the arm 2 using four (4) hexagon socket head bolts (M4 x 16).
8) Fix the ball screw nut on the axis 3 bracket using six (6) hexagon socket head bolts (M4 x 12).

9) Move the ball screw in the vertical direction by ten reciprocating motions and align the center. After that, while holding the axis 3 bracket to prevent it from being displaced from the current position, tighten four (4) hexagon socket head bolts (M4 x 16) clamped temporarily, and fix the axis 3 bracket in position.

10) Hang the timing belt to the axis 3 motor assembly which was previously removed in 3), and temporarily secure it to Arm 2 with four (4) flange head bolts (M4 x 10, application of Loctite not necessary). Move the ball screw spline unit up and down to make it fit in. Apply tension using two (2) axis 3 tension adjustment bolts (M4 x 16, application of Loctite not necessary).
While pulling the axis 3 motor plate with the axis 3 tension adjustment bolts, measure the tension with a tension meter. Tighten the flange head bolt at the location where the tension reaches a value slightly smaller than 49 N (because the tension gets larger by retightening of the flange head bolt). Adjust so that the tensile value becomes between 49 and 57 N when the flange head bolt is retightened. Then, secure with the hex nut.

The values to be used for the tension meter are as shown in the table below.

Table 4.4 Value of tension of axis 3 timing belt (THL800, THL900, THL1000, THL1200)

Value of tension [N]	Unit mass [g/m]	Belt width [mm]	Span [mm]
49 to 57	2.2	10	260

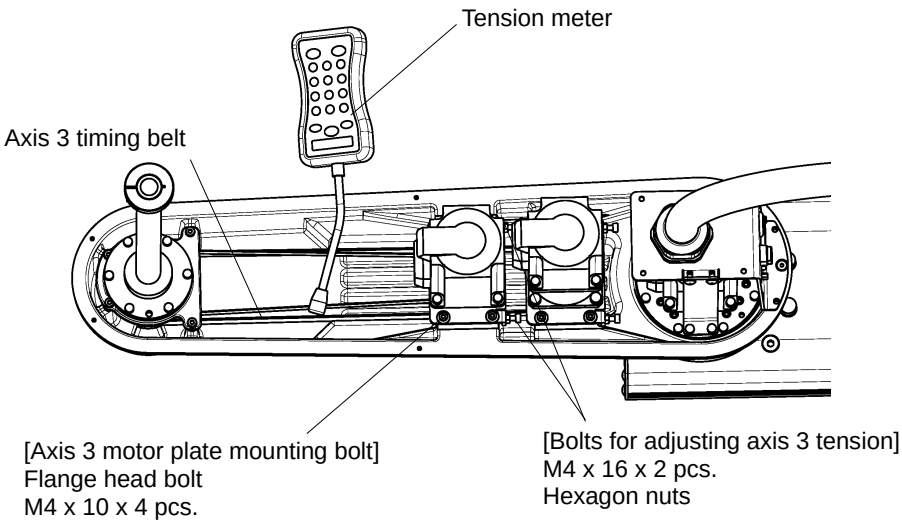


Fig. 4.38 Adjusting axis 3 tension (THL800, THL900, THL1000, THL1200)

11) Connect the connector and set the zero points of Axes 3 and 4. After that, while pressing the brake release switch, manually move the ball screw in the vertical direction. Check if you feel a catch in the movement or if the movement of the ball screw is less smooth. If such a problem has been found, the ball screw center may be misaligned. Go back to where the center can be aligned, and go through the work procedure again.

12) If there is no problem, put cables back to where there were and mount the axis 2 arm cover. Replacement of the timing belt has been completed.

13) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

4.5.4 Replacing Axis 4 Timing Belt

NOTICE

– Never pull out the ball screw nut from the ball screw spline shaft.
Otherwise, the ball screw spline unit from which balls have come out cannot be used.
– After the timing belt is replaced, be sure to re-set the zero position.
Because a mechanical zero position is shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– When axis-4 timing belt is replaced, the axis-3 motor and timing belt must also be removed.
Therefore, the precautions for the replacement of the axis-3 motor and timing belt must be also observed.

1) Remove the arm 2 cover, the axis 3 motor assembly, the axis 3 timing belt, the ball screw nut and the ball screw spline shaft.
For the disconnecting procedures, see Steps 1) through 6) of “Para. 4.5.3, Replacing Axis 3 Timing Belt.”

2) Loosen two (2) tension adjustment bolts (M4 x 16) for the axis 4, remove four (4) flange head bolts (M4 x 10) securing the axis 4 motor plate, and then pull out the axis 4 motor assembly. Pull out the axis 4 timing belt upward.

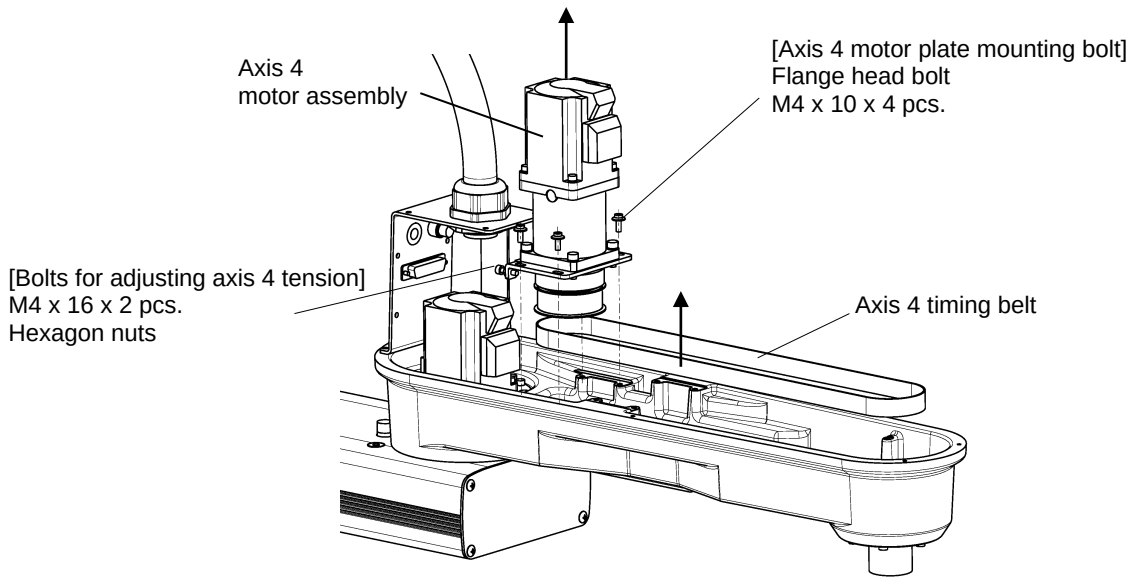


Fig. 4.39 Dismounting harness guide and support plate

(THL800, THL900, THL1000, THL1200)

3) Mount a new belt. Temporarily clamp axis 3 bracket on the axis 2 arm using the four (4) hexagon socket head bolts (M4 x 16). After the ball nut and ball screw spline shaft removed in step 1) has been passed through the axis 3 timing belt, fix them in position. Then after routing the axes 3 and 4 timing belts, mount the ball screw spline shaft. For the mounting of the ball screw spline shaft, see "4.6.5 Mounting of ball screw spline shaft".

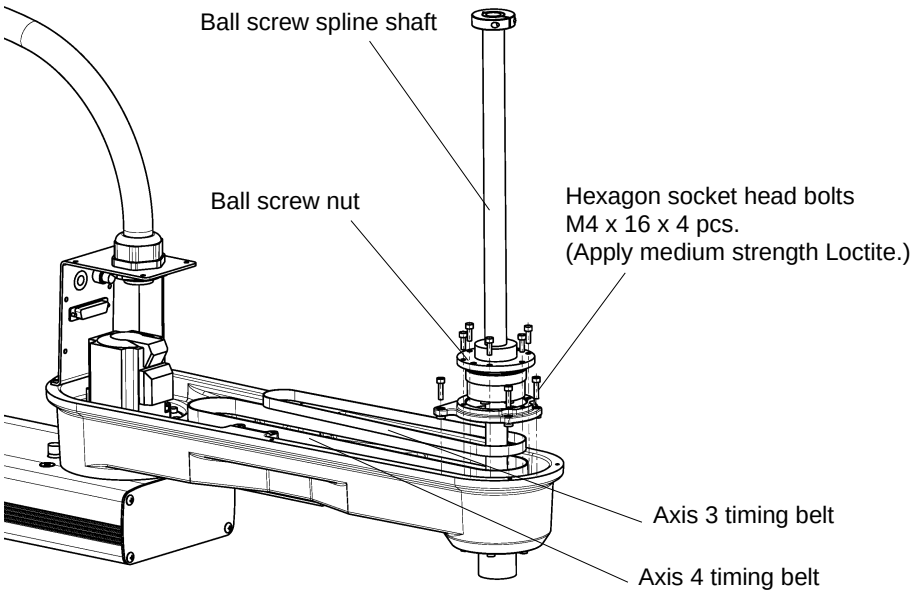


Fig. 4.40 Mounting ball screw shaft (THL800, THL900, THL1000, THL1200)

4) Mount the lower stopper on the ball screw shaft. For the stopper mounting procedure, see "4.6.5 Mounting of ball screw spline shaft".

5) Temporarily set the axis 4 motor assembly which was disconnected in Step 2) above to Arm 2 with the four (4) flange head bolts (M4 x 10 Loctite is not necessary) after setting the axis 4 timing belt to the axis 4 motor pulley. Be careful with the motor assembly mounting phase at this time.

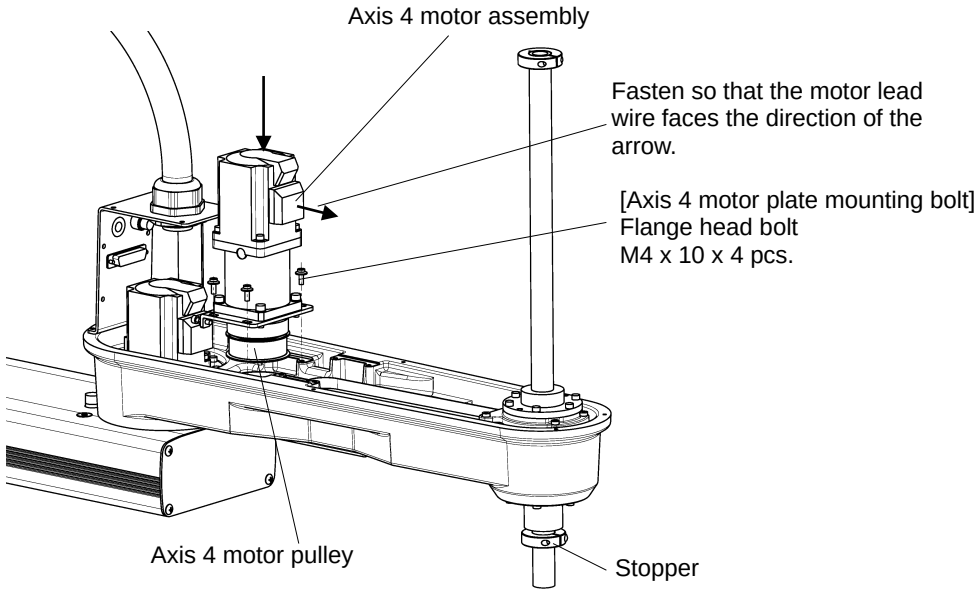


Fig. 4.41 Mounting axis 4 motor assembly (THL800, THL900, THL1000, THL1200)

6) While pulling the axis 4 motor plate with the axis 4 tension adjustment bolts, measure the tension with a tension meter. Tighten the flange head bolt at the location where the tension reaches a value slightly smaller than 150 N (because the tension gets larger by retightening of the flange head bolt). Adjust so that the tensile value becomes between 150 and 180 N when the flange head bolt is retightened. After that, slightly tighten the tension adjusting bolt, and fix it in position using a hexagon nut. If the tension adjusting bolt is not tightened, tension adjusting bolt may be disconnected during robot operation. The values to be used for the tension meter are as shown in the table below.

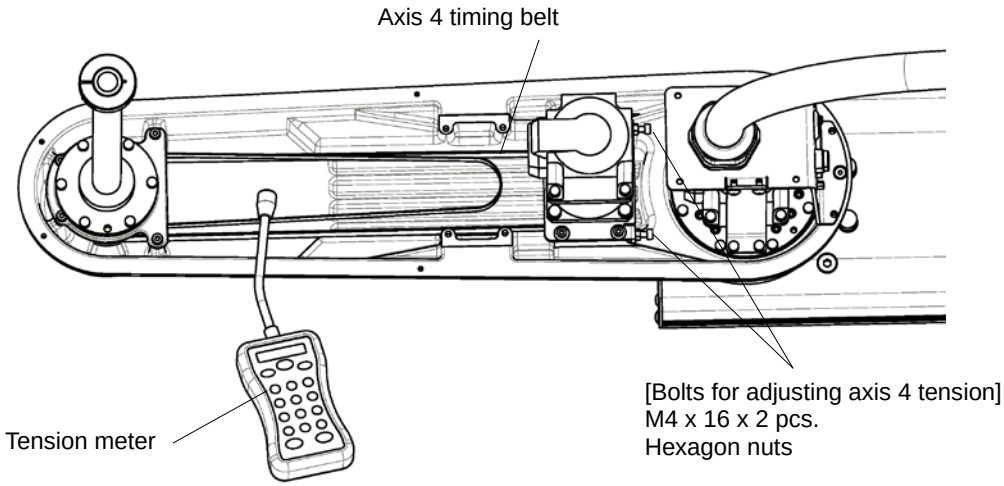


Fig. 4.42 Ball screw and axis 4 motor assembly (THL800, THL900, THL1000, THL1200)

Table 4.5 Value of tension of axis 4 timing belt (THL800, THL900, THL1000, THL1200)

Value of tension [N]	Unit mass [g/m]	Belt width [mm]	Span [mm]
150 to 180	4.1	15	340

7) Mount the axis 3 motor assembly. For the mounting procedures, see “Para. 4.4.8, Mounting Axis 3 Motor” and “Para. 4.5.3, Replacing Axis 3 Timing Belt” above.

8) Arrange the connectors and cables as originally set. (See the pictures you took beforehand.)

9) Perform home setting for Axes 3 and 4.

10) Press the brake release switch and move Axis 3 in the vertical direction to make sure that there is no problem.

11) Carry out a test operation of Axes 3 and 4 and make sure that the belt tension is appropriate.

12) Attach the arm 2 cover. Now replacement of the axis 4 timing belt completes.

4.5.5 Check Timing Belt Adjustment

Perform inspection and adjustment of each timing belt semi-annually (every six (6) months).

1) Remove the arm 2 cover. (See “Para. 4.3.1, Arm 2 Cover.”)
2) If the belt is worn out, replace it. (See "4.5.3 Replacement of axis 3 timing belt.) If a crack or severe abrasion has been detected, the timing belt must be replaced. If the belt is scratched, contact us to see if it should be replaced or not. (See "4.5.3 Replacement of axis 3 timing belt" and "4.5.4 Replacement of axis 4 timing belt".) If the belt is loosened heavily, adjust its tension, referring the following table. Note that the value of tension at replacement of timing belt (i.e., when mounting a new timing belt) is the same as the value of tension at adjustment. If the tension is appropriate, mount the arm 2 cover. Now the inspection is complete.
3) When adjustment of the tension is required, loosen the four (4) flange head bolts (M4 x 10) securing the axis 3 and axis 4 motor plates, respectively. Measure the tension on the tension meter while adjusting the tension adjustment bolt (axis 3: M4 x 16 x 2 bolts, axis 4: M4 x 16 x 2 bolts and hexagonal nut).
4) When the tension has fallen under the appropriate range, tighten the flange head bolts.
5) Tighten the tension adjustment bolts further and fasten with the hexagon nuts. If the bolt is not tightened any further, the bolt can come out while the robot is in operation.
6) Mount the arm 2 cover. Now the adjustment completes.

4.6 Filling Grease to Ball Screw Spline Unit and Replacement

The ball screw spline unit is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠DANGER

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– After the ball screw spline unit is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

4.6.1 Type of Ball Screw Spline Unit

The ball screw spline unit used in this robot is shown below.
When you place an order for the ball screw spline unit for replacement, specify the robot model (THL800, THL900, THL1000, THL1200) and our drawing number.

Table 4.6 Ball screw spline unit (THL800, THL900, THL1000, THL1200)

Description	Stroke	Our drawing No.	Unit code
Ball screw spline unit	300 mm	H852702	Y610A3ME0

4.6.2 Ball Screw Spline Unit Location

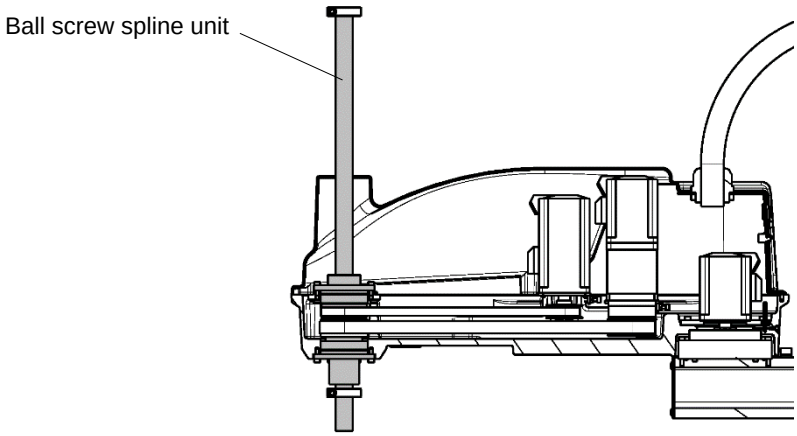


Fig. 4.43 Ball screw spline unit location (THL800, THL900, THL1000, THL1200)

4.6.3 Greasing Ball Screw Spline Unit and Applying Anticorrosive

⚠ CAUTION

– When moving the ball screw spline shaft by hand, never insert your hand or finger inside the mechanical stopper or under the robot hand.
Otherwise, your hand or finger may get caught in the mechanical stopper and hand.

NOTICE

– Never touch the ball screw spline unit with bare hands.
Touching it with bare hands may cause the ball screw spline unit to rust.
- Be sure to wear gloves.

– When greasing up the ball screw spline unit and apply corrosion inhibitor, be sure to cover the peripheral equipment, etc.
Grease and corrosion inhibitor may drip on the peripheral equipment.
- Wipe out excess grease and corrosion inhibitor with waste cloth.

– Be sure to check the application state of grease daily.
A shortage of grease and development of rust may cause flaws in the sliding surface, resulting in a reduction in the performance and lifespan of the robot. The robot may malfunction.

– Be sure to use the specified type of grease.
Otherwise, the performance and lifespan of the robot may be reduced. The robot may malfunction.

– Be sure to check the application state of corrosion inhibitor daily.
Otherwise, rust may develop on the ball screw spline unit.
- Corrosion inhibitor must be applied only when it is used up.

In principle, grease should be applied to the ball screw spline unit every three months. Be sure to apply grease whenever there is a shortage of grease.
At daily inspection also, make sure that the ball screw spline unit is filled with a sufficient volume of grease.
Check the condition of anticorrosive in daily inspection. If anticorrosive is dried up, apply it.

Table 4.7 Recommended anticorrosive (THL800, THL900, THL1000, THL1200)

Recommended anticorrosive	Maker
KLUBER A20	NOK
WD-40	ST TRADING

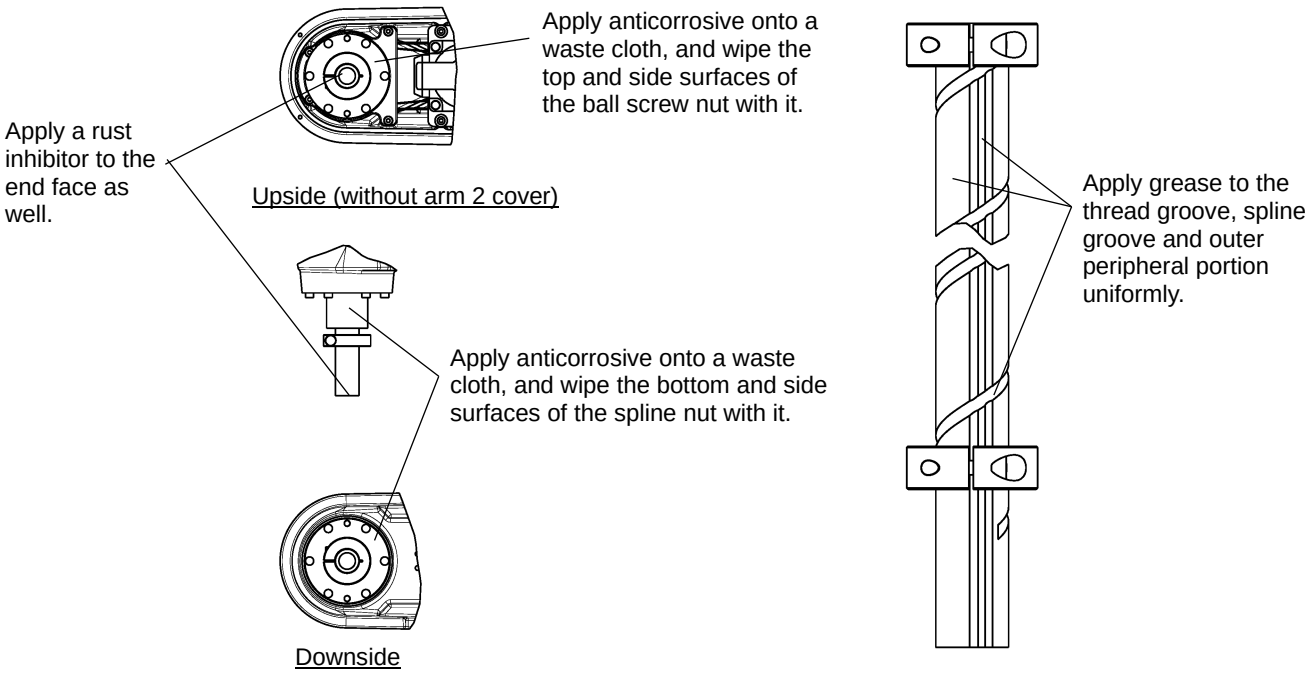


Fig. 4.41 Applying anticorrosive and grease to ball screw spline unit (THL800, THL900, THL1000, THL1200)

1) Remove the arm 2 cover (see “Para. 4.3.1, Arm 2 Cover.”)

2) Connect the controller power supply plug and turn off the servo system.

3) Move the arm to a position where Axis 3 can be moved over the full stroke by hand.

4) Push down the ball screw spline shaft to the lower limit while pressing the axis 3 brake release switch.

5) Directly apply the grease to the exposed shaft area by brash.
Apply the grease to such an extent that the shaft groove can be filled. Wipe out the surplus grease.

Table 4.8 Recommended grease (THL800, THL900, THL1000, THL1200)

Recommended grease	Maker
AFF grease	THK

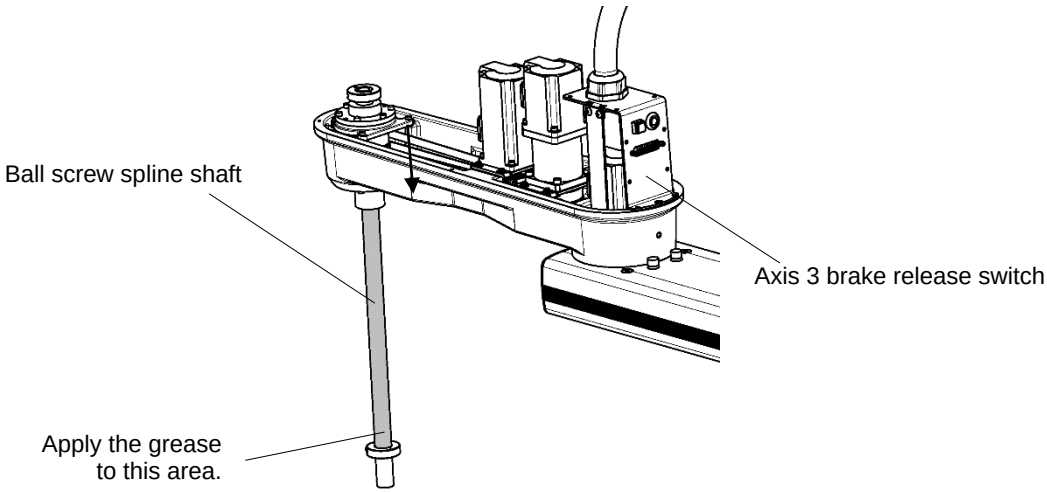


Fig. 4.45 Greasing lower side of ball screw spline unit
(THL800, THL900, THL1000, THL1200)

6) Push up the shaft up to the upper limit while pressing the axis 3 brake release switch.

7) Directly apply the grease by brash to the shaft area sticking up from the cover. Apply the grease to such an extent that the shaft groove can be filled. Wipe out the surplus grease.

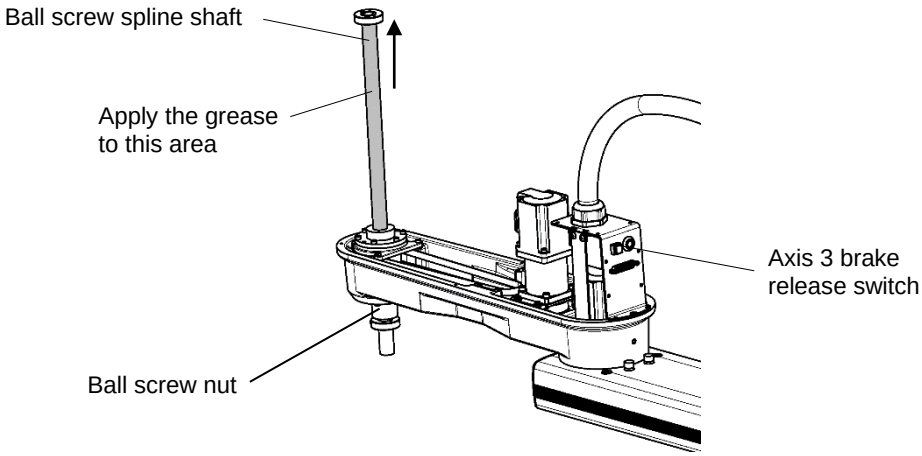


Fig. 4.46 Greasing upper side of ball screw spline unit (THL800, THL900, THL1000, THL1200)

8) Move the shaft up and down repeatedly while pressing the axis 3 brake release switch to fill the grease uniformly. Wipe out the surplus grease. Now the work completes.

4.6.4 Dismounting Ball Screw Spline Unit

⚠DANGER

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– Never touch the ball screw spline unit with bare hands.
Touching it with bare hands causes the ball screw spline unit to rust.
- Be sure to wear gloves.
– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The robot may malfunction if you replace it with the power on.
– Be sure to handle the ball screw spline unit carefully.
Drop impact or application of excessive external force could ruin the unit.
– After the ball screw spline unit is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

In the descriptions on replacing the ball screw spline unit, the procedures for changing the axis 3 and 4 motors and timing belts are not included. For details, refer to the descriptions on replacement of each part (see “Para. 4.4.7, Dismounting Axis 3 Motor”, “Para. 4.4.9, Dismounting Axis 4 Motor”, “Para. 4.5.3, Replacing Axis 3 Timing Belt” and “Para. 4.5.4, Replacing Axis 4 Timing Belt.”)

1) Remove the arm 2 cover. (See “Para. 4.3.1, Arm 2 Cover.”)
2) Disconnect the hand, tool, etc. followed by the upper and lower stoppers.
3) Disconnect the axis 3 motor assembly . Keep the axis 4 belt loose.

4) Remove six (6) hexagon socket head bolts (M4 x 12) securing the ball screw nut, and pull out the ball screw nut and ball screw spline shaft together as an integral structure. During this work, make sure that the ball screw nut and shaft will not be disconnected.

NOTICE

– Never pull out the ball screw nut from the ball screw spline shaft.
Otherwise, the ball screw spline unit from which balls have come out cannot be used.

5) Remove four (4) hexagon socket head bolts (M4 x 16). Then remove the axis 3 bracket.

6) Remove the timing belts from Axes 3 and 4.

7) Remove four (4) hexagon socket head bolts (M5 x 20) and washers, and then remove the axis 3 nut pulley which is secured to the ball screw nut.

8) Also remove the upper stopper from the ball screw shaft.

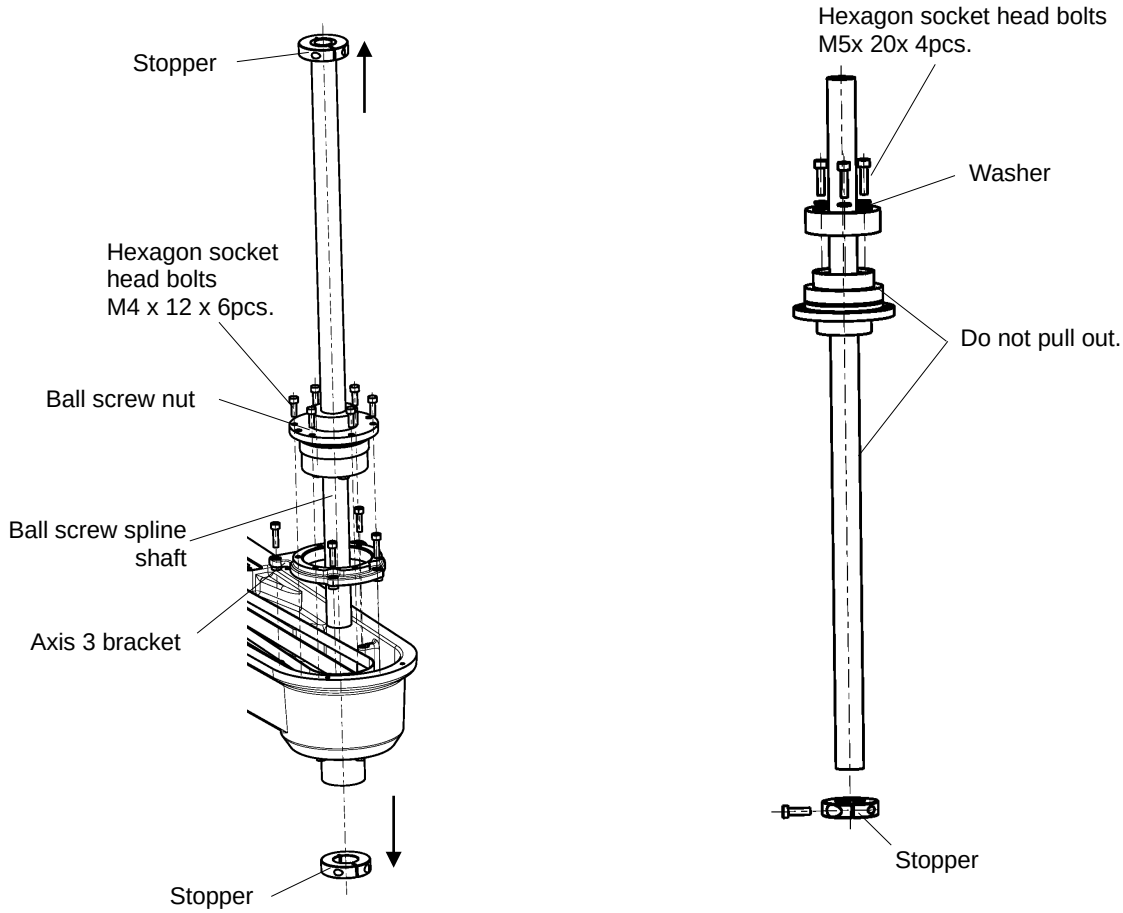


Fig. 4.47 Dismounting ball screw spline unit (THL800, THL900, THL1000, THL1200)

9) Remove six (6) hexagon socket head bolts (M4 x 12) securing the ball spline nut, and pull out the ball spline nut downward.
If the ball spline nut is hard to remove, screw a bolt into the ball spline nut tap (M4) to remove it.

10) Disconnect the axis 4 nut pulley secured to the ball spline nut with six (6) hexagon socket head bolts (M5 x 12). When it is hard to disconnect, screw the bolt into the tap for disassembly (M4) machined on the pulley and disconnect.

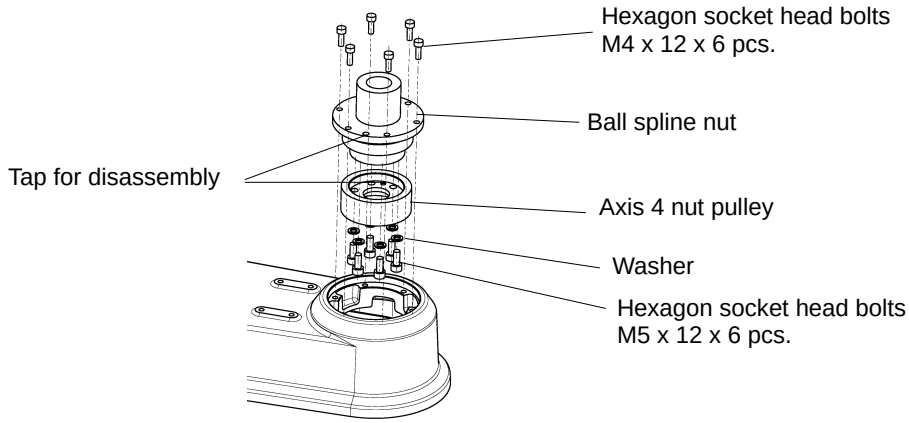


Fig. 4.48 Dismounting ball screw spline nut (THL800, THL900, THL1000, THL1200)

11) Insert the disconnected ball spline nut into the ball screw spline shaft.
A marking for phase adjustment is stamped on both the ball screw spline shaft and the ball spline nut. Align the two markings when inserting the ball spline nut.
To prevent the nut from slipping off, wind cable ties around near the top and bottom of the shaft.

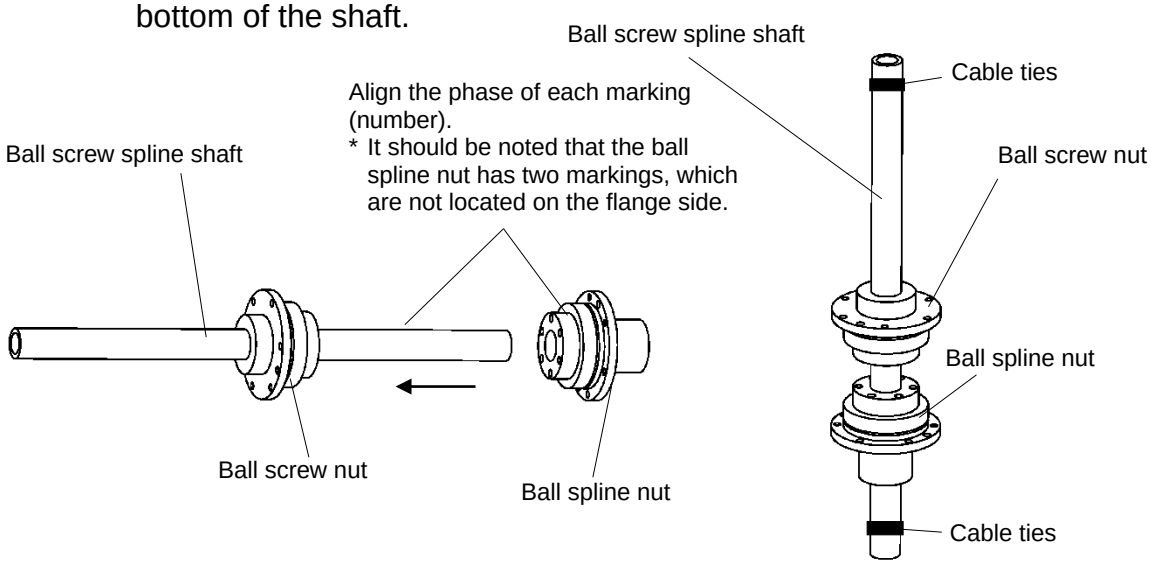


Fig. 4.49 Ball screw spline nut (THL800, THL900, THL1000, THL1200)

4.6.5 Mounting Ball Screw Spline Unit

⚠ DANGER

– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– Never touch the ball screw spline unit with bare hands.
Touching it with bare hands causes the ball screw spline unit to rust.
- Be sure to wear gloves.
– When replacing the ball screw spline unit, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– Be sure to handle the ball screw spline unit carefully.
Drop impact or application of excessive external force could ruin the unit.
– After the ball screw spline unit is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”
– The replacement of the ball screw spline unit requires the axis-3 and axis-4 timing belts to be also removed.
Therefore, the precautions for the replacement of the axis-3 and axis-4 timing belts must be also observed.

1) Disconnect the ball spline nut from the new ball screw spline unit.

2) Mount the axis 4 nut pulley on the ball spline nut with the hexagon socket head bolts (M5 x 12 x 6 pcs.) and washers.
The bolts are difficult to tighten because the pulley rotates. Thus, tighten them while holding the pulley with pliers or the likes. In doing so, use a waste cloth to protect the pulley from scratches.

3) Mount the ball spline nut on Arm 2 with the hexagon socket head bolts (M4 x 12 x 6 pcs.) and washers. When doing so, be careful with the orientation of the removal tap.

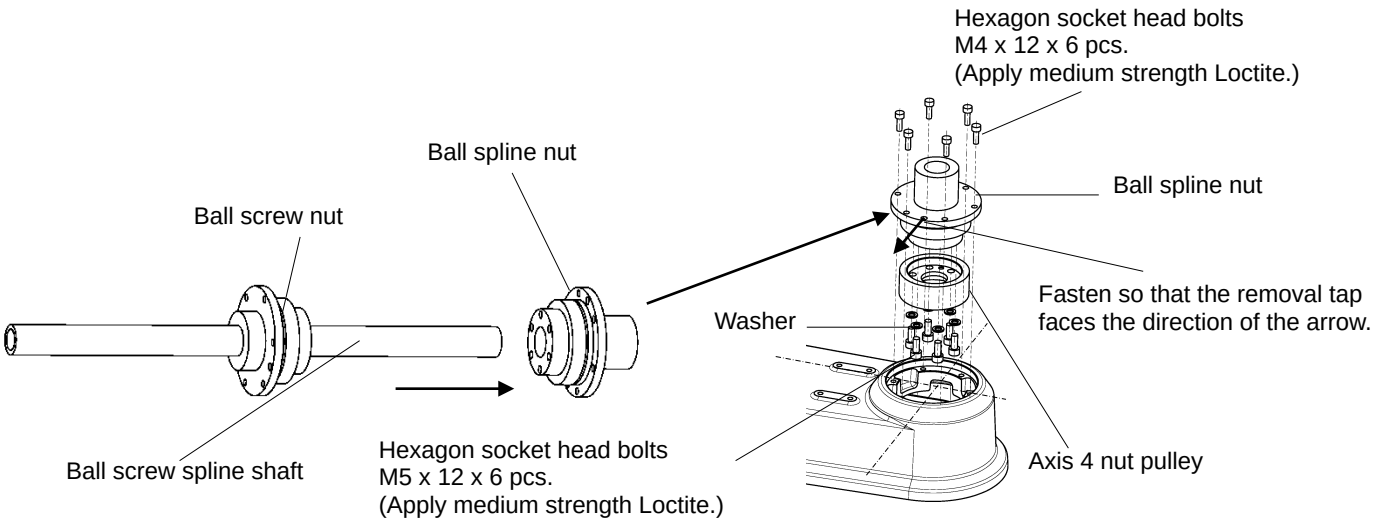


Fig. 4.50 Mounting ball spline nut (THL800, THL900, THL1000, THL1200)

4) Mount the axis 4 timing belt and axis 4 motor assembly, then adjust the belt tension. (See “Para. 4.5.4, Replacing Axis 4 Timing Belt.”)

5) Temporarily tighten the axis 4 bracket to Arm 2 with four (4) hexagon socket head bolts (M4 x 16).

6) Secure the axis 3 nut pulley to the ball screw nut with four (4) hexagon socket head bolts (M5 x 20) and washers. Mount the upper stopper. For the stopper mounting position, see Fig. 4.52

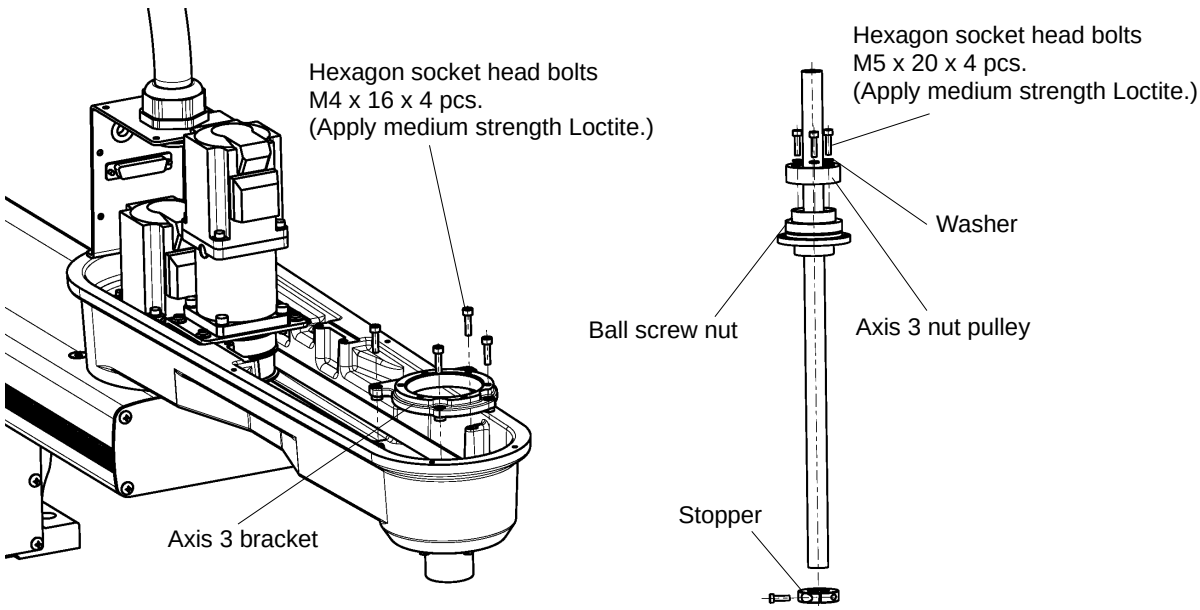


Fig. 4.51 Axis 3 bracket and pulley (THL800, THL900, THL1000, THL1200)

NOTICE

– Never pull out the ball screw nut from the ball screw spline shaft.
Otherwise, the ball screw spline unit from which balls have come out cannot be used.

7) Mount the ball screw nut and the ball screw spline shaft together through the ball spline nut, and then mount it to the axis 3 bracket with six (6) hexagon socket head bolts (M4 x 12). At this time, align the phases of both markings put on the tip of the ball spline nut and the ball screw spline shaft. In this case, take care of the direction of the tap for draw out.

8) Mount a stopper at a position 40 mm from the bottom of the ball screw spline shaft.
When mounting each stopper, align its split line with the left end of the spline groove.

9) Manually move the shaft several times in the vertical direction and make sure that the movement is smooth. After that, tighten the bolt temporarily clamping the axis 3 bracket. In this case, fix it by holding the axis 3 bracket to prevent it from moving.

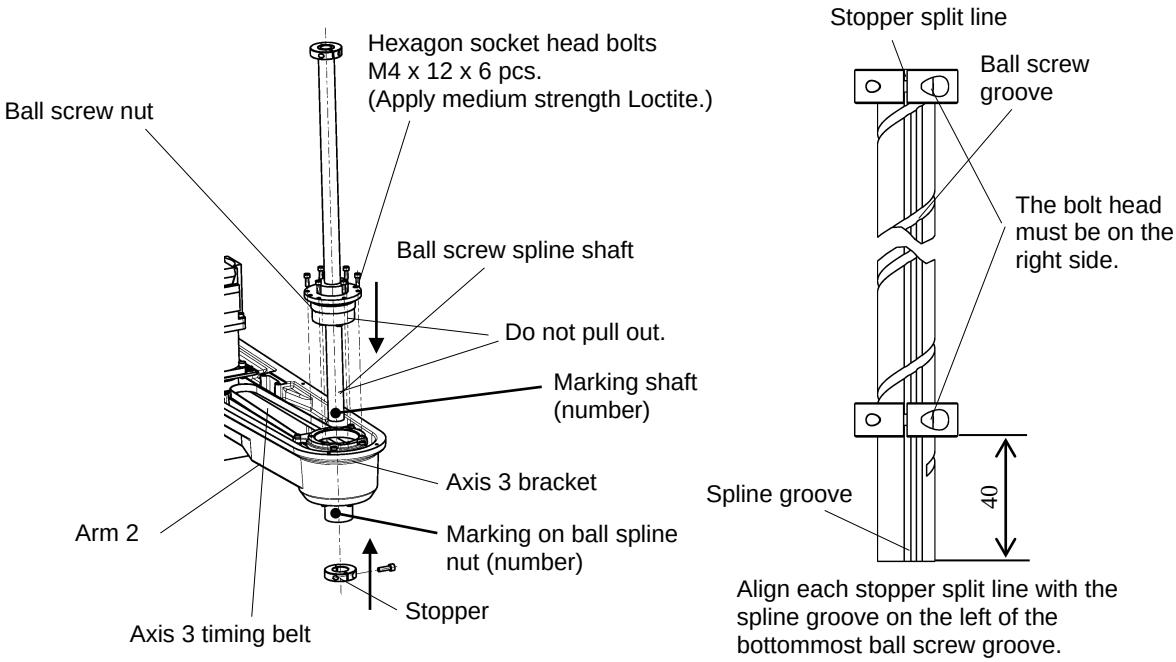


Fig. 4.52 Mounting ball screw spline nut (THL800, THL900, THL1000, THL1200)

10) Mount the axis 3 motor assembly, then adjust the belt tension. (See “Para. 4.5.3, Replacing Axis 3 Timing Belt.”)

11) Arrange the connectors and cables as originally set. (See the pictures you took beforehand.)

12) Perform home setting for Axes 3 and 4.
When performing home setting for Axis 4, align the stopper split line with the mating mark on the ball spline nut.
After home setting, remove the mating mark from the old ball spline nut, and attach it to the new one.

13) Carry out a test operation of Axes 3 and 4 and make sure that each part operates properly.

14) Mount the arm 2 cover. Now replacement of the ball screw spline unit completes.

4.7 Reduction Gear Replacement

The reduction gear is to be replaced by our service engineer. If it is replaced by the customer, we will not guarantee any consequential trouble or accident.

⚠DANGER

– When replacing the reduction gear, be sure to remove the power plug of the controller.
The removed cable terminal may be touched to result in an electric shock. A worker may be entangled in rotation of the motor that has operated unpredictably.

NOTICE

– When replacing the reduction gear, be sure to remove the power plug of the controller.
If replacement work is done with the power turned on, the robot may malfunction.
– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.
– After the reduction gear is replaced, be sure to re-set the zero position.
Because a mechanical zero position may be shifted with replacement, the robot cannot perform normal operation.
- For how to re-set the zero position, see “8. Robot Home Point and Position Detector Error.”

4.7.1 Type of Reduction Gear

The reduction gears used in this robot are shown below.
When you place an order for the reduction gear for replacement, specify the robot model (THL800, THL900, THL1000, THL1200), axis name, type and our drawing number.

Table 4.9 Type of reduction gear (THL800, THL900, THL1000, THL1200)

Description	Axis name	Our drawing No.	Unit code
Reduction gear	Axis 1	S891110	Y610A3PS0
	Axis 2	S891111	Y610A3PT0
	Axis 4	S875239	Y610A3LY0

4.7.2 Reduction Gear Locations

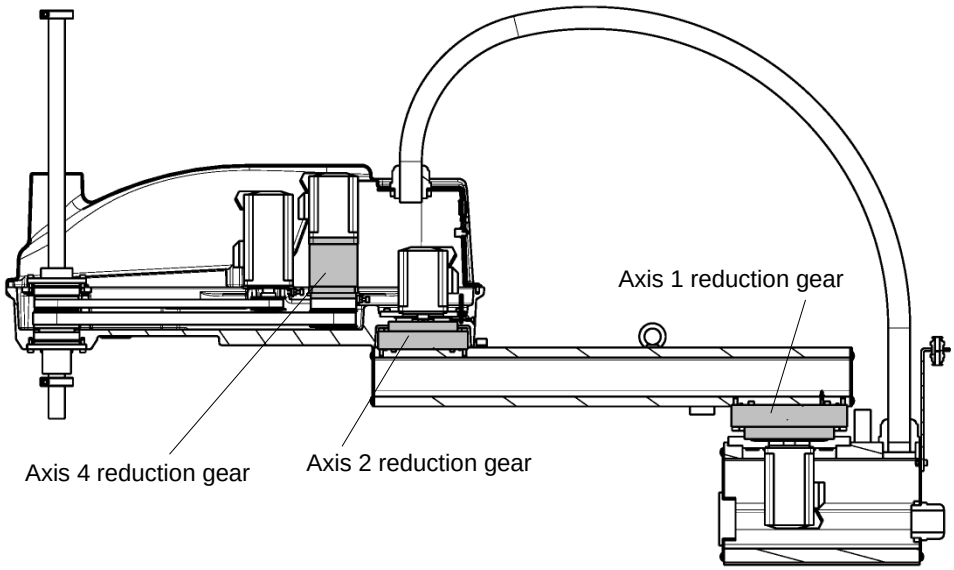


Fig. 4.53 Reduction gear locations (THL800, THL900, THL1000, THL1200)

4.7.3 Dismounting Axis 1 Reduction Gear

1) Remove the base front cover. (See “Para. 4.3.2, Base Covers.”)
2) Remove the parts up to the axis 1 motor assembly according to "4.3.2 Base cover".
3) Remove sixteen (16) hexagon socket head bolts (M4 x 20) that secure the base, and then remove Arm 1.

⚠ CAUTION

– When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

– Be sure to handle the arm carefully.
Drop impact or application of excessive external force could ruin the arm.

4) Remove twelve (12) hexagon socket head bolts (M5 x 35) that secure the axis 1 reduction gear to Arm 1, and then remove the axis 1 reduction gear. Also remove the O-ring mounted in the groove of Arm 1 at this time. Also, remove the liquid gasket attached to the base mounting surface of the axis 1 reduction gear with the back of a cutter or a similar tool without damaging.

5) Remove the wave generator from the axis 1 motor. (See “Para. 4.4.3, Dismounting Axis 1 Motor.”)

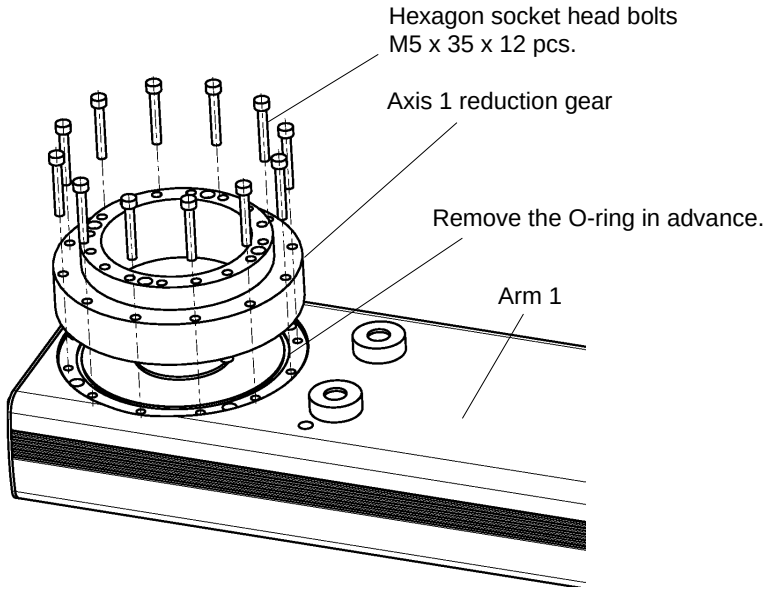


Fig. 4.54 Dismounting axis 1 reduction gear (THL800, THL900, THL1000, THL1200)

4.7.4 Mounting Axis 1 Reduction Gear

NOTICE

– Be sure to use wave generator provided for new reduction gear to be replaced.
The combination of the reduction gear and wave generator is determined. If they are used in a wrong combination, abnormal sound will occur. The positioning accuracy and lifespan will be reduced.
– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.
– Never forget fitting the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
- Use the new O ring provided for the reduction gear.

1) Mount a new wave generator provided as an accessory with a new reduction gear. For the mounting procedure, see “Para. 4.4.4, Mounting Axis 1 Motor.”)

2) Cleanly wipe off dusts and stains on the axis 1 reduction gear of Arm 1. Apply grease to an O-ring attached to the new reduction gear, and then mount it in the O-ring groove of Arm 1.

3) Secure the axis 1 reduction gear with twelve (12) hexagon socket head bolts (M5 x 35).

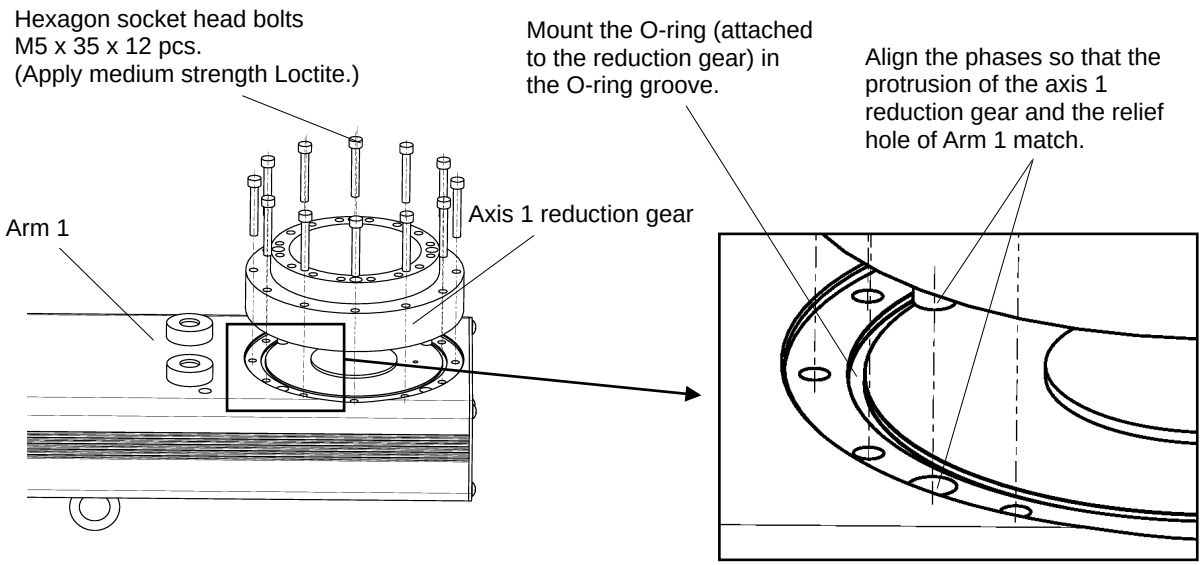


Fig. 4.55 Mounting axis 1 reduction gear (THL800, THL900, THL1000, THL1200)

4) Apply grease inside the main body of the axis 1 reduction gear as thick as the thickness of the attached wave generator. Also apply grease to the wave generator. Additionally, apply grease to an O-ring attached to the new reduction gear, and then mount it in the O-ring groove of the reduction gear.

Amount of grease filled in Axis 1
53 g

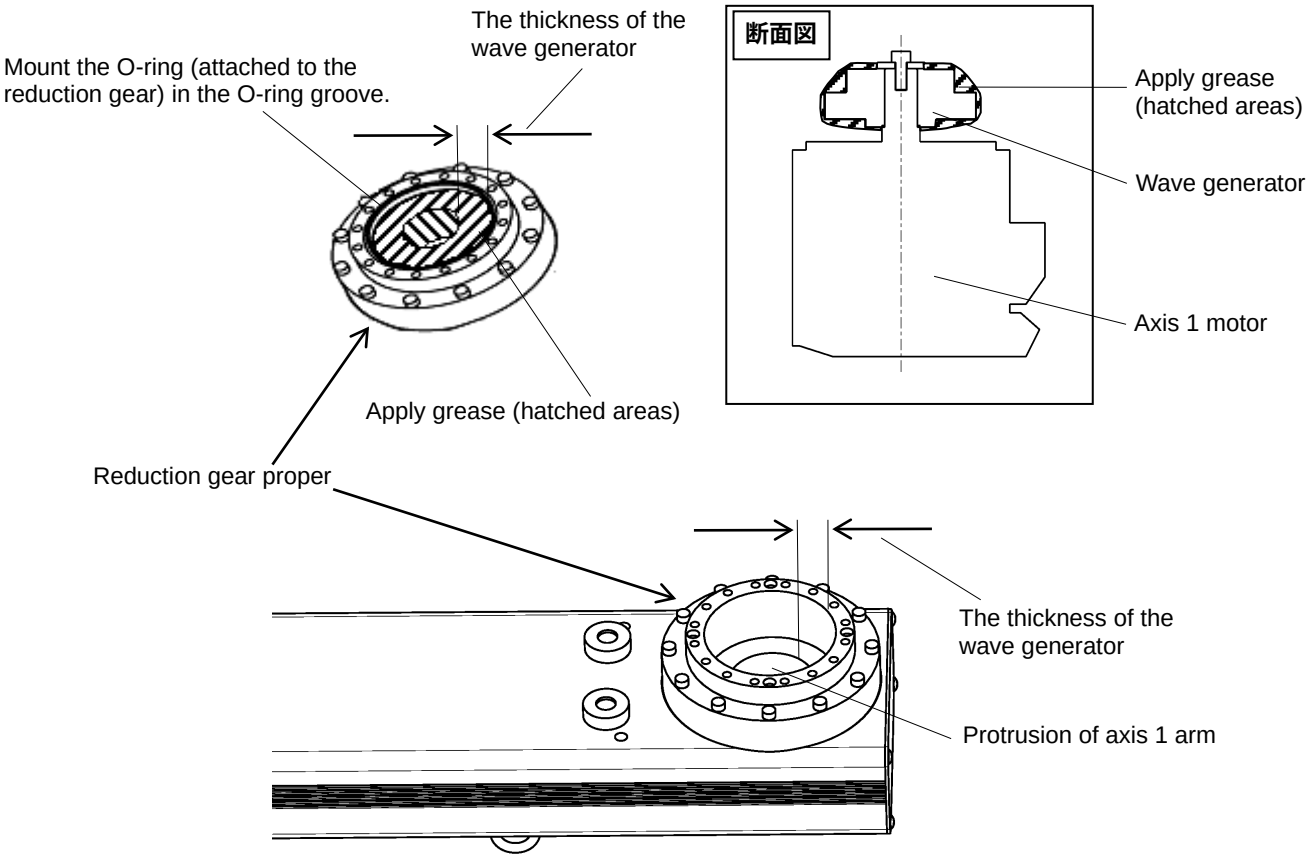


Fig. 4.56 Application of grease to axis 1 reduction gear (THL800, THL900, THL1000, THL1200)

NOTICE

– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

– Be sure to observe the filling amount of grease.
If the filling amount is tool large, the starting torque increases and the internal seal is broken due to rise in the internal pressure in the reduction gear. If the filling amount is too small, malfunction is caused by poor lubrication.

– Never forget fitting the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
- Use the new O ring provided for the reduction gear.

5) Mount the axis 1 motor assembly according to "4.4.4 Mounting of axis 1 motor", and fix the Arm 1 on the base.

6) Mount the base front cover. (See “Para. 4.3.2, Base Covers.”)

7) After changing the axis 1 motor, move Arm 1 by hand and check that there is no abnormal sound.

8) Set up the axis 1 home position to complete axis 1 reduction gear replacement. Because the reduction gear is changed, home position settings of Axes 2 through 4 are required.

9) Carry out a test operation of Axis 1 to make sure that each part operates properly.

4.7.5 Dismounting Axis 2 Reduction Gear

1) Remove the arm 2 cover. (See “Para. 4.3.1, Arm 2 Cover.”)
Remove the harness guide and the support plate, and then remove the axis 2 motor assembly. Also remove the O-ring (CO0545A) of the O-ring groove. (See “Para. 4.4.5, Dismounting Axis 2 Motor.”)

2) Remove ten (10) hexagon socket head bolts (M4 x 20) that secure Arm 2 and the reduction gear.

3) Remove twelve (12) hexagon socket head bolts (M4 x 30) that secure the axis 2 reduction gear mounted to Arm 1, and then remove the axis 2 reduction gear. At this time, remove the O-ring mounted in the O-ring groove of Arm 1.

4) Remove the wave generator from the axis 2 motor. (See “Para. 4.4.5, Dismounting Axis 2 Motor.”)

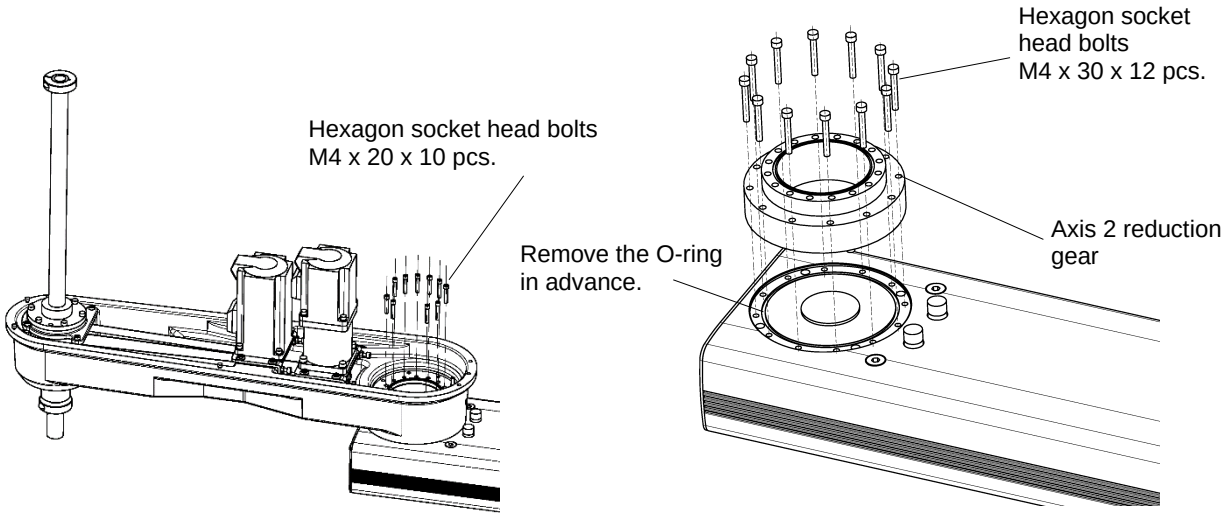


Fig. 4.57 Dismounting axis 2 reduction gear (THL800, THL900, THL1000, THL1200)

⚠ CAUTION

– When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

– Be sure to handle the arm carefully.
Drop impact or application of excessive external force could ruin the arm.

4.7.6 Mounting Axis 2 Reduction Gear

NOTICE

– Be sure to use wave generator provided for new reduction gear to be replaced.
The combination of the reduction gear and wave generator is determined. If they are used in a wrong combination, abnormal sound will occur. The positioning accuracy and lifespan will be reduced.
– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.
– Never forget fitting the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
- Use the new O ring provided for the reduction gear.

1) Mount the wave generator attached to the new reduction gear to the axis 2 motor. (See “Para. 4.4.6, Mounting Axis 2 Motor.”)

2) Cleanly wipe off dusts and stains on the reduction gear of Arm 1.
Apply grease to an O-ring attached to the new reduction gear, and then mount it in the O-ring groove of Arm 1.

3) Secure the axis 2 reduction gear to Arm 1 with twelve (12) hexagon socket head bolts (M4 x 30).

4) Apply grease to the axis 2 reduction gear. (See “Para. 4.7.4, Mounting Axis 1 Reduction Gear.” However, the amount of grease filled is different from that for the axis 1 reduction gear. Regarding the filled amount for the axis 2 reduction gear, see the following.)

Amount of grease filled in Axis 2
24 g

5) Apply grease to the other O-ring attached to the new reduction gear, and then mount it in the O-ring groove of the axis 2 reduction gear.

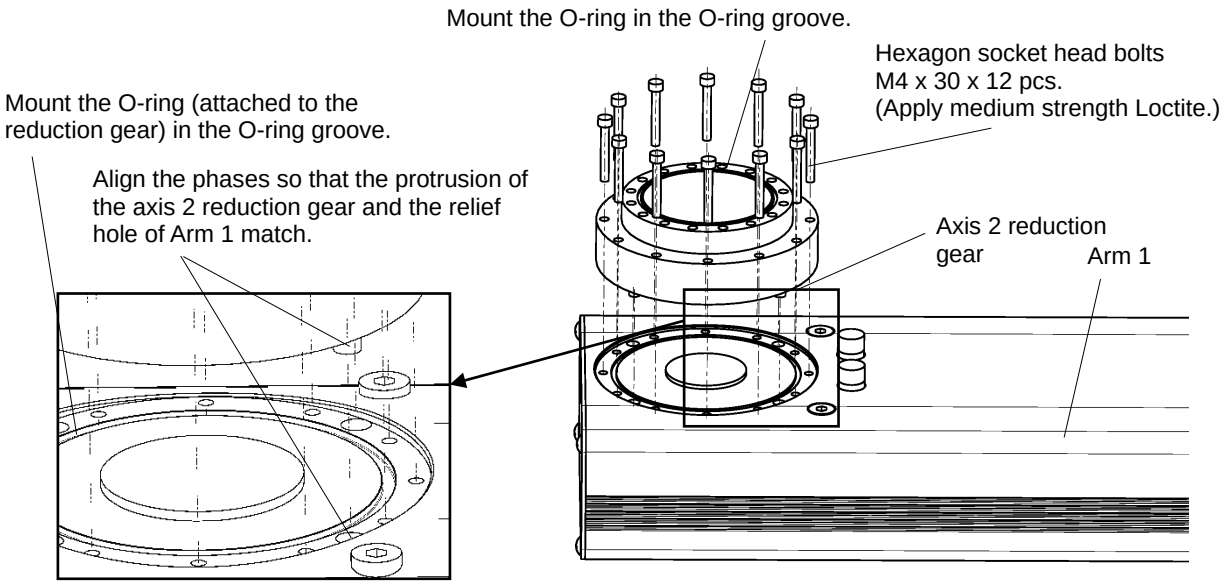


Fig. 4.58 Mounting axis 2 reduction gear (THL800, THL900, THL1000, THL1200)

NOTICE

– Never forget fitting the O ring, and never use the cut or damaged O ring.
Otherwise, grease will leak from the fitting surface.
- Use the new O ring provided for the reduction gear.

– Be sure to use the specified types of grease.
Using grease of other than specified types causes malfunction due to poor lubrication.

– Be sure to observe the filling amount of grease.
If the filling amount is tool large, the starting torque increases and the internal seal is broken due to rise in the internal pressure in the reduction gear. If the filling amount is too small, malfunction is caused by poor lubrication.

6) Secure Arm 2 to the new reduction gear with ten (10) hexagon socket head bolts (M4 x 20).

7) Apply grease to the O-ring (CO0545A), and then mount it in the O-ring groove of Arm 2.

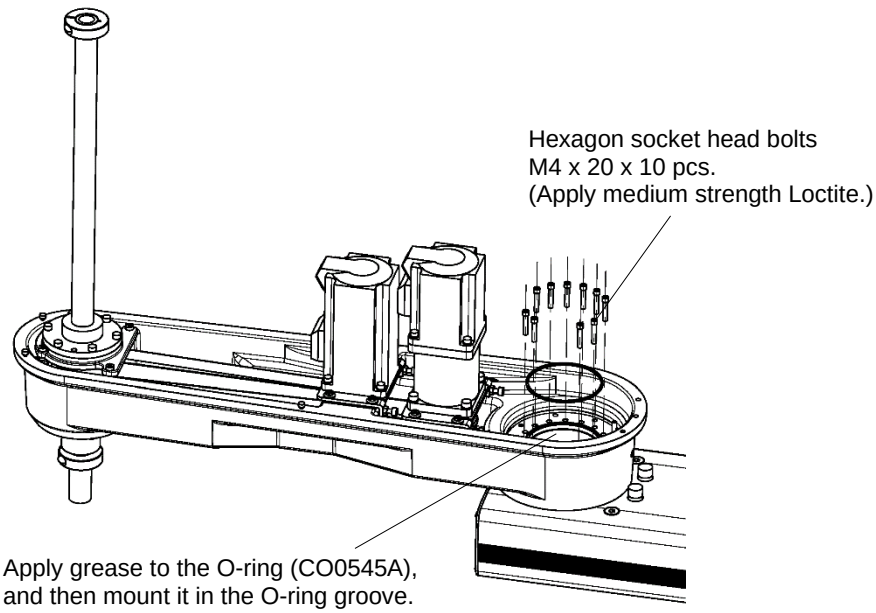


Fig. 4.59 Mounting arm 2 (THL800, THL900, THL1000, THL1200)

⚠ CAUTION

– When fitting or removing the arm, one worker should loosen the clamp bolts and other worker should support the arm.
If work is done by one worker, the arm cannot be supported and may drop.

NOTICE

– Be sure to handle the arm carefully.
Drop impact or application of excessive external force could ruin the arm.

8) Mount the axis 2 motor assembly, the harness guide and the support plate. Put back the cables to the original locations.

9) Move Arm 2 by hand and check that there is no abnormal sound before turning on the power.

10) Mount the arm 2 cover.

11) Set up the home positions of axes 2 to complete axis 2 reduction gear replacement.

4.7.7 Replacing Axis 4 Reduction Gear

NOTICE

– Be sure to handle the reduction gear carefully.
Drop impact or application of excessive external force could ruin the reduction gear.
– Be careful not to forget to loosen the coupling that fastens the reduction gear and motor shaft when dismounting reduction gear.
– Be careful not to forget tightening the coupling that fastens the reduction gear and motor shaft when dismounting reduction gear.

1) Remove the arm 2 cover. (See “Para. 4.3.1, Arm 2 Cover.”)
2) Remove only the axes 3 and 4 motor assemblies from the arm 2 according to steps (1) and (2) of "4.5.4 Replacement of axis 4 timing belt".
3) Remove the cap attached to the lateral side of the axis 4 reduction gear, then loosen the coupling bolt (M4) securing the axis 4 motor shaft and the input shaft of the axis 4 reduction gear.
4) Remove four (4) hexagon socket head bolts (M4 x 12) and washers securing the motor, and draw out the axis 4 motor upward.
5) While holding the axis 4 motor pulley with pliers or the likes, remove six (6) hexagon socket head bolts (M4 x 12) to remove the pulley. In doing so, use a waste cloth (or the like) as a cushion to protect the pulley from scratches.
6) Remove four (4) hexagon socket head bolts (M5 x 16) and washers securing the axis 4 motor plate and reduction gear to the motor base, then dismount the reduction gear.

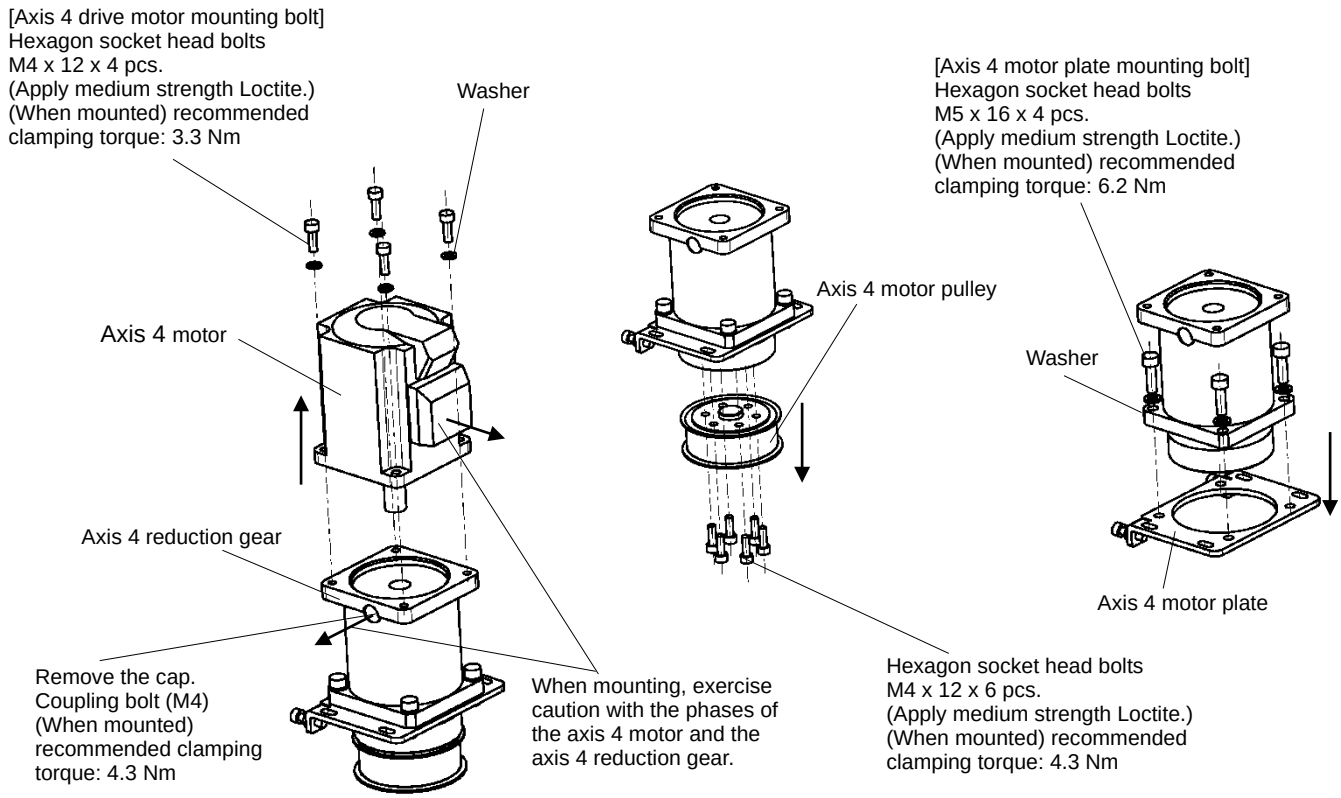


Fig. 4.60 Replacing axis 4 reduction gear (THL800, THL900, THL1000, THL1200)

7) Assemble the new reduction gear, motor and pulley in the reverse order of Steps 3) to 6) above. Be careful with the phases of the axis 4 motor, the axis 4 reduction gear and the axis 4 motor plate at this time.

8) Assemble the axis 4 motor assembly and timing belt according to Steps 4) through 10) of “Para. 4.5.4, Replacing Axis 4 Timing Belt” above. Mount the arm 2 cover.

9) Set the zero points of Axes 3 and 4 to complete the replacement of axis 4 reduction gear.

5. Maintenance of Controller (TSL3000)

5.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of the controller, follow the items given below so that the work can be carried out safely.

⚠DANGER

– When opening the cover of the controller, do not allow water or other fluids or foreign objects to get inside. This could cause a short-circuit inside the controller, which could lead to a fire or an electric shock due to ground leakage.
– Use SHIBAURA MACHINE special products for replacing certain maintenance parts. Wiring of unauthorized connectors and terminals could lead to a fire or an electric shock due to ground leakage.
– Be sure to request the SHIBAURA MACHINE servicing company if any maintenance or replacement work not described in this instruction manual is required. (A fee will be charged for this work.) SHIBAURA MACHINE will not be liable for any accidents or damage that occurs due to replacement work performed by the customer or its results.

⚠ WARNING

– Do not touch the controller terminals or internal board with your hands for about two minutes after the power cable is disconnected. A residual voltage is still in the large-capacity capacitor inside the controller, and this could result in an electric shock.
– Use SHIBAURA MACHINE authorized spare parts when replacing parts. Use of unauthorized parts could result in a fire or a collision with a robot in unexpected motion.
– Before starting any work, be sure to always wear protective gear, and unplug the power cable. If the power is turned on, the servo power module, servo amplifier module, switching power supply, and other components will carry a charge, and an electric shock could result.
– Turn off the power before opening the controller cover. If the cover is opened while the power is on, your hands or conductive tools could touch the power-carrying part, resulting in an electric shock.
– Perform the following actions whenever entering the movable range of the robot.
  - Wear protective gear
  - Place the teach pendant in an accessible location to the worker.
  - Change to MANUAL mode (remove the mode selector key so that the mode cannot be changed)
  ⇒ If the above operations are not performed, the emergency stop could be cleared by another person during the daily inspection work within the movable range, and a collision could occur with a robot that was set to continuous operation.
– Use voltage measurement and the LEDs to check that no PN voltage remains in the large-capacity capacitor. If a PN voltage remains, an electric shock could occur.
– Do not put conductive objects or fingers while the power is on. Touching the power-carrying part could result in an electric shock.
– All parts are special specifications parts. Please request these parts from SHIBAURA MACHINE when purchasing and ordering.

5.2 Layout of Controller Parts

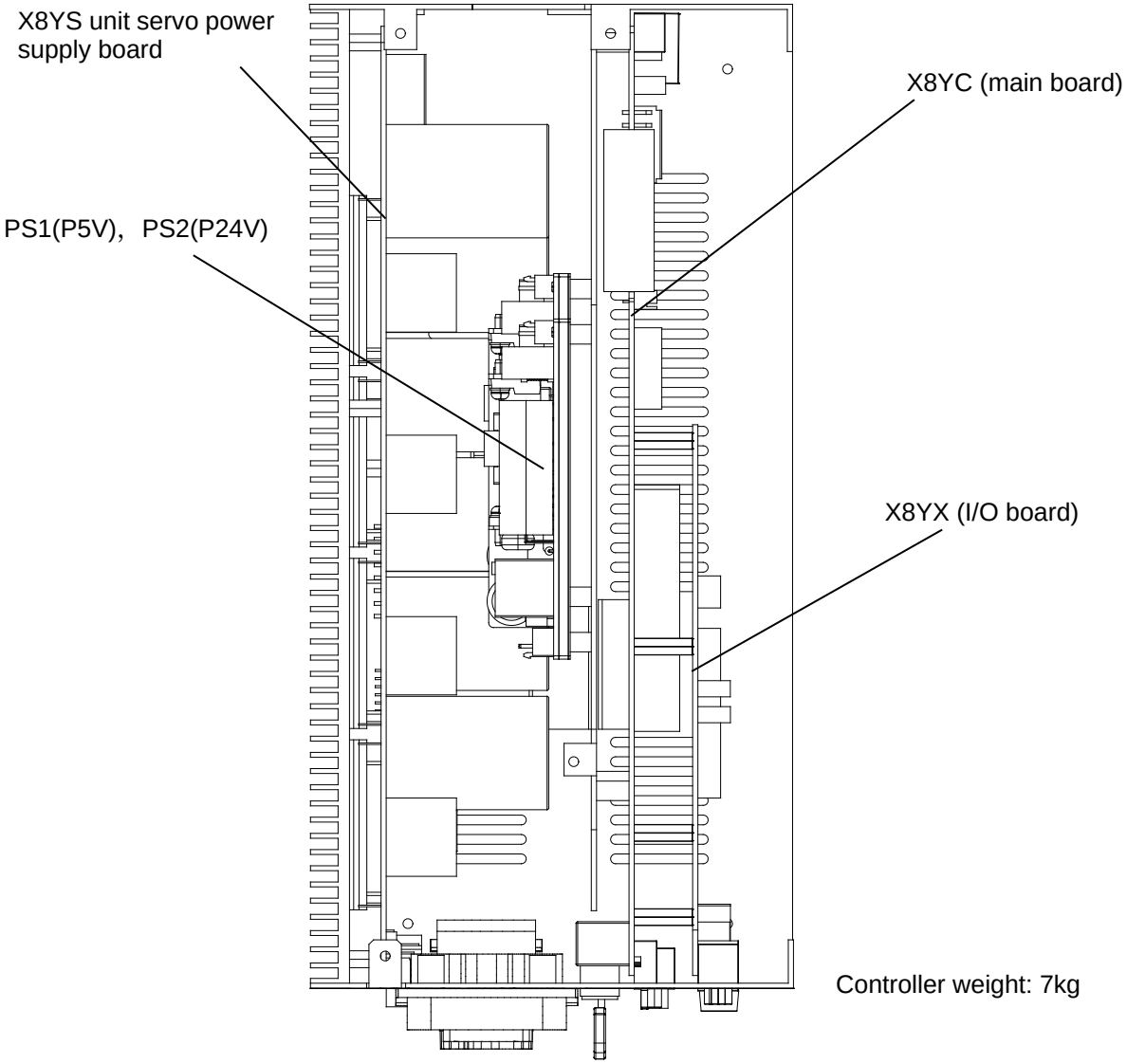


Fig. 5.1 Layout of controller parts (TSL3000)

Table 5.1 Controller parts (TSL3000)

Part name	Descriptions
PS1, PS2	PS1 (P5V), PS2 (P24V) output switching power supply
X8YC	Main control printed board
X8YX	I/O output printed board
X8YS	Servo power module

5.3 Maintenance Procedures

5.3.1 Check of Controller Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction.
To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.

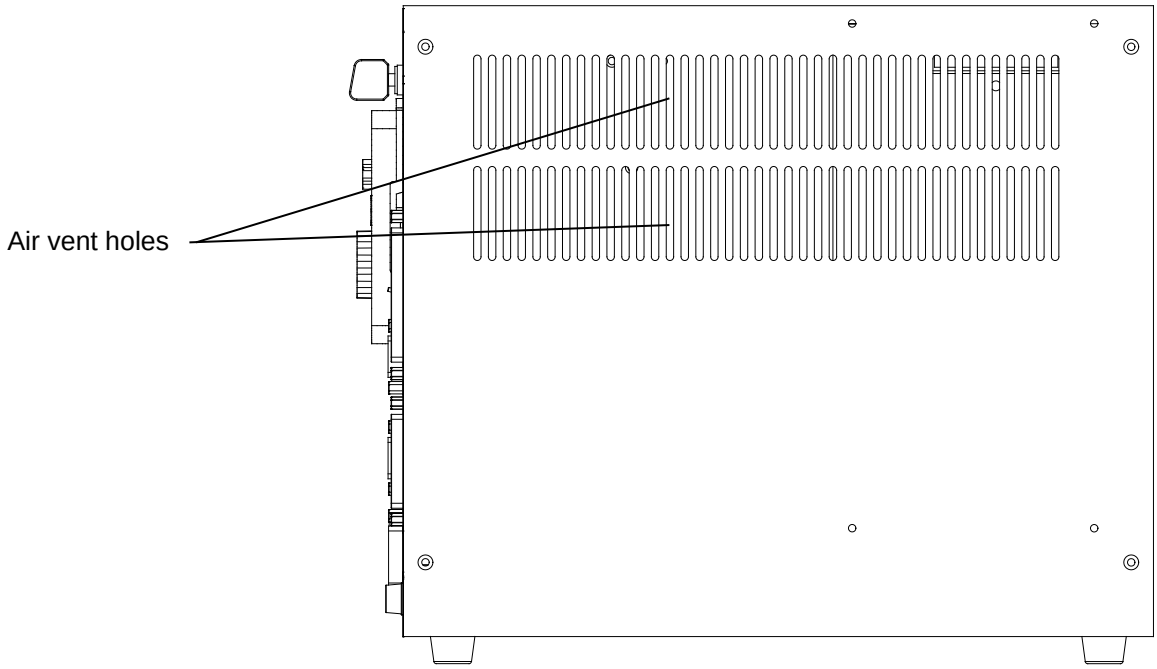


Fig. 5.2 Controller side views (TSL3000)

1) If there is any obstacle, move it away from the air vent hole so that air flow is not blocked.

2) Make sure that no contaminant is left on the air vent holes. If any contaminant is left on the vent hole, remove it.

5.3.2 Check of Safety Devices for Function

Make sure that the EMERGENCY stop pushbutton switches equipped on the teach pendant work properly. Also make sure that the safety devices controlled by the external operation input signals work correctly.

1) Make sure that the EMERGENCY (emergency stop) switch provided on the teach pendant functions properly.
   a) Turn around the key at front of controller for selecting the TEACH.
   b) Turn the EMERGENCY switch [1] clockwise and make sure that the EMERGENCY switch [1] turns off.
   c) Press the SERVO ON switch [2] on the teach pendant to turn on the servo power while keeping the ENABLE switch [3] at the neutral position, and make sure that the SERVO ON LED [2] is illuminated.

      Note: The servo power cannot be turned on in the TESTRUN mode.

   d) Press the EMERGENCY STOP switch [1] on the teach pendant and make sure that the servos are turned off. The SERVO ON LED [2] goes out.

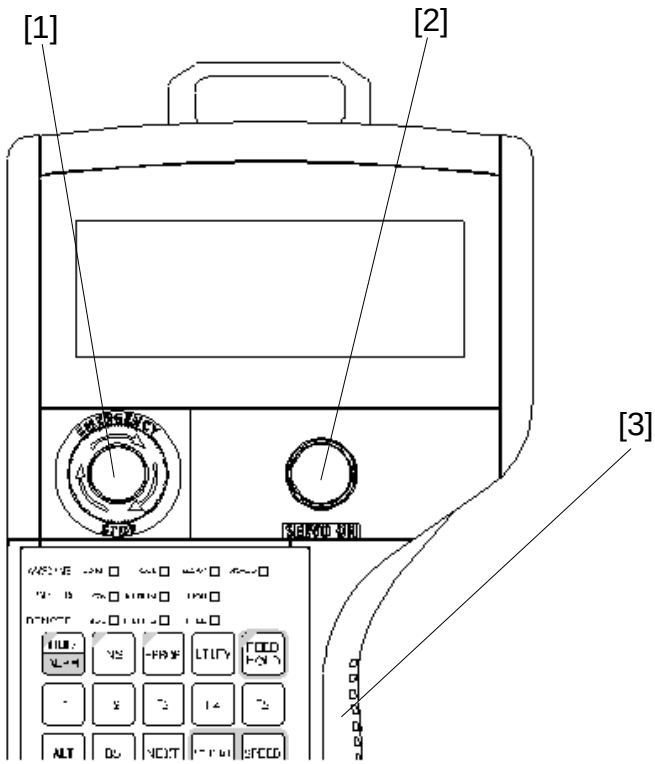


Fig. 5.3 EMERGENCY STOP switch provided on teach pendant (TSL3000)

At this time, make sure that the EMERGENCY STOP switch [1] remains depressed.

e) Turn the EMERGENCY STOP switch [1] clockwise and make sure that the EMERGENCY switch [1] turns off.

2) Make sure that the safety devices controlled by the external operation input signals work properly.

a) Connect the power supply plug to the power source to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal “emergency stop” (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

b) Set ON the safety devices connected with external operation input signal “low speed command”, and make sure that each signal functions properly.

5.3.3 Battery Replacement

DANGER

– Do not attempt to incinerate, disassemble, short-circuit, or charge the battery. This could cause the battery to rupture, resulting in a fire. Dispose of the battery in compliance with the regulations of the customer.

NOTICE

– If the battery is to be kept removed for more than one (1) minute, copy all programs and various parameters stored in the internal memory to the personal computer. In the worst case, all memory data are lost.

The memory equipped on the X8YC printed board of the robot controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the memory data, resulting in the data being lost and faults caused by leakage of the battery liquid.

If the following alarm has occurred, replace the battery with a new one.

- 1–145 MAIN Battery alarm

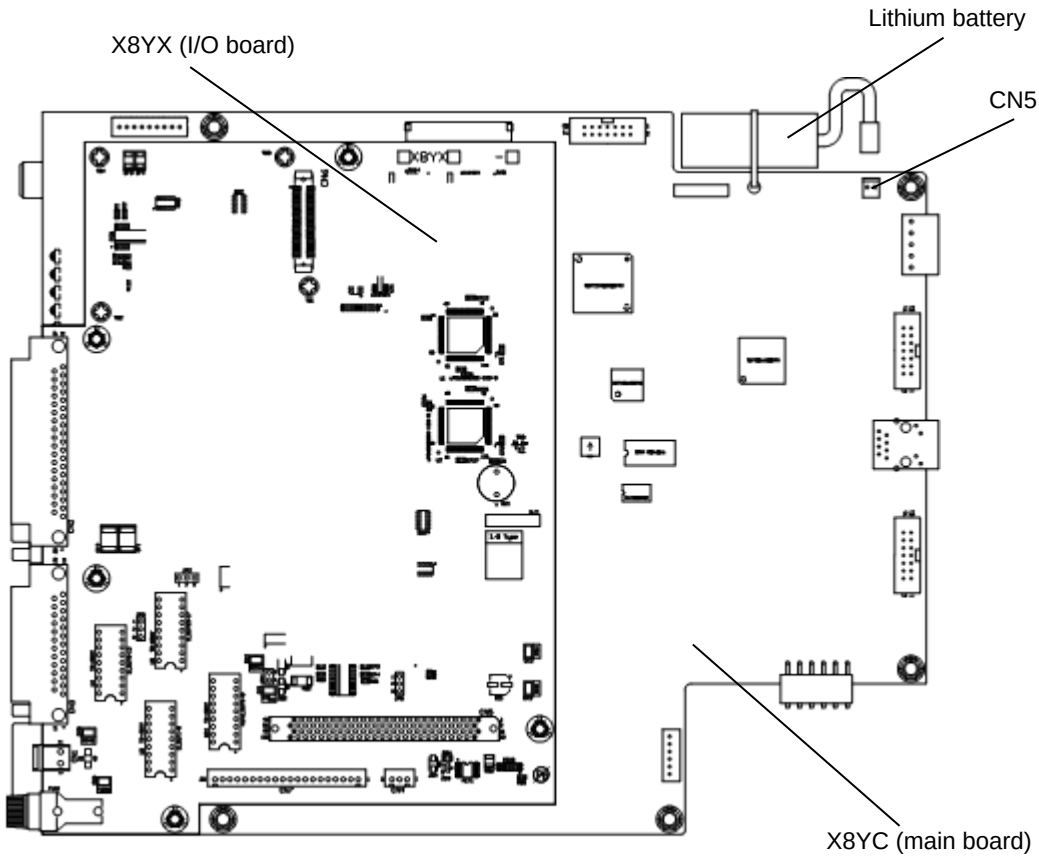


Fig. 5.4 Location of lithium battery (TSL3000)

Table 5.2 Battery replacement (TSL3000)

Type of battery	Manufacturer	Battery service life	Remarks
ER6C WK27	Hitachi Maxell	10 years (at room temperature)	Discontinued in 2022
ER6V C4	TOSHIBA	10 years (at room temperature)	Adopted starting in 2022

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.
Only the battery shown in the table above should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

[Battery replacement procedures]

1) If the battery is to be kept removed for more than one (1) minute, copy all programs and various parameters stored in the internal memory to the personal computer. In the worst case, all memory data are lost.

2) Disconnect the controller's power plug from the power supply.

3) Have a new battery ready, remove eight (8) countersunk screws (M3 x 6) that secure the cover, and then remove the cover from the main body.

4) Remove the battery from the cable ties securing the battery.
(For the battery position, see Fig. 5.4.)

5) Remove the battery connector from CN5 of the X8YC board. Immediately connect the connector of the new battery to CN5. (If the battery is kept removed for more than one (1) minute, all memory data are lost.)
Even if not used, the lithium battery capacity decreases by natural discharge. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.

6) Secure the battery connected to CN5 with cable ties.

7) Using countersunk screws (M3 x6 x 8 pcs.), mount the cover on the controller.

8) If the battery is kept removed for more than one (1) minute, connect the power supply plug of the controller to the power source and load the USB memory or personal computer in which the TSPC (option) is already installed.
If the controller has detected the battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

5.3.4 Replacement of Switching Power Supply Unit

⚠ DANGER

– Install the switching power supply in the location shown in the instruction manual. If the up and bottom or right and left sides are backwards, or the connector is different, this could cause a short-circuit or broken wire, resulting in a fire or an electric shock due to ground leakage.

The life of the switching power supply unit (5 VDC/24 VDC) used in the robot controller differs with the operating conditions. This unit uses an aluminum electrolytic capacitor, and if the load current is large, the running time is long and the ambient temperature is high, the life will reduce.
As the life varies largely with the user's operating conditions (5 to 10 years), replace the power supply unit every five (5) years for preventive maintenance.

[Replacement procedures]

1) Turn off the power breaker and disconnect the power supply plug from the power source.
2) Disconnect the ACIN connector from the controller.
3) Remove the key from the controller.
4) Remove eight (8) countersunk screws (M3 x 6) that secure the cover, and then remove the cover from the main body.

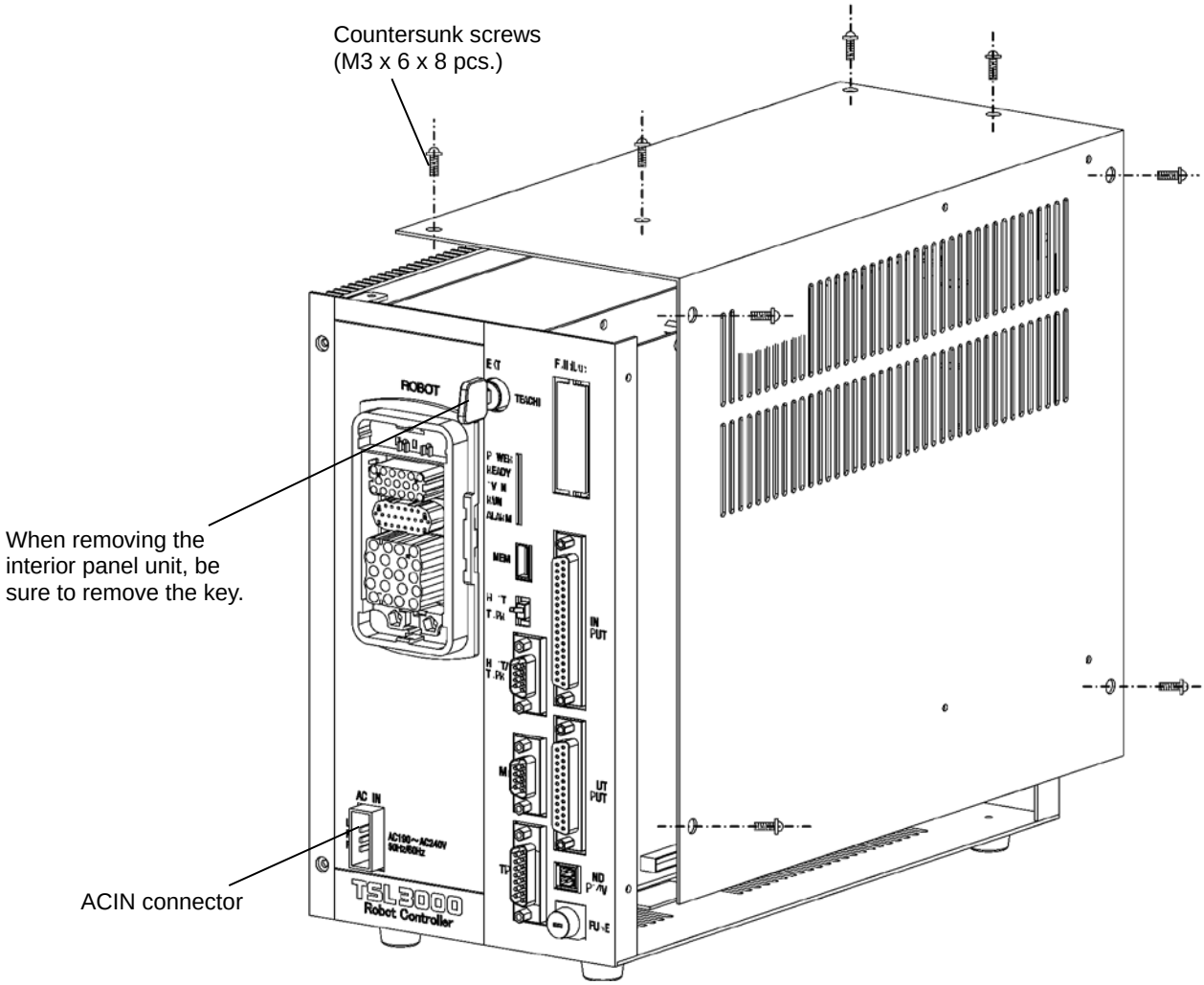


Fig. 5.5 Removing cover (TSL3000)

5) Remove the DC cable (blue and red), the SVIF cable (blue) and the encoder cable (blue) connected to the X8YC (main board) as well as the hand I/O cable (blue) and the brake cable (blue) connected to the X8YX (I/O board).

6) Remove the cables (red x 1 line, blue x 1 line, black x 2 lines) connected to the PS1 and the PS2.
(See Fig. 5.9.)

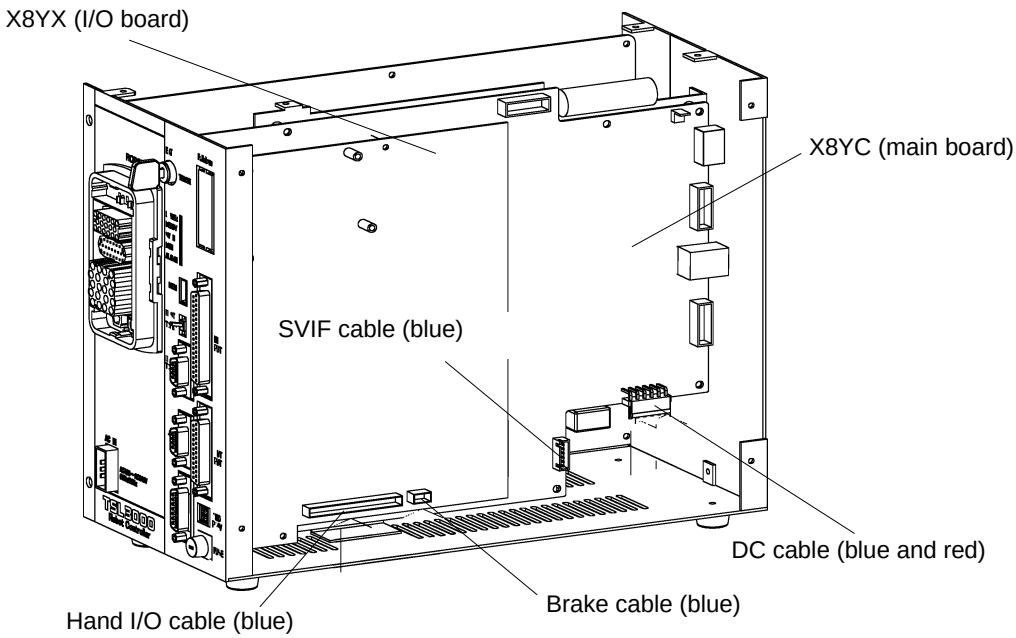
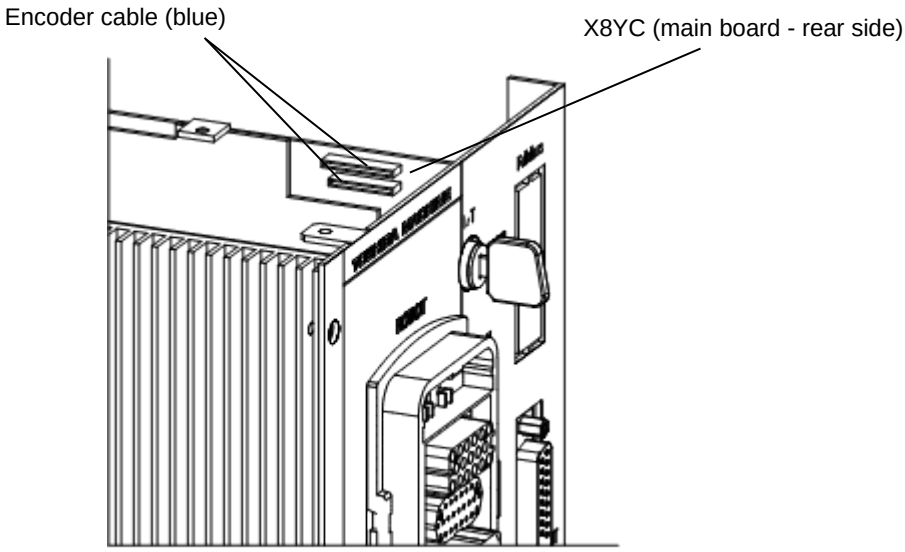


Fig. 5.6 Controller interior (TSL3000)



Controller front side view

Fig. 5.7 Controller interior (TSL3000)

7) Remove one (1) countersunk screw (M3 x 6) that secures the interior panel unit. Pull out the interior panel unit toward the rear of the controller.
When removing the interior panel unit, be sure to remove the key.

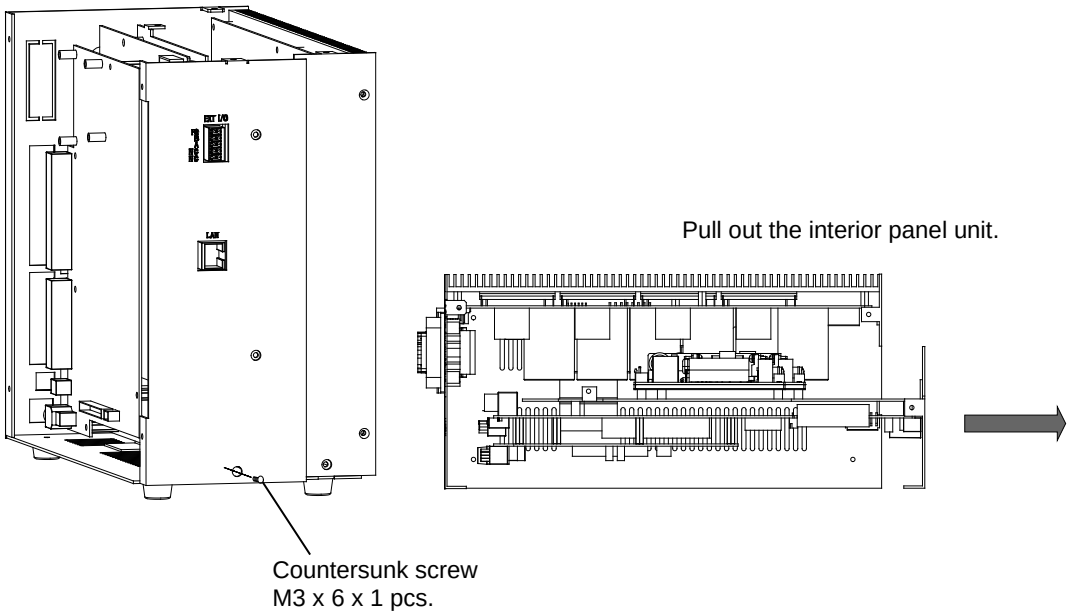


Fig. 5.8 Removing interior panel unit (TSL3000)

8) Remove four (4) sems screws (M3 x 8) that secure each of the switching power supplies (PS1, PS2), and then remove the switching power supplies from the interior panel unit.

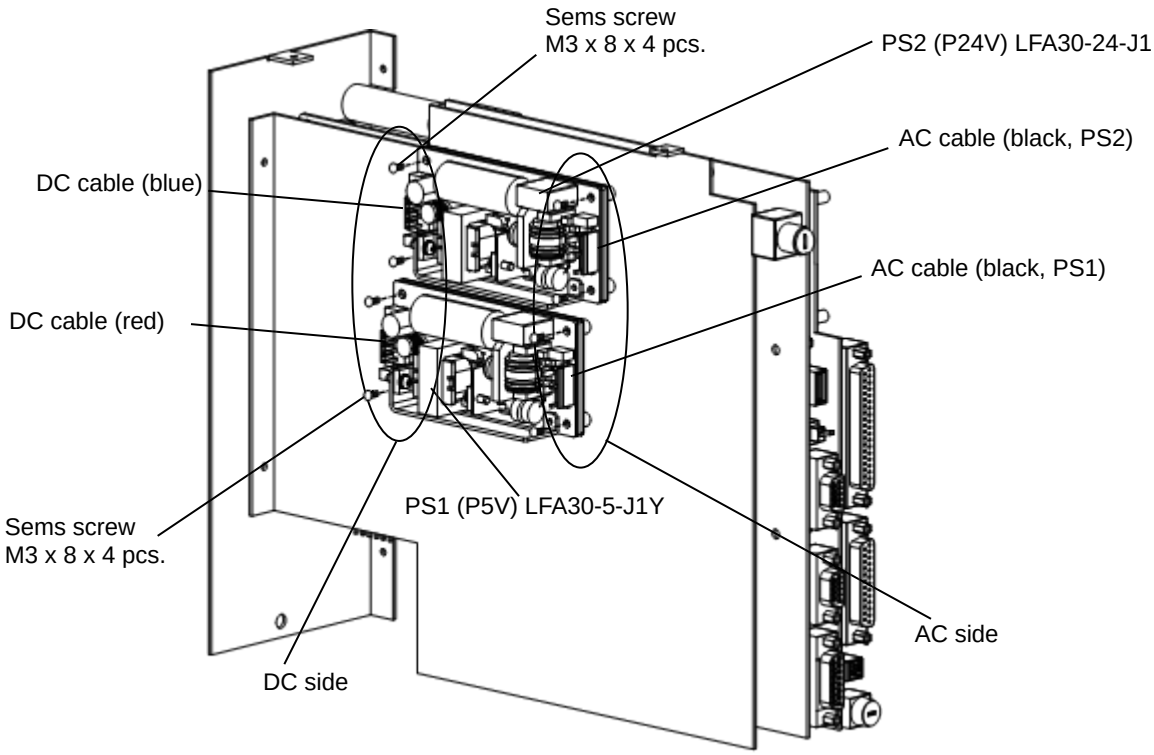


Fig. 5.9 Side view of interior panel unit assembly (TSL3000)

9) Using four (4) sems screws (M3 x 8), secure each of the new switching power supplies to the main body.
Note) The PS1 and PS2 switching power supplies have the same installation pitch. Never install the PS1 and the PS2 upside down or right and left in reverse.

10) Attach a cable to each of the PS1 and the PS2. (See Fig. 5.9.)
Note) On the PS1, connect the DC cable (red) to the DC side and the AC cable (black, indicated as PS1) to the AC side and, on the PS2, connect the DC cable (blue) to the DC side and the AC cable (black, indicated as PS2).

11) Insert the removed interior panel unit into the controller, and then secure it with one (1) countersunk screw (M3 x 6).

12) Connect the DC cable (blue and red), the SVIF cable (blue) and the encoder cable (blue) to the X8YC (main board) and, also, connect the hand I/O cable (blue) and the brake cable (blue) to the X8YX (I/O board). (See Fig. 5.6.)

13) Mount the cover to the main body using eight (8) countersunk screws (M3 x 6).

5.3.5 Replacement of Fuse (X8YX Printed Board)

If the current exceeding the specified current has run through the I/O unit, the fuse of the front of the controller is blown out. If the alarm saying “I/O Fuse Broken (8–273)” is displayed, replace the fuse with a new one.
If the fuse of the output module has been blown out, examine and identify a fault circuit, remedy the cause, then replace the fuse.

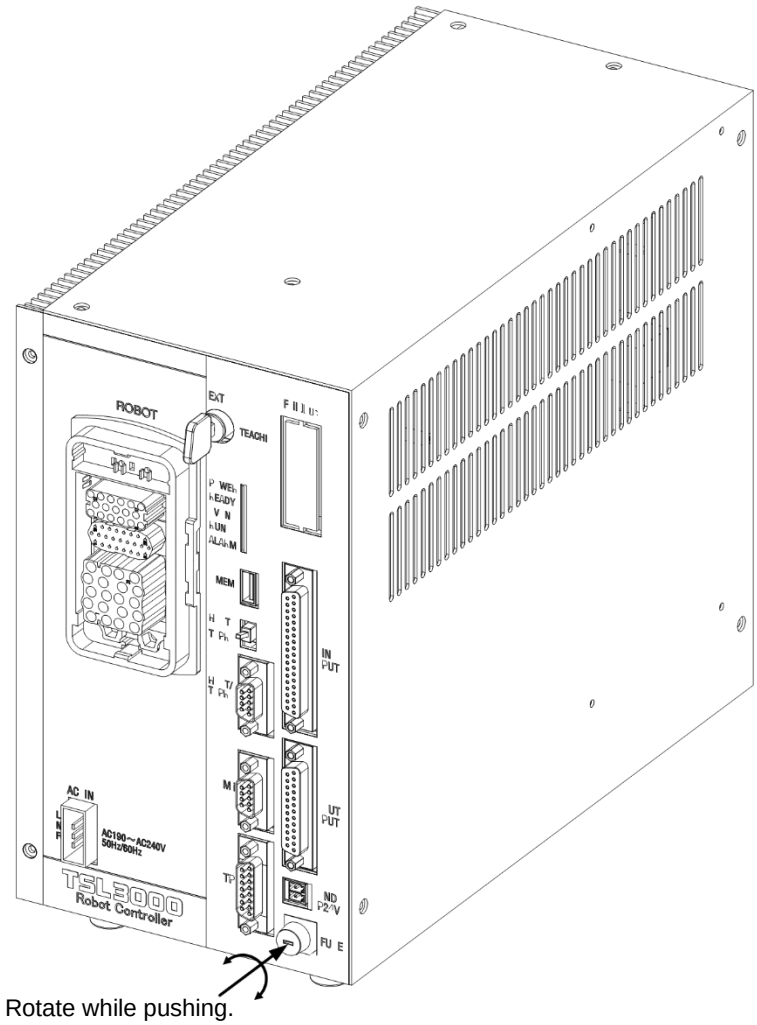


Fig. 5.10 Changing fuse (TSL3000)

Table 5.3 Fuse replacement (TSL3000)

Type of fuse	Manufacturer
51NM030H	PICO

[Replacement procedures]

1) Remove the power supply plug of the controller from the power source.
2) Remove the fuse holder shown in the above figure. (To release the lock, push and turn the holder 90° counterclockwise.)
3) Remove the fuse and set a new fuse in the fuse holder.
4) Mount the fuse holder. (To lock the holder, push and turn the holder 90° clockwise.)
5) Connect the power supply plug of the controller to the power source and make sure that no error will occur.

5.3.6 Replacement of Output ICs (X8YX Printed Board)

If the current exceeding the specified current has run through the output unit, the transistor output ICs on the X8YX printed board are damaged.
When this happens, replace the ICs.
Before the replacement, examine and identify an fault circuit, remedy the cause, then replace the ICs.
The relationship between the output ICs and output signals is shown below.

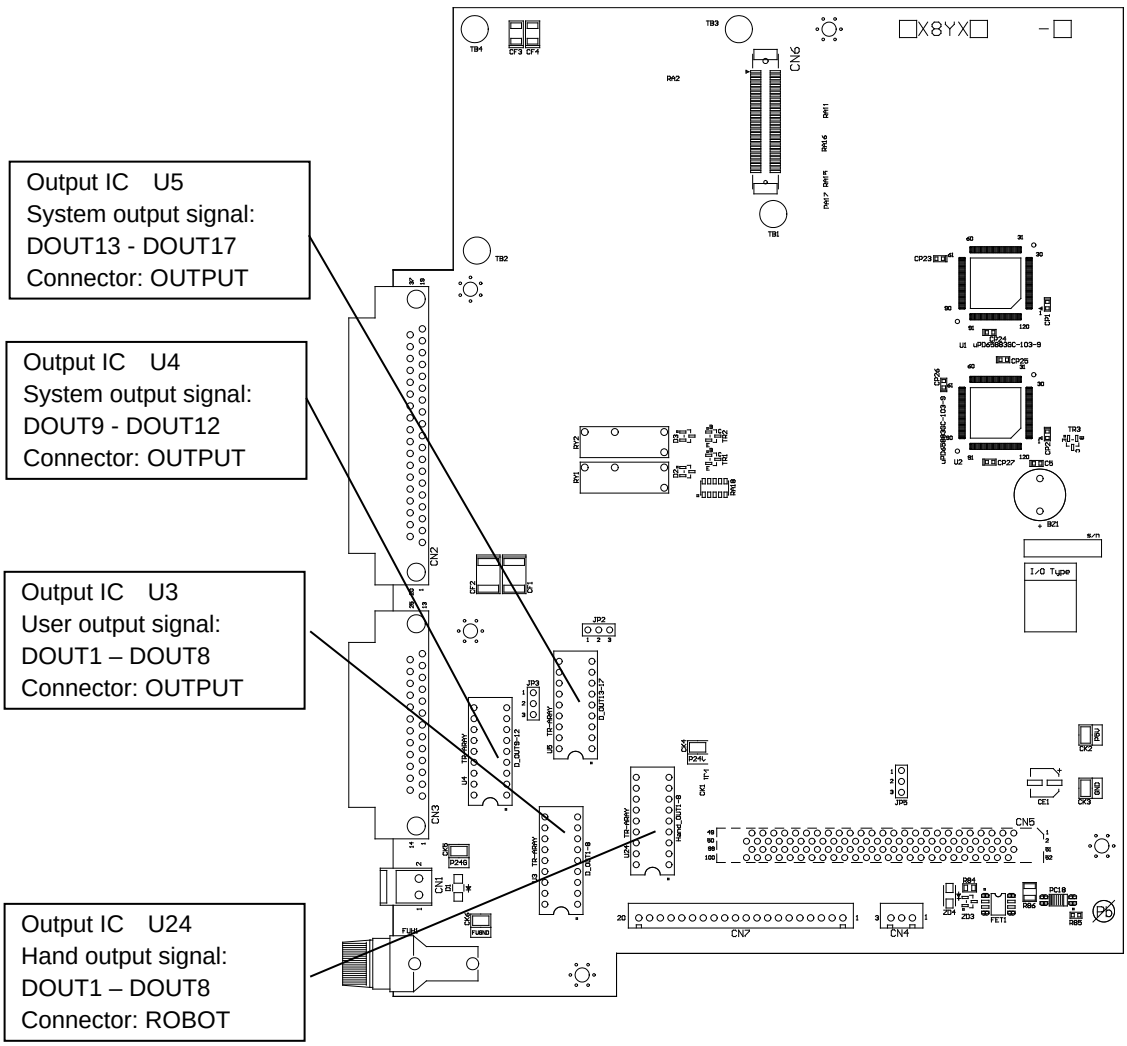


Fig. 5.11 I/O Board (TSL3000)

There are two types of transistor output ICs. Refer to the following for the output types.

Table 5.4 Transistor output ICs (TSL3000)

	Output type	Transistor output IC type	Maker
(1)	Type-N	TD62084APG	Toshiba
(2)	Type-P	M54562WP	Mitsubishi Electric

[Replacement procedures]

1) Turn off the breaker for power supply.

2) Disconnect the ACIN connector from the controller.

3) Remove eight (8) countersunk screws (M3 x 6) that secure the cover, and then remove the cover from the main body.

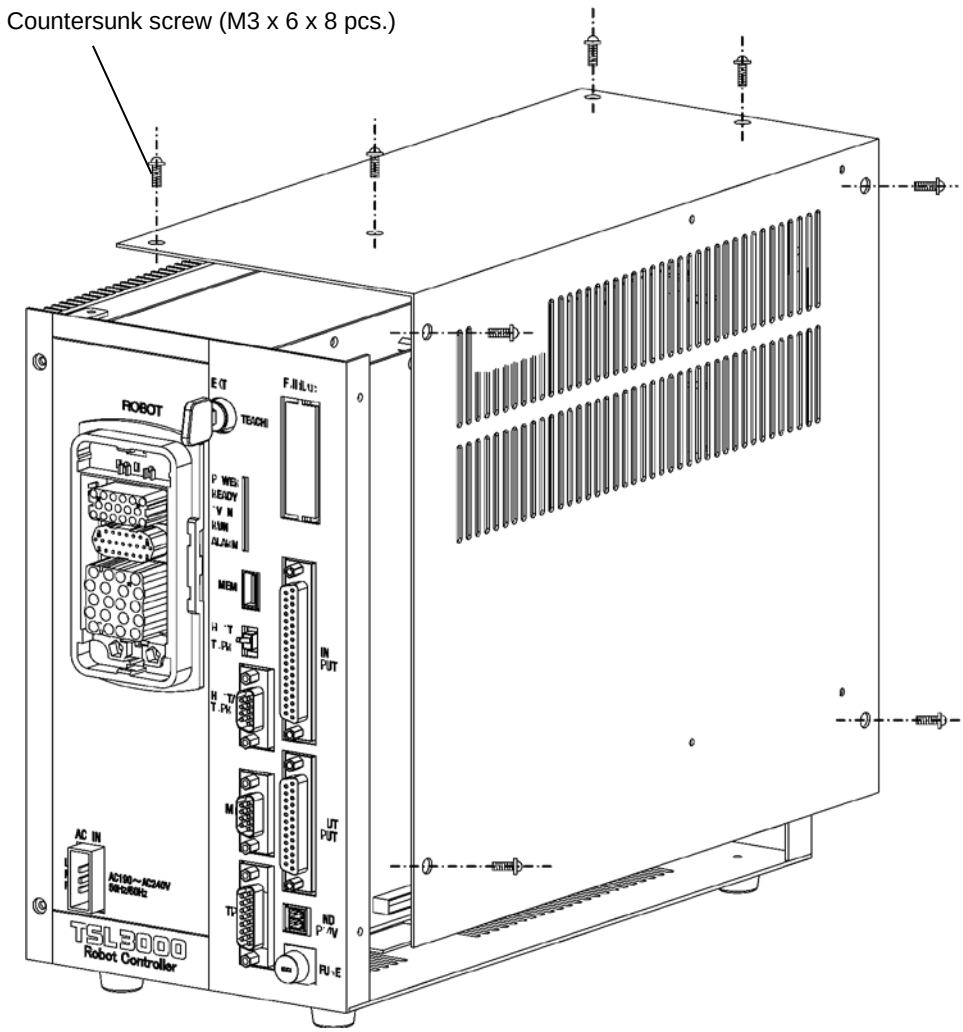


Fig. 5.12 Removing cover (TSL3000)

4) Disconnect the hand I/O cable (blue) and the brake cable (blue) that are connected to the X8YX (I/O board).

5) Remove five (5) sems screws (M3 x 8) that secures the X8YX board, and then remove the X8YX board.

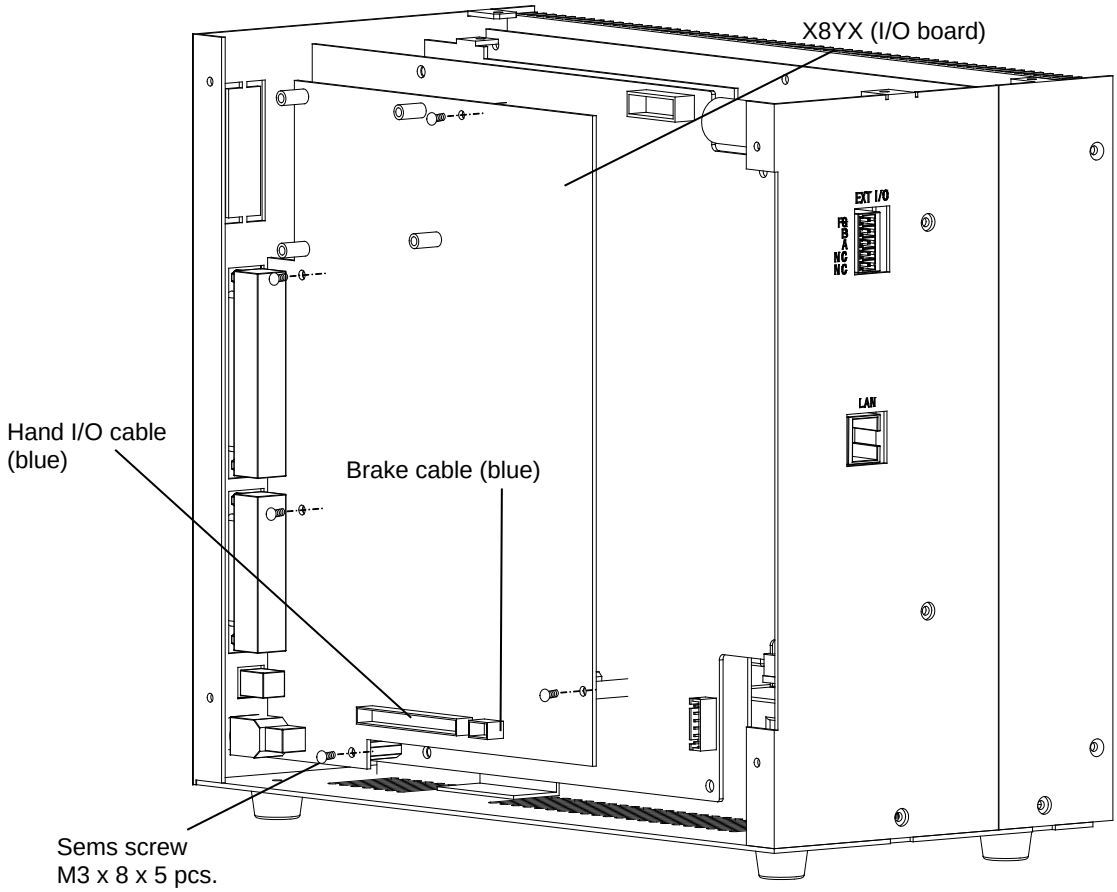


Fig. 5.13 Removing X8YX board (TSL3000)

6) Remove the transistor output IC shown in Fig. 5.11 from the outlet.

7) Mount a new transistor output IC to the outlet.
Note) Pay attention to the model number and orientation of the transistor output IC.

8) Mount the X8YX board to the main body using five (5) sems screws (M3 x 8).

9) Connect the hand I/O cable (blue) and the brake cable (blue) to the X8YX (I/O board), and then mount the main body cover to the main body with eight (8) countersunk screws (M3 x 6).

10) Connect the controller main body power supply plug.

6. Maintenance of Controller (TSL3000E)

6.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of the controller, follow the items given below so that the work can be carried out safely.

⚠DANGER

– When opening the cover of the controller, do not allow water or other fluids or foreign objects to get inside. This could cause a short-circuit inside the controller, which could lead to a fire or an electric shock due to ground leakage.
– Use SHIBAURA MACHINE special products for replacing certain maintenance parts. Wiring of unauthorized connectors and terminals could lead to a fire or an electric shock due to ground leakage.
– Be sure to request the SHIBAURA MACHINE servicing company if any maintenance or replacement work not described in this instruction manual is required. (A fee will be charged for this work.) SHIBAURA MACHINE will not be liable for any accidents or damage that occurs due to replacement work performed by the customer or its results.

⚠ WARNING

– Do not touch the controller terminals or internal board with your hands for about two minutes after the power cable is disconnected. A residual voltage is still in the large-capacity capacitor inside the controller, and this could result in an electric shock.
– Use SHIBAURA MACHINE authorized spare parts when replacing parts. Use of unauthorized parts could result in a fire or a collision with a robot in unexpected motion.
– Before starting any work, be sure to always wear protective gear, and unplug the power cable. If the power is turned on, the servo power module, servo amplifier module, switching power supply, and other components will carry a charge, and an electric shock could result.
– Turn off the power before opening the controller cover. If the cover is opened while the power is on, your hands or conductive tools could touch the power-carrying part, resulting in an electric shock.
– Perform the following actions whenever entering the movable range of the robot.
  - Wear protective gear
  - Place the teach pendant in an accessible location to the worker.
  - Change to MANUAL mode (remove the mode selector key so that the mode cannot be changed)
  ⇒ If the above operations are not performed, the emergency stop could be cleared by another person during the daily inspection work within the movable range, and a collision could occur with a robot that was set to continuous operation.
– Use voltage measurement and the LEDs to check that no PN voltage remains in the large-capacity capacitor. If a PN voltage remains, an electric shock could occur.
– Do not put conductive objects or fingers while the power is on. Touching the power-carrying part could result in an electric shock.

6.2 Layout of Controller Parts

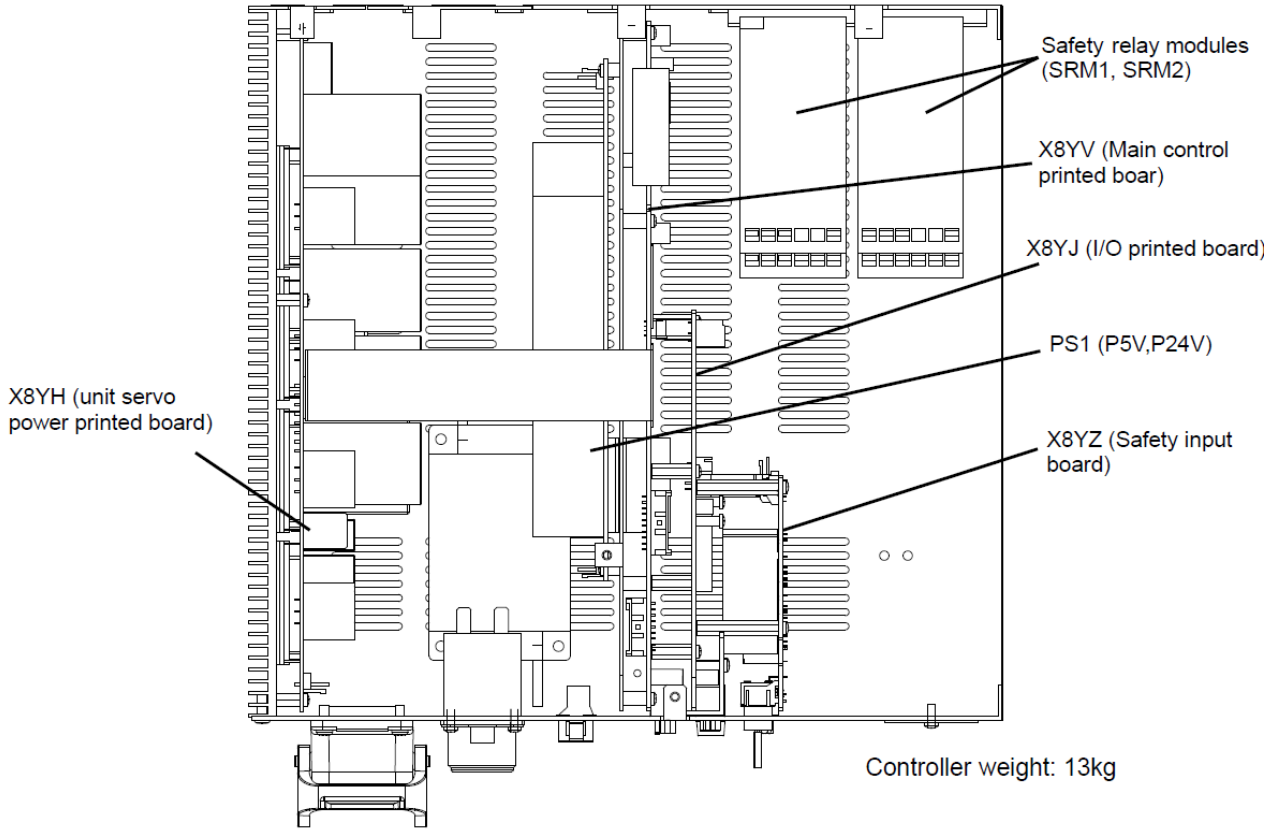


Fig. 6.1 Layout of controller parts (TSL3000E)

Table 6.1 Controller parts (TSL3000E)

Part name	Descriptions
PS1	PS1 (P5V,P24V) output switching power supply
X8YV	Main control printed board
X8YJ	I/O output printed board
X8YS	Servo power module (for 4 axes)
SRM1,SRM2	Safety relay modules

6.3 Maintenance Procedures

6.3.1 Check of Controller Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction.
To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.

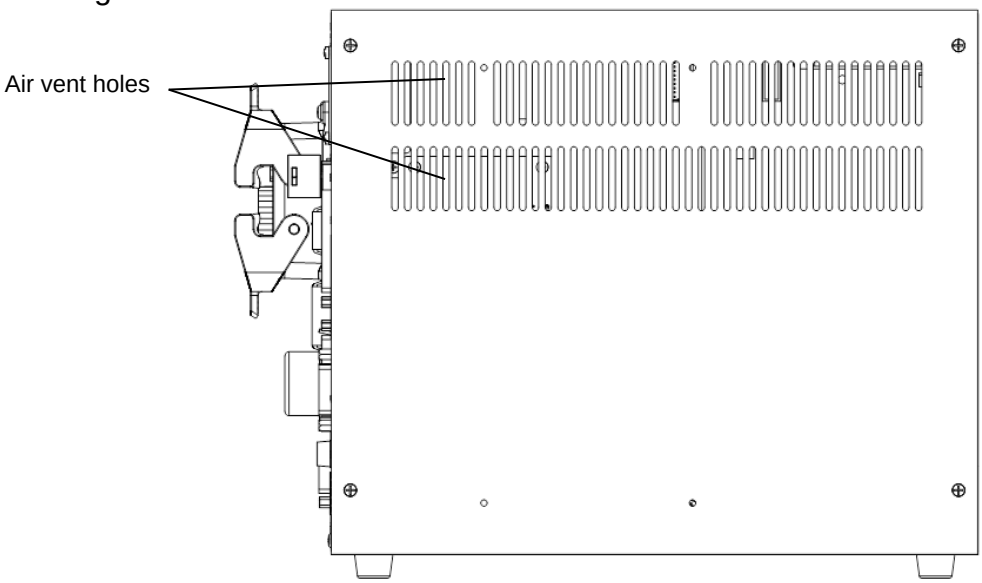


Fig. 6.2 Controller side views (TSL3000E)

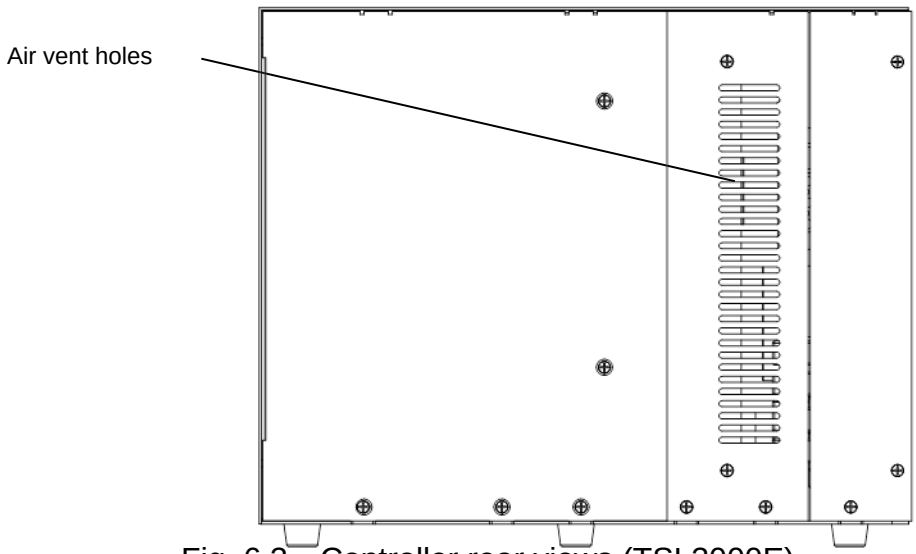


Fig. 6.3 Controller rear views (TSL3000E)

1) If there is any obstacle, move it away from the air vent hole so that air flow is not blocked.

2) Make sure that no contaminant is left on the air vent holes. If any contaminant is left on the vent hole, remove it.

6.3.2 Check of Safety Devices for Function

Make sure that the EMERGENCY stop pushbutton switches equipped on the teach pendant work properly. Also make sure that the safety devices controlled by the external operation input signals work correctly.

1) Make sure that the EMERGENCY (emergency stop) switch provided on the teach pendant functions properly.
   a) Turn around the key at front of controller for selecting the TEACH.
   b) Turn the EMERGENCY switch [1] clockwise and make sure that the EMERGENCY switch [1] turns off.
   c) Press the SERVO ON switch [2] on the teach pendant to turn on the servo power while keeping the ENABLE switch [3] at the neutral position, and make sure that the SERVO ON LED [2] is illuminated.

      Note: The servo power cannot be turned on in the TESTRUN mode.

   d) Press the EMERGENCY STOP switch [1] on the teach pendant and make sure that the servos are turned off. The SERVO ON LED [2] goes out.

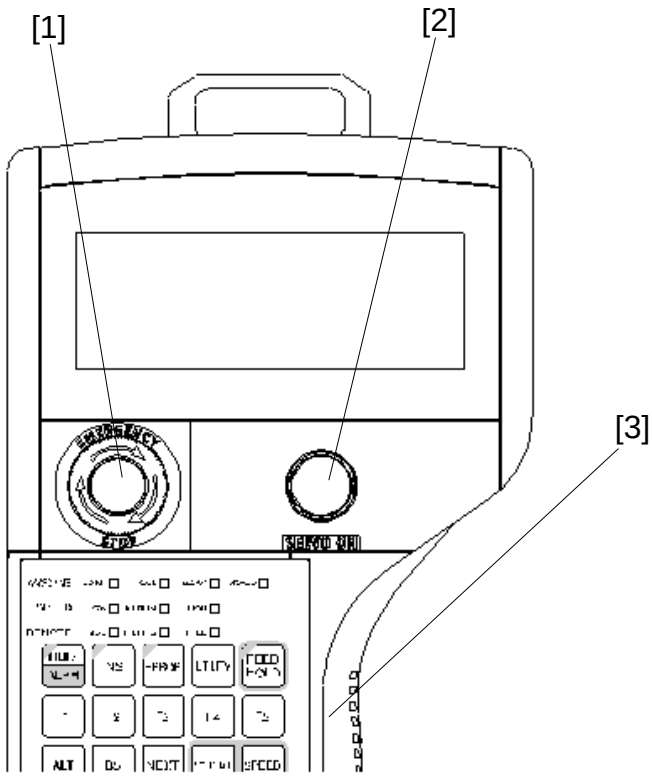


Fig. 6.4 EMERGENCY STOP switch provided on teach pendant (TSL3000E)

At this time, make sure that the EMERGENCY STOP switch [1] remains depressed.

e) Turn the EMERGENCY STOP switch [1] clockwise and make sure that the EMERGENCY switch [1] turns off.

2) Make sure that the safety devices controlled by the external operation input signals work properly.

a) Connect the power supply plug to the power source to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal “emergency stop” (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

b) Set ON the safety devices connected with external operation input signal “low speed command”, and make sure that each signal functions properly.

6.3.3 Battery Replacement

⚠ DANGER

– Do not attempt to incinerate, disassemble, short-circuit, or charge the battery. This could cause the battery to rupture, resulting in a fire. Dispose of the battery in compliance with the regulations of the customer.

NOTICE

– If the battery is to be kept removed for more than one (1) minute, copy all programs and various parameters stored in the internal memory to the personal computer. In the worst case, all memory data are lost.

The memory equipped on the X8YC printed board of the robot controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the memory data, resulting in the data being lost and faults caused by leakage of the battery liquid.

If the following alarm has occurred, replace the battery with a new one.

- 1–145 MAIN Battery alarm

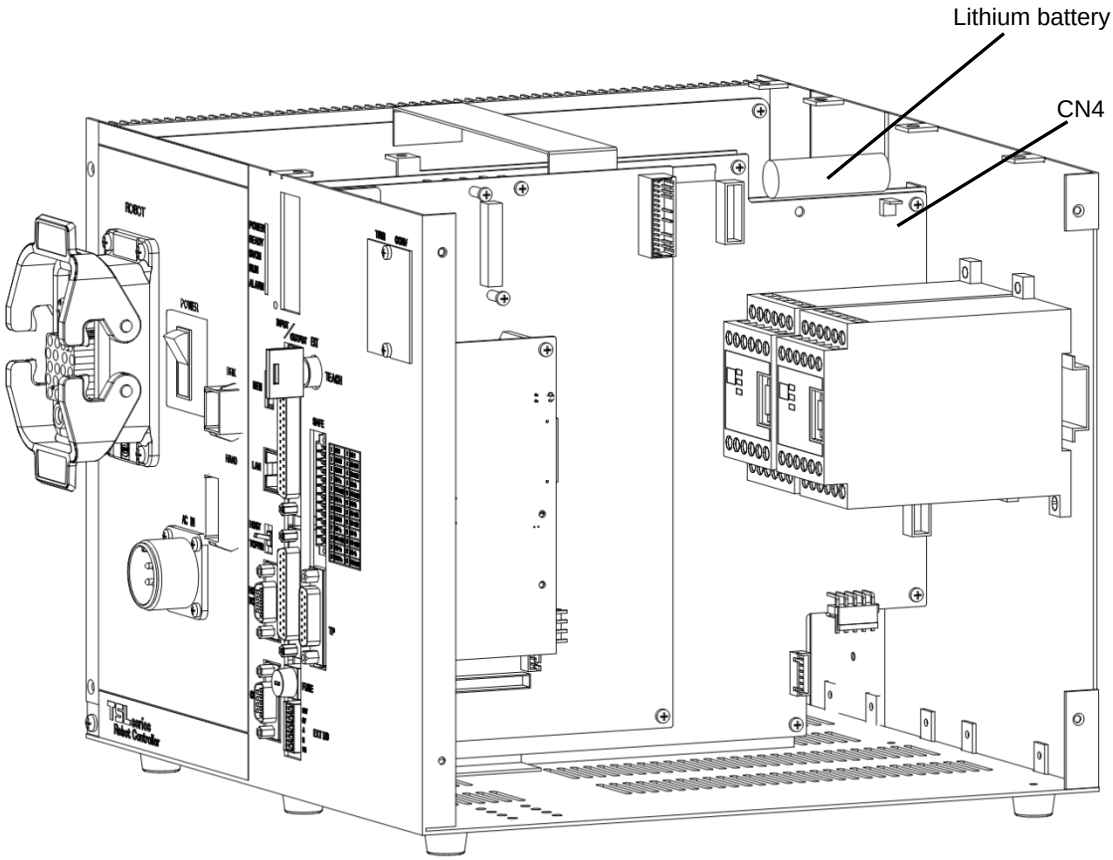


Fig. 6.5 Location of lithium battery (TSL3000E)

Table 6.2 Battery replacement (TSL3000E)

Type of battery	Manufacturer	Battery service life	Remarks
ER6C WK27	Hitachi Maxell	10 years (at room temperature)	Discontinued in 2022
ER6V C4	TOSHIBA	10 years (at room temperature)	Adopted starting in 2022

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.
Only the battery shown in the table above should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

[Battery replacement procedures]

1) If the battery is to be kept removed for more than one (1) minute, copy all programs and various parameters stored in the internal memory to the personal computer. In the worst case, all memory data are lost.
2) Disconnect the controller's power plug from the power supply.
3) Have a new battery ready, remove ten (10) countersunk screws (M3 x 6) that secure the cover, and then remove the cover from the main body.
4) Remove the battery from the cable ties securing the battery.
(For the battery position, see Fig. 6.5.)
5) Remove the battery connector from CN4 of the X8YV board. Immediately connect the connector of the new battery to CN4. (If the battery is kept removed for more than one (1) minute, all memory data are lost.)
Even if not used, the lithium battery capacity decreases by natural discharge. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.
6) Secure the battery connected to CN4 with cable ties.
7) Using countersunk screws (M3 x6 x 10 pcs.), mount the cover on the controller.
8) If the battery is kept removed for more than one (1) minute, connect the power supply plug of the controller to the power source and load the USB memory or personal computer in which the TSPC (option) is already installed.
If the controller has detected the battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

6.3.4 Replacement of Switching Power Supply Unit

⚠ DANGER

– Install the switching power supply in the location shown in the instruction manual. If the up and bottom or right and left sides are backwards, or the connector is different, this could cause a short-circuit or broken wire, resulting in a fire or an electric shock due to ground leakage.

The life of the switching power supply unit (5 VDC/24 VDC) used in the robot controller differs with the operating conditions. This unit uses an aluminum electrolytic capacitor, and if the load current is large, the running time is long and the ambient temperature is high, the life will reduce.
As the life varies largely with the user's operating conditions (5 to 10 years), replace the power supply unit every five (5) years for preventive maintenance.

[Replacement procedures]

1) Turn off the power breaker and disconnect the power supply plug from the power source.
2) Disconnect the ACIN connector from the controller.
3) Remove the key from the controller.
4) Remove ten (10) countersunk screws (M3 x 6) that secure the cover, and then remove the cover from the main body.

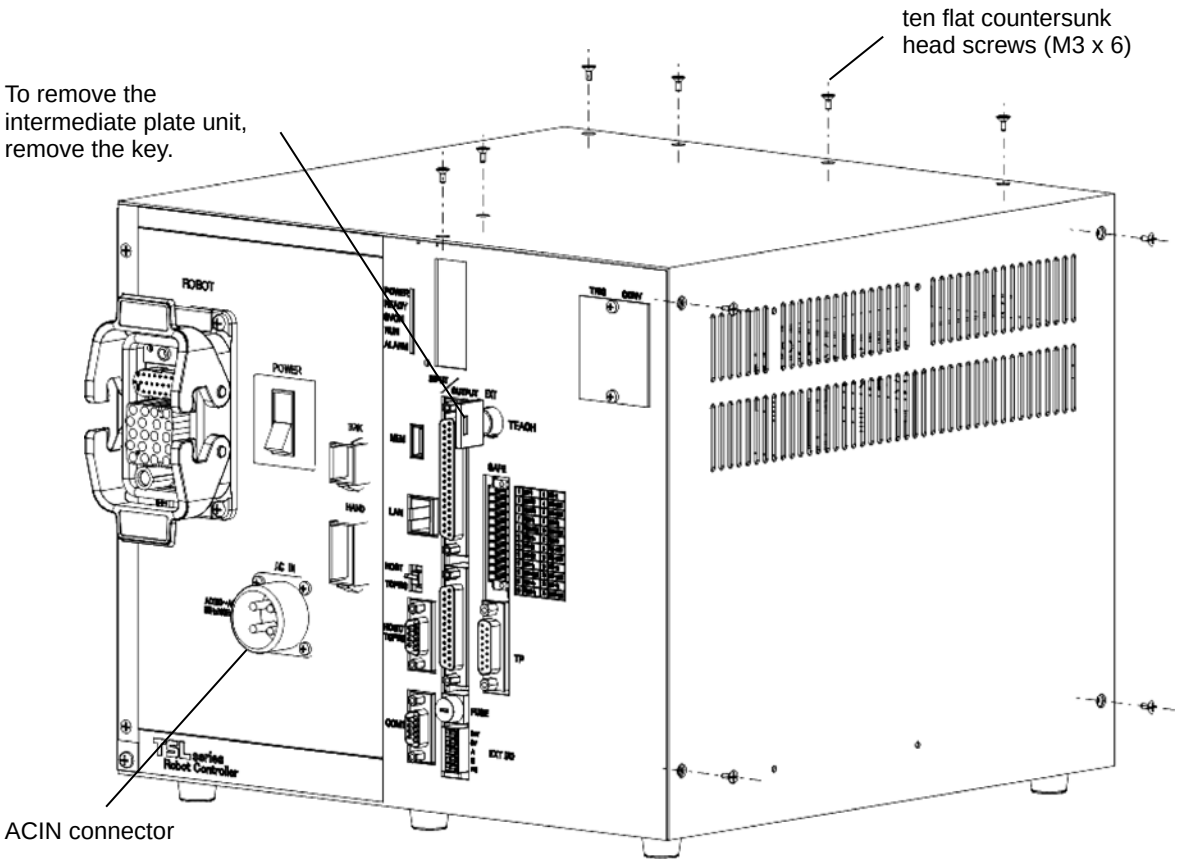


Fig. 6.6 Removing cover (TSL3000E)

5) Remove the DC cable (blue and red), the SVIF cable (blue) and the encoder cable (blue) connected to the X8YV (main board) as well as the hand I/O cable (blue) and the brake cable (blue) connected to the X8YJ (I/O board).
6) Remove the cables (blue x 1 line, black x 1 lines) connected to the PS1. (See Fig. 6.10.)

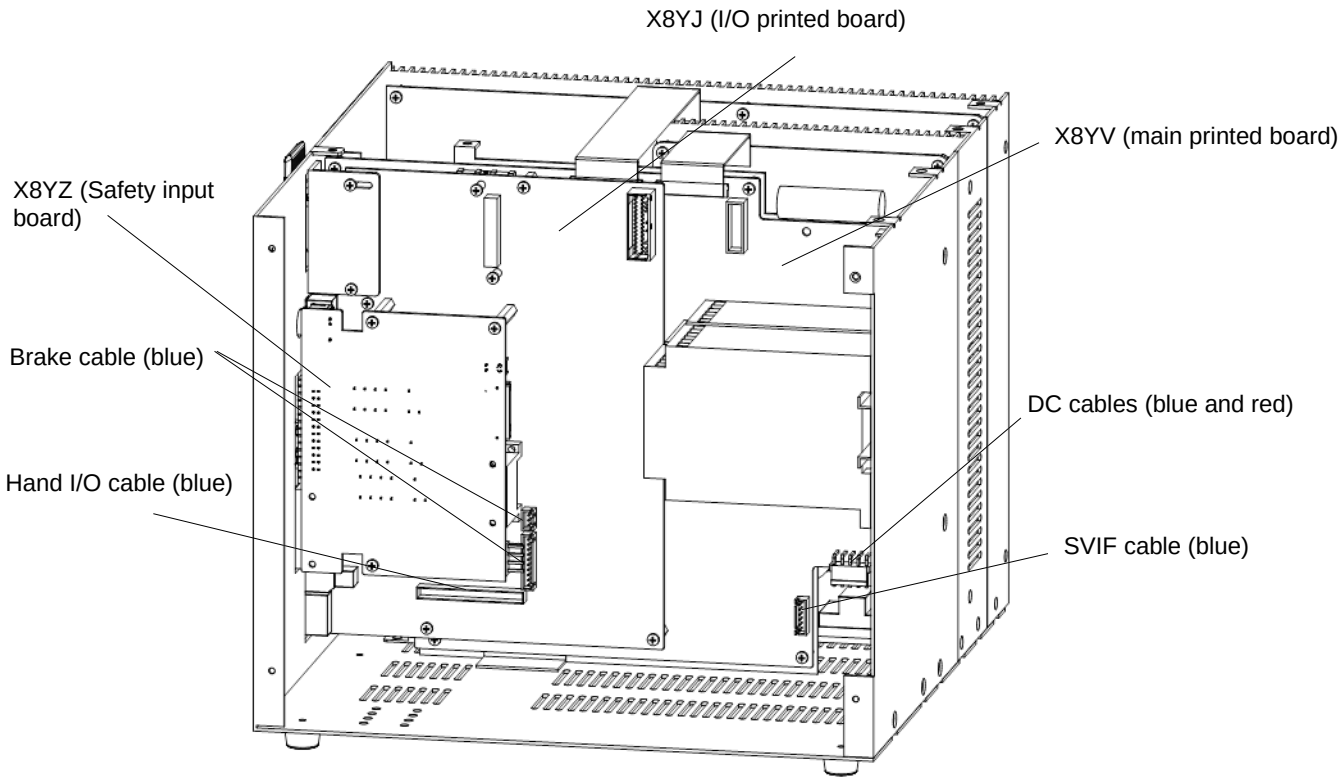


Fig. 6.7 Controller interior side view (TSL3000E)

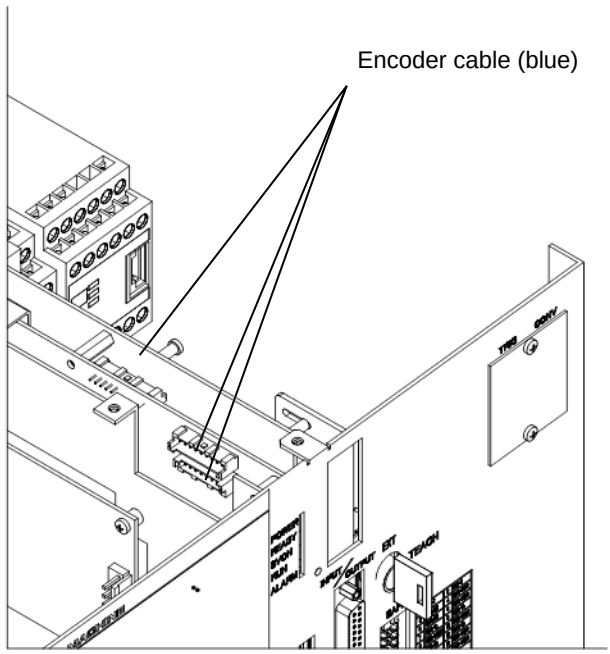
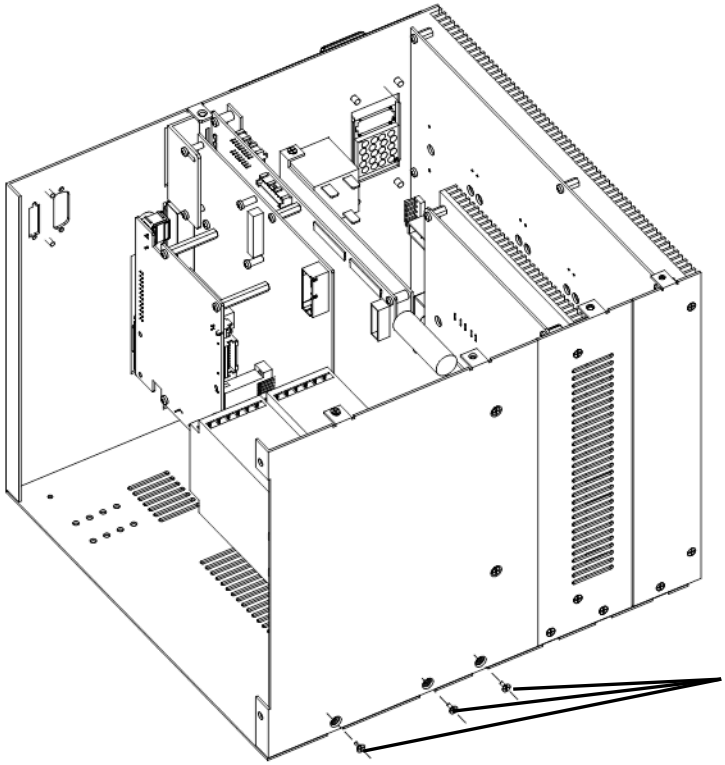


Fig. 6.8 Controller interior front view (TSL3000E)

7) Remove three (3) countersunk screw (M3 x 6) that secures the interior panel unit.
Pull out the interior panel unit toward the rear of the controller.
When removing the interior panel unit, be sure to remove the key.



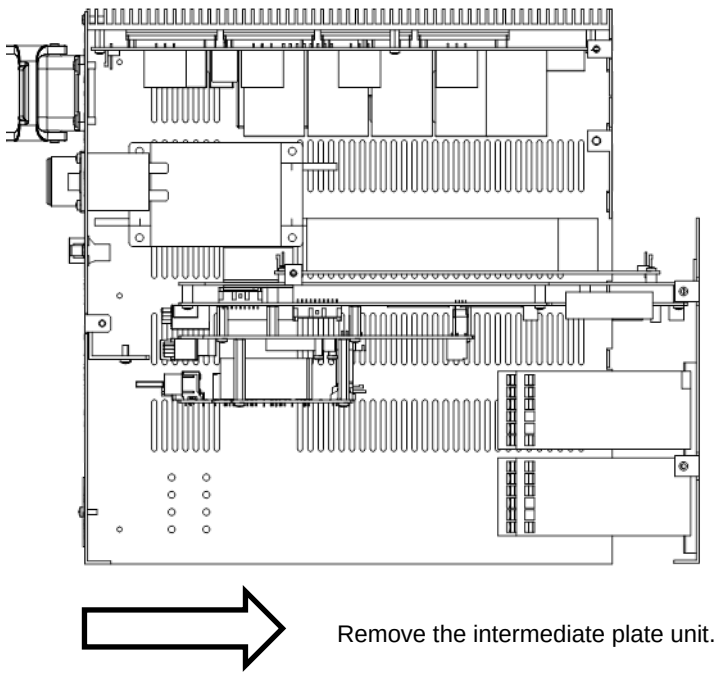


Fig. 6.9 Removing interior panel unit (TSL3000E)

8) Remove four (4) sems screws (M3 x 8) that secure each of the switching power supplies (PS1), and then remove the switching power supplies from the interior panel unit.

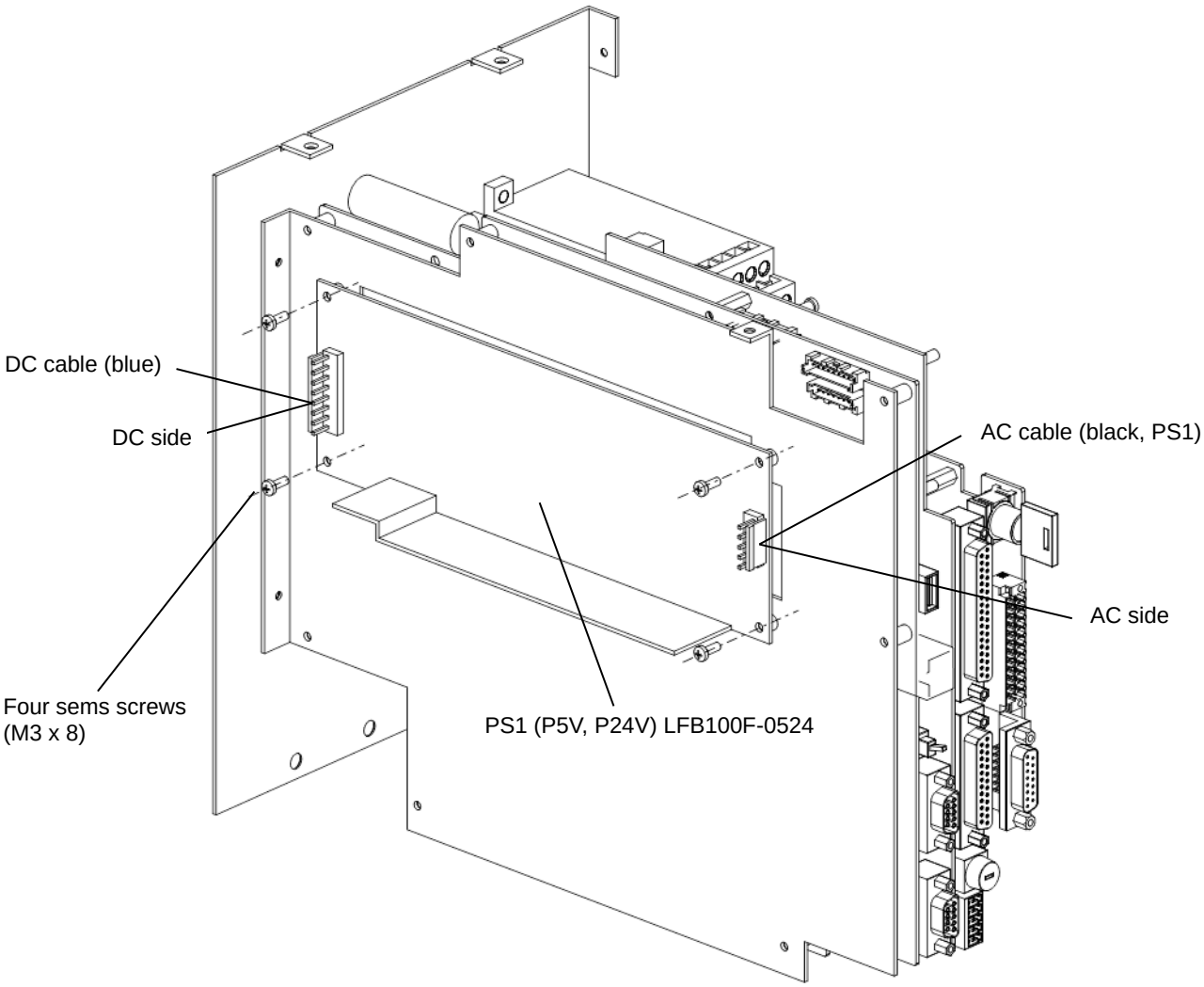


Fig. 6.10 Side view of interior panel unit assembly (TSL3000E)

9) Using four (4) sems screws (M3 x 8), secure the new switching power supplies to the main body.
   Note) Never install the PS1 switching power supplies right and left in reverse.

10) Attach a cable to the PS1.

11) Insert the removed interior panel unit into the controller, and then secure it with three (3) countersunk screw (M3 x 6).

12) Connect the DC cable (blue), the SVIF cable (blue) and the encoder cable (blue) to the X8YV (main board) and, also, connect the hand I/O cable (blue) and the brake cable (blue) to the X8YJ (I/O board). (See Fig. 6.7 and 6.8)

13) Mount the cover to the main body using ten (10) countersunk screws (M3 x 6).

6.3.5 Replacement of Fuse (X8YJ Printed Board)

If the current exceeding the specified current has run through the I/O unit, the fuse of the front of the TSL3000 controller is blown out. If the alarm saying “I/O Fuse Broken (8–273)” is displayed, replace the fuse with a new one.
If the fuse of the output module has been blown out, examine and identify a fault circuit, remedy the cause, then replace the fuse.

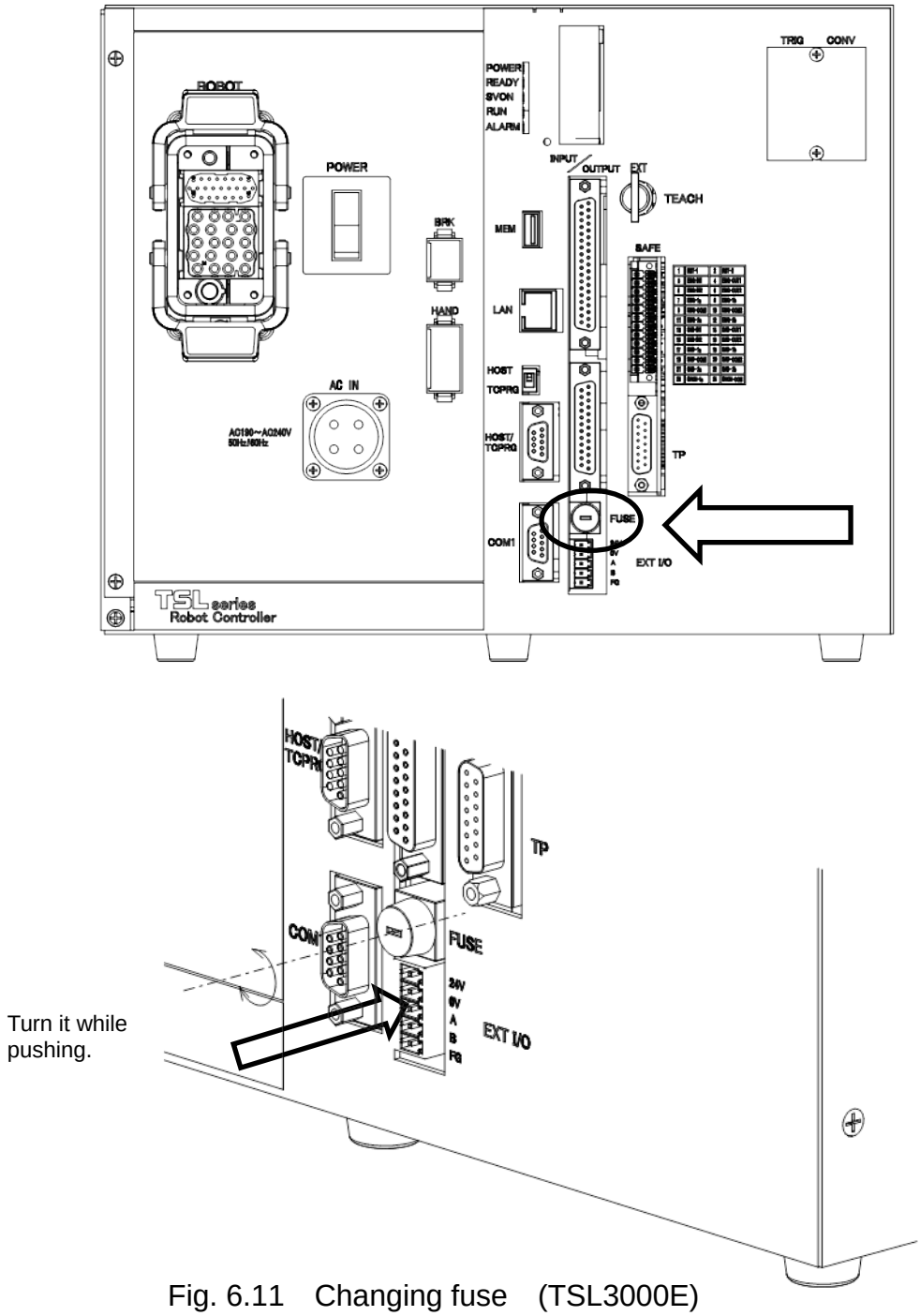


Fig. 6.11 Changing fuse (TSL3000E)

Table 6.3 Fuse replacement (TSL3000E)

Type of fuse	Manufacturer
51NM030H	PICO

[Replacement procedures]

1) Remove the power supply plug of the controller from the power source.
2) Remove the fuse holder shown in the above figure. (To release the lock, push and turn the holder 90° counterclockwise.)
3) Remove the fuse and set a new fuse in the fuse holder.
4) Mount the fuse holder. (To lock the holder, push and turn the holder 90° clockwise.)
5) Connect the power supply plug of the controller to the power source and make sure that no error will occur.

7. Maintenance of Controller (TS3000, TS3000E)

7.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of the controller, follow the items given below so that the work can be carried out safely.

⚠DANGER

– When opening the cover of the controller, do not allow water or other fluids or foreign objects to get inside. This could cause a short-circuit inside the controller, which could lead to a fire or an electric shock due to ground leakage.
– Use SHIBAURA MACHINE special products for replacing certain maintenance parts. Wiring of unauthorized connectors and terminals could lead to a fire or an electric shock due to ground leakage.
– Be sure to request the SHIBAURA MACHINE servicing company if any maintenance or replacement work not described in this instruction manual is required. (A fee will be charged for this work.) SHIBAURA MACHINE will not be liable for any accidents or damage that occurs due to replacement work performed by the customer or its results.

⚠ WARNING

– Do not touch the controller terminals or internal board with your hands for about two minutes after the power cable is disconnected. A residual voltage is still in the large-capacity capacitor inside the controller, and this could result in an electric shock.
– Use SHIBAURA MACHINE authorized spare parts when replacing parts. Use of unauthorized parts could result in a fire or a collision with a robot in unexpected motion.
– Before starting any work, be sure to always wear protective gear, and unplug the power cable. If the power is turned on, the servo power module, servo amplifier module, switching power supply, and other components will carry a charge, and an electric shock could result.
– Turn off the power before opening the controller cover. If the cover is opened while the power is on, your hands or conductive tools could touch the power-carrying part, resulting in an electric shock.
– Perform the following actions whenever entering the movable range of the robot.
  - Wear protective gear
  - Place the teach pendant in an accessible location to the worker.
  - Change to MANUAL mode (remove the mode selector key so that the mode cannot be changed)
  ⇒ If the above operations are not performed, the emergency stop could be cleared by another person during the daily inspection work within the movable range, and a collision could occur with a robot that was set to continuous operation.
– Use voltage measurement and the LEDs to check that no PN voltage remains in the large-capacity capacitor. If a PN voltage remains, an electric shock could occur.
– Do not put conductive objects or fingers while the power is on. Touching the power-carrying part could result in an electric shock.

7.2 Layout of Controller Parts

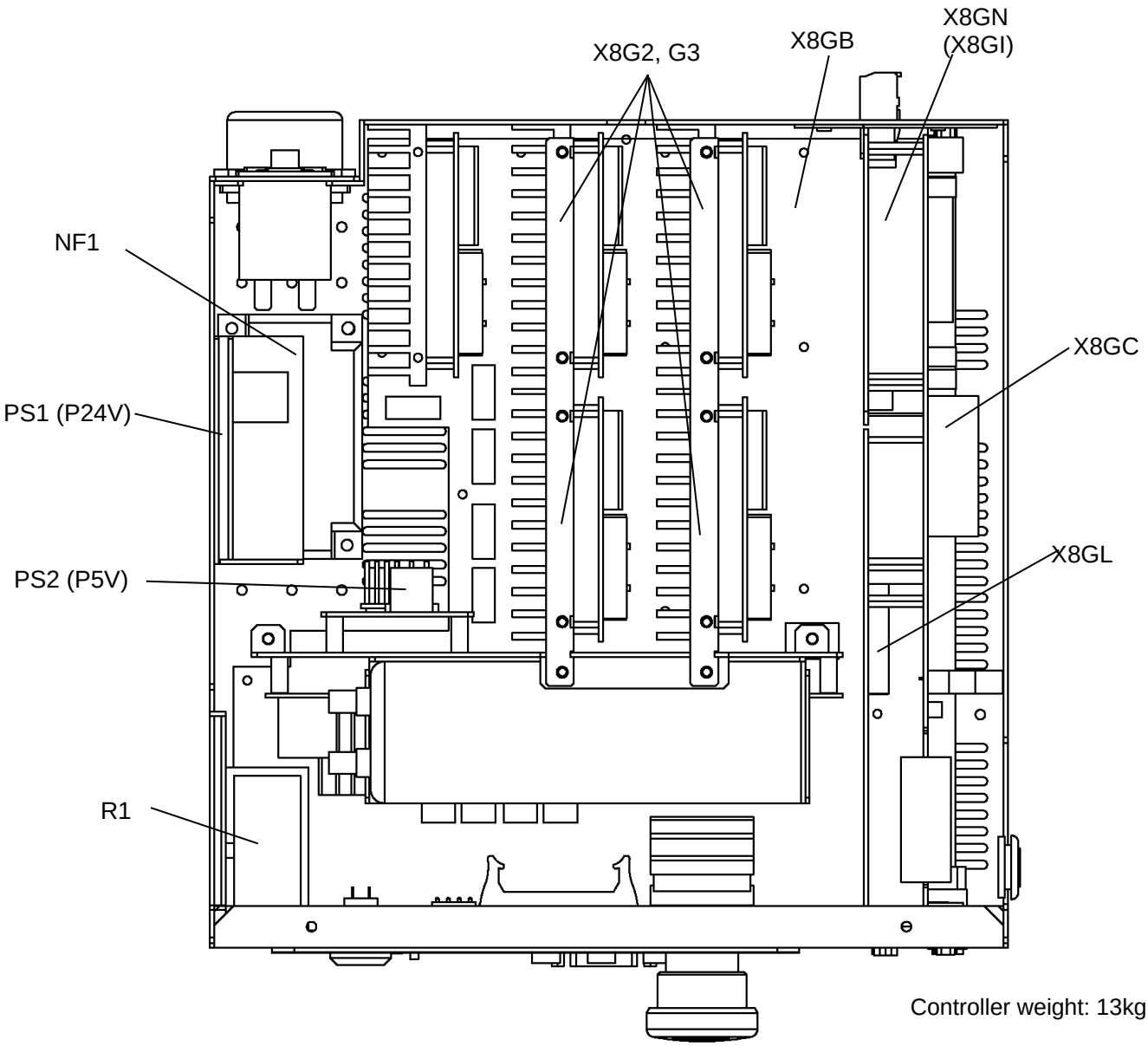


Fig. 7.1 Layout of controller parts (TS3000, TS3000E)

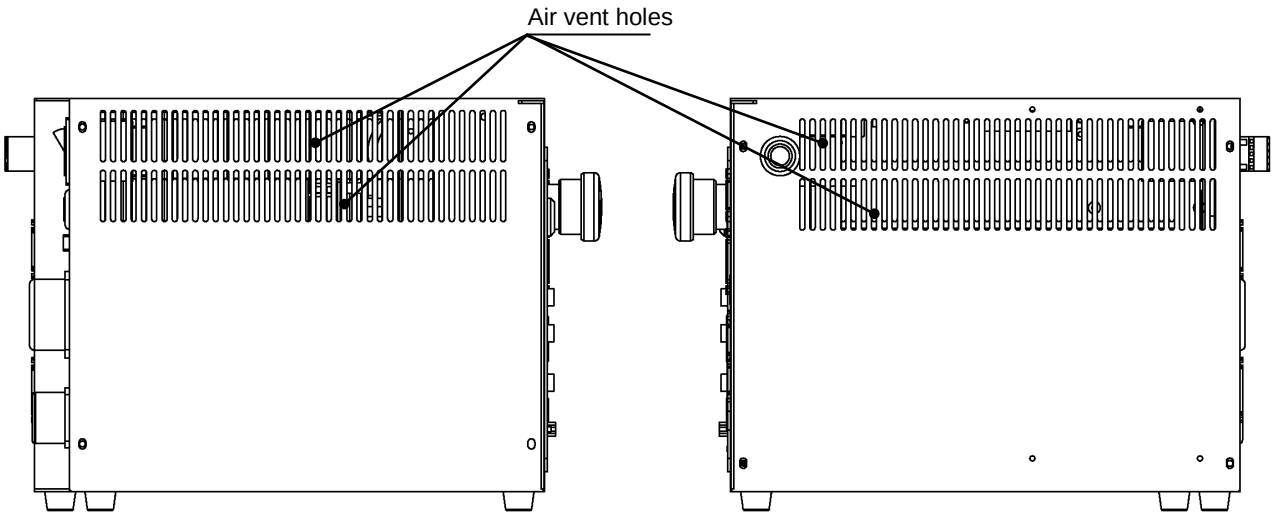
Table 7.1 Controller parts (TS3000, TS3000E)

Part name	Descriptions
PS1	P24V output switching power supply
PS2	P5V output switching power supply
X8GC	Main control printed board
X8GN (X8GI)	I/O output printed board
X8GL	Servo logic printed board
X8GI - X8G3	Servo amplifier printed board
X8GB	Mother board
R1	Regenerative resistor
NF1	Noise filter

7.3 Maintenance Procedures

7.3.1 Check of Controller Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction.
To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.



Left side view Right side view

Fig. 7.2 Controller side views (TS3000, TS3000E)

a) If there is any obstacle, move it away from the air vent hole so that air flow is not blocked.

b) Make sure that no contaminant is left on the air vent holes. If any contaminant is left on the vent hole, remove it.

c) If the controller is placed sideways, make sure that the air vent holes on the bottom surface of the controller are more than 11 mm away from the floor surface. Always keep the space of 11 mm or over. The length of rubber shoe is approximately 11 mm.

7.3.2 Check of Safety Devices for Function

Make sure that the EMERGENCY stop pushbutton switches equipped on the control panel and teach pendant work properly. Also make sure that the safety devices controlled by the external operation input signals work correctly.

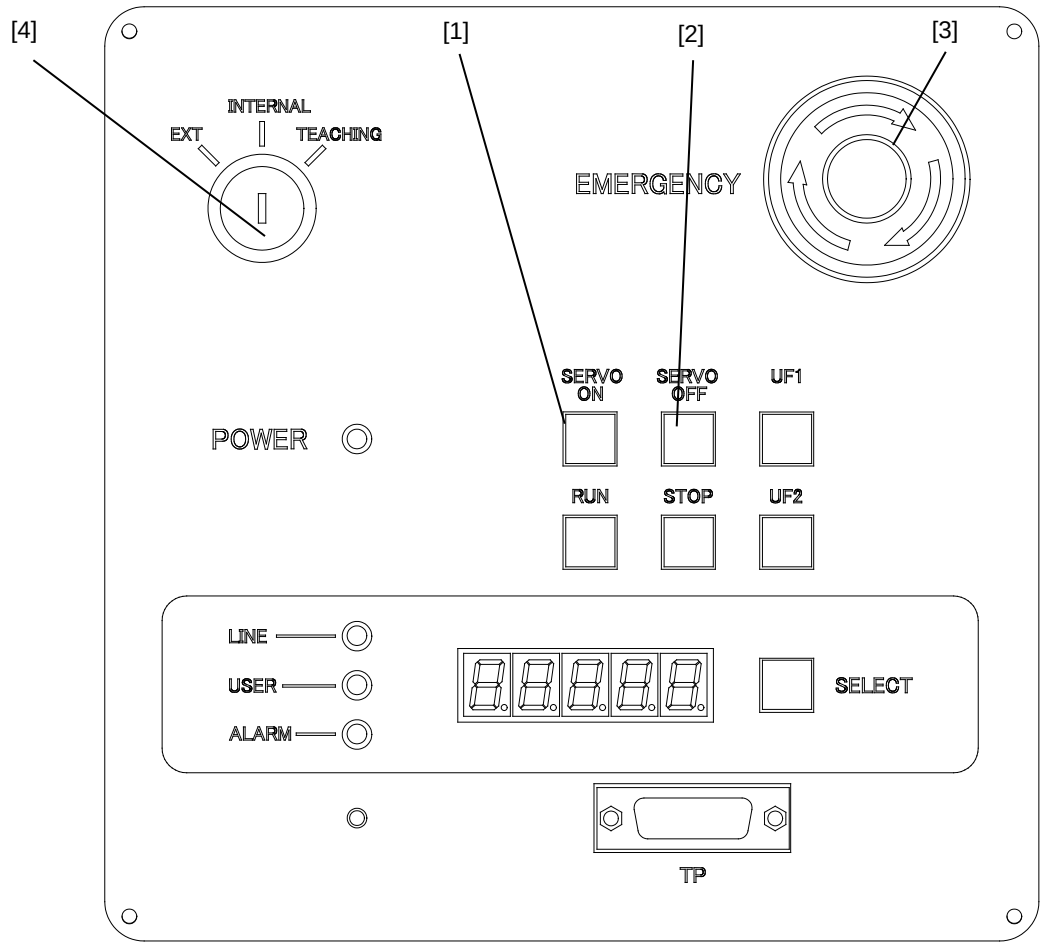


Fig.7.3 Layout of control panel switches (TS3000, TS3000E)

a) Make sure that the EMERGENCY (emergency stop) switch [3] provided on the control panel functions properly.

1) Set the key switch <4> to the INTERNAL mode.

2) Press the SERVO ON switch [1] provided on the control panel to turn the servos on and make sure that the SERVO ON switch [1] LED is illuminated.

3) Press the EMERGENCY switch [3] and make sure that the servos are turned off. The SERVO ON switch [1] LED goes out and the SERVO OFF switch [2]

LED is illuminated. At this time, make sure that the EMERGENCY switch [3] remains depressed.

4) Turn the EMERGENCY switch [3] clockwise and make sure that the same switch turns off.

b) Make sure that the EMERGENCY STOP switch [6] provided on the teach pendant functions properly.

1) Turn around the key at front of controller for selecting the TEACHING.

2) To turn on the servo power, press the SERVO ON switch [5] on the teach pendant while keeping the ENABLE switch at the neutral position. Make sure that the SERVO ON switch LED [5] is illuminated.

3) Press the EMERGENCY STOP switch [4] on the teach pendant and make sure that the servos are turned off. The SERVO ON switch LED [5] goes out.

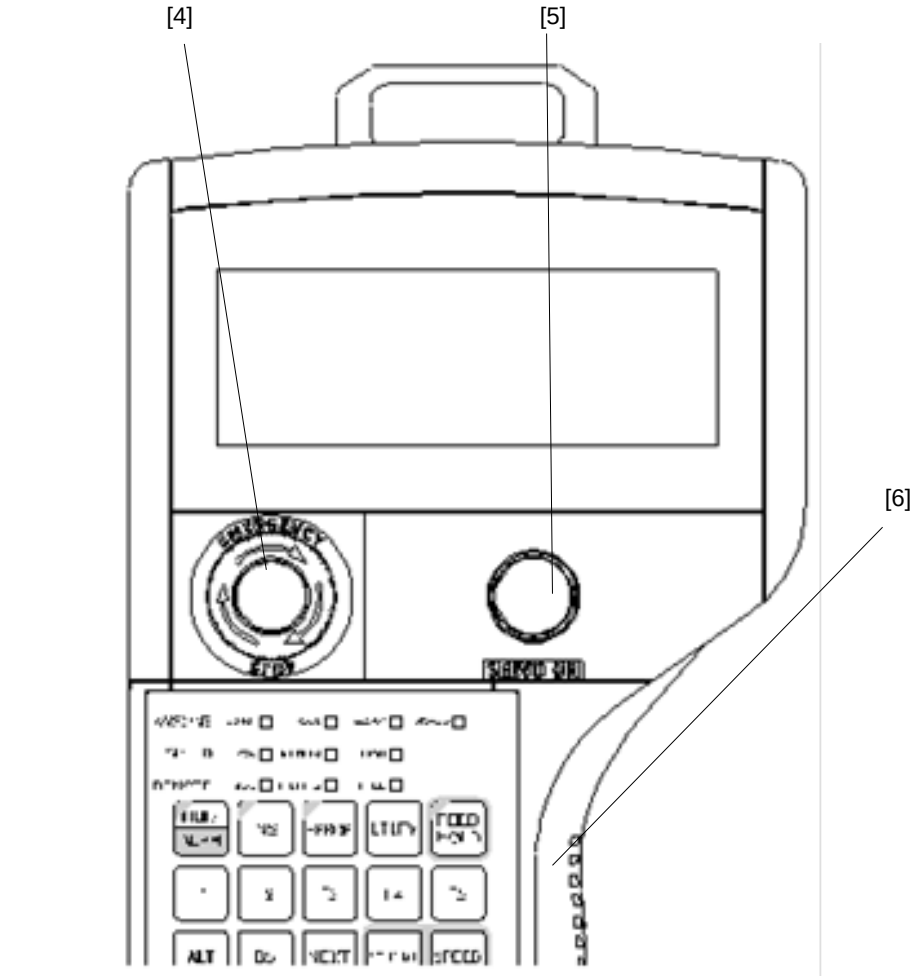


Fig. 7.4 EMERGENCY STOP switch provided on teach pendant (TS3000, TS3000E)

At this time, make sure that the EMERGENCY STOP switch [4] remains depressed.

3) Turn the EMERGENCY STOP switch [4] clockwise and make sure that the same switch [4] turns off.

c) Make sure that the safety devices controlled by the external operation input signals work properly.

1) Press the SERVO ON switch [1] equipped on the control panel to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal “emergency stop” (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

2) Set ON the safety devices connected with external operation input signal “low speed command”, and make sure that each signal functions properly.

7.3.3 Battery Replacement

DANGER

– Do not attempt to incinerate, disassemble, short-circuit, or charge the battery. This could cause the battery to rupture, resulting in a fire. Dispose of the battery in compliance with the regulations of the customer.

NOTICE

– If the battery is to be kept removed for more than one (1) minute, copy all programs and various parameters stored in the internal memory to the personal computer. In the worst case, all memory data are lost.

The memory equipped on the X8GC printed board of the robot controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the memory data, resulting in the data being lost and faults caused by leakage of the battery liquid.

If the following alarm has occurred, replace the battery with a new one.

- 1–145 MAIN Battery alarm

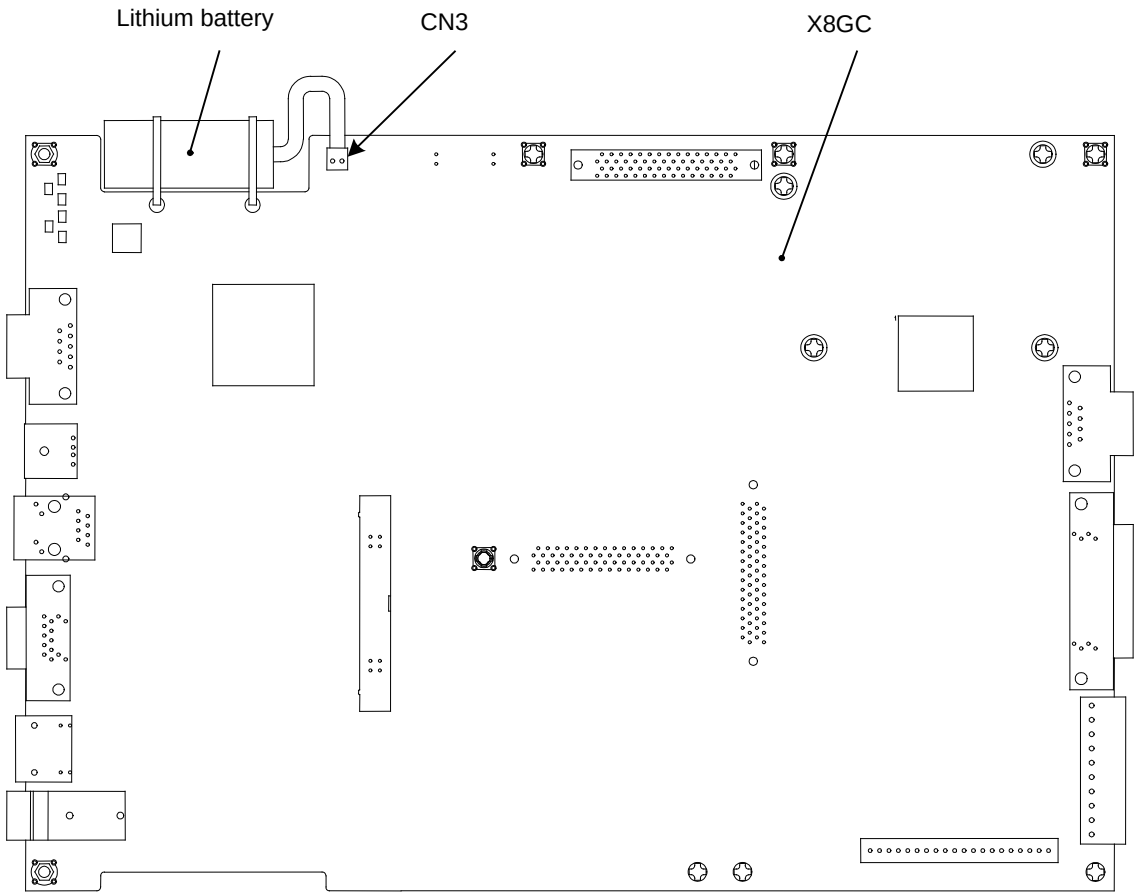


Fig. 7.5 Location of lithium battery (TS3000, TS3000E)

Table 7.2 Lithium ion battery (TS3000, TS3000E)

Type of battery	Manufacturer	Battery service life	Remarks
ER6C WK27	Hitachi Maxell	10 years (at room temperature)	Discontinued in 2022
ER6V C4	TOSHIBA	10 years (at room temperature)	Adopted starting in 2022

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.
Only the battery shown in the table above should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

[Battery replacement procedures]

1) If the battery is to be kept removed for more than one (1) minute, copy all programs and various parameters stored in the internal memory to the personal computer. In the worst case, all memory data are lost.

2) Have a new battery ready, remove eleven (11) cross truss head screws (M3 x 6) that secure the cover, and then remove the cover from the main body.

3) Prepare a new battery, then remove the cover from the controller.

4) Disconnect the battery connector from CN3 of the X8GC printed board, and remove the battery from the cable ties securing the battery.
(For the battery position, see Fig. 7.5.)

5) Set the new battery immediately. (If the battery is kept removed for more than one (1) minute, all memory data are lost.)
Even if not used, the lithium battery capacity decreases by natural discharge. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.

6) Mount the cover to the main body using eleven(11) cross truss head screws (M3 x 6).

7) If the battery is kept removed for more than one (1) minute, turn on the main power switch and load the programs and various parameters from the floppy disk unit or personal computer in which the TSPC (option) is already installed.
If the controller has detected the battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

7.3.4 Replacement of Switching Power Supply Unit

⚠ DANGER

– Install the switching power supply in the location shown in the instruction manual. If the up and bottom or right and left sides are backwards, or the connector is different, this could cause a short-circuit or broken wire, resulting in a fire or an electric shock due to ground leakage.

The life of the switching power supply unit (DC5 V/DC24 V) used in the robot controller differs with the operating conditions. This unit uses an aluminum electrolytic capacitor, and if the load current is large, the running time is long and the ambient temperature is high, the life will reduce.
As the life varies largely with the user's operating conditions (5 ~ 10 years), replace the power supply unit every five (5) years for preventive maintenance.

[PS1 replacement procedures]

1) Turn off the main power switch.
2) Turn off the power breaker.
3) Remove eleven(11) cross truss head screws (M3 x 6) that secure the cover, and then remove the cover from the main body.

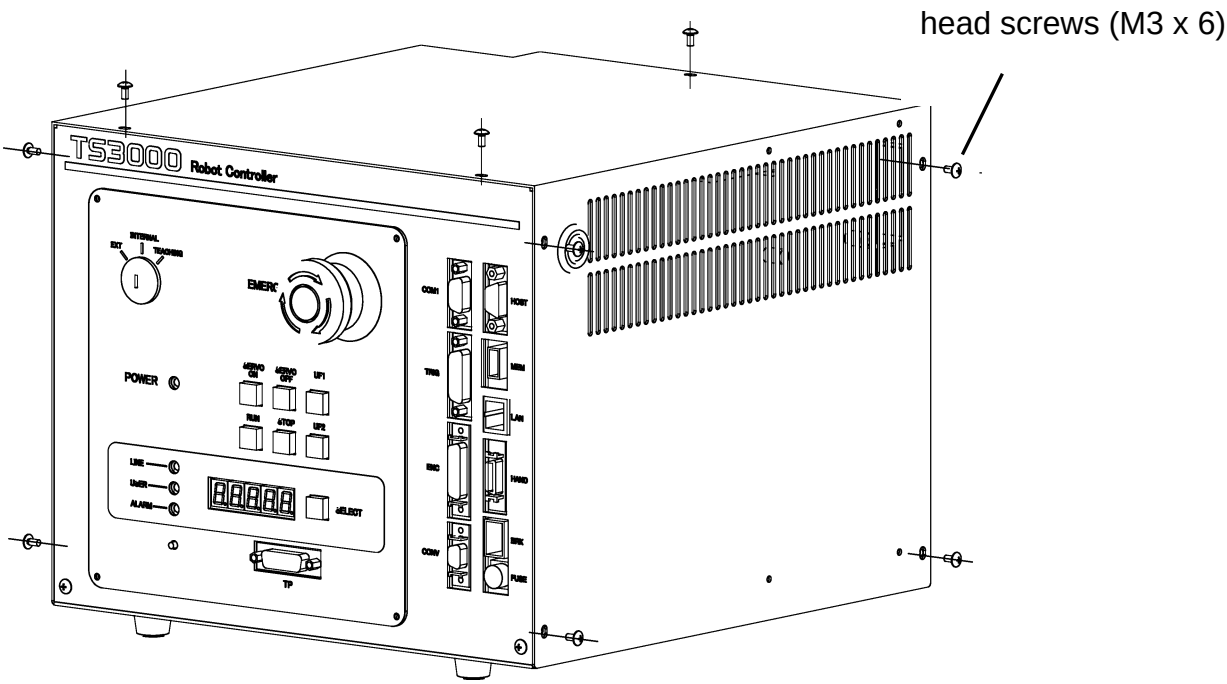


Fig. 7.6 Removal of cover (TS3000, TS3000E)

4) Remove the two (2) countersunk screws (M3 x 8) securing the switching power supply (PS1).

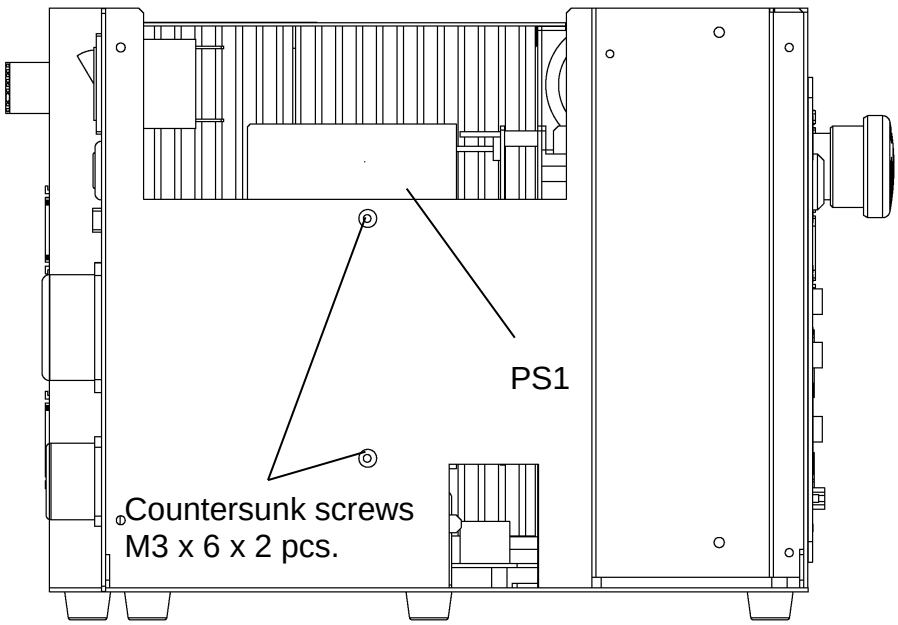


Fig. 7.7 Removal of PS1 Switching Power Supply Unit (TS3000, TS3000E)

5) Mount the new power supply unit.
6) Mount the cover to the main body.

[PS2 replacement procedures]

1) Turn off the main power switch.

2) Turn off the power breaker.

3) Remove eleven (11) cross truss head screws (M3 x 6) that secure the cover, and then remove the cover from the main body.

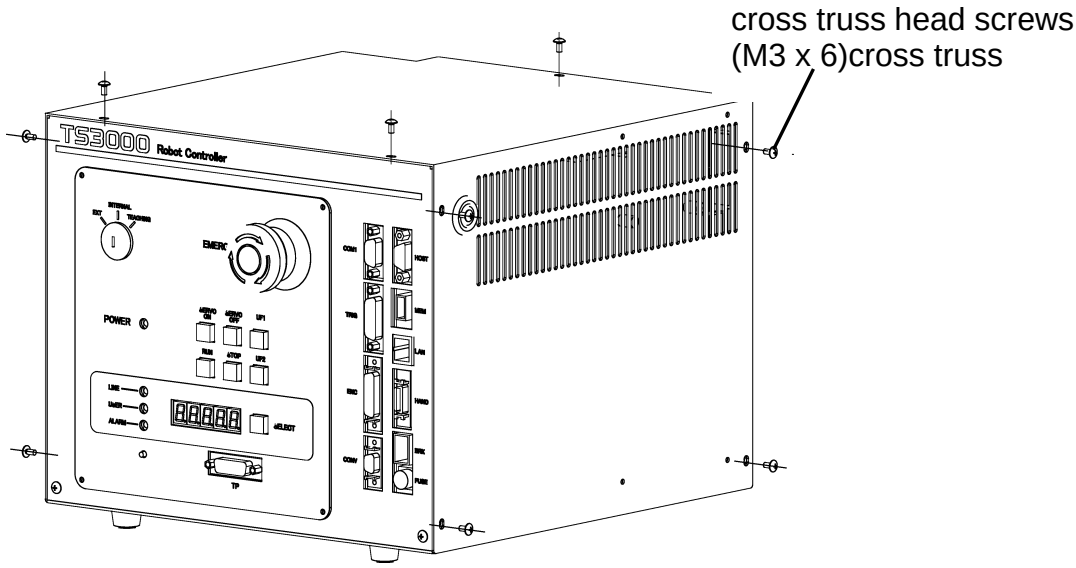


Fig. 7.8 Removal of cover (TS3000, TS3000E)

4) Remove servo driver.

Note: Mount servo driver in same position as before. When mounting mistake, the robot don't move.

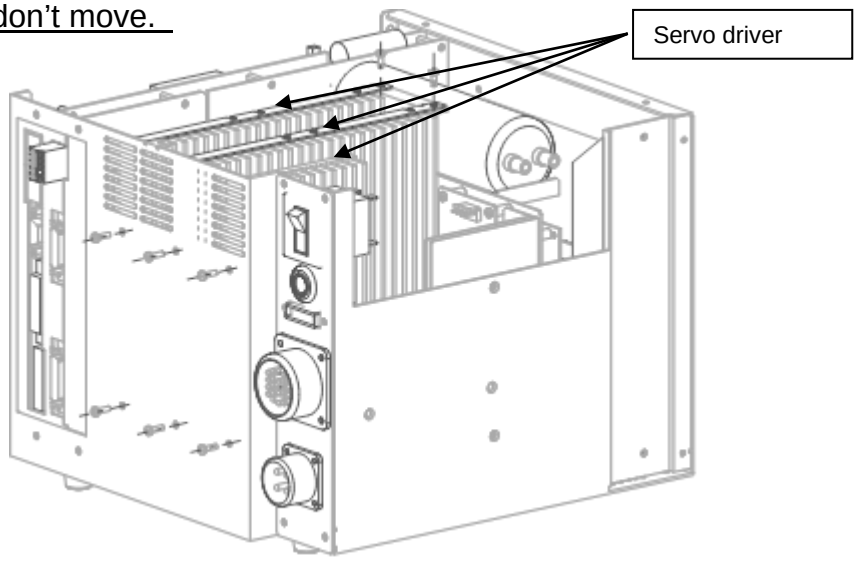


Fig. 7.9 Removal of servo driver (TS3000, TS3000E)

5) Remove the four (4) countersunk screws (M3 x 6) securing the switching power supply (PS2).

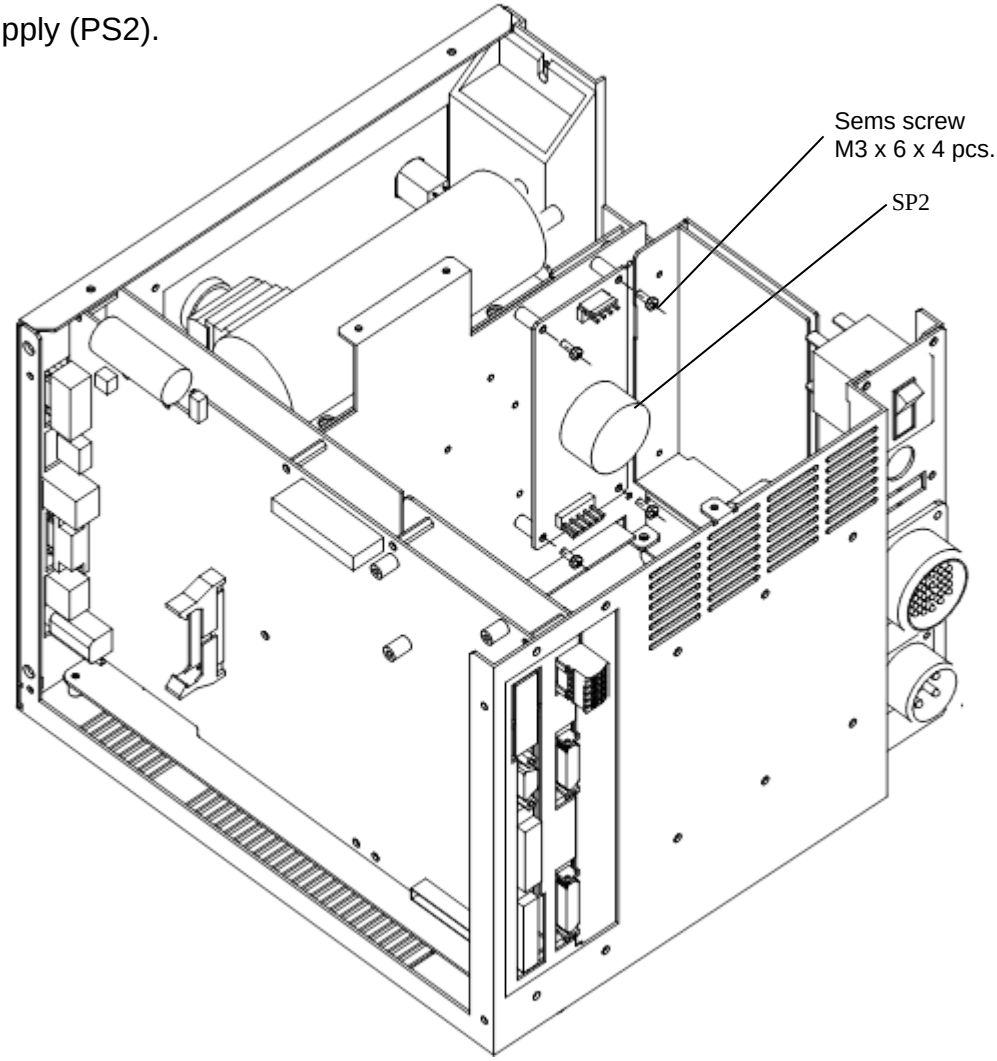


Fig. 7.10 Removal of SP2 switching power unit (TS3000, TS3000E)

6) Mount the new power supply unit.
7) Mount the cover to the main body.

7.3.5 Replacement of Fuse (X8GN, X8GI Printed Board)

If the current exceeding the specified current has run through the I/O unit, the fuse in the front of controller is blown out. If the alarm saying “I/O Fuse Broken (8–273)” is displayed, replace the fuse with a new one.
If the fuse of the output module has been blown out, examine and identify an fault circuit, remedy the cause, then replace the fuse.

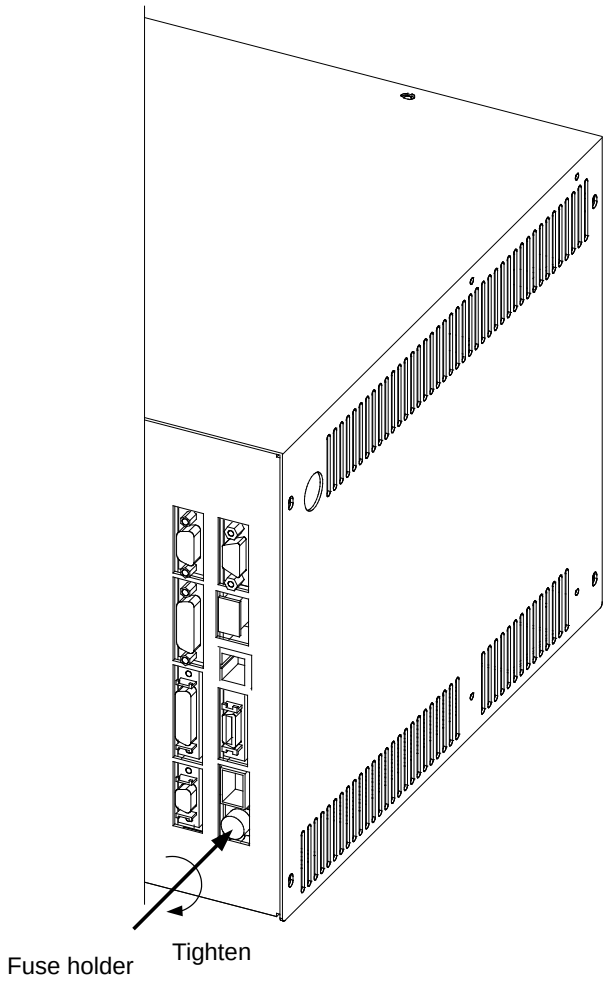


Fig. 7.11 Replacement of fuse (TS3000, TS3000E)

Table 7.3 Fuse (TS3000, TS3000E)

Type of fuse	Manufacturer
51NM030H	PICO

[Replacement procedures]

1) Turn off the controller main power.

2) Remove the fuse holder in the above figure. (Push and turn the holder to the left by 90°. It will be unlocked.)
3) Remove the fuse and set a new fuse.
4) Mount the fuse holder. (Push and turn the holder to the right by 90°. It will be locked.)
5) Turn on the controller main power and make sure that no error will occur.

7.3.6 Replacement of Output ICs (X8GN, X8GI Printed Board)

If the current exceeding the specified current has run through the output unit, the ICs on the X8GC, X8GN (X8GI) printed board is damaged.
When this happens, replace the ICs.
Before the replacement, examine and identify an fault circuit, remedy the cause, then replace the ICs.
The relationship between the output ICs [2] and output signals is shown below.

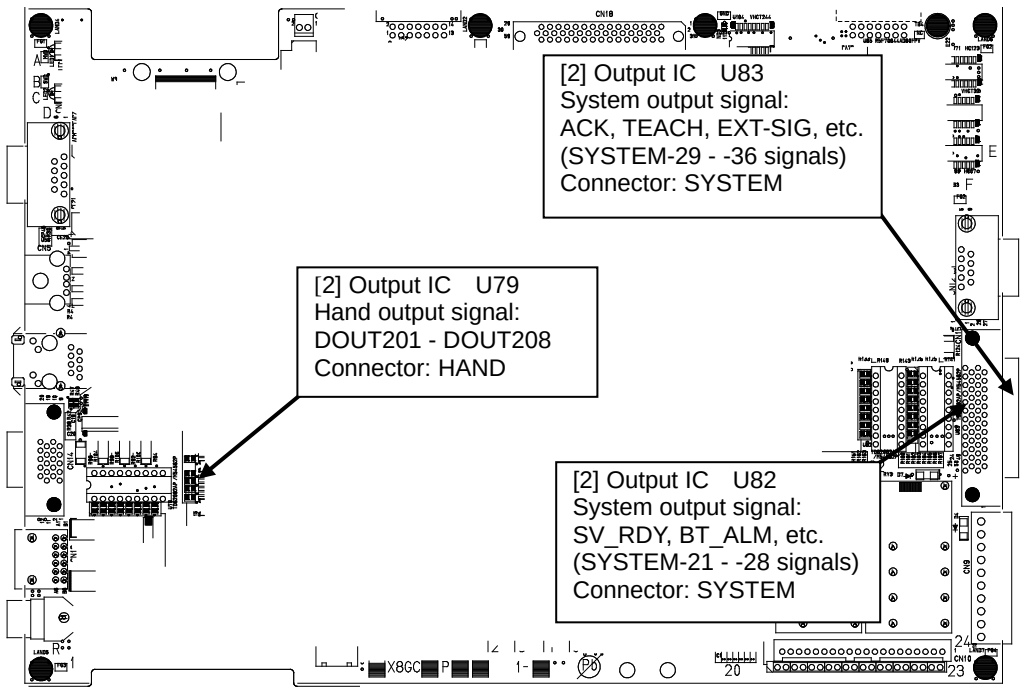


Fig. 7.12 Main printed board (X8GC) (TS3000, TS3000E)

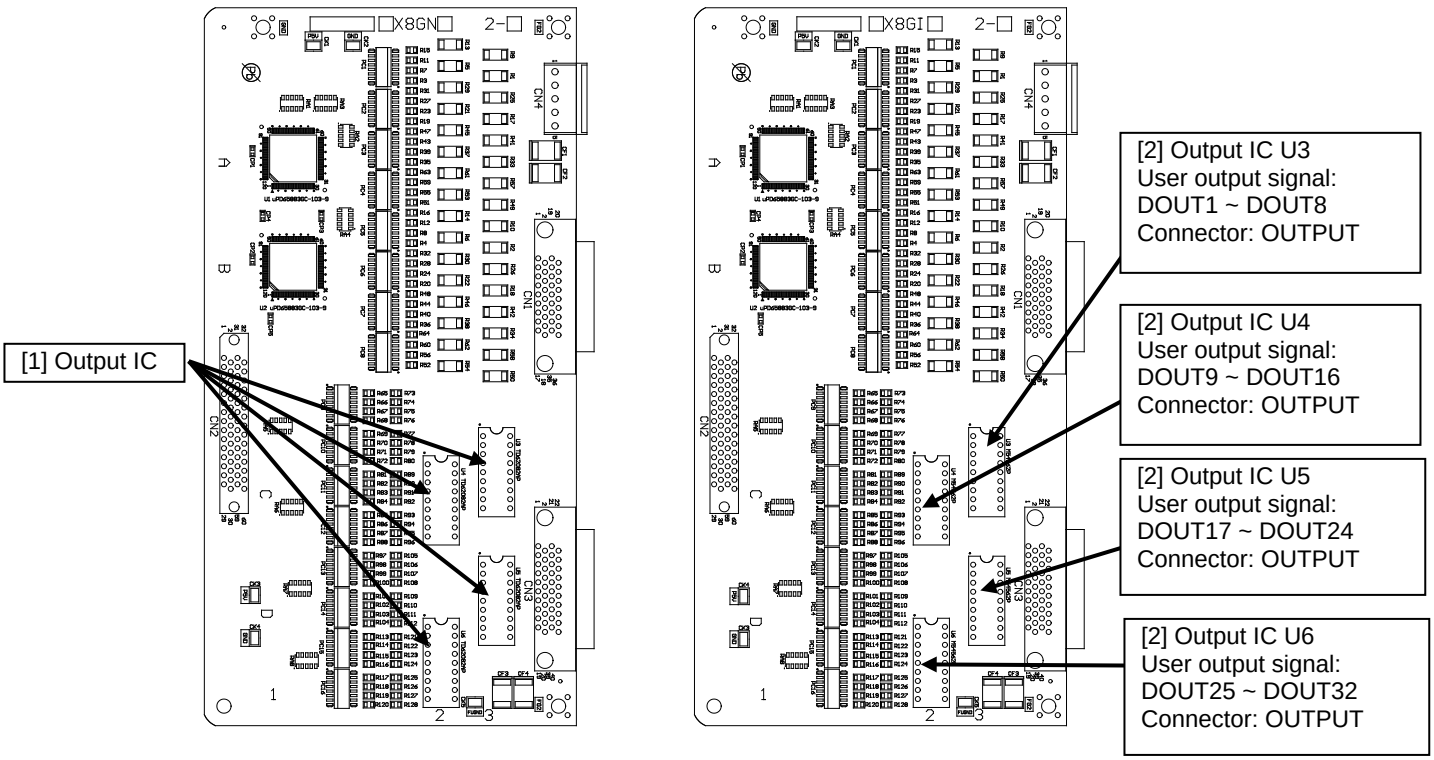


Fig. 7.13 I/O printed board(X8GN/X8GI) (TS3000, TS3000E)

Table 7.4 Output ICs (TS3000, TS3000E)

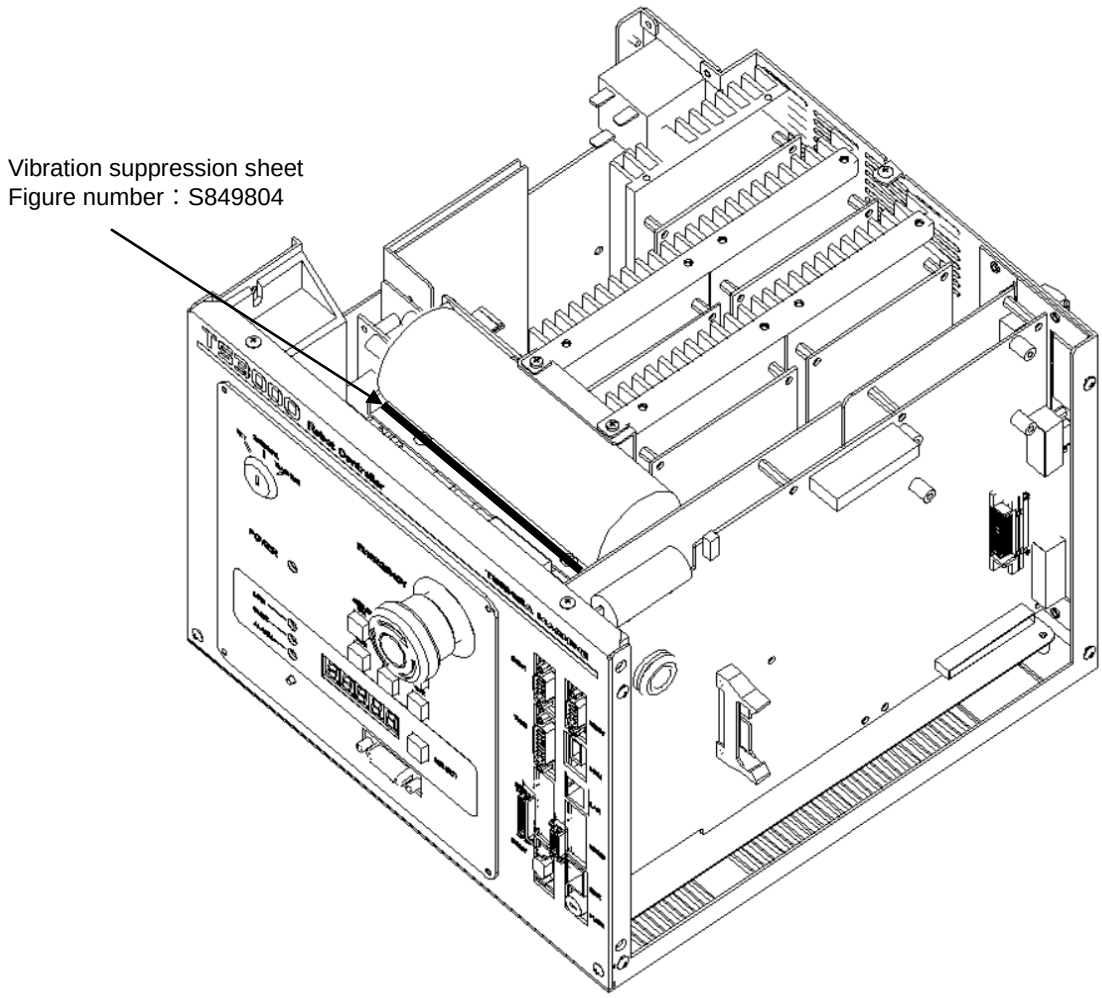
	Name of board	Type of IC	Manufacturer
[1]	X8GN	TD62082AP	Toshiba Corp.
[2]	X8GI	M54562WP	Mitsubishi Electric

[Replacement procedures]

1) Turn off the controller main power.
2) Remove eleven(11) cross truss head screws (M3 x 6) that secure the cover, and then remove the cover from the main body.
3) Disconnect all connectors connected to the X8GC, X8GN (X8GI) printed board in the controller.
4) Disconnect the X8GC, X8GN (X8GI) printed boards from the controller.
5) Remove the ICs shown above from respective sockets.
6) Attach the new ICs to the sockets.
7) Mount the X8GC, X8GN (X8GI) printed boards on the controller.
8) The connector taken off connect X8GC, and mount the cover to the main body using eleven(11) cross truss head screws (M3 x 6).
9) Turn on the controller main power.

7.3.7 Replacement of large capacitor vibration suppression sheet

If the large capacity capacitor vibration suppression sheet (Vibration suppression sheet) is used for more than 5 years, the sheet may deteriorate. Please replace them during overhaul. Since it is difficult for the customer to perform this replacement work, please contact our service department.



8. Robot Home Point and Position Detector Error

8.1 Setting Robot Home Point

Before delivery from our plant, home point setting is performed for the robot after its arm is secured with clamp for home point setting.
At the time of home point setting, position data of the motor position detector (i.e., encoder) is backed up by batteries, and coordinates of the robot need not be set each time the power is connected.
The position data of the motor position detector comes in the two (2) types; Servo offset and multi-turn data.

Servo offset
At home point setting operation of the robot (i.e., ZEROP operation and REORG operation), the data are written into the parameter file. As the data is backed up by batteries, the parameters should be loaded again from the attached system disk at replacement of the main control printed board.

Multi-turn data:
This data is kept in the encoder by battery backup, which differs from the backup battery of the main control printed board. When the battery voltage drops, correct multi-turn data cannot be guaranteed and the encoder itself outputs an alarm. This data is set to zero (0) at home point setting operation of the robot (i.e., ZEROP operation and REORG operation). Likewise, it is set to zero (0) also by multi-turn data reset operation.

8.2 Position Detector Error

If a position detector error (encoder error) occurs, either of the following errors is shown on the error display of the robot controller.

Error code	Error content	Remarks
8–601	Axis1 Enc multi count err	
8–602	Axis2 Enc multi count err	
8–603	Axis3 Enc multi count err	
8–604	Axis4 Enc multi count err	
8–605	Axis5 Enc multi count err	Case of TS3000 or TS3000E (When you use axis addition function)

The position detector error is an error detected by the encoder itself. Even if the data in the parameter file is lost (or changed) to change the home point coordinates of the robot, for instance, a position detector error will not occur. If the machine coordinates have changed without causing a position detector error, the mechanical connecting position of the servo motor and mechanical unit shifted or the data in the parameter file described above was changed. (Ex.: Tooth skip of timing belt, etc.)

Errors detected by the encoder itself include the battery voltage drop, error caused by temperature rise in the encoder, counter overflow, internal counter data inconsistency, etc. Among these errors, the error that occurs most frequently is the battery voltage drop that is caused by the absence of maintenance at specified change intervals due to a long-term shutdown or cable breakage.
To prevent the position detector error, therefore, replace the batteries on a regular basis. For the battery replacement, see “Para. 8.3 , Replacing Position Detector (or Encoder) Batteries”.
Also, a heavy vibration should not be exerted on the robot or the robot should not be moved suddenly by hand when the power is disconnected. Especially, when transporting and storing the robot, secure the robot in a posture for shipment, using the attached clamps.

Restoration from position detector error:
The restoring operation from the position detector error comes in the following four (4) manners. Make sure on the error display that a position detector error is generated. Identify the type of the position detector error and position data on the encoder status display, then perform each restoring operation. For encoder status display, see “Para. 8.5, Encoder Status Display.”

[1] Encoder error reset operation: See Para. 8.6
[2] Multi-turn data reset operation: See Para. 8.7.
[3] ZEROP operation: See Para. 8.8.
[4] HOME operation: See Para. 8.9.

(1) When a position detection error has occurred during normal operation with the mechanical connection position unchanged:

A position detector error may occur after battery voltage drop, or battery or cable replacement. When this happens, execute the following restoring operation.

1) Execute "[1] Encoder error reset operation."

2) Make sure that the error has been reset, referring to the error display.

After resetting the error, effect an emergency stop, manually move each axis to the motion limit and make sure that a soft limit error generates.

1.1) **When the error has been reset:**

Connect the controller power (DO NOT turn on the servo power), and make sure by moving the arm, etc. by hand that the position detector error will not occur again. Then disconnect the controller power and turn on again to make sure that the same error will not recur.

1.1.1) **When the error occurs again:**

The position detector has possibly malfunctioned.

1.1.2) **When the error will not occur again:**

Turn on the servo power. Select the TEACHING mode and gradually move each axis at LOW-SPEED to make sure of its soft stroke. If the soft stroke or machine coordinate is abnormal, perform "[4] Multi-turn data reset operation."

1.2) **When the error is not reset:**

Replace the batteries, then perform "[1] Encoder error reset operation" again. Make sure again that the error has been reset, referring to the error display.

1.2.1) **When the error has been reset:**

Execute the same operation as in Para. 1.1) above.

1.2.2) **When the error cannot be reset:**

The position detector has possibly malfunctioned, or the encoder cable has broken.

(2) When the mechanical connecting position with the motor has changed after motor or belt replacement:

A position detector error may occur after replacement and adjustment of the servo motor or timing belt. When this happens, execute the following restoring operation.

1) Execute “[2] HOME operation.”

2) Execute “8.9.5 Transfer of Home Position Data.”

3) Make sure that the error has been reset, referring to the error display.

After resetting the error, effect an emergency stop, manually move each axis to the motion limit and make sure that a soft limit error generates. If the mechanical interference is caused before generation of the soft stroke limit error, repeat the restoring operation in the manner described above.

⚠ DANGER

– When performing maintenance with the power turned on in the dangerous range of the robot, be sure to switch the master mode key inserted to the controller to “TEACH” to pull out the key and carry it with the teach pendant.
Otherwise, the robot that has operated unpredictably may collide with a worker.
- The worker should receive safety training regarding industrial robot inspection-related work defined by each country’s laws and regulations.
- Display the fact that work is in progress near the controller.

– When approaching the robot being stopped, be sure to check the robot operation status.
If the robot is under operation, the robot that has restarted operation may collide with the worker.
- Check the operation status indicated on the indicator lamp of the controller and on the teach pendant.
- The robot operation status is output by signals (servo operation preparation completion, external mode, operation preparation completion, automatic operation, failure and emergency stop). Connect to the display equipment, etc.

– When performing maintenance with the power turned on in the dangerous range of the robot, be sure to arrange a work observer outside the dangerous range of the robot.
Otherwise, when a worker in the dangerous range of the robot suffers damage, the worker cannot be rescued.
- The observer should secure safety of the worker and the third party who approaches the dangerous range of the robot.
- The observer should also receive safety training regarding industrial robot inspection-related work defined by each country’s laws and regulations.

⚠ WARNING

– When rotating the axis 4 by hand, never insert your hand or finger inside the upper and lower mechanical stoppers or under the robot hand.
Otherwise, your hand or finger may get caught in the mechanical stopper or robot hand.
- When the axis 4 rotates, the ball screw spline shaft will move up and down.
- The axis-4 motor is not equipped with the brake. If the robot hand, etc. is touched by hand when the servo power is off, the axis 4 will rotate.

– When releasing the motor brake, be sure to support the ball screw spline shaft.
Otherwise, the shaft will drop by its own weight concurrently with release of the brake.
- A worker should support the shaft to prevent hand or finger from getting caught.
- When something strange is felt during work, release the brake release switch immediately. The brake will be also applied by turning off the power to the controller.
- The brake release switch is located on the arm-2 top face of the robot. Do not press the switch by mistake.
- The worker should receive safety training regarding industrial robot teaching- or inspection-related work defined by each country's laws and regulations.

⚠ CAUTION

– Be sure to start maintenance when the surface temperature becomes less than 50°C after power off.
Otherwise, touching hot motor or periphery of the motor results in burn injury.

NOTICE

– If zero point is re-set using the ZEROP function or REORG function, be sure to perform re-teaching of the teaching point.
The zero position shifts slightly due to error in setting. If the zero position is used as is, the robot will collide with the peripheral equipment.

8.3 Replacing Position Detector (or Encoder) Batteries

⚠ DANGER

– Never touch liquid that leaked from the battery.
If liquid adheres to your skin or eye, inflammation results.
- When liquid adheres to your skin, wash it away with flowing water immediately.
- When liquid adheres to your eye, wash it away with flowing water immediately and then see a doctor.
- Replace the parts to which liquid has adhered.

NOTICE

– Be sure to replace the position detector battery during annual inspection.
If position detector data is lost, the robot cannot be operated.
- When the battery voltage has dropped, a “battery alarm” will occur. At this stage, data is not lost. If the batteries are replaced, the alarm is automatically released.
- When the battery voltage has dropped further, a “battery error” will occur. At this stage, data is lost. A position detector error alarm will also occur.
- If data is lost, re-set the zero position. For how to set the zero position, see “8. Robot Home Point and Position Detector Error.”

– When the robot is not used with the power turned off, be sure to replace the position detector battery every two (2) months.
Because the battery consumption becomes large in the power off condition, position detector data will be lost before periodic inspection.

8.3.1 Battery Box Location

The battery box for the position detector is provided inside the battery box cover of the base unit.

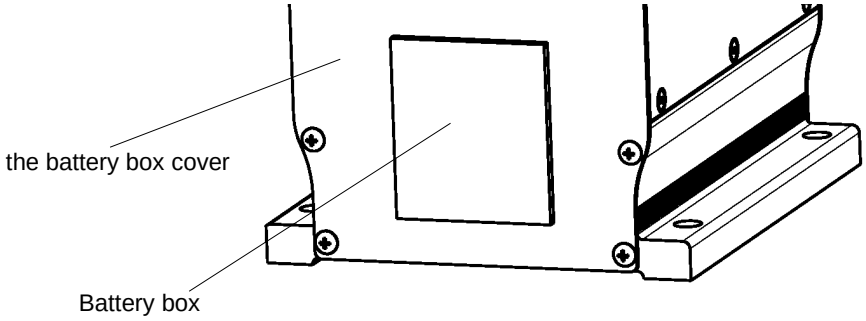


Fig. 8.1 Battery box location

8.3.2 Replacing Batteries

⚠ DANGER

– When replacing the battery with the power kept on, be sure to emergency-stop the robot.
Otherwise, the robot that has operated unpredictably will collide with a worker.
- The worker should receive special training, etc. regarding industrial robot inspection-related work defined by each country's laws and regulations.

1) Assure the safe work. Then keep the power ON and press the EMERGENCY pushbutton switch to effect an emergency stop on the robot. Unless the safe work can be assured, turn the power off.

2) Open the battery box cover and remove the batteries.
Set the three (3) new batteries, taking care of the "+" and "–" polarities.
To replace the batteries while the power is turned off, finish the replacement within five (5) minutes.

3) If alarms don't clear in table 8.1 "1-40x Axis x Enc Battery low (Battery alarm)" after you reset alarm, please try to the method of 4). When alarms are cleared by this method, finish work after closing the cover.

4) Please remove the battery box cover “Fig.8.1”. Eject connector by reference of the next Fig and check electric voltage about 4.5V by tester. If electric voltage is low, it has possibilities of bad fit. Please remove the battery and return the battery after cleaning the bracket of the battery box. If electric voltage is about 4.5V for tester, return to method of 3). And please don’t remove connector.

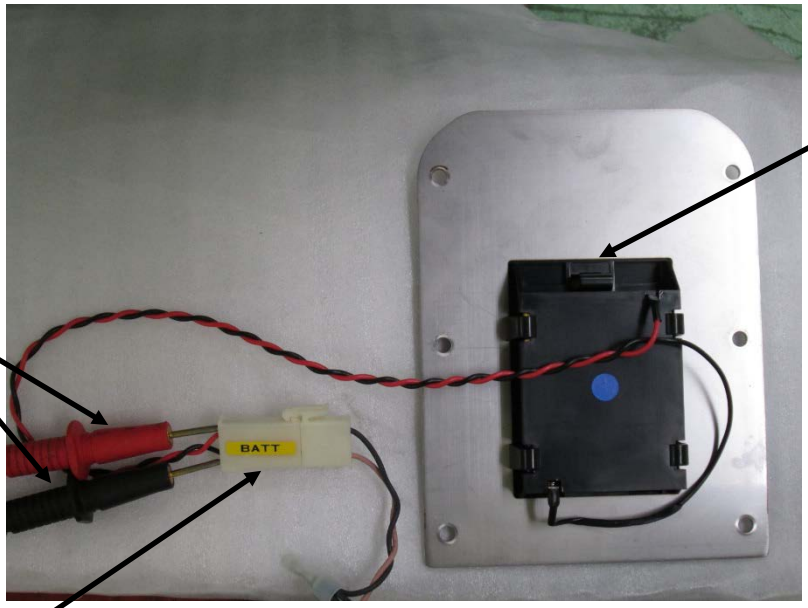


Fig. 8.2 check of battery bad fit

8.3.3 Battery Error Code

When a position detector error including battery error has occurred, the following error code is shown on the error display.

When electric voltage decreased, the following error code is shown on the error display. When the following error code is shown in the error history at generation of a position detector error, the error can be identified as the battery error.

Table 8.1 1-40 x battery error code

Error code	Error content	Remarks
1-401	Axis1 Enc Battery low (Battery alarm)	
1-402	Axis2 Enc Battery low (Battery alarm)	
1-403	Axis3 Enc Battery low (Battery alarm)	
1-404	Axis4 Enc Battery low (Battery alarm)	
1-405	Axis5 Enc Battery low (Battery alarm)	Case of TS3000 or TS3000E (When you use axis addition function)

If the error in table 8.1 occurred, please replace battery immediately. If unattended, level 8 errors such as listed in Table 8.2 to 8.4 occur. In addition, it is possible to lost robot origin position information.

Table 8.2 8-60 x battery error code

Error code	Error content	Remarks
8-601	Axis1 Enc multi count err	
8-602	Axis2 Enc multi count err	
8-603	Axis3 Enc multi count err	
8-604	Axis4 Enc multi count err	
8-605	Axis5 Enc multi count err	Case of TS3000 or TS3000E (When you use axis addition function)

If the error in table 8.2 displayed, counter skipping may be occurring due to insufficient battery power. Change the battery. If changing the battery does not correct the problem and the error is still displayed, the encoder may be faulty. In such a case, it is necessary to change the motor. Please consult our service representative.

Table 8.3 8-61 x battery error code

Error code	Error content	Remarks
8-611	Axis1 Enc battery empty	
8-612	Axis2 Enc battery empty	
8-613	Axis3 Enc battery empty	
8-614	Axis4 Enc battery empty	
8-615	Axis5 Enc battery empty	Case of TS3000 or TS3000E (When you use axis addition function)

If this error is displayed, the battery power may be insufficient. Change the battery. And it is possible to lose robot origin position information because the battery power was insufficient. If alarm does not clear after you replace battery, Please set origin position by referring to “Section 8, Robot Home Point and Position Detector Error.”

Table 8.4 8-40 x battery error code

Error code	Error content	Remarks
8-401	Axis1 Encoder abnormal	
8-402	Axis2 Encoder abnormal	
8-403	Axis3 Encoder abnormal	
8-404	Axis4 Encoder abnormal	
8-405	Axis5 Encoder abnormal	Case of TS3000 or TS3000E (When you use axis addition function)

In the case of level 8 battery errors, the above two types errors in table 8.2 and 8.3 are basically displayed; however, the errors in table 8.4 may be displayed in rare cases. If this error is displayed, change the battery. And it is possible to lose robot origin position information because the battery power was insufficient. If alarm does not clear after you replace battery, Please set origin position by referring to “Section 8, Robot Home Point and Position Detector Error.”.

The battery error is one of the position detector errors (i.e., encoder errors), and there is other cause leading to the position detector error. The restoring method from the position detector error varies with the error contents generated. To properly execute the restoring method from the position detector error, therefore, read through and

completely understand the descriptions carried in “Section 8, Robot Home Point and Position Detector Error.”

8.4 About Home Position

Home position what need at the moment of home position setting is shown. When the robot is moved to home position about 1 axis, 2 axis or 4 axis, suit to home point match-marks. See “Para. 8.4.1 to 8.4.3, Locations of Robot Home Point Match-Marks.” The 3 axis of robot don’t have home point match-marks. See “Para. 8.4.4 to 8.4.6, Home Position of 3 axis” and move to that position.

8.4.1 Locations of Robot Home Point Match-Marks (THL300,THL400)

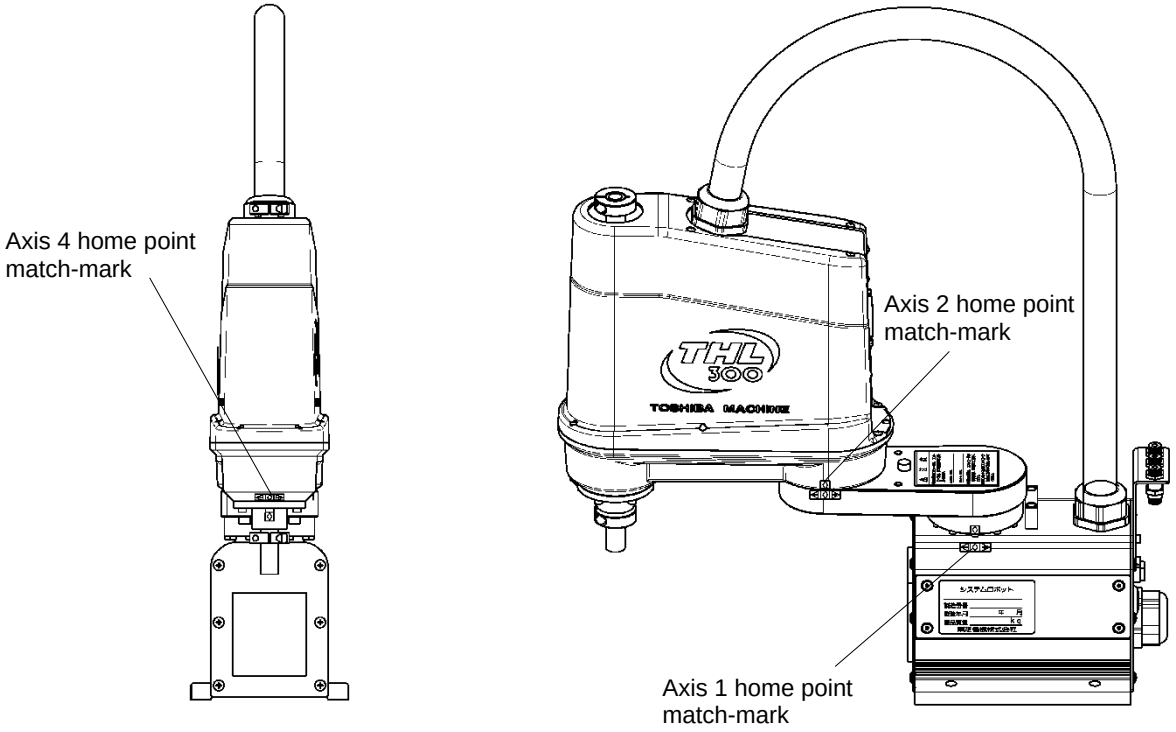


Fig. 8.3 Locations of home point match-marks (THL300, THL400)

8.4.2 Locations of Robot Home Point Mach-Mark (THL500, THL600 and THL700)

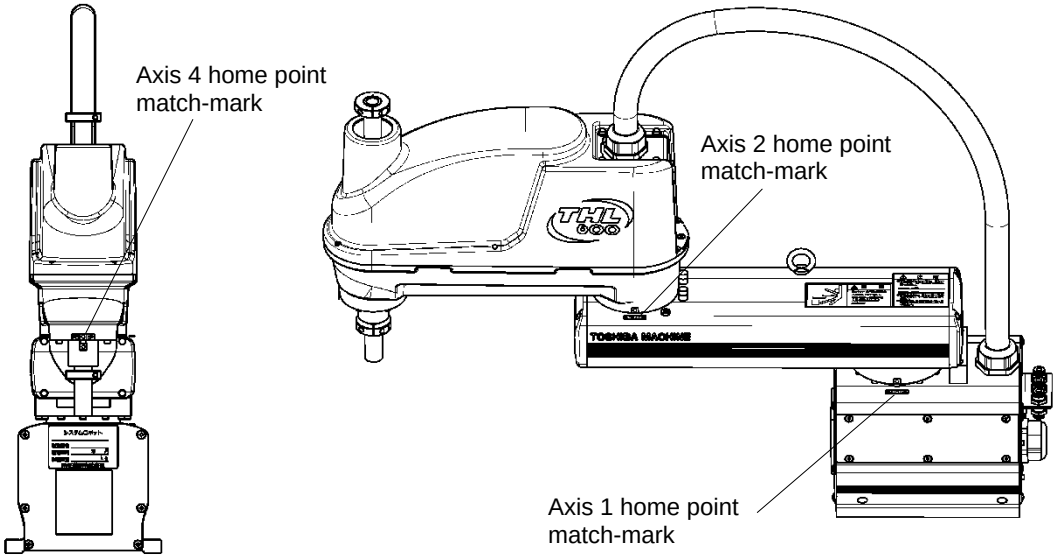


Fig. 8.4 Locations of home point match-marks (THL500, THL600 and THL700)

8.4.3 Locations of Robot Home Point Mach-Mark (THL800, THL900, THL1000 and THL1200)

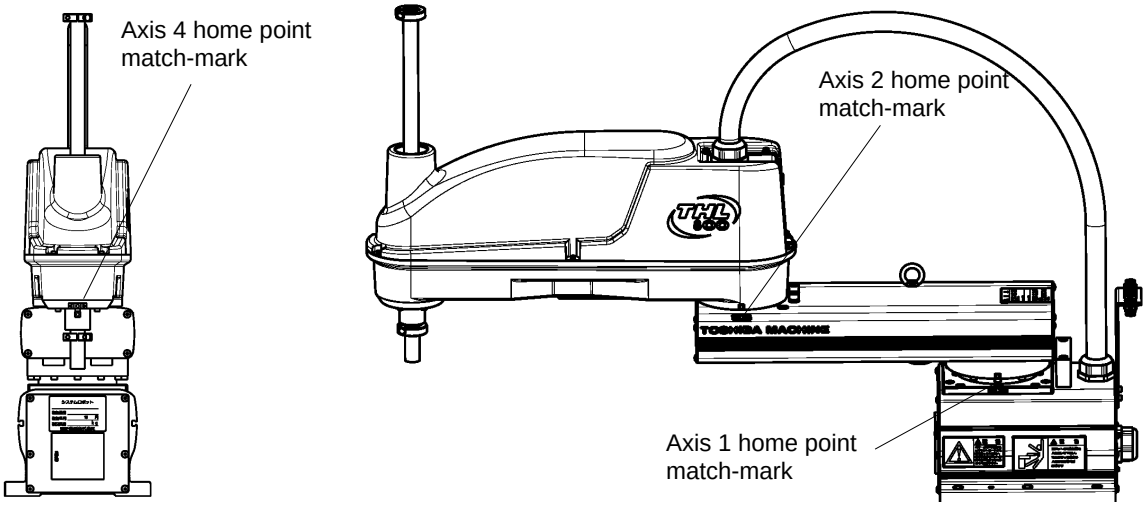


Fig. 8.5 Locations of home point match-marks
(THL800, THL900, THL1000, and THL1200)

8.4.4 Home Position of 3 axis (THL300 and THL400)

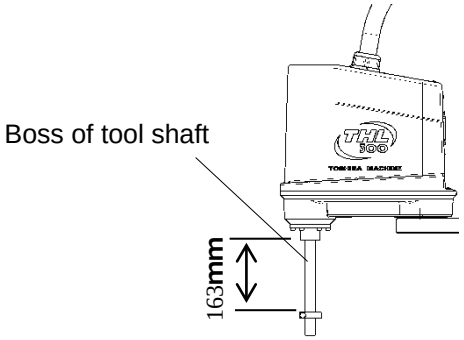
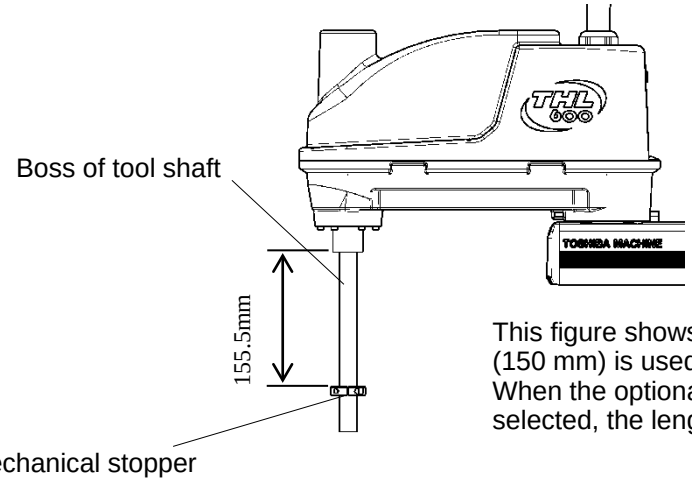


Fig. 8.6 Dimension of axis 3 home point (THL300, THL400)

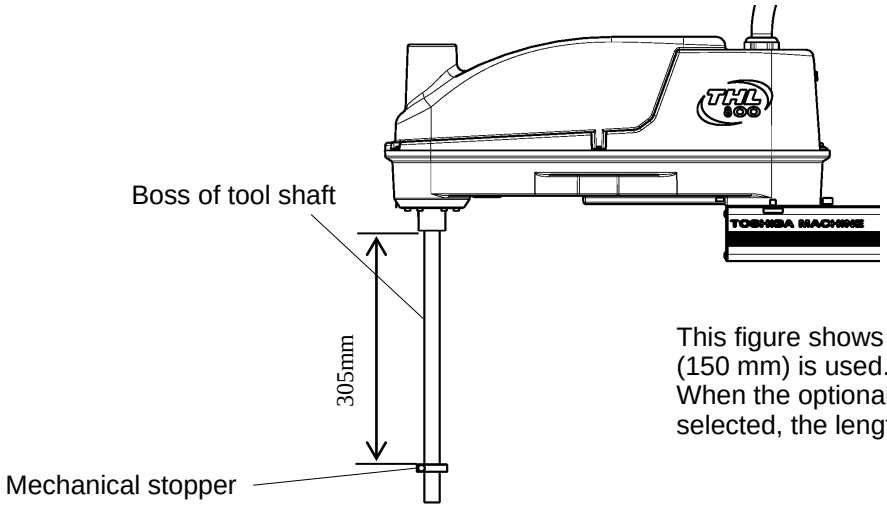
8.4.5 Home Position of 3 axis (THL500, THL600 and THL700)



This figure shows when the standard axis stroke (150 mm) is used.
When the optional axis stroke (300 mm) is selected, the length is 305.5 mm.

Fig. 8.7 Dimension of axis 3 home point (THL500, THL600 and THL700)

8.4.6 Home Position of 3 axis (THL800, THL900, THL1000 and THL1200)



This figure shows when the standard axis stroke (150 mm) is used.
When the optional axis stroke (300 mm) is selected, the length is 305.5 mm.

Fig. 8.8 Dimension of axis 3 home point (THL800, THL900, THL1000 and THL1200)

8.5 Encoder Status Display

Call the encoder status screen on the teach pendant display, and make sure of the position data and the type of position detector error there. The type of the position detector error (i.e., error status) is expressed as shown below. The multi-turn data, home position data within one (1) full turn of motor and error status are displayed under the column of [MULTI], [SINGLE] and [Err-df], respectively.
To call the encoder status screen, observe the following procedures.

Table 8.5 Error status table

Error status	Description
0000	Signifies the normal condition.
0100	This error occurs when a heavy vibration has been exerted on the robot or when the robot has been moved fast by hand while the power drive cable was disconnected while the power was off. The multi-turn data has possibly shifted.
0200	Because axis feed was not executed after the power was connected, the encoder resolution has deteriorated. During axis feed, the encoder resolution returns to normal. No alarm will generate.
8000	The battery voltage has dropped. Replace the batteries.
4000	The battery voltage has dropped further than the above. An error has possibly occurred in the multi-turn data.
4200	Errors 4000 and 0200 have occurred at the same time.
C200	Errors 8000, 4000 and 0200 have occurred at the same time.

[1] Press the “UTILITY” key provided on the teach pendant. Press the [NEXT] key until [ENC] is displayed on the function menu.

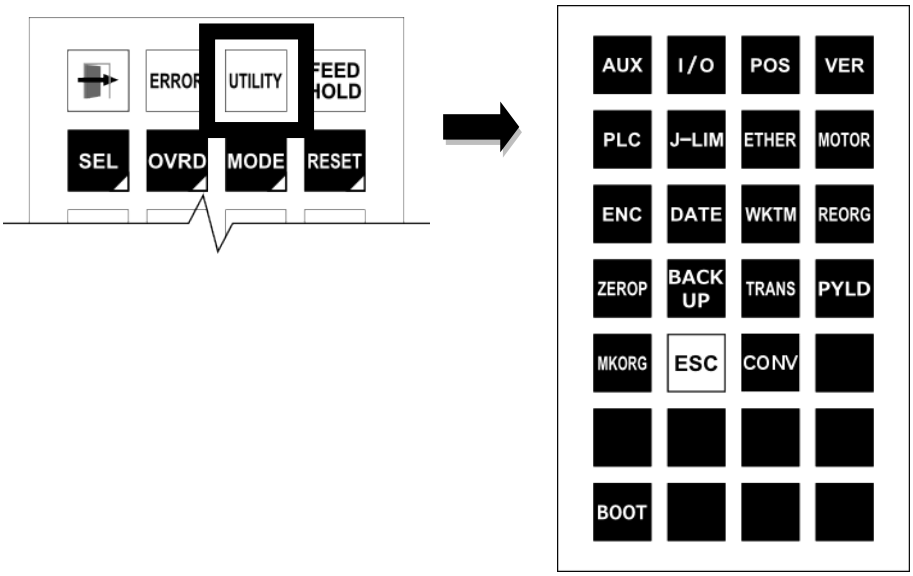
[2] Press the [ENC] (F5) key, and the encoder status screen appears on the display.



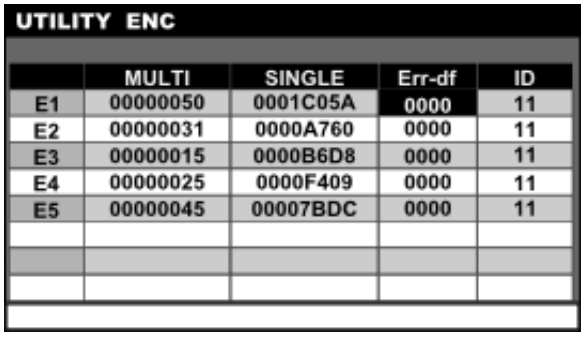
If the error status shown in the error status table above is displayed, you can restore from the position detector error.

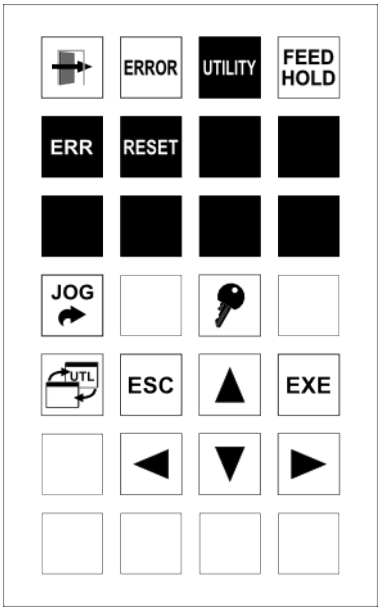
＜Case of TP3000＞

[1] Press the “UTILITY” key provided on the teach pendant. The display of the key sheet changes to the utility function selection keys.



[2] Press the [ENC] key among the utility function selection keys. The encoder Status Display and key sheet are displayed.





8.6 [1] Encoder Error Reset Operation

When the mechanical connecting position of the servo motor and mechanical unit remains unchanged, that is, after battery replacement due to battery voltage drop or cable replacement due to cable breakage, execute this operation.

1) Call the encoder status screen on the teach pendant display, referring to Para. 8.5 above.

2) Just after the encoder status screen is called, the cursor points out the [Err-df] column on the [E1] line. Move the cursor to a line for the system you wish to execute the encoder error reset operation by means of the cursor keys (i.e., [-] and [¯] keys).



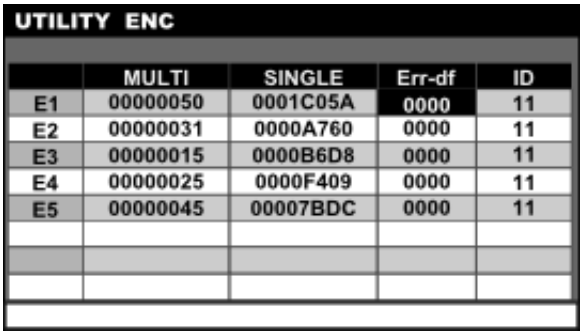
3) To reset the axis 3 encoder error, for instance, move the cursor to the [Err-df] column on the [E3] line and press the [RESET] (F3) key. If everything is OK, press the “EXE” key. Now the axis 3 encoder error has been reset.

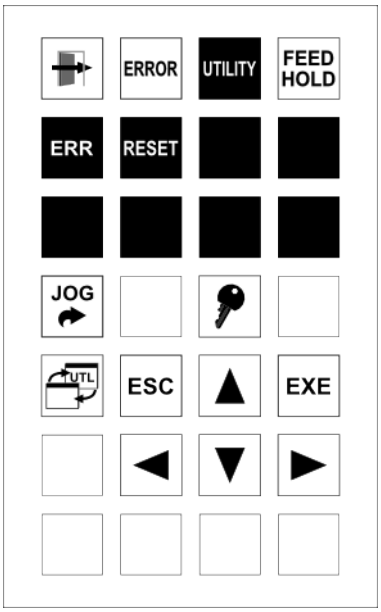
4) Call the normal error screen, press the [RESET] (F3) key to reset the error of “8-603 Axis3 Enc multi count err.”

＜Case of TP3000＞ Indicates the different part against the above TP1000 operation.

3) To reset the axis 3 encoder error, for instance, move the cursor to the [Err-df] column on the [E3] line and press the [RESET] key. If everything is OK, press the “EXE” key. Now the axis 3 encoder error has been reset.

4) Call the normal error screen, press the [RESET] key to reset the error of “8-603 Axis3 Enc multi count err.”





8.7 [2] Restoring Home Position Data by Multi-Turn Data Clear

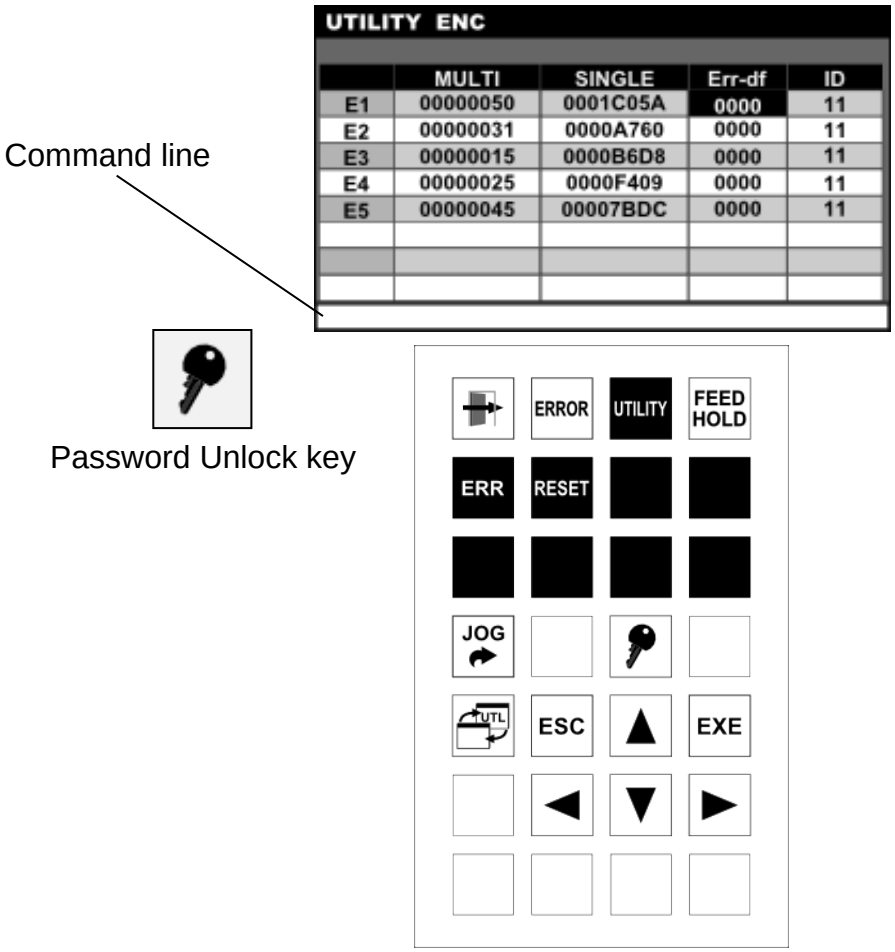
When the mechanical connecting position of the servo motor and mechanical unit remains unchanged, that is, after battery replacement due to battery voltage drop or cable replacement due to cable breakage, execute this operation after encoder error reset operation.

1) See “Para. 8.4 About Home Position” and move to that position.
2) Select the TEACHING mode by means of the master mode switch. And see “Para. 8.5. Encoder Status Display”, and print Encoder Status Display.
3) When the encoder information screen appears, press the [ALT] and [0] keys at the same time to cancel the editing lock mode.
4) After editing lock is disabled, press the [←] key to move to the MULTI column.
5) Set the cursor to an axis whose multi-turn data is to be reset to zero (0), and press the [F3] function key corresponding to the [RESET] key menu.
6) Turn off the controller power, then turn it on again.
7) Execute “Para. 8.10. affirmation” before moving robot. Move the robot to the teach points, etc. and make sure that the robot can be located at them exactly. (Thus, the robot can be restored completely to the original condition.)

＜Case of TP3000＞ Indicates the different part against the above TP1000 operation.

3) When the encoder information screen appears, press the Password Unlock key on the key sheet to cancel the editing lock mode. Enter number “1048” on the command and then press the [EXE] key.

5) Set the cursor to an axis whose multi-turn data is to be reset to zero (0), and press the [RESET] key on the key sheet, press the [EXE] key in succession.



8.8 [3] Restoring Home Position Data by ZEROP Function

When mechanical joint positions have deviated due to motor change and so on, and when re-teaching is assumed, execute this operation.

NOTICE

– If the zero point is re-set using the ZEROP function or REORG function, be sure to perform re-teaching of teaching points.
The zero position shifts slightly due to the error in setting. If the zero position is used as is, the robot will collide with the peripheral equipment, etc.

1) Select the TEACHING mode by means of the master mode switch.

2) See “Para. 8.4, About Home Position” and move to that position.

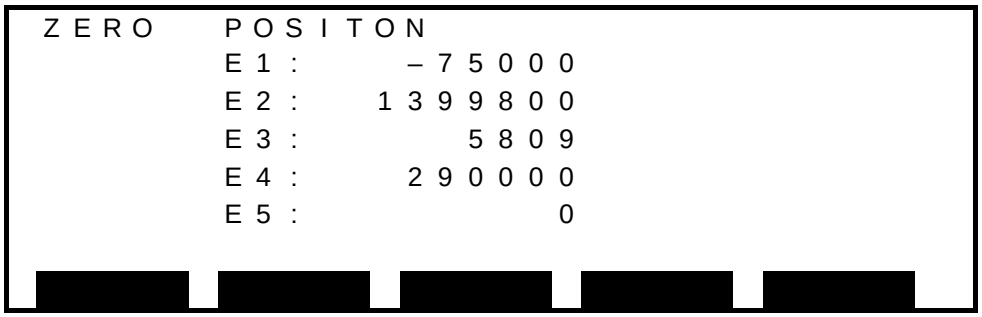
3) Press the [UTILITY] key equipped on the teach pendant.

4) Press the [NEXT] (F6) key on the teach pendant twice.

5) Turn off the servo power.

6) Press the [F3] key on the teach pendant to select the ZEROP mode.

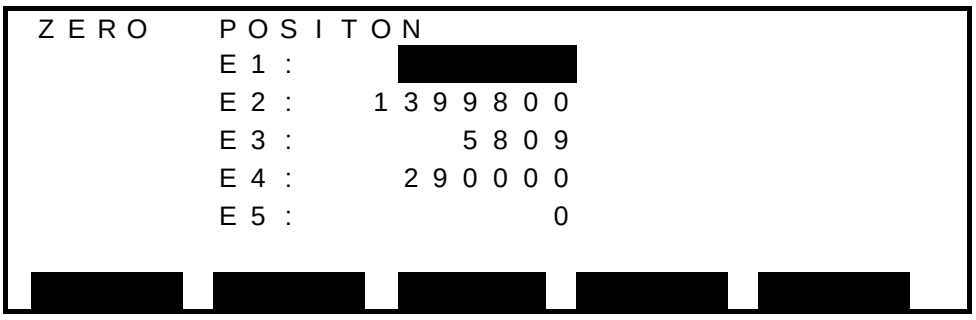
7) The following home point setting screen is called.



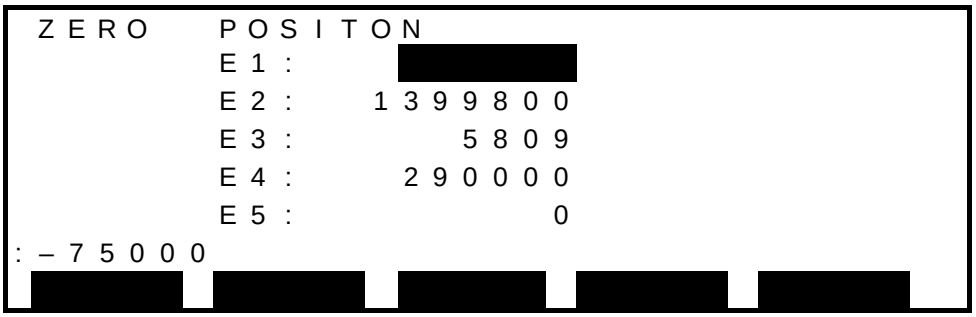
8) Press the [ALT] and [0] keys on the teach pendant at the same time.

9) The following home point editing screen appears.

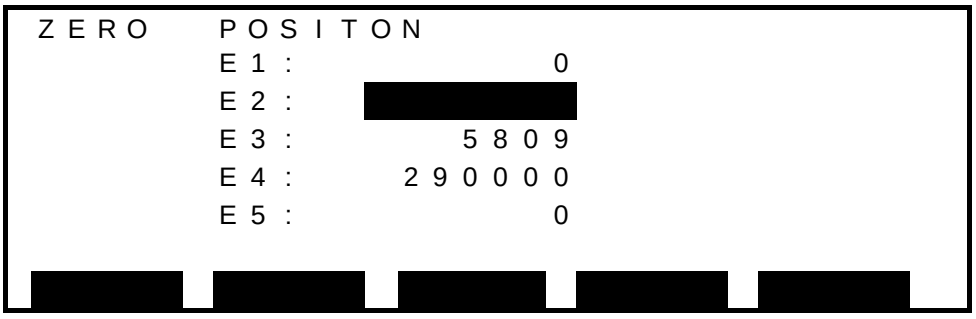
10) The numerical value of E1 (axis 1) is inversely displayed.



11) Press the “EXE” key on the teach pendant, and the value of “E1” is displayed at the lower left side. Set “0” in lieu of this value and press the “EXE” key. Then the E1 value becomes “0”.



12) When the “↓” key on the teach pendant is pressed, the following E2 value is displayed by reversed image. Perform setting in the same manner as above.



NOTICE

– If re-setting the zero point using the ZEROP function or REORG function, be sure to observe the operation procedure.

The axis 3 operates simultaneously with the operation of the axis 4. Therefore, the zero point is not restored when the order of zero setting is wrong.

Operation procedure

1) Move the robot to the specified position.
2) Move the cursor to E4 on the screen, enter data, and restore the zero point of the axis 4.
3) Move the cursor to E3 on the screen, enter data, and restore the zero point of the axis 3.
   - E1 (axis 1) and E2 (axis 2) have no order of zero re-setting.

13) Turn off the controller power, then turn it on again.

14) Execute “Para. 8.10. affirmation” before moving robot. Move the robot to the teach points, etc. and make sure that the robot can be located at them exactly.

＜Case of TP3000＞

1) Select the TEACHING mode by means of the master mode switch.

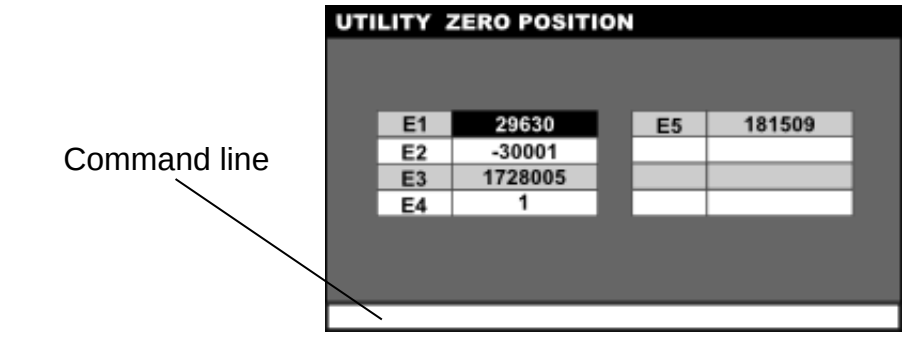
2) See “Para. 8.4, About Home Position” and move to that position.

3) Press the [UTILITY] key equipped on the teach pendant.

4) The display of the key sheet changes to the utility function selection keys.

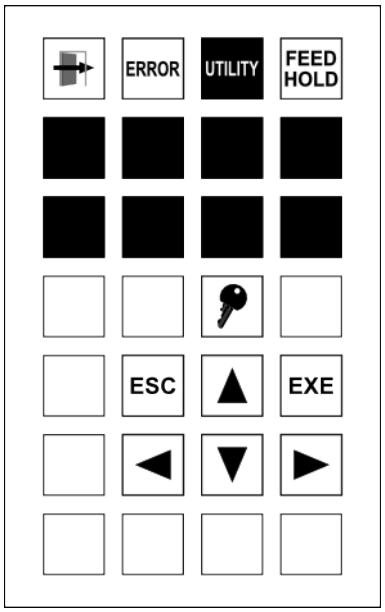
5) Press the [ZEROP] key among the utility function selection keys.

6) The following home point setting screen and key sheet are called.





Password Unlock key



7) Press the Password Unlock key on the key sheet to cancel the editing lock mode. Enter number “1048” on the command and then press the [EXE] key.

8) The home point editing screen appears.

9) The numerical value of E1 (axis 1) is inversely displayed.

10) When the “ ↓ ” key is pressed, the E4 value is displayed by reversed image.

11) Press the “EXE” key on the teach pendant, and the value of “E4” is displayed at the command line. Set “0” in lieu of this value and press the “EXE” key. Then the E4 value becomes “0”.

12) When the “↑” key on the teach pendant is pressed, the following E3 value is displayed by reversed image. Perform setting in the same manner as above.

NOTICE

– If re-setting the zero point using the ZEROP function or REORG function, be sure to observe the operation procedure.

The axis 3 operates simultaneously with the operation of the axis 4. Therefore, the zero point is not restored when the order of zero setting is wrong.

Operation procedure

1) Move the robot to the specified position.
2) Move the cursor to E4 on the screen, enter data, and restore the zero point of the axis 4.
3) Move the cursor to E3 on the screen, enter data, and restore the zero point of the axis 3.
   - E1 (axis 1) and E2 (axis 2) have no order of zero re-setting.

13) Turn off the controller power, then turn it on again.
14) Execute “Para. 8.10. affirmation” before moving robot. Move the robot to the teach points, etc. and make sure that the robot can be located at them exactly.

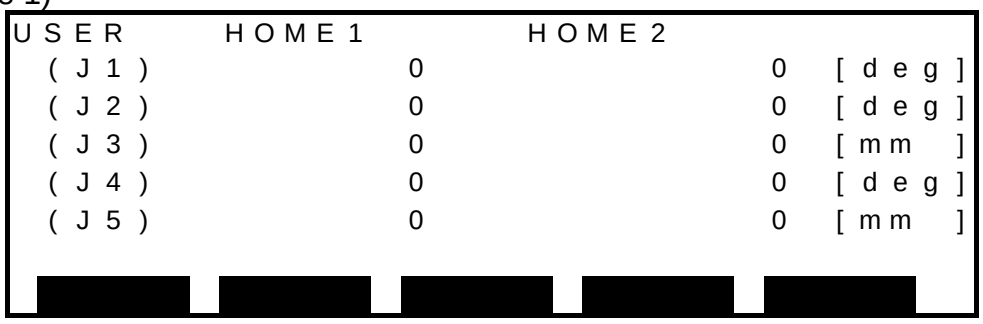
8.9 [4] Restoring Home Position Data by HOME Function

8.9.1 Outline of HOME Function

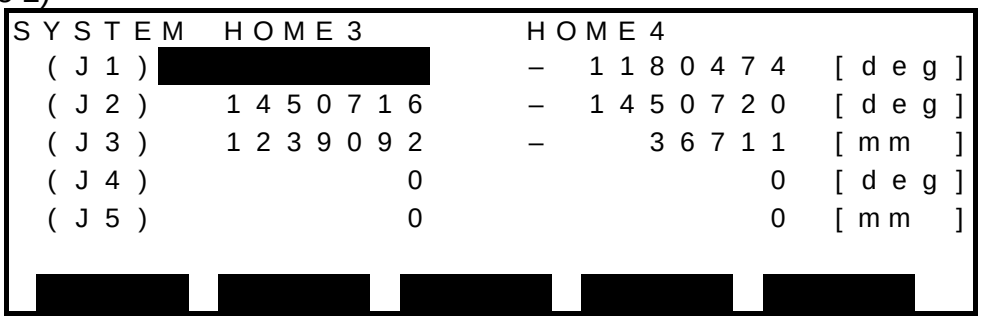
This function memorizes a specific coordinate value in each axis and restores the original home position data by using the HOME function when the machine home position data has been destroyed.

The set points are provided for HOME1 to HOME4, and Axis 1 to Axis 3 plus (+) side mechanical stopper positions are set in HOME3, and Axis 1 to Axis 3 minus (–) mechanical stopper positions in HOME4. Axis 4 is set at the 0° position according to the home point match-mark.
For HOME1 and HOME2, you can specify any values. It is recommended that when building up a system, you specify the place for securing the robot end.

(Page 1)



(Page 2)



Caution: This function is used to restore the machine home point of each axis. NEVER use the function at other than the setting for restoration.

＜Case of TP3000＞

(Page 1)

UTILITY REORG

1/2

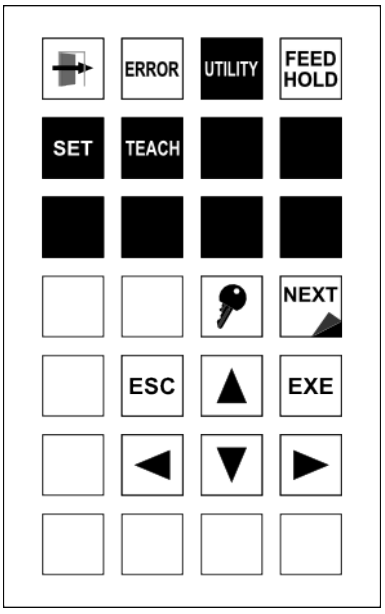
	HOME 1	HOME 2	
J1	0	0	deg
J2	0	0	deg
J3	0	0	mm
J4	0	0	deg
J5	0	0	deg

(Page 2)

UTILITY REORG

2/2

	HOME 3	HOME 4	
J1	0	0	deg
J2	0	0	deg
J3	0	0	mm
J4	0	0	deg
J5	0	0	deg



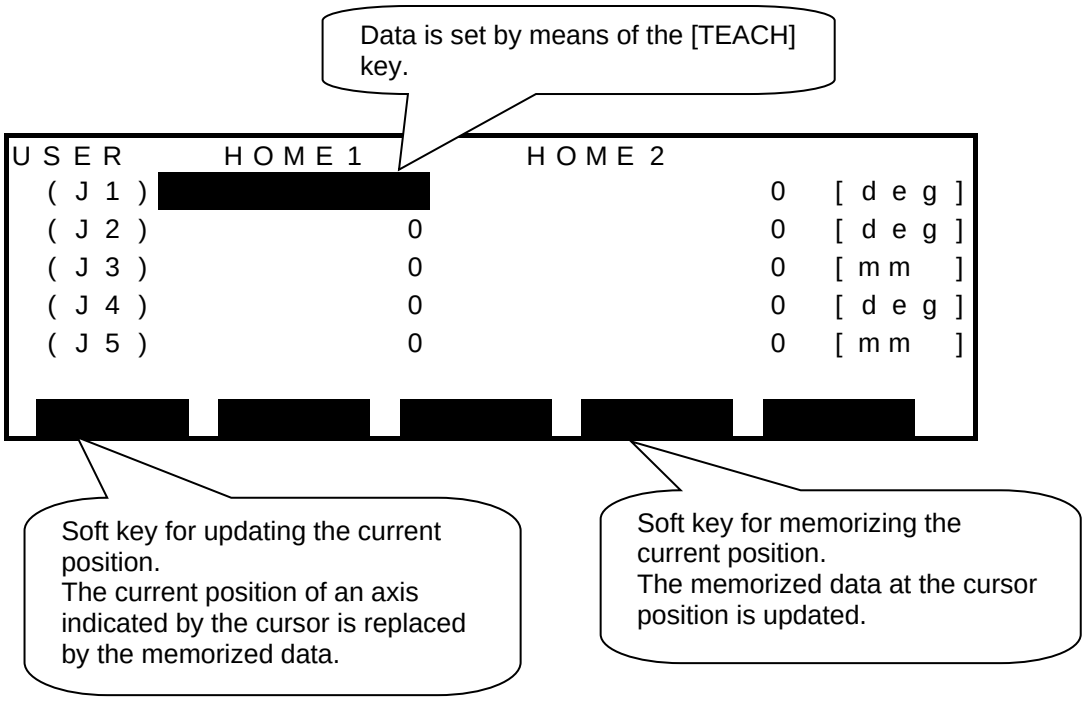
Caution: This function is used to restore the machine home point of each axis. NEVER use the function at other than the setting for restoration. HOME3 and HOME4 are the initial values which are set when shipped form the factory, they cannot be edited in principle. If they must be edited, it is necessary to unlock the password.

8.9.2 Setting HOME1 and HOME2

NOTICE

– Be sure to enter data to HOME1 and HOME2 for the REORG function when installing the robot. Even if data is entered after the zero point has shifted, it cannot be used as data for restoring the zero point.

1) Guide the robot to secure the tool shaft at a desired position.
2) Turn off the servo power.
3) Press the [UTILITY] key equipped on the teach pendant.
4) Press the [NEXT] key until the [REORG] (F1) menu appears, then press the [REORG] key.

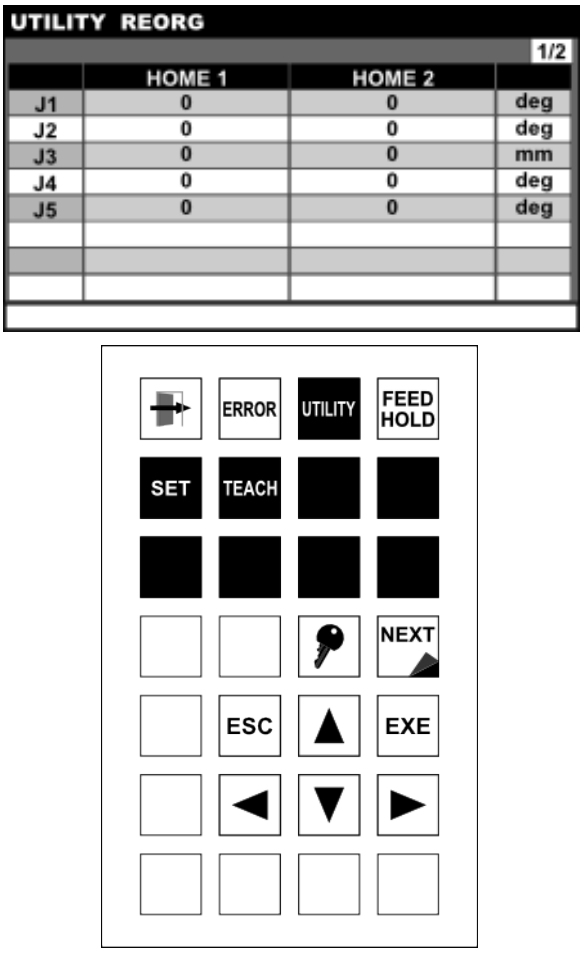


5) When the above screen appears, memorize the position where the robot was secured for each axis by moving the cursor and using the [TEACH] (F4) key mode.

＜Case of TP3000＞ Indicates the different part against the above TP1000 operation.

4) Press the [REORG] key among the utility function selection keys. The home point setting screen and key sheet are called.

5) Memorize the position where the robot was secured for each axis by moving the cursor and using the [TEACH] key mode.



8.9.3 How to Restore Data by HOME1 or HOME2

When mechanical joint positions have deviated due to motor change and so on, and when re-teaching is assumed, execute this operation.

NOTICE

– If the zero point is re-set using the ZEROP function or REORG function, be sure to perform re-teaching of teaching points.
The zero position shifts slightly due to the error in setting. If the zero position is used as is, the robot will collide with the peripheral equipment, etc.

1) Guide the robot and secure it at the position set in HOME1 or HOME2 above.
2) Turn off the servo power.
3) Press the [UTILITY] key equipped on the teach pendant.
4) Press the [NEXT] key until the [REORG] (F1) menu appears, then press the [REORG] key.
5) Move the cursor to each axis under HOME1 of the REORG screen, then press the [SET] (F1) key to update the coordinate value set in HOME1 to the position where the robot was secured.
6) Can used robot immediately after execute to restore data by HOME1 or HOME2. But the format of home position data differ from data are factory-set. In case of you continue to use without change, when battery empty, When you execute to restore home position data by multi-turn data clear, home position data can't restore by right. See "Para. 8.9.5, Change for Home Position Data Form" and execute this operation.
7) Execute "Para. 8.10. affirmation" before moving robot. Move the robot to the teach points, etc. and make sure that the robot can be located at them exactly.

＜Case of TP3000＞ Indicates the different part against the above TP1000 operation.

4) Press the [REORG] key among the utility function selection keys. The home point setting screen and key sheet are called.

5) Move the cursor to each axis under HOME1 of the REORG screen, then press the [SET] key to update the coordinate value set in HOME1 to the position where the robot was secured.

NOTICE

– If re-setting the zero point using the ZEROP function or REORG function, be sure to observe the operation procedure.

The axis 3 operates simultaneously with the operation of the axis 4. Therefore, the zero point is not restored when the order of zero setting is wrong.

Operation procedure

1) Move the robot to the specified position.
2) Move the cursor to E4 on the screen, enter data, and restore the zero point of the axis 4.
3) Move the cursor to E3 on the screen, enter data, and restore the zero point of the axis 3.
   - E1 (axis 1) and E2 (axis 2) have no order of zero re-setting.

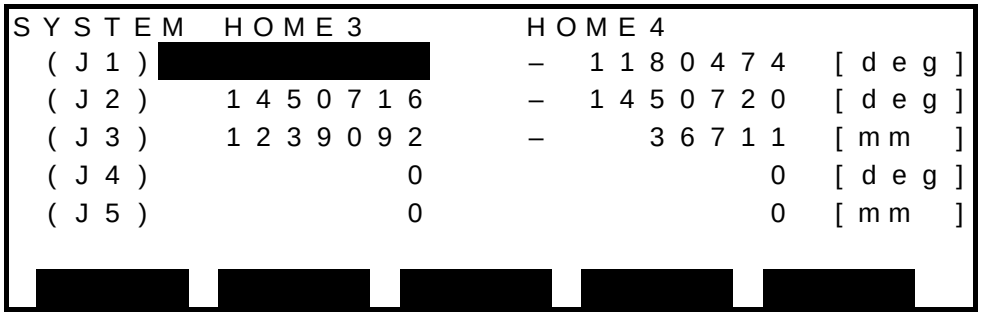
8.9.4 How to Restore Data by HOME3 or HOME4

When mechanical joint positions have deviated due to motor change and so on, and when re-teaching is assumed, execute this operation. The plus (+) side mechanical stopper positions and minus (–) side mechanical stopper positions of Axes 1 to 3 are factory-set in HOME3 and HOME4, respectively. Axis 4 is set to the home point match-mark. (The 4 axis have the potential to differ from existing home position. Because the robot don't have the mechanical stopper of 4 axis.)
The machine home point can be restored, using the above.
This paragraph deals with the restoration procedures, using HOME3.

NOTICE

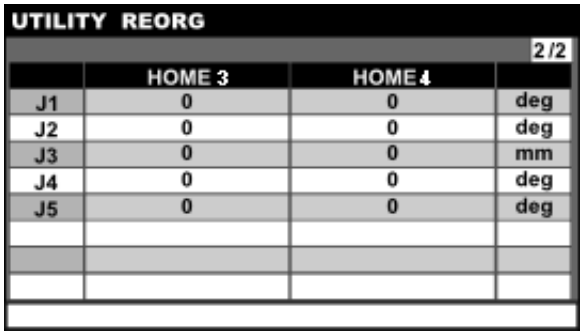
– If the zero point is re-set using the ZEROP function or REORG function, be sure to perform re-teaching of teaching points.
The zero position shifts slightly due to the error in setting. If the zero position is used as is, the robot will collide with the peripheral equipment, etc.

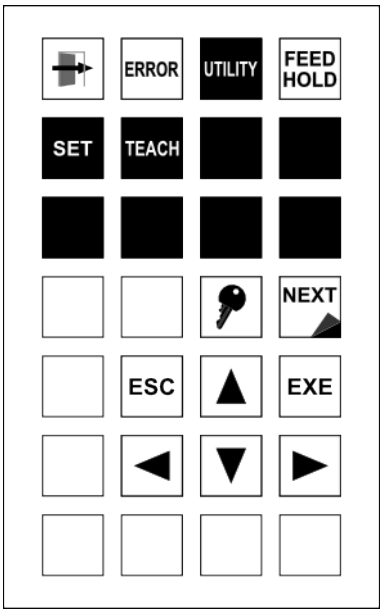
1) Turn off the servo power.
2) Press the [UTILITY] key provided on the teach pendant.
3) Press the [NEXT] key until the [REORG] menu appears, then press the [REORG] (F1) key.
4) Press the [NEXT] key to call Page 2.



＜Case of TP3000＞ Indicates the different part against the above TP1000 operation.

3) Press the [REORG] key among the utility function selection keys. The home point setting screen and key sheet are called.





5) Contact Axis 1 of the robot to the plus (+) side mechanical stopper by hand.

6) Move the cursor to (J1) of HOME3, then press the [SET] (F1) key. Now, the machine home point of Axis 1 is restored.

7) Likewise, contact Axis 2 of the robot to the plus (+) side mechanical stopper by hand.

8) Move the cursor to (J2) of HOME3, then press the [SET] ([F1]) key. Now, the machine home point of Axis 2 is restored.

9) Set Axis 4 to the home point match-mark.

10) Contact Axis 3 of the robot to the plus (+) side mechanical stopper by hand.

11) Move the cursor to (J4) of HOME3, then press the [SET] ([F1]) key. Now, the machine home point of Axis 4 is restored.

12) Move the cursor to (J3) of HOME3, then press the [SET] ([F1]) key. Now, the machine home point of Axis 3 is restored.

13) Press the [ESC] key to escape from the HOME screen.

14) Can used robot immediately after execute to restore data by HOME3 or HOME4. But the format of home position data differ from data are factory-set. In case of you continue to use without change, when battery empty and you execute to restore home position data by multi-turn data clear, home position data can’t restore by right. See “Para. 8.9.5, Change for Home Position Data Form” and execute this operation.

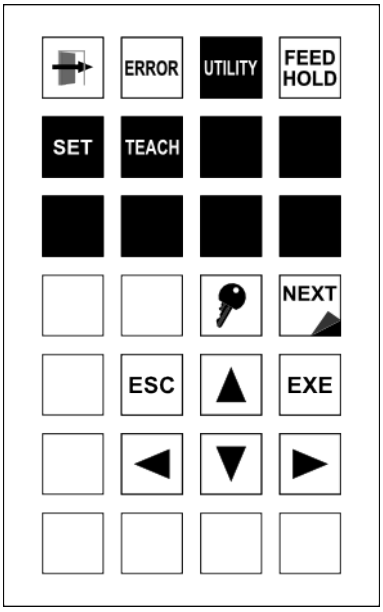
15) Execute “Para. 8.10. Affirmation” before moving robot. Move the robot to the teach points, etc. and make sure that the robot can be located at them exactly.

＜Case of TP3000＞ Indicates the different part against the above TP1000 operation.

If they must be edited to HOME3 and HOME4, it is necessary to unlock the password. Press the Password Unlock key on the key sheet to cancel the editing lock mode. Enter number “1048” on the command and then press the [EXE] key.The other operations the same as that of TP1000.



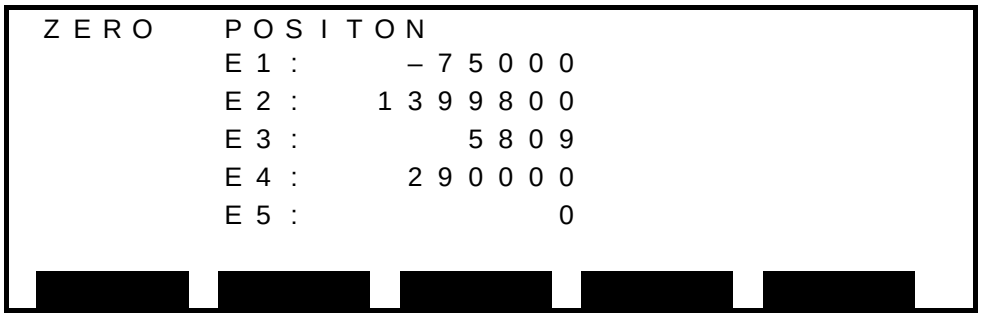
Password Unlock key



8.9.5 Change for Home Position Data Form

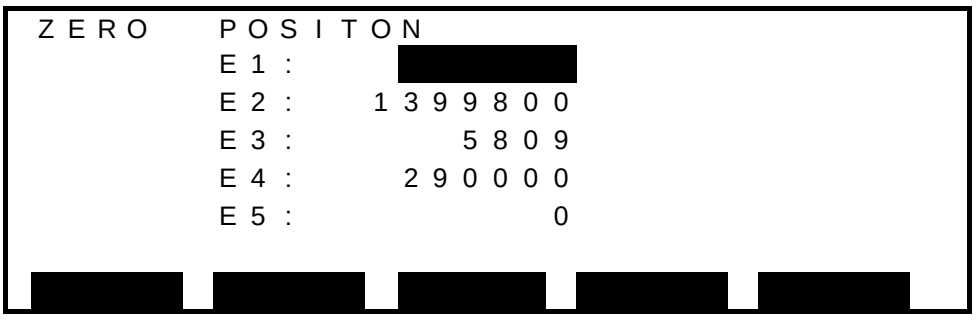
The format of servo offset and multi-turn data differ from format are factory-set after HOME operation. In case of you continue to use without change, when battery empty and you execute to restore home position data by multi-turn data clear, home position data can't restore by right. Restore format of home position data by this operation after HOME operation.

1) Select the TEACHING mode by means of the master mode switch.
2) Each axis of robot move zero position by DO function. Be careful setting low speed of OVRD. About DO function, See language manual.
3) Press the [UTILITY] key equipped on the teach pendant from initial screen.
4) Press the [NEXT] (F6) key on the teach pendant twice.
5) Turn off the servo power.
6) Press the [F3] key on the teach pendant to select the ZEROP mode.
7) The following home point setting screen is called..

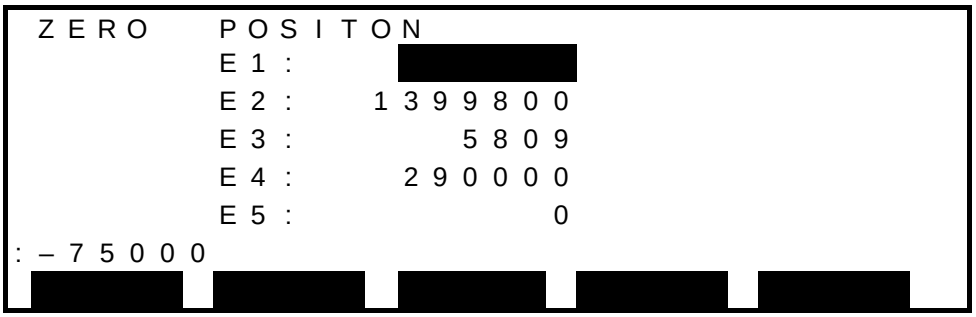


8) Press the [ALT] and [0] keys on the teach pendant at the same time.
9) The following home point editing screen appears.

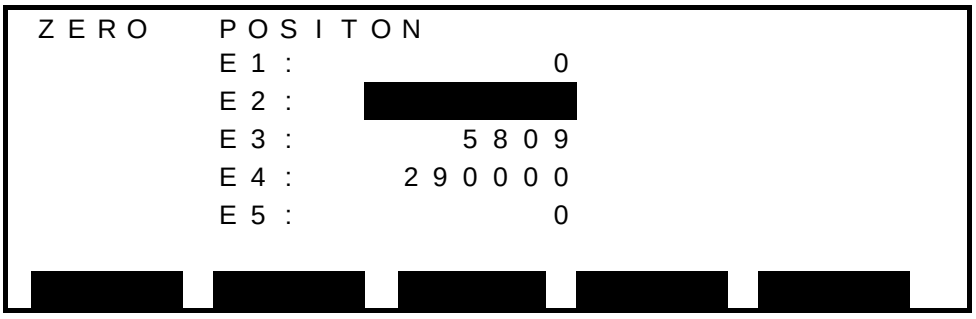
10) The numerical value of E1 (axis 1) is inversely displayed.



11) Press the [EXE] key on the teach pendant, and the value of “E1” is displayed at the lower left side. Set “0” in lieu of this value and press the [EXE] key. Then the E1 value becomes “0”.



12) When the [↓] key on the teach pendant is pressed, the following E2 value is displayed by reversed image. Perform setting in the same manner as above.



NOTICE

– If re-setting the zero point using the ZEROP function or REORG function, be sure to observe the operation procedure.

The axis 3 operates simultaneously with the operation of the axis 4. Therefore, the zero point is not restored when the order of zero setting is wrong.

Operation procedure

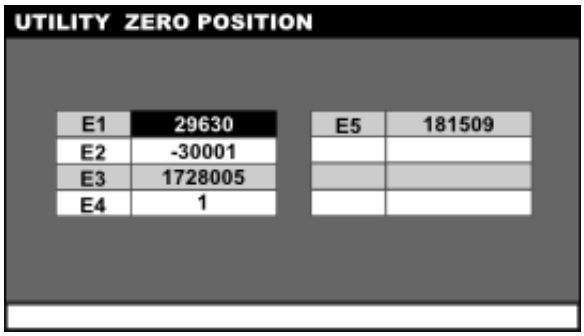
1) Move the robot to the specified position.
2) Move the cursor to E4 on the screen, enter data, and restore the zero point of the axis 4.
3) Move the cursor to E3 on the screen, enter data, and restore the zero point of the axis 3.
   - E1 (axis 1) and E2 (axis 2) have no order of zero re-setting.

13) Turn off the controller power, then turn it on again.

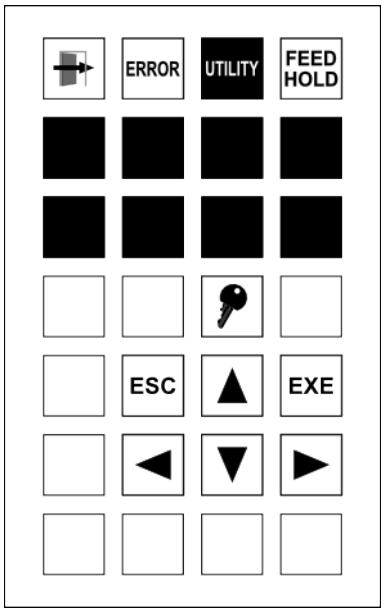
14) Execute “Para. 8.10. Affirmation” before moving robot. Move the robot to the teach points, etc. and make sure that the robot can be located at them exactly.

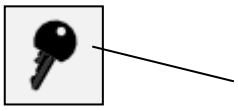
＜Case of TP3000＞

1) Select the TEACHING mode by means of the master mode switch.
2) Each axis of robot move zero position by DO function. Be careful setting low speed of OVRD. About DO function, See language manual.
3) Press the [UTILITY] key equipped on the teach pendant from initial screen.
4) The display of the key sheet changes to the utility function selection keys.
5) Press the [ZEROP] key among the utility function selection keys.
6) The following home point setting screen and key sheet are called.



Command line





Password Unlock key

7) Press the Password Unlock key on the key sheet to cancel the editing lock mode. Enter number “1048” on the command and then press the [EXE] key.
8) The home point editing screen appears.
9) The numerical value of E1 (axis 1) is inversely displayed.
10) When the “ ↓ ” key is pressed, the E4 value is displayed by reversed image.

11) Press the “EXE” key on the teach pendant, and the value of “E4” is displayed at the command line. Set “0” in lieu of this value and press the “EXE” key. Then the E4 value becomes “0”.

12) When the “↑” key on the teach pendant is pressed, the following E3 value is displayed by reversed image. Perform setting in the same manner as above.

NOTICE

– If re-setting the zero point using the ZEROP function or REORG function, be sure to observe the operation procedure.

The axis 3 operates simultaneously with the operation of the axis 4. Therefore, the zero point is not restored when the order of zero setting is wrong.

Operation procedure

1) Move the robot to the specified position.
2) Move the cursor to E4 on the screen, enter data, and restore the zero point of the axis 4.
3) Move the cursor to E3 on the screen, enter data, and restore the zero point of the axis 3.
   - E1 (axis 1) and E2 (axis 2) have no order of zero re-setting.

13) Turn off the controller power, then turn it on again.
14) Execute “Para. 8.10. affirmation” before moving robot. Move the robot to the teach points, etc. and make sure that the robot can be located at them exactly.

8.10 Affirmation

Effect an emergency stop, manually move each axis to the motion limit and make sure that a soft limit error generates. Make sure of the robot current position according to the POS data of the UTILITY mode. Make sure of teaching point, too.

NOTICE

– After the zero point has been re-set, be sure to perform check work.
Unless the zero point is restored correctly, the robot will collide with the mechanical stopper or peripheral equipment when it operates.

8.11 Backup of Data

Connect the personal computer and controller using TSPC (option), then save the “MACHINE.PAR” file from the robot controller to the hard disk, etc. of the personal computer. Load the home position data to the system parameter file.
The home position data is the data characteristic of the robot, which differs with the robot you use. This data is necessary when you effect a cold start and restore the home position data again. Be sure to save the data.

Now, the operation is complete.

9. Replacement Parts for Maintenance

9.1 Cautions for maintenance replacement parts

⚠ DANGER

– Always replace with SHIBAURA MACHINE authorized maintenance replacement parts.
Use of unauthorized parts could result in a fire or other unexpected accident.
- All parts except for the battery for the position detector are special specifications parts. Please request these parts from SHIBAURA MACHINE when purchasing and ordering.

– Except for replacement of batteries and fuses, please request the replacement work from the SHIBAURA MACHINE servicing company. A fee will be charged for this work.
SHIBAURA MACHINE will not be liable for any accidents or damage that occurs due to replacement work performed by the customer or its results.

NOTICE

– Always replace with SHIBAURA MACHINE authorized maintenance replacement parts.
Use of unauthorized parts could result in reduced robot performance and lifespan. A breakdown could also occur.

9.2 List of Replacement Parts – Main Robot (THL300, THL400)

	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
1	AC servo motor		S890967	Y610A3NL0	SHIBAURA MACHINE	1	Axis 1
2			S890968	Y610A3NM0		2	Axis 2, 4
3			S777296	Y610A37A0		1	Axis 3
4	Reduction gear		S890907	Y610A3NG0	SHIBAURA MACHINE	1	Axis 1 (A type)
5			S890906	Y610A3NH0		1	Axis 2 (B type)
6			S958148	Y610D06N0		1	Axis 1 (A type)
7			S958147	Y610D06M0		1	Axis 2 (B type)
8			S890969	Y610A3NJ0		1	Axis 4
9	Timing pulley		S777399 (ball screw side)	Y610A3NK0	SHIBAURA MACHINE	1	Axis 3
10			S777400 (motor side)			1	Axis 3
11			S777401 (ball screw side)			1	Axis 4
12			S777402 (motor side)			1	Axis 4
13	Timing belt		S890882			1	Axis 3
14			S890883			1	Axis 4
15	Ball screw spline unit		H852810	Y610A3NE0	SHIBAURA MACHINE	1	Axis 3
16	Main body harness For THL300		F113098	Y610A3Q60	SHIBAURA MACHINE	1	
17	Main body harness For THL400		F113089	Y610A3Q70	SHIBAURA MACHINE	1	
18	Grease	SK-2 (For reduction gear)			HDS		Axis 1, 2 (A type)
19		SFB No.1 (For reduction gear)			Nidec-Shimpo Corp.		Axis 1, 2 (A type)
20		AFF grease (Lithium-based for ball screw)			THK		Axis 3
21	Air tube for hand I/O	TUZ0425R-20			SMC	*1	Color (Red)
22		TUZ0425BU-20			SMC		Color (Blue)
23		TUZ0425W-20			SMC		Color (White)
24	Encoder backup battery	Size AA alkali battery				3	All axes

*1 The minimum purchase unit of air tube is 20 m

9.3 List of Replacement Parts – Main Robot (THL500, THL600, THL700)

	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
1	AC servo motor		S875289	Y610A3M20	SHIBAURA MACHINE	1	Axis 1
2			S875290	Y610A3M30		1	Axis 2
3			S746337	Y610A3430		1	Axis 3
4			S875291	Y610A3M40		1	Axis 4
5	Reduction gear		S875237	Y610A3LW0	SHIBAURA MACHINE	1	Axis 1 (A type)
6			S875238	Y610A3LX0		1	Axis 2 (B type)
7			S994595	Y610D0FW0		1	Axis 1 (A type)
8			S994596	Y610D0FX0		1	Axis 2 (B type)
9			S875239	Y610A3LY0		1	Axis 4
10	Timing pulley		S875171 (motor side)	Y610A3LZ0	SHIBAURA MACHINE	1	Axis 3
11			S875170 (ball screw side)			1	Axis 3
12			S875173 (motor side)			1	Axis 4
13			S875172 (ball screw side)			1	Axis 4
14	Timing belt		S875174			1	Axis 3
15			S875175			1	Axis 4
16	Ball screw spline unit		H852666(Z150)	Y610A3LT0	SHIBAURA MACHINE	1	Axis 3
17			H852702(Z300)	Y610A3ME0		1	
18	Main body harness For THL500		F112965	Y610A3LP0	SHIBAURA MACHINE	1	
19	Main body harness For THL600		F112954	Y610A3LQ0	SHIBAURA MACHINE	1	
20	Main body harness For THL700		F112966	Y610A3LR0	SHIBAURA MACHINE	1	
21	Grease	SK-1A (For reduction gear)			HDS		Axis 1, 2 (A type)
22		SFB No.1			Nidec-Shimpo Corp.		Axis 1, 2 (B type)
23		AFF grease (Lithium-based for ball screw)			THK		Axis 3
24	Air tube for hand I/O	TUZ0604R-20			SMC	*1	Color (Red)
25		TUZ0604BU-20			SMC		Color (Blue)
26		TUZ0604W-20			SMC		Color (White)
27	Encoder backup battery	Size AA alkali battery				3	All axes

*1 The minimum purchase unit of air tube is 20 m.

9.4 List of Replacement Parts – Main Robot (THL800, THL900, THL1000,THL1200)

	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
1	AC servo motor		S746328	Y610A34C0	SHIBAURA MACHINE	1	Axis 1
2			S875290	Y610A3M30		1	Axis 2
3			S746337	Y610A3430		1	Axis 3
4			S875291	Y610A3M40		1	Axis 4
5	Reduction gear		S891110	Y610A3PS0	SHIBAURA MACHINE	1	Axis 1
6			S891111	Y610A3PT0		1	Axis 2
7			S875239	Y610A3LY0		1	Axis 4
8	Timing pulley		S875171 (motor side)	Y610A3PV0	SHIBAURA MACHINE	1	Axis 3
9			S875170 (ball screw side)			1	Axis 3
10			S875173 (motor side)			1	Axis 4
11			S875172 (ball screw side)			1	Axis 4
12	Timing belt		S891108			1	Axis 3
13			S891109			1	Axis 4
14	Ball screw spline unit		H852702	Y610A3ME0	SHIBAURA MACHINE	1	Axis 3
15	Main body harness For THL800		F113044	Y610A3PM0	SHIBAURA MACHINE	1	
16	Main body harness For THL900		F113045	Y610A3PN0	SHIBAURA MACHINE	1	
17	Main body harness For THL1000		F113046	Y610A3PP0	SHIBAURA MACHINE	1	
18	Main body harness For THL1200		F113318	Y610D0150	SHIBAURA MACHINE	1	
19	Grease	SK-1A (For reduction gear)			HDS		Axis 1, 2
20		AFF grease (Lithium-based for ball screw)			THK		Axis 3
21	Air tube for hand I/O	TUZ0604R-20			SMC	*1 *2	Color (Red)
22		TUZ0604BU-20			SMC		Color (Blue)
23		TUZ0604W-20			SMC		Color (White)
24	Encoder backup battery	Size AA alkali battery				3	All axes

*1 The minimum purchase unit of air tube is 20 m.

*2. In the case of THL 1200, the air cables are wired in a flexible duct

9.5 List of Replacement Parts – Controller (THL300, THL400/TSL3000)

No.	Part name	Type	Unit code	Maker	Q'ty	Remarks
1	PS1 (Switching power supply)	LFA30F-5-J1Y		COSEL	1	P5V power supply
2	PS2 (Switching power supply)	LFA30F-24-J1			1	P24V power supply
3	Lithium battery	ER6V C4 (ER6C WK27 was obsolete)		TOSHIBA	1	For X8YC printed board
4	Fuse	51NM030H		PICO	1	For X8YX printed board
5	Printed board	X8YCB (Main control)	Y610A90B0	SHIBAURA MACHINE	1	
6		X8YSC (Servo)	Y610A9050		1	
7		X8YXA (I/O board)	Y610A9020		1	Type-N
8		X8YXB (I/O board)	Y610A9030		1	Type-P
9	Transistor output IC	TD62084APG		Toshiba	4	Type-N
10		TD62783APG			4	Type-P
11	Teach pendant	TP1000	Y610A2600	SHIBAURA MACHINE	1	With 5 m-long cable
12		TP3000	Y610A43A0		1	With 5 m-long cable High-performance TP (Teaching Pendant)
13	System disk	TS3000SYS	Y610A3HC0	SHIBAURA MACHINE	1	CD-ROM

9.6 List of Replacement Parts – Controller (THL500～THL1200/TSL3000)

No.	Part name	Type	Unit code	Maker	Q'ty	Remarks
1	PS1 (Switching power supply)	LFA30F-5-J1Y		COSEL	1	P5V power supply
2	PS2 (Switching power supply)	LFA30F-24-J1			1	P24V power supply
3	Lithium battery	ER6V C4 (ER6C WK27 was obsolete)		TOSHIBA	1	For X8YC printed board
4	Fuse	51NM030H		PICO	1	For X8YX printed board
5	Printed board	X8YCB (Main control)	Y610A90B0	SHIBAURA MACHINE	1	
6		X8YSB (Servo)	Y610A9040		1	
7		X8YXA (I/O board)	Y610A9020		1	Type-N
8		X8YXB (I/O board)	Y610A9030		1	Type-P
9	Transistor output IC	TD62084APG		Toshiba	4	Type-N
10		TD62783APG			4	Type-P
11	Teach pendant	TP1000	Y610A2600	SHIBAURA MACHINE	1	With 5 m-long cable
12		TP3000	Y610A43A0		1	With 5 m-long cable High-performance TP (Teaching Pendant)
13	System disk	TS3000SYS	Y610A3HC0	SHIBAURA MACHINE	1	CD-ROM

9.7 List of Replacement Parts – Controller (THL300, THL400/TSL3000E)

No.	Part name	Type	Unit code	Maker	Q'ty	Remarks
1	PS1 (Switching power supply)	LEB100F-0524		COSEL	1	P5V, P24V power supply
2	Lithium battery	ER6V C4 (ER6C WK27 was obsolete)		TOSHIBA	1	For X8YV printed board
3	Main power switch	IR-11-A8-25-1BF		Nippon Thermo	1	
4	Fuse	51NM030H		PICO	1	For X8YJ printed board
5	Printed board	X8YVA (Main control)	Y610A90E0	SHIBAURA MACHINE	1	
6		X8YHC (Servo)	Y610A90G0		1	1 axis to 4 axis
7		X8YJA (I/O board)	Y610A90J0		1	
8		X8YZA (Safety board)	Y610A90Z0		1	
9	Teach pendant	TP1000	Y610A2600	SHIBAURA MACHINE	1	With 5 m-long cable
10		TP1000 remodeling CE	Y610A2250		1	
11		TP3000	Y610A43A0		1	With 5 m-long cable High-performance TP (Teaching Pendant)
12	System disk	TS3000SYS	Y610A3HC0	SHIBAURA MACHINE	1	CD-ROM

9.8 List of Replacement Parts – Controller (THL500～THL1200/TSL3000E)

No.	Part name	Type	Unit code	Maker	Q'ty	Remarks
1	PS1 (Switching power supply)	LEB100F-0524		COSEL	1	P5V, P24V power supply
2	Lithium battery	ER6V C4 (ER6C WK27 was obsolete)		TOSHIBA	1	For X8YV printed board
3	Main power switch	IR-11-A8-25-1BF		Nippon Thermo	1	
4	Fuse	51NM030H		PICO	1	For X8YJ printed board
5	Printed board	X8YVA (Main control)	Y610A90E0	SHIBAURA MACHINE	1	
6		X8YHB (Servo)	Y610A90F0		1	1 axis to 4 axis
7		X8YJA (I/O board)	Y610A90J0		1	
8		X8YZA (Safety board)	Y610A90Z0		1	
9	Teach pendant	TP1000	Y610A2600	SHIBAURA MACHINE	1	With 5 m-long cable
10		TP1000 remodeling CE	Y610A2250		1	
11		TP3000	Y610A43A0		1	With 5 m-long cable High-performance TP (Teaching Pendant)
12	System disk	TS3000SYS	Y610A3HC0	SHIBAURA MACHINE	1	CD-ROM

9.9 List of Replacement Parts – Controller (THL300, THL400/TS3000, TS3000E)

No.	Part name	Type	Unit code	Maker	Q'ty	Remarks
1	PS1 (Switching power supply)	PBA75F-24		COSEL	1	
2	PS2 (Switching power supply)	LDA30F-5-Y			1	
3	Main power switch	IR-11-A8-10-1BF		Nippon Thermo	1	
4	Lithium battery	ER6V C4 (ER6C WK27 was obsolete)		TOSHIBA	1	
5	Fuse	51NM030H		Hitachi Maxell	1	For X8GC printed boards
6	Printed board	X8GCA (Main control)	Y610A4000	SHIBAURA MACHINE	1	Type N
7		X8GCC(Main control)	Y610A4010		1	Type N, CE/Category3
8		X8GCA (Main control)	Y610A4020		1	Type P
9		X8GCC (Main control)	Y610A4030		1	Type P, CE/Category3
10		X8GLA (Servo logic)	Y610A4040		1	
11		X8GNA (I/O board)	Y610A40A0		1	Type N
12		X8GIA (I/O board)	Y610A40B0		1	Type P
13		X8G2A (10A) (Servo)	Y610A40H0		1	Axis 1
14		X8G3A(5A) (Servo)	Y610A40K0		3	Axis 2, 3, 4
15		X8GHA (Servo power supply)	Y610A40S0		1	
16		X8GBA (Mother board)	Y610A40R0		1	
17	Operation panel unit	X8GOA	Y610A4080	SHIBAURA MACHINE	1	Normal
18		X8GOD	Y610A40U0		1	CE/Category3
19	Teach pendant	TP1000	Y610A2600	SHIBAURA MACHINE	1	With 5 m-long cable
20		TP1000 remodeling CE	Y610A2250		1	
21		TP3000	Y610A43A0		1	With 5 m-long cable High-performance TP (Teaching Pendant)
22	System disk	TS3000SYS	Y610A3HC0	SHIBAURA MACHINE	1	CD-ROM
23	Replacement of Output ICs	TD62084APG		Toshiba	1	Type N
24		M54562WP		Mitsubishi Electric	1	Type P
25	Vibration suppression sheet	S849804		SHIBAURA MACHINE	1	For large capacity capacitor vibration suppression

9.10 List of Replacement Parts – Controller (THL500～THL1200/TS3000, TS3000E)

No.	Part name	Type	Unit code	Maker	Q'ty	Remarks
1	PS1 (Switching power supply)	PBA75F-24		COSEL	1	
2	PS2 (Switching power supply)	LDA30F-5-Y			1	
3	Main power switch	IR-11-A8-10-1BF		Nippon Thermo	1	
4	Lithium battery	ER6V C4 (ER6C WK27 was obsolete)		TOSHIBA	1	
5	Fuse	51NM030H		Hitachi Maxell	1	For X8GC printed boards
6	Printed board	X8GCA (Main control)	Y610A4000	SHIBAURA MACHINE	1	Type N
7		X8GCC(Main control)	Y610A4010		1	Type N, CE/Category3
8		X8GCA (Main control)	Y610A4020		1	Type P
9		X8GCC (Main control)	Y610A4030		1	Type P, CE/Category3
10		X8GLA (Servo logic)	Y610A4040		1	
11		X8GNA (I/O board)	Y610A40A0		1	Type N
12		X8GIA (I/O board)	Y610A40B0		1	Type P
13		X8G2A (20A) (Servo)	Y610A40F0		1	Axis 1
14		X8G3A(10A) (Servo)	Y610A40H0		3	Axis 2, 3, 4
15		X8GHA (Servo power supply)	Y610A40S0		1	
16		X8GBA (Mother board)	Y610A40R0		1	
17	Operation panel unit	X8GOA	Y610A4080	SHIBAURA MACHINE	1	Normal
18		X8GOD	Y610A40U0		1	CE/Category3
19	Teach pendant	TP1000	Y610A2600	SHIBAURA MACHINE	1	With 5 m-long cable
20		TP1000 remodeling CE	Y610A2250		1	
21		TP3000	Y610A43A0		1	With 5 m-long cable High-performance TP (Teaching Pendant)
22	System disk	TS3000SYS	Y610A3HC0	SHIBAURA MACHINE	1	CD-ROM
23	Replacement of Output ICs	TD62084APG		Toshiba	1	Type N
24		M54562WP		Mitsubishi Electric	1	Type P
25	Vibration suppression sheet	S849804		SHIBAURA MACHINE	1	For large capacity capacitor vibration suppression